Supplemental Material to "A dictionary between finite spectral triples and Kaluza-Klein data"

Draft, 3 November 2026

This is the supplement of the paper *A dictionary between finite spectral triples and Kaluza-Klein data*. Its parts keep the numbers of the full Supplemental Material, so that a citation of a part in the paper refers to the part of the same name here.

Parts contained. Lemma parts: 05, 16, 22, 23, 24, 25, 28, 29, 30. Eleven-dimensional files: README, 01, 02, 07, 08, 09, 10, 11. Other parts: R3, R8, R9, R10, R11, R15, R16, R17, R18, R19, R21 (the letters H, L and R mark the history of the field, the literature check and the referee reports).

The full Supplemental Material, with all parts of the project, exists as `supplemental.pdf` and `supplemental_mobile.pdf` in the same folder as this file. A reference inside a part to a part not listed here points to the full version.

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Part 05

The geometric identification exists: the minimal grading-preserving spectral truncation of a suitable chiral interval is exactly Connes–Lott, with

$$|m| = \frac{\pi}{2l}, \qquad d_{\mathrm{trunc}} = \frac{2l}{\pi}.$$

The truncated endpoint states are smeared states on the original interval. Moreover, this interval has no zero mode, which limits the physical interpretation of both extremes.

Part 1. Deconstructed chain

Statement. There is a natural family with opposite endpoint chiralities, an exact Connes–Lott member at $N = 2$, and endpoint distance converging to l. The answer depends on the representation of the site algebra; the familiar lattice-distance anomaly is not universal.

Hypotheses. We consider complex even spectral triples, without imposing additional real-structure or KO-dimension axioms. Let $N \geq 2$, $n = N - 1$, and

$$A_N = \mathbb{C}^N, \qquad H_N = \mathrm{span}\{L_1, \dots, L_{N-1}, R_2, \dots, R_N\}.$$

Thus interior sites carry both components, site 1 only L, and site N only R. Define

$$\pi_N(a) = \mathrm{diag}(a_1, \dots, a_{N-1}, a_2, \dots, a_N),$$
$$\Gamma_N = \mathrm{diag}(I_n, -I_n).$$

If S_n is the lower shift, set

$$B_N = h_N(I_n - S_n), \qquad D_N = \begin{pmatrix} 0 & B_N \\ B_N^* & 0 \end{pmatrix}.$$

These are forward/backward differences:

$$(D_N\psi)_{L_i} = h_N(R_{i+1} - R_i), \qquad (D_N\psi)_{R_j} = h_N(L_{j-1} - L_j),$$

with the missing components interpreted as

$$R_1 = 0, \qquad L_N = 0.$$

There are nearest-neighbour couplings and onsite L-R terms at interior sites.

For compatibility with the spectral truncation in Part 3, choose

$$\theta_N = \frac{\pi}{4N-2}, \qquad h_N = \frac{\pi}{4l\sin\theta_N} \sim \frac{N}{l}. \tag{1}$$

This fixes the smallest positive eigenvalue to $\pi/(2l)$ at every N.

Proof.

The representation is faithful, $[\Gamma_N, \pi_N(a)] = 0$, and $\Gamma_N D_N = -D_N\Gamma_N$. At $N = 2$,

$$A_2 = \mathbb{C}^2, \quad H_2 = \mathbb{C}^2, \quad D_2 = \begin{pmatrix} 0 & h_2 \\ h_2 & 0 \end{pmatrix}, \quad h_2 = \frac{\pi}{2l}.$$

This is exactly the stated two-point triple; a diagonal unitary supplies any desired phase of m.

Ordering the basis as

$$L_1, R_2, L_2, R_3, \dots, L_{N-1}, R_N$$

makes D_N a path adjacency matrix with alternating signs. A diagonal unitary removes those signs. The usual sine eigenvectors of a path therefore give

$$\operatorname{spec}(D_N) = \left\{ 2h_N \cos\frac{j\pi}{2N-1} : 1 \le j \le 2N-2 \right\} = \{\pm\lambda_{N,r} : 1 \le r \le N-1\},$$

where

$$\boxed{\lambda_{N,r} = 2h_N \sin((2r-1)\theta_N).} \tag{2}$$

Consequently,

$$\boxed{K_N(t) = \operatorname{Tr} e^{-tD_N^2} = 2\sum_{r=1}^{N-1} e^{-4th_N^2 \sin^2((2r-1)\theta_N)}.} \tag{3}$$

For a real site function a, the onsite terms commute with $\pi_N(a)$, while

$$B_N\pi_R(a) - \pi_L(a)B_N = h_N \,\mathrm{diag}(a_2 - a_1, \dots, a_N - a_{N-1}).$$

Hence

$$\|[D_N, \pi_N(a)]\| = h_N \max_i |a_{i+1} - a_i|.$$

The triangle inequality bounds $|a_j - a_i|$ by $|j-i|/h_N$; the function $a_i = (i-1)/h_N$ attains this bound. Thus

$$\boxed{d_N(i,j) = \frac{|i-j|}{h_N}.} \tag{4}$$

In particular,

$$\boxed{d_N(1,N) = \frac{4l(N-1)}{\pi} \sin\frac{\pi}{4N-2} \longrightarrow l.} \tag{5}$$

One can instead choose the geometric normalization

$$h_N = \frac{N-1}{l}. \tag{6}$$

Then $d_N(1,N) = l$ exactly for every N, with the same continuum limit. However, its $N = 2$ mass is $1/l$, rather than the continuum lowest eigenvalue $\pi/(2l)$. Equations (1) and (6) distinguish spectral calibration from metric calibration at finite resolution.

Two other natural choices clarify the literature:

- **One-component staggered path.** Take $A = \mathbb{C}^M$, $H = \mathbb{C}^M$, alternating grading, and $D = h$ times path adjacency. Opposite endpoint chiralities require even M. Its spectrum is $2h\cos(j\pi/(M+1))$. For $k = |i-j|$,

$$\boxed{d(i,j) = \begin{cases} (k+1)/(2h), & k \text{ odd}, \\ \sqrt{k(k+2)}/(2h), & k \text{ even}. \end{cases}} \tag{7}$$

  In particular $d(1,M) = M/(2h)$. Thus $h = M/(2l)$ gives endpoint distance l, whereas $h = M/l$ gives $l/2$.

Here is a direct proof of (7). Restrict the commutator to the connecting subpath and extend the function constantly outside it. After removing phases, its norm is that of weighted adjacency, with dimensionless edge weights $w_i = h|\Delta a_i|$. If its two bipartition sizes are p, q, the off-diagonal block C satisfies

$$\sum_i w_i = \mathbf{1}_p^T C \mathbf{1}_q \le \sqrt{pq}\,\|C\|.$$

For $p = q$, alternating weights $1, 0, 1, \dots, 1$ attain the bound. For $p = q + 1$, the weights

$$\frac{(q, 1, q-1, 2, \dots, 1, q)}{\sqrt{q(q+1)}}$$

attain it: the normalized constant vectors on the two bipartitions are singular vectors with singular value 1, and positivity gives the norm 1. This proves (7).

These are the nonadditive distances studied by Bimonte–Lizzi–Sparano, §5 and given explicitly by Atzmon, Eq. (30). Reduction to the connecting path also agrees with Iochum–Krajewski–Martinetti, Proposition 4.

- **Vertex-edge incidence Dirac.** Let $\partial : \mathbb{C}^N \to \mathbb{C}^{N-1}$ be the path difference matrix and use

$$D_{\mathrm{inc}} = h \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}.$$

Its spectrum and heat trace are

$$\{0\} \cup \left\{ \pm 2h \sin \frac{k\pi}{2N} : 1 \le k < N \right\},$$

$$K_{\mathrm{inc}}(t) = 1 + 2 \sum_{k=1}^{N-1} e^{-4th^2 \sin^2(k\pi/(2N))}.$$

These follow by substituting the cosine eigenvectors into the path Laplacian $\partial^*\partial$. Representing a site function on each edge by its value at the tail gives distance $|i-j|/h$. But H has dimension $2N-1$, has a zero mode, and both endpoint vertex components have the same grading. At $N = 2$, its spectrum is $0, \pm\sqrt{2}h$, so it is not Connes–Lott.

Thus the quoted integer deconstruction tower belongs to the incidence construction. Opposite chiral boundary conditions produce the shifted tower (2).

Status: PROVED

Part 2. Continuum limit

Statement. The chain above converges to an interval Dirac operator in three precise senses: ordered low eigenvalues, heat traces locally uniformly for $t > 0$, and pure-state metric spaces in Gromov–Hausdorff distance. It also has a controlled one-dimensional heat window.

Hypotheses. On $H_l = L^2([0,l],\mathbb{C}^2)$, define

$$D_l = \begin{pmatrix} 0 & \partial_x \\ -\partial_x & 0 \end{pmatrix}, \qquad \operatorname{Dom} D_l = \{(L,R) \in H^1 : R(0) = 0,\ L(l) = 0\}, \tag{8}$$

with scalar multiplication by $C^\infty([0,l])$ and grading $\Gamma = \operatorname{diag}(1,-1)$.

These are opposite local chiral conditions. Reflection parity is Γ at 0 and $-\Gamma$ at l; their composition gives antiperiodicity on the doubled interval.

Proof.

Integration by parts gives the boundary form

$$[\overline{L}\,\widetilde{R} - \overline{R}\,\widetilde{L}]_0^l.$$

The conditions in (8) form a maximal isotropic boundary subspace, so D_l is self-adjoint. Its resolvent is compact by the compact embedding $H^1 \hookrightarrow L^2$.

Solving $D_l\psi = \lambda\psi$ gives

$$\psi_\lambda(x) = c(\cos\lambda x, \sin\lambda x), \qquad \cos(\lambda l) = 0.$$

Therefore

$$\boxed{\operatorname{spec}(D_l) = \left\{ \pm\frac{(r-\frac{1}{2})\pi}{l} : r = 1,2,... \right\}.} \tag{9}$$

The kernel is zero. Squaring gives mixed Neumann–Dirichlet conditions on L, and Dirichlet–Neumann conditions on R.

For scalar f,

$$\|[D_l, f]\| = \|f'\|_\infty, \qquad d_l(x,y) = |x-y|. \tag{10}$$

Under normalization (1), with $m_1 = \pi/(2l)$,

$$\lambda_{N,r} = m_1 \frac{\sin((2r-1)\theta_N)}{\sin\theta_N} \longrightarrow (2r-1)m_1. \tag{11}$$

This proves low-eigenvalue convergence. Since

$$\sin y \geq \frac{2y}{\pi} \quad (0 \leq y \leq \pi/2),$$

we also have

$$\lambda_{N,r} \geq \frac{2r-1}{l}. \tag{12}$$

A summable Gaussian majorant now proves, uniformly on every $t \in [\varepsilon, T]$,

$$K_N(t) \longrightarrow K_l(t) = 2\sum_{r=1}^{\infty} e^{-t((r-\frac{1}{2})\pi/l)^2}. \tag{13}$$

Poisson summation gives

$$\boxed{K_l(t) = \frac{l}{\sqrt{\pi t}}\left(1 + 2\sum_{q=1}^{\infty}(-1)^q e^{-q^2 l^2/t}\right).} \tag{14}$$

In particular, $K_l(t) \sim l/\sqrt{\pi t}$ as $t/l^2 \to 0$.

The finite-chain heat window has the following rigorous meaning. For every sequence $t_N > 0$ satisfying

$$\frac{t_N}{l^2} \to 0, \qquad \frac{N^2 t_N}{l^2} \to \infty, \tag{15}$$

one has

$$\boxed{K_N(t_N) \sim \frac{l}{\sqrt{\pi t_N}}, \qquad -2t_N \frac{K_N'(t_N)}{K_N(t_N)} \to 1.} \tag{16}$$

To prove this, put $u_r = \pi\sqrt{t_N}(r-\frac{1}{2})/l$. On bounded u_r-intervals, the sine argument in (11) tends uniformly to zero, so $\sqrt{t_N}\lambda_{N,r}/u_r \to 1$. The mesh tends to zero and its upper endpoint tends to infinity. Bound (12) controls the tails. Gaussian Riemann sums, and then the same sums weighted by $t_N\lambda_{N,r}^2$, prove both assertions.

Finally, identify site i with $x_i = (i-1)l/(N-1)$. Equation (4) gives

$$d_N(i,j) = \rho_N |x_i - x_j|, \qquad \rho_N = \frac{N-1}{l h_N} = 1 + O(N^{-1}).$$

The metric distortion and the grid mesh are both $O(l/N)$, proving Gromov–Hausdorff convergence to $[0, l]$.

The endpoint chirality here concerns allowed boundary components. It does not supply localized chiral zero modes.

Status: PROVED

Part 3. Minimal spectral truncation

Statement. The minimal nonzero grading-preserving spectral truncation of (8) is unitarily equivalent to Connes–Lott, with

$$\boxed{|m| = m_1 = \frac{\pi}{2l}, \qquad c = \frac{\pi}{2}.}$$

Its natural truncated endpoint states have distance $1/m_1 = 2l/\pi$. Literal endpoint evaluations do not descend through compression.

Hypotheses. Let

$$P = \mathbf{1}_{[0,E]}(|D_l|), \qquad m_1 \le E < 3m_1,$$

and give $S = P\,C([0,l])\,P$ its inherited operator-system structure. This is the compression construction described in Connes–van Suijlekom, §1.

Proof.

An orthonormal chiral basis of PH_l is

$$u_L(x) = \sqrt{\frac{2}{l}} \begin{pmatrix} \cos(\pi x/(2l)) \\ 0 \end{pmatrix}, \qquad u_R(x) = \sqrt{\frac{2}{l}} \begin{pmatrix} 0 \\ \sin(\pi x/(2l)) \end{pmatrix}.$$

In this basis,

$$PD_lP = \begin{pmatrix} 0 & m_1 \\ m_1 & 0 \end{pmatrix}, \qquad P\Gamma P = \mathrm{diag}(1,-1). \quad (17)$$

Compression of multiplication is

$$Q(f) = \begin{pmatrix} \frac{2}{l}\int_0^l f(x)\cos^2\frac{\pi x}{2l}\,dx & 0 \\ 0 & \frac{2}{l}\int_0^l f(x)\sin^2\frac{\pi x}{2l}\,dx \end{pmatrix}. \tag{18}$$

Since

$$Q(1) = I, \qquad Q(\cos(\pi x/l)) = \tfrac{1}{2}\,\mathrm{diag}(1,-1),$$

its range is exactly the full diagonal algebra:

$$\boxed{S = \mathbb{C} \oplus \mathbb{C}.} \tag{19}$$

Here the compressed operator system happens to be an algebra, although Q is not multiplicative.

Evaluation on the finite-dimensional space PH_l is bounded. At 0, only u_L has a nonzero boundary value; at l, only u_R does. The normalized evaluation vectors therefore define

$$\omega_0(\mathrm{diag}(a,b)) = a, \qquad \omega_l(\mathrm{diag}(a,b)) = b.$$

Using (17),

$$\|[PD_lP, \mathrm{diag}(a,b)]\| = m_1|a-b|,$$

and hence

$$\boxed{d_{\mathrm{trunc}}(\omega_0,\omega_l) = \frac{1}{m_1} = \frac{2l}{\pi} < l.} \tag{20}$$

Minimality follows because there is no zero mode and the grading exchanges the eigenspaces at $+m_1$ and $-m_1$. A nonzero grading-preserving spectral subspace must retain both.

There is an essential distinction between these states and literal evaluations. For

$$g(x) = \cos(3\pi x/l),$$

equation (18) gives $Q(g) = 0$, while $g(0) = 1$ and $g(l) = -1$. Thus neither endpoint evaluation factors through Q.

The pullbacks of the truncated endpoint states are instead the probability measures

$$d\mu_L = \frac{2}{l}\cos^2\frac{\pi x}{2l}\,dx, \qquad d\mu_R = \frac{2}{l}\sin^2\frac{\pi x}{2l}\,dx.$$

Their distance using the **original** continuum seminorm is

$$d_{D_l}(\mu_L,\mu_R) = \int_0^l \left|\int_0^x (d\mu_L - d\mu_R)\right| dx = \frac{4l}{\pi^2}. \quad (21)$$

The equality follows by integration by parts and is attained by $f = -x$. Equations (10), (20), and (21) concern three different state/metric comparisons.

Status: PROVED

Part 4. The two limits

Amendment (Terra 6.1, 3 Nov 2026; report). The statements of this part are correct for the stated bare chain, with one precision: the condition N/l -> infinity gives uniform convergence of the dispersion on bounded energy windows and of normalized heat traces and counting densities, and does not by itself give convergence of the unnormalized difference of heat traces (for N = floor(l^(5/4)) it diverges). Escapes for the empty light sector: a tuned bulk mass or a scaled boundary coupling.

Statement. At fixed finite rank, $m \to 0$ produces infinitely separated finite geometry, not an additional heat dimension. A resolved interval requires a growing tower. As $m \to \infty$, the metric collapses, but the light Hilbert space is zero.

Hypotheses. Identify

$$m = m_1 = \frac{\pi}{2l},$$

use normalization (1), and hold physical energies or heat times fixed unless stated otherwise.

Proof.

For fixed N, $D_N = mT_N$ with T_N independent of m. Therefore, as $m \to 0$,

$$D_N \to 0, \qquad K_N(t) \to 2(N-1), \qquad d_N(i,j) \to \infty \quad (i \neq j).$$

For the minimal truncation,

$$K_2(t) = 2e^{-tm^2}.$$

Every finite N has ultraviolet heat dimension zero. Making its finitely many modes light does not change that conclusion.

For the full interval, a fixed energy E is in the resolved one-dimensional regime when

$$\frac{1}{l} \ll E.$$

For its finite-chain approximation, one additionally needs

$$\boxed{\frac{1}{l} \ll E \ll \frac{N}{l}.} \tag{22}$$

Thus, as $l \to \infty$, an exact continuum limit throughout a fixed positive energy window follows if $N/l \to \infty$. Equivalently, as $m \to 0$, one needs $Nm \to \infty$.

The continuum counting function and heat density make the emergence explicit:

$$\#\{|\lambda| \le E\} = 2\left\lfloor \frac{lE}{\pi} + \frac{1}{2} \right\rfloor, \qquad \frac{K_l(t)}{l} \longrightarrow \frac{1}{\sqrt{\pi t}} \quad (l \to \infty). \tag{23}$$

Based at an endpoint, the metric limit is a half-line; the opposite endpoint recedes to infinity.

The order of limits matters even for the normalized heat trace:

$$\lim_{l\to\infty}\lim_{N\to\infty} \frac{K_N(t)}{l} = \frac{1}{\sqrt{\pi t}}, \qquad \lim_{N\to\infty}\lim_{l\to\infty} \frac{K_N(t)}{l} = 0. \tag{24}$$

Likewise, taking $t \to 0$ at fixed N gives dimension zero, whereas taking $N \to \infty$ first gives the interval's dimension one.

As $m \to \infty$, all distances tend to zero. At fixed N, the algebra elements with uniformly bounded commutators reduce to constants. But every eigenvalue has magnitude at least m, so

$$\|(D_N - i)^{-1}\| \le \frac{1}{\sqrt{m^2+1}} \longrightarrow 0. \tag{25}$$

The same holds for D_l, after identifying the varying intervals by dilation. Its heat trace tends to zero at every fixed $t > 0$.

Thus the metric limit is a point, while the embedded resolvent limit has

$$H_{\text{light}} = \ker T_N = 0.$$

There is no finite operator limit on the original nonzero Hilbert space, and no surviving fermion pair at fixed energy.

Finally, the two-state spectral truncation exists for

$$m_1 \leq E < 3m_1.$$

A genuine spectral cutoff $E \ll 1/l$ retains no modes. The ratio between the first two positive levels is always 3, so the minimal pair has no parametrically large separation from the next level.

Status: PROVED

Verdict on hypothesis H (theorem T1)

Supported for T1's geometric assertion. Connes–Lott is exactly the minimal nonzero grading-preserving spectral compression of the opposite-chirality interval above, with

$$l = \frac{\pi}{2|m|}.$$

The chain recovers its spectrum, heat window, and metric.

The precise replacement for the stronger interpretation of H is:

> The two sheets are truncated endpoint evaluation states, whose pullbacks are smeared over the interval. Refinement restores the interval. Decompactification produces a one-dimensional heat regime only when the number of retained modes grows sufficiently rapidly. Collapse identifies the metric points but removes all fermionic modes of this particular interval at fixed energy.

Accordingly, a nonempty two-state sector at $E \ll 1/l$, or surviving light fermions after collapse, is disproved by this model. The construction supplies no KO-6 or Standard Model identification.

Independent numerical check (Claude, 2 Nov 2026)

For l = 1 and N = 2, 3, 5, 9 we checked the following. The spectrum of D_N agrees with +/- 2 h_N sin((2r-1) theta_N) to 1e-14. The lowest positive eigenvalue equals pi/(2l) for every N. For random real site functions, |||[D_N, pi_N(a)]||| equals h_N max abs(a_{i+1} - a_i), which is the identity behind d_N(i,j) = abs(i-j)/h_N. The endpoint distance d_N(1,N) = 0.6366, 0.7869, 0.8844, 0.9398 tends to l. For the minimal truncation, Q(cos(3 pi x/l)) = 0, and the smeared-state distance 0.405290 matches 4l/pi^2. The truncated distance is 2l/pi = 0.6366, identical to d_2(1,2).

Literature (../LITERATURE.md, section A). The Kaluza-Klein reading of the two-point space is in Connes-Lott (sections I and III: "two copies of Z separated by a distance L"); the two points as points of a fifth dimension are in Viet-Wali 1996; the two-point model as an approximation of a manifold, to be related to dimensional reduction, is in Dimakis-Muller-Hoissen 1994 (section III, Example 4, and section VII). Connes-van Suijlekom (section 1) note that compressions P A P are operator systems, of which the chiral interval is an exception. The truncation theorem with |m| = pi/(2l) was not found in print.

Part 16

Noncommutative finite real triples from commutative data: truncating the coarse correspondence

Proved by Claude Opus 5.5 without Astra or Fable, in response to the challenge that the T0 note overread lemma 15 and the Farsi-Latrémolière limits. Signs and the level-two algebra were checked with numpy on random data (section "Numerical check"). The term "coarse correspon-

dence" is that of Connes, *Noncommutative Geometry* (1994), chapter V, for the bimodule of Hilbert-Schmidt operators, and the form D = Delta + J Delta J^-1 is in Krajewski (arXiv:hep-th/9701081) and Paschke-Sitarz (arXiv:q-alg/9612029); all checked against the sources. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): the examples and Corollary 16.4 are correct; amendments to Proposition 16.1, Theorem 16.2, Corollary 16.3 and the caveats are incorporated below; Remark 16.5 and Example 16.6 come from the referee's assessment of what the lemma achieves. The two-sheet section was refereed separately (Terra 6.1, 3 Nov 2026; report), with one error in Corollary 16.8 and one in Example 16.9, corrected below.

Outcome. Compressing functions on a connected commutative space to a fixed spectral window produces a noncommutative finite algebra with no limit taken. With the standard reality J a J^-1 = a*, the inherited structure fails order zero exactly when that algebra is noncommutative (Proposition 16.1). With the parent Hilbert space doubled to H_0 (x) conj(H_0), order zero and order one hold for every such algebra (Theorem 16.2, Corollary 16.3); these two conditions are formal consequences of the tensor square, available for any finite even representation (Remark 16.5). The KO-6 signs and the count dim H_F = 2 k^2 are reached by a product with a Clifford point, which is an input and fails finite orientability (Corollary 16.4), and more faithfully by a parent with two sheets, which gives the form of the Chamseddine-Connes classification (Theorem 16.7, Corollary 16.8, Example 16.9). This selects no particular algebra and gives no dynamics.

Setting

Let (A_0, H_0, D_0, gamma_0) be an even spectral triple, A_0 commuting with gamma_0, with D_0 of compact resolvent. Let Sigma = -Sigma be a finite subset of R and P_0 = 1_Sigma(D_0), of rank k, so that P_0 commutes with gamma_0. Put V = P_0 H_0, graded by gamma_V, with d = D_0 restricted to V, and

```
Q(a) = P_0 a P_0 on V,     O_F = Q(A_0),     A_F =
↪  C*(O_F) in B(V).
```

O_F is an operator system, in general not an algebra, and A_F consists of even operators. Compression is stan-

dard: for functions it is the Toeplitz compression, and its use in truncating spectral triples is due to D'Andrea-Lizzi-Martinetti and Connes-van Suijlekom. The latter note that the C*-algebra generated by PAP is in many examples a noninformative full matrix algebra. Since Q(1) = 1, A_F is the bicommutant of O_F, so its structure is set by the operators commuting with all compressed functions, such as the grading in example (b) and j in Example 16.9. That is where the information lies, and the fact that A_F is noncommutative is modest.

Examples of A_F.

(a) *Chiral interval, lowest level.* For condition (c) with M = 0 and Sigma = {+/- pi/(2l)}, O_F = C + C = A_F (lemma 05).

(b) *Chiral interval, next level.* For Sigma = {+/- pi/(2l), +/- 3 pi/(2l)}, V has basis (c_1, 0), (c_2, 0), (0, s_1), (0, s_2), with c_r = (2/l)^{1/2} cos((r - 1/2) pi x/l) and s_r = (2/l)^{1/2} sin((r - 1/2) pi x/l). With C_n = (1/l) int f cos(n pi x/l),

Q(f) = (Q_L, Q_R), Q_L = [[C_0 + C_1, C_1 + C_2], [C_1 + C_2, C_0 + C_3]], Q_R = [[C_0 - C_1, C_1 - C_2], [C_1 - C_2, C_0 - C_3]].

The values C_0, C_1, C_2, C_3 are independent and determine Q(f), so O_F has dimension 4. Each of Q_L and Q_R alone ranges over all symmetric 2 x 2 matrices, which generate M_2(C), so A_F projects onto each summand. For f_1 = 1/2 + cos(pi x/l) + cos(2 pi x/l) + cos(3 pi x/l) and f_2 = 1/2 - cos(pi x/l) + cos(2 pi x/l) - cos(3 pi x/l),

```
Q(f_1) = ([[1, 1], [1, 1]], 0),     Q(f_2) = (0,
↪  [[1, -1], [-1, 1]]).
```

M_2(C) is simple, so multiplying Q(f_1) and Q(f_2) by elements of A_F gives M_2(C) + 0 and 0 + M_2(C). Hence A_F = M_2(C) + M_2(C), of dimension 8. The finite algebra of the interval itself is noncommutative at the second level.

(c) *Twistor space.* For the level-one Toeplitz compression on CP^3, Q(x_a) = Gamma_a/5 (Proposition 10.1), and these generate A_F = M_4(C) on V = C^4. Here V is the kernel of the Dolbeault-Dirac operator twisted by O(1), so d = 0 on V.

The standard correspondence

Proposition 16.1 (order zero fails with inherited reality). Let J_0 be a real structure on H_0 (antiunitary, J_0^2 = +/-1, with the grading sign), with J_0 a J_0^-1 = a* for all a in A_0 and J_0 D_0 = eps' D_0 J_0, and let J_F = J_0 restricted to V. Then J_F Q(a)* J_F^-1 = Q(a), and the data (A_F, V, d, J_F) satisfy order zero if and only if A_F is commutative. The same holds for a product with a fixed finite factor that itself satisfies order zero, with the compatible product real structure.

Proof. J_0 preserves V because Sigma = -Sigma. Then J_0 P_0 a* P_0 J_0^-1 = P_0 a P_0. Order zero would make Q(a) commute with Q(b)° = Q(b) for all a, b, so O_F and hence A_F would be commutative; the converse is immediate, since J_F A_F J_F^-1 = A_F. For the product, take the unit in the finite factor: [1 (x) Q(a), 1 (x) Q(b)] = 1 (x) [Q(a), Q(b)]. ■

Together with Amendment 2 of lemma 07 (first order fails for C + C with inherited J_2), this is the correct form of the obstruction. It concerns the standard correspondence L^2(A_0) with the pointwise real structure.

The coarse correspondence

Let conj(H_0) be the conjugate Hilbert space, with canonical antiunitary xi -> xi^c, and with gamma^c, D^c the conjugate operators, gamma^c xi^c = (gamma_0 xi)^c and D^c xi^c = (D_0 xi)^c. On H = H_0 (x) conj(H_0), which is the coarse correspondence of A_0, equivalently the Hilbert-Schmidt operators on H_0, define for homogeneous xi, eta

```
pi(a) = a (x) 1,
DD = D_0 (x) 1 + gamma_0 (x) D^c,
Gamma = gamma_0 (x) gamma^c,
J (xi (x) eta^c) = (-1)^{|xi| |eta|} eta (x) xi^c.
```

Theorem 16.2. (A_0, H, DD, J, Gamma) is a real even spectral triple with signs J^2 = 1, J DD = DD J, J Gamma = Gamma J (KO-dimension 0). For even a, b with bounded commutators,

```
a° := J pi(a)\* J^-1 = 1 (x) (a\*)^c,     [pi(a),
 ↪  b°] = 0,     [[DD, pi(a)], b°] = 0,
```

and norm([DD, pi(a)]) = norm([D_0, a]). So if the original triple is metric, the new triple is metric with the same Connes distance on A_0. If the Weyl exponent p = lim log N_0(R)/log R of D_0 exists, that of DD is 2 p, since N_0(R/sqrt(2))^2 <= N_DD(R) <= N_0(R)^2.

Proof. J is antilinear, and J^2 = 1 because the sign is applied twice with the degrees exchanged. Gamma has eigenvalue (-1)^{|xi| + |eta|}, symmetric under the exchange. For the Dirac operator,

```
J DD (xi (x) eta^c) = (-1)^{(|xi| + 1) |eta|} eta
 ↪  (x) (D_0 xi)^c + (-1)^{|xi|} (-1)^{|xi| (|eta| +
 ↪  1)} D_0 eta (x) xi^c
                     = (-1)^{|xi| |eta|} [D_0 eta (x)
                      ↪  xi^c + (-1)^{|eta|} eta (x)
                      ↪  (D_0 xi)^c] = DD J (xi (x)
                      ↪  eta^c).
```

For even a, a* preserves degree, so J pi(a)* J^-1 (eta (x) xi^c) = eta (x) (a* xi)^c, which gives the formula for a° and order zero. Since a is even, [DD, pi(a)] = [D_0, a] (x) 1, which commutes with 1 (x) (b*)^c. Finally DD^2 = D_0^2 (x) 1 + 1 (x) (D^c)^2, because gamma_0 anticommutes with D_0. An eigenbasis of D_0 decomposes DD into finite-dimensional invariant blocks, so DD is essentially self-adjoint on the algebraic tensor product, with compact resolvent, and the spectrum is +/- (lambda_i^2 + lambda_j^2)^{1/2}. ■

Geometrically, when H_0 = L^2(S_X) on an even-dimensional spin manifold X, a charge conjugation commuting with D_X identifies conj(H_0) with H_0, H is L^2(S_{X x X}) and DD is the ordinary product Dirac operator. (The grading may acquire an overall sign under this identification.) So the data are a product-manifold Dirac operator on X x X with a restricted observable algebra C(X) (x) 1, and a J covering the flip of the two factors, which differs from the pointwise spin real structure. They count as commutative Kaluza-Klein data only with that restriction and that J: they are neither the ordinary spin triple of X x X with its full function algebra nor a finite-rank twist over X. The restriction is essential: on all of C(X x X), the compressions generate A_F (x) conj(A_F), and order zero generally fails again.

Corollary 16.3 (truncation). Let P = P_0 (x) conj(P_0). It commutes with DD, Gamma and J. It

is a joint cutoff in the two factors, and need not be a spectral projection of DD: for absolute eigenvalues 1, 5, 7, sqrt(73) of D_0 with P_0 selecting the first three, the selected pair (5, 7) and the excluded pair (1, sqrt(73)) both give DD^2 = 74. The compression of pi(A_0) is O_F (x) 1; passing to the generated algebra gives the finite real even triple

```
T_F = (A_F, V (x) conj(V), D_F, J_F, Gamma_F),
 ↪  D_F = d (x) 1 + gamma_V (x) d^c = Delta + J_F
 ↪  Delta J_F^-1,     Delta = d (x) 1,
```

with A_F acting on the left. It has KO-dimension 0, and satisfies order zero and order one for every A_F, non-commutative or not, with dim H_F = k^2. The family is exact at each truncation; no limit is taken.

Proof. The proof of Theorem 16.2 applies verbatim on V with arbitrary even a, b in B(V), since it uses only the grading and the tensor structure. For J_F Delta J_F^-1, apply J to (d (x) 1) J (xi (x) eta^c): the signs combine to (-1)^{|xi|}, giving gamma_V (x) d^c. ■

So H_F has the bimodule form of Connes' finite geometries, with A_F on the left and its opposite on the right, and D_F has the first-order form of Krajewski and Paschke-Sitarz.

Example (a) gives (C + C, C^2 (x) C^2, m (sigma_1 (x) 1 + sigma_3 (x) sigma_1)). There [[D_F, e], e°] = 0 for e = diag(1, 0), which repairs the first-order failure of Counterexample 08.4 at the price of doubling. Example (b) gives (M_2(C) + M_2(C), C^16, D_F).

Corollary 16.4 (KO-6 by a Clifford point). The product of T_F with (C, C^2, 0, sigma_1 K, sigma_3) is

```
(A_F, V (x) conj(V) (x) C^2, D_F (x) 1, J_F (x)
 ↪  sigma_1 K, Gamma_F (x) sigma_3),
```

with signs (+, +, -), order zero, order one and dim H = 2 k^2. It fails the usual zero-dimensional orientability axiom: every zero-cycle built from the left and opposite actions acts as T (x) 1_2, while the grading is Gamma_F (x) sigma_3.

Proof. (sigma_1 K)^2 = 1 and sigma_1 K sigma_3 = -sigma_3 sigma_1 K, while the D term is untouched. a° is a° of T_F tensored with 1. ■

Remark 16.5 (the order conditions are formal). For every finite even representation (A, V, d, gamma), the map V -> V (x) conj(V), d -> d (x) 1 + gamma (x) d^c gives a real bimodule with order zero and order one, using no prior finite real structure. So these two conditions follow from choosing the tensor square, and say nothing specific about A_F. The construction does not recover an arbitrary prescribed finite (H, D, J). What is specific to T0 is that A_F is generated by compressed functions, that the tensor square is itself the truncation of a commutative parent in the restricted sense above, and families such as the following.

Example 16.6 (a tower-decoupling family, from the referee). Let D_0 be the Dolbeault-Dirac operator on CP^3 twisted by O(1), with V = ker D_0 = C^4, and let DD_l = DD/l on the doubled parent. Its kernel is V (x) conj(V), and its gap grows like 1/l. Here P is a spectral projection. Compression followed by algebra generation gives the constant finite data A_F = M_4(C), dim H_F = 16, D_F = 0, and the Clifford point gives dim H = 32 with the KO-6 signs. Order one is automatic, since D_F = 0. The whole triple converges to (C, C^16, 0) by Theorem 15.3, consistently with lemma 15: the two statements concern the whole algebra and the truncation.

Two sheets: the form of the classification

Chamseddine and Connes (*Why the Standard Model*, arXiv:0706.3688, sections 2-3) classify irreducible pairs (A, J) with order zero and a separating vector: the centre of the complexified algebra is C, with A_C = M_k(C) on itself and J x = x*, or C + C, with A_C = M_k(C) + M_k(C) on itself and J(x, y) = (y*, x*). With centre C, J gamma = -gamma J is impossible. The pair of Corollary 16.4 is therefore reducible: the projection 1 (x) (1 + sigma_1)/2 commutes with A_F and J. The irreducible form comes from a parent with two sheets. At the finite level the bimodule (W (x) conj(V)) (+) (V (x) conj(W)) with its adjoint-flip J is already stated in Remark 2.9 of that paper; what we add is its realization from parents and compressed functions.

Theorem 16.7 (two-sheet coarse parent). Let (A_X, H_X, D_X, gamma_X) and (A_Y, H_Y, D_Y,

gamma_Y) be even spectral triples. On H = (H_X (x) conj(H_Y)) (+) (H_Y (x) conj(H_X)), for even a in A_X and b in A_Y, put

```
pi(a, b) = (a (x) 1) (+) (b (x) 1),
DD = (D_X (x) 1 + gamma_X (x) D_Y^c) (+) (D_Y (x) 1
 ↪  + gamma_Y (x) D_X^c),
Gamma = (gamma_X (x) gamma_Y^c) (+) (-gamma_Y (x)
 ↪  gamma_X^c),
J (xi (x) eta^c) = (-1)^{|xi| |eta|} eta (x) xi^c,
```

where J maps each summand to the other. Then J^2 = 1, J DD = DD J and J Gamma = -Gamma J, the KO-6 signs. Moreover

```
(a, b)° = (1 (x) (b\*)^c) (+) (1 (x) (a\*)^c),
```

so the first summand is an (A_X, A_Y)-bimodule and the second an (A_Y, A_X)-bimodule, and order zero and order one hold.

Proof. The computation of Theorem 16.2 applies with D_Y in place of the second factor: on xi (x) eta^c in the first summand, J DD and DD J both give (-1)^{|xi| |eta|} [D_Y eta (x) xi^c + (-1)^{|eta|} eta (x) (D_X xi)^c]. The sign in Gamma on the second summand gives J Gamma = -Gamma J. For a°, apply J, then pi(a, b)*, then J, as in Theorem 16.2: an element of the first summand goes to the second, where b* acts on the left, and returns with b* on the right factor. Since a and b are even, [DD, pi(a, b)] = ([D_X, a] (x) 1) (+) ([D_Y, b] (x) 1), which commutes with the opposite actions. ■

Corollary 16.8 (truncation; the form of the classification). For symmetric windows with V_X, V_Y of dimensions k_X, k_Y, and even subalgebras A_{F,X} in B(V_X), A_{F,Y} in B(V_Y), the compression

```
T_F = (A_{F,X} + A_{F,Y}, (V_X (x) conj(V_Y)) (+)
 ↪  (V_Y (x) conj(V_X)), D_F, J_F, Gamma_F)
```

has the KO-6 signs, order zero and order one, and dim H_F = 2 k_X k_Y. If gamma_{V_X} lies in A_{F,X} and gamma_{V_Y} in A_{F,Y}, the real orientability relation holds: with c = (gamma_{V_X}, gamma_{V_Y}) and z = (1, -1), Gamma_F = pi(z c) c°, a zero-cycle with coefficients in the algebra and its opposite. This is weaker than Gamma_F in pi(A_F). If V_X and V_Y are ungraded with k_X = k_Y = k and the generated algebras are full, then (A_F, H_F,

J_F) is exactly the irreducible pair with centre C + C: identify both summands with M_k(C), so that J_F(x, y) = (y*, x*); an invertible matrix in each summand is a separating vector, and a projection commuting with both full matrix actions is scalar on each summand, with J equating the scalars. In all cases D_F commutes with the two sheet projections, so the sheets are not connected. It need not commute with the whole centre: for V_X = C^2, gamma_X = sigma_3, d_X = sigma_1, A_{F,X} = C + C acting diagonally and V_Y = C, the central element (diag(1, 0), 0) does not commute with D_F.

Proof. Theorem 16.7 applies on the truncation with arbitrary even operators. For orientability, pi(c) c° = (gamma_{V_X} (x) gamma_{V_Y}^c) (+) (gamma_{V_Y} (x) gamma_{V_X}^c), and z flips the sign of the second summand. The last statements are direct. ■

Example 16.9 (twistor level one, and M_2(H) + M_4(C)). Let D_0 be the Dolbeault-Dirac operator on CP^3 twisted by O(1), with kernel H^0(CP^3, O(1)) = C^4, and let j be the antiunitary with j^2 = -1 commuting with the Gamma_a of Proposition 10.1, which makes C^4 a quaternionic space H^2. Use this space twice, as W (with the functions of S^4 pulled back along the twistor projection) and as V (with the functions of CP^3).

1. A real function f on S^4 pulls back to a function invariant under w -> j w. So j Q(f) j^-1 = Q(f), and Q(f) is a Hermitian element of End_H(W). Since Q(x_a) = Gamma_a/5, the real algebra generated is End_H(W) = M_2(H), of real dimension 16.
2. For real functions on CP^3, Q(w* B w / w* w) = (B + tr(B) 1)/5 for Hermitian B, so Q is onto the Hermitian matrices, and the real algebra generated is End_C(V) = M_4(C), of real dimension 32.

Hence Corollary 16.8, with the parent gradings, which are trivial on the kernels, gives the real algebra A = M_2(H) + M_4(C) on H_F = Hom(V, W) (+) Hom(W, V) = E (-) E* of dimension 32, with J(x, y) = (y*, x*). These are the algebra, bimodule and real structure of Chamseddine and Connes. The generation is of real algebras from real functions: complex generation on the S^4 side would give M_4(C).

The nontrivial grading of W in that construction is not obtained from this parent. An even parent has [P gamma P, P a P] = 0 for every compressed function, while Gamma_5 = 5 Q(x_5) anticommutes with Q(x_1). So the grading has to be added after truncation, as abstract finite data, fixing an axis. The full algebra is then a graded prealgebra rather than the algebra of an even triple, with even part H + H + M_4(C), and Gamma_F(xi, eta) = (Gamma_5 xi, -eta Gamma_5), which is Proposition 3.2 of that paper; the real orientability relation holds on the even part. Similarly d = mu n^a Gamma_a with n orthogonal to the axis is the compression of a bounded real multiplication operator, but adding that operator to the parent Dirac operator keeps neither the kernel as a spectral projection nor the parent grading as an odd perturbation, so d too is abstract data. It commutes with the two sheet projections and with the centre R + C of the full prealgebra, and not with the centre of the even algebra.

This bears on the step left open in canonical section 4, where M_2(H) in M_4(C) is a nested pair on one four-space, with "the independent E, F module roles" to be derived. The example closes that abstract finite-module problem after adding a second sheet: the nested actions on one four-space become independent summands acting on two Hom spaces. It does not turn four global sections into rank-four fibres over S^4, derive the two sheets from the fibration, or produce the Dirac operator connecting the two pieces that the selection of A_F = C + H + M_3(C) requires. The fibre-rank problem remains open.

The inputs are the second sheet, the restriction of one algebra to base functions, real algebra generation, the four-dimensional window, the grading axis and d. The content beyond the formal bimodule is the concrete compression calculation in items 1 and 2, already in the corpus, and its explicit match with the classified data.

Caveats

These results should be read as narrowly as the note they correct.

1. **Metric.** T_F is usually not a metric spectral triple. For A_F = B(V_+) + B(V_-) with V_+ of dimension at least 2 and d odd, the commutant

of d in A_F contains the pairs (x, B^-1 x B) with x commuting with B B*, when B is the invertible block of d. In example (b), (diag(1, 0), diag(1, 0)) has L = 0, so d leaves a residual U(1) x U(1) of U(2) x U(2) unbroken. The Standard Model finite triple has the same feature, through M_3(C).

2. **Spectrum.** The eigenvalues of D_F are +/- (lambda_i^2 + lambda_j^2)^{1/2}. In example (a) they are +/- sqrt(2) m with Connes distance 1/m, so the doubled spectrum differs from the masses in d, and the mass relations of Theorem 9.6 and lemma 11 would have to be redone.
3. **Parent.** The doubled parent is a choice. Its algebra sees one factor of X x X, its J covers the flip, which is outside the spin real structure of X x X, and generating A_F from O_F is an additional operation. We have no dynamical reason for it. For an ordinary compact spin X of dimension n, its Weyl exponent is 2 n with metric dimension n, which bears on the counting of 9, 10 and 11 and has not been analyzed.
4. **KO-6.** The Clifford point is an input, as F_0 is in Theorem D, and it fails finite orientability. In Theorem 16.7 the sign comes from the block structure of two sheets, which is itself an input. The documents checked here establish neither a construction of two sheets from one connected parent nor a no-go theorem. The grading-reversing structure of lemma 08 is a candidate we have not tested.
5. **Count.** For k = 4, dim H = 32, the value in the classification of Chamseddine-Connes, and k = 4 occurs in examples (b) and (c). For k = 4 in examples (b) and (c) alone this is a count and nothing more: the classification's algebra is M_2(H) + M_4(C), acting differently, and those examples do not produce it. Example 16.9 does, as abstract data with the listed inputs.

Consequence for T0 and T0b

T0 asks for finite real triples from commutative Kaluza-Klein data under specified notions of convergence. For one specified truncation procedure (compression, then generation, on the doubled parent with restricted algebra and flip J), this lemma gives a noncommutative finite al-

gebra with an explicit bimodule, and a family with the tower decoupled (Example 16.6). It does not establish the target under every meaning of ordinary Kaluza-Klein data. For T0b, it gives examples with the requested signs and first-order relations; with two sheets it reaches the form of the classification, and in Example 16.9 the real algebra M_2(H) + M_4(C), bimodule and real structure of Chamseddine and Connes, with the grading added as abstract data. It leaves open an operator connecting the two pieces and hence the selection of A_F, a parent that yields the grading and nonzero mass data, and the beyond-first-order and twisted frameworks. See the background note.

Numerical check

Python (numpy), run on 3 Nov 2026 in a temporary directory, not kept. On random odd d and even a, b for dim V_+ = dim V_- = 2 with J = M K: J^2 = 1, J DD = DD J, J Gamma = Gamma J, DD Gamma = -Gamma DD, order zero and order one all hold to machine precision, with norm([a, b]) = 7.67; the KO-6 signs of Corollary 16.4 hold. For Theorem 16.7 and Example 16.9, with k = 4, W graded by Gamma_5 and V ungraded: J^2 = 1, J Gamma = -Gamma J, J D = D J, D Gamma = -Gamma D, D = Delta + J Delta J^-1, order zero and order one for random a, b, and Gamma_F = pi(z c) c° all hold to machine precision; j^2 = -1 commutes with the Gamma_a, and x_a(j w) = x_a(w); the real algebras generated by Gamma_a, by i Gamma_a and by all Hermitian matrices have real dimensions 16, 32 and 32; and sum_a x_a Gamma_a = 2 (P_w + P_{jw}) - 1. For example (b), by quadrature with 23 test functions: dim O_F = 4 and the generated algebra has dimension 8; the two displayed values Q(f_1) and Q(f_2) were reproduced by quadrature.

Part 22

The finite Dirac operator as a boundary condition on an interval

For STRATEGY T0, T0b, T10 and T13, following lemma 12 and lemma 20. Proved by Claude Opus 5.5, with the identities checked in Python (section “Numerical check”).

Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors and one gap as first written, and amendments; all are incorporated below.

Outcome. Let (H_F, D_F, J_F, Gamma_F) be finite data of signs (+, +, -), and let l ||D_F|| < pi/2. Then D_F, with J_F and Gamma_F, is exactly the lowest spectral window of the pure derivative on an interval of length l, with no bulk mass and no Yukawa term, on the domain fixed by the boundary coupling tan(l D_F). So all the eigenvalues of D_F, Yukawa and Majorana blocks alike, are quantized internal momenta k with tan(k l) equal to a singular value of the coupling. The input is the finite triple: the content is that a consistent parent exists, with these properties:

- Preservation of the domain forces the observables at the wall into C_F, the commutant of D_F in A_F, while A_F acts freely in the bulk.
- The parent satisfies order one for any finite data with order zero.
- The compression is the exact average Q_l(a) = (2l)^-1 int_{-l}^{l} e^{ixD_F} a e^{-ixD_F} dx, so it tends to the finite representation as l -> 0, with defects of order l^2.
- The scalar inner fluctuations of the parent are pure gauge under its allowed endpoint gauge group. On the moving window, they act by the gauge orbit of D_F through the transport along the interval. This reproduces the quadratic term of Chamseddine-Connes-van Suijlekom on the orbit, and it leaves the singular values fixed, so the radial mode of the Higgs field is absent from this parent.

With the family X(l) = tan(l M) and l -> 0, the couplings of the light fermions vanish as m l at fixed mass, the tower decouples, and the broken gauge directions go to pi/(2l). The parent algebra is matrix-valued, so T0 for a commutative parent remains open.

Setting

Let H_F = H_+ (+) H_- be finite-dimensional, Gamma_F = diag(1, -1), let c : H_+ -> H_- be antiunitary, and put J_F(a, b) = (c^-1 b, c a), so that J_F^2 = 1 and J_F Gamma_F = -Gamma_F J_F. Let D_F = [[0, M*], [M, 0]] with M : H_+ -> H_-. Then J_F

D_F = D_F J_F is equivalent to M = c M* c. By the Takagi factorization, in bases related by c, we may write M = V |M| with a unitary V satisfying V = c V* c.

Let l > 0 with l ||D_F|| < pi/2, and define X : H_- -> H_+ by

```
tan(l D_F) = [[0, X], [X\*, 0]],   that is,   X =
↪  tan(l |M|) V\*.
```

Since J_F commutes with real functions of D_F, c X = X* c^-1. Conversely, every X with c X = X* c^-1 arises from exactly one such D_F, namely D_F = l^-1 arctan of the block operator. In the limit, X = l M* + O(l^3).

The parent. On H_l = L^2([0, l]; C^2) (x) (H_+ (+) H_-), write psi = (Q_L, Q_R; U_L, U_R), with Q valued in H_+ and U in H_-. Let (D psi)_L = psi_R' and (D psi)_R = -psi_L' on both fields, the operator of lemma 12 with no bulk mass, on the domain

```
Q_R(0) = 0,   U_L(0) = 0,   Q_R(l) = X U_R(l),
↪  U_L(l) = -X\* Q_L(l).
```

Call it D_X. Put Gamma^ psi = (Q_L, -Q_R; U_L, -U_R) and J^ psi = (c^-1 U_R, -c^-1 U_L; -c Q_R, c Q_L). Let (E psi)_L = psi_R and (E psi)_R = -psi_L, so that D_X = E d/dx on its domain.

In the Standard Model form of the classification, with E = Hom(V, W) and W = W_+ (+) W_-, one takes H_+ = E_+ (+) E_-* and H_- = E_- (+) E_+*, with c the adjoint map. At X = 0, the Q-fields have boundary type (a) and the U-fields type (b), in the terms of lemma 09; for X != 0 the modes continue from those zero modes, which generally lift. A single matrix X encodes the Yukawa blocks and the Majorana block together. Its entries need not keep the original block pattern, since the tangent brings in higher powers of D_F: for M = [[a, b], [b, 0]] real symmetric, X_22 = (l^3/3) a b^2 + O(l^5).

The window

Theorem 22.1. D_X is self-adjoint with compact resolvent, and J^^2 = 1, J^ D_X = D_X J^, J^ Gamma^ = -Gamma^ J^, Gamma^ D_X = -D_X Gamma^. Let R(x) = exp(x Gamma_F D_F), and define W : H_F -> H_l so that for phi = (a, b), the L-components (Q_L,

U_L) of W phi at x are l^{-1/2} R(x)(a, 0), and the R-components (Q_R, U_R) are l^{-1/2} R(x)(0, b). Then W is an isometry and

```
D_X W = W D_F,    J^ W = W J_F,    Gamma^ W = W
↪ Gamma_F.
```

With sigma_k the singular values of M and theta_k = l sigma_k, the signed spectrum of D_X is (n pi +- theta_k)/l, n in Z, with multiplicities added when values coincide; a zero singular value gives a zero mode of multiplicity two. So the spectrum of |D_X| consists of the sigma_k and of (n pi +- theta_k)/l for n >= 1, and ran W is the spectral subspace of |D_X| <= ||D_F||, and the next level is at least (pi - l ||D_F||)/l.

Proof. Self-adjointness is Lemma 12.1. Since Gamma_F D_F is anti-Hermitian, R(x) is unitary, and with C(x) = cos(x |M|), C'(x) = cos(x |M*|), S(x) = V sin(x |M|),

```
R(x)(a, 0) = (C(x) a, -S(x) a),    R(x)(0, b) =
↪ (S(x)\* b, C'(x) b).
```

Isometry follows from the unitarity of R(x) and the orthogonality of the L- and R-components. The conditions at 0 hold since S(0) = 0. At l, using V* cos(l |M*|) = cos(l |M|) V*, we get X C'(l) = S(l)* and X* C(l) = S(l), which are the conditions at the wall. For the derivative, the L-part of D W phi is d/dx R(x)(0, b) = R(x) Gamma_F D_F (0, b) = R(x)(M* b, 0), and the R-part is -d/dx R(x)(a, 0) = R(x)(0, M a); together these are W D_F phi. The relations for Gamma^ are direct. For J^, use c |M| c^-1 = |M*| and V = c V* c, which give c^-1 C' = C c^-1 and c S(x)* = S(x) c^-1. The signs of the parent follow in the same way, using c X = X* c^-1 for the domain. For the spectrum, constant unitaries on Q and on U commute with E d/dx and bring X to diag(tan theta_k), which reduces to Example 12.5 in each channel. ■

Corollary 22.2 (conjugated form). D_X = R (D_0 + E Gamma_F D_F) R*, with R acting pointwise on the pair (Q, U) and D_0 the uncoupled operator, X = 0. So after this unitary change of domain, D_F is a constant matrix connection, odd for the grading, with relative endpoint rotation R(l). This is equation (21) of lemma 20, for general D_F.

Proof. D(R phi) = R(D phi + E R* R' phi), and R* R' =

Gamma_F D_F. For phi in the uncoupled domain, the computation at the wall in the proof above shows that R phi satisfies the coupled conditions. ■

So in this family a fermion mass is a momentum: D_X has no mass term, and the light eigenvalue k = theta/l is fixed by tan(k l) = tan(theta), a singular value of X. In the finite description, the condition relating k to l is lost and the value appears as free data in D_F.

The parent as a real spectral triple

Let (A_F, H_F, D_F, J_F, Gamma_F) be finite data as above, with an even representation pi = pi_+ (+) pi_- of a unital finite-dimensional real or complex *-algebra A_F, faithful, unital and satisfying order zero, with pi°(b) = J_F pi(b)* J_F^-1. Order one is not assumed. Let

```
C_F = {a in A_F : [pi(a), D_F] = 0},    A_1 = {f in
 ↪  C^infinity([0, 1], A_F) : f(1) in C_F},
```

with f acting pointwise by pi_+(f(x)) on Q and pi_-(f(x)) on U.

Theorem 22.3.

1. A function f in C^infinity([0, 1], A_F) preserves Dom D_X if and only if f(1) lies in C_F.
2. (A_1, H_1, D_X, J^, Gamma^) is a real even spectral triple, with [D_X, pi(f)] = E pi(f') and J^ pi(g)* J^^-1 = pi°(g) pointwise.
3. It satisfies order zero and order one, with signs (+, +, -).

So A_F acts freely in the bulk, and the boundary condition restricts the observables at the wall to the commutant of D_F. In the Standard Model data with nonzero Yukawa blocks, this is the algebra {(lambda, lambda, m)} of the photon and gluons; with vanishing Yukawa blocks the commutant is larger. Order one holds in this parent whatever the finite data, for its pointwise algebra: the finite condition appears only on the truncation, and arbitrary compressions need not preserve either order condition.

Proof. Item 1: at 1, the traces Q_L(1) and U_R(1) are free, and applying f to the conditions gives pi_+(f(1)) X = X pi_-(f(1)) and pi_-(f(1)) X* = X* pi_+(f(1)), as in

Result 3 of lemma 20. That is [pi(f(l)), tan(l D_F)] = 0, equivalent to [pi(f(l)), D_F] = 0 because l ||D_F|| < pi/2 makes D_F a function of tan(l D_F). At 0 the conditions are preserved since pi is even. Item 2: the commutator is the Leibniz rule. Conjugation by J^ acts on (Q, U) as J_F does, and pi(g) acts identically on the two spin components, which gives the second formula. This preserves the domain since J_F commutes with D_F. Item 3: pi(f(x)) and pi°(g(x)) commute at each x by finite order zero, and E commutes with both; so [E pi(f'), pi°(g)] = E [pi(f'), pi°(g)] = 0. ■

This is consistent with Counterexample 4 of lemma 20, whose finite data A_F = C + C, D_F = m sigma_1 fail order one while the parent satisfies it.

The compression

Let Q_l(T) = W* T W for a bounded operator T on H_l. For T(x) a bounded function with values in the even operators on H_F, acting pointwise and identically on both spin components, we write T for the multiplication operator.

Theorem 22.4.

```
Q_l(T) = l^-1 int_0^l cos(x ad_{D_F})(T(x)) dx,
 ↪  Q_l(E T) = -l^-1 int_0^l sin(x ad_{D_F})(T(x))
 ↪  dx.
```

For constant a in A_F,

```
Q_l(a) = sinc(l ad_{D_F})(pi(a)) = (2l)^-1
 ↪  int_{-l}^{l} e^{ix D_F} pi(a) e^{-ix D_F} dx,
```

so in an eigenbasis of D_F with eigenvalues mu_i, the matrix elements are pi(a)_{ij} sin(l (mu_i - mu_j))/(l (mu_i - mu_j)). Moreover:

1. Q_l(A_l) = Q_l(C^infinity([0, l], A_F)), so the image of the parent algebra contains Q_l(a) for every a in A_F.
2. ||Q_l(a) - pi(a)|| <= (l^2/6) |||[D_F, [D_F, pi(a)]]|| <= (2/3) l^2 ||D_F||^2 ||a||.
3. Q_l([D_X, f]) = [D_F, Q_l(f)] for f in A_l.
4. With a' = Q_l(a) and b' = J_F Q_l(b)* J_F^-1, |||[a', b']|| <= (8/3) l^2 ||D_F||^2 ||a|| ||b||, and |||[[D_F, a'], b'] - [[D_F, pi(a)], pi°(b)]|| <= (16/3) l^2 ||D_F||^3 ||a|| ||b||.

So as l -> 0 with D_F fixed, D_F, J_F and Gamma_F stay exact, and the compressed observables converge to pi(A_F) with rate l^2. The finite order conditions hold in the limit exactly when they hold for the finite data.

Proof. From the definition of W, the expectation of T splits into the L- and R-parts, so Q_l(T) is l^-1 int of the Gamma_F-even part of R(x)* T(x) R(x). Put K = Gamma_F D_F. For even Z, ad_K Z = Gamma_F [D_F, Z], which is odd, and ad_K (Gamma_F [D_F, Z]) = -[D_F, [D_F, Z]]. So the even part of e^{-xK} T e^{xK} is cos(x ad_{D_F})(T). Likewise, E exchanges the L- and R-parts with a sign, so Q_l(E T) is l^-1 int of Gamma_F times the odd part of R* T R, and that odd part is -Gamma_F sin(x ad_{D_F})(T). The formula for constant a follows from cos(x ad_D)(Z) = (e^{ixD} Z e^{-ixD} + e^{-ixD} Z e^{ixD})/2.

Item 1: the map f -> Lambda(Q_l(f)), for a linear functional Lambda, is integration against a smooth density. If it vanishes on A_l, which contains all f of compact support in [0, l), the density vanishes on [0, l), hence everywhere by continuity. Item 2: by Taylor's formula with integral remainder, ||cos(x ad_D)(Z) - Z|| <= (x^2/2) ||ad_D^2 Z||, since cos(x ad_D) has norm at most 1 by the average formula; averaging over x gives l^2/6. Item 3: P commutes with D_X. It also follows from the two formulas by integration by parts, where the boundary term at l vanishes because f(l) is in C_F. Item 4: J_F commutes with D_F, so b' is the same average of pi°(b). Expand both commutators with the bound of item 2 and |||[D_F, Z]|| <= 2 ||D_F|| ||Z||. ■

The convergence is a statement about the maps Q_l, on the operator systems, in the setting of Connes and van Suijlekom (arXiv:2004.14115, sections 1 and 3). The generated algebra is not continuous at l = 0. An analytic example, from the referee: let M = [[0, 0, 1], [0, 0, 1], [1, 1, 0]], D = [[0, M], [M, 0]], with A_F = C^3 acting diagonally and equally on the two blocks. With P_1, P_2 the first two channel projections, [Q_l(P_1), Q_l(P_2)]_{12} = -l^2/3 + O(l^4), so the compressions of commuting projections generate a noncommutative algebra for small nonzero l. On the one-generation Standard Model data, a numerical closure with tolerance 10^-7 gives complex dimension 15 for the algebra generated

by pi(B), with B = C + H + M_3(C), and 30 for Q_l(B) at l = 0.3; these are numerical values with no certified rank.

Scalar fluctuations

Theorem 22.5. Let h be smooth with values in the anti-Hermitian elements of A_F. The self-adjoint one-forms of the parent are exactly the E pi(h). Let u' = -h u, u(0) = 1, and let w = u(l), a unitary of A_F. Then

```
D_X + E pi(h) = pi(u) D_{X_w} pi(u)\*,      X_w =
↪  pi_+(w)\* X pi_-(w),
```

and the window of D_{X_w} carries pi(w)* D_F pi(w). With the real structure, for A = E pi(h) we have J^ A J^^-1 = E J_F pi(h) J_F^-1, and D_X + A + J^ A J^^-1 has the window U_w* D_F U_w, with U_w = pi(w) J_F pi(w) J_F^-1, and this commutes with J_F. Moreover, with u~(x) = u(x) w*, which lies in A_l since u~(l) = 1, and U~ = pi(u~) J_F pi(u~) J_F^-1,

```
D_X + A + J^ A J^^-1 = U~ D_X U~\*.
```

So every scalar connection in this model is gauge equivalent to the original parent under the allowed endpoint gauge group (referee).

Proof. A one-form is sum f_1 [D_X, f_2] = E pi(sum f_1 f_2'), and E pi(h) is self-adjoint exactly when h is anti-Hermitian, since E is anti-Hermitian and commutes with pi(h) and pi is faithful. Every smooth such h arises: with b(x) = int_l^x h, we have b(l) = 0, so b is in A_l, and [D_X, b] = E pi(h). Next, pi(u)* D pi(u) = D + E pi(u* u'), and pi(u)* E pi(h) pi(u) = E pi(u* h u), so the sum is D with no connection term. At 0 the conditions are unchanged. At l, the substitution psi = pi(u) phi gives the conditions with X_w. Then tan(l pi(w)* D_F pi(w)) has the block X_w. For general w, this operator need not commute with J_F, so we use the spectrum and the isometry of Theorem 22.1 without its real-structure part. For the real fluctuation, J^ commutes with E, u(x) is replaced by pi(u) J_F pi(u) J_F^-1, using finite order zero, and U_w commutes with J_F, so the real window follows. The last formula follows by differentiating, with the endpoint condition. ■

Corollary 22.6 (the quadratic term as trans-

port). By Chamseddine-Connes-van Suijlekom (arXiv:1304.7583, section III), for unitary v in A_F, with U = v J v J^-1, one has U D U* = D + A_(1) + A~*(1)* + *A*(2), for the universal form v delta(v*). Taking v = w*, on the gauge orbit the fluctuation of the parent, which satisfies order one and has no quadratic term, gives on the moving window exactly the fluctuated operator of that framework (equations (10)-(11) of section III.2 there), quadratic term included, for finite data that fail order one. This answers the first open point of lemma 19 on these gauge orbits and with the moving window. For the two-point data with J = sigma_1 K and w = diag(e^{i alpha}, e^{i beta}), it gives the upper link m e^{-2 i (alpha - beta)}, which is m e^{2 i psi} of Corollary 19.2 with psi defined from v = w*. Theorem 22.5 and the covariant projection of lemma 14 share a transport principle. The fixed compression, as in lemma 13, is a different object: by Theorem 22.4, it gives Q_l(E pi(h)) = -[D_F, l^-1 int_0^l x pi(h) dx] + O(l^3 ||D_F||^3 ||h||) for h fixed, so that for constant h it is -(l/2)[D_F, h], while the first derivative of the transported operator is -l [D_F, h]. The difference is the contribution of the moving spectral projection.

No-go theorem 22.7 (the radial mode). *Hypotheses:* the parent above, on one interval, with fixed X, no bulk masses, and scalar inner fluctuations from the even algebra A_l, with the moving spectral window. *Conclusion:* the fluctuations leave all singular values of the finite Dirac operator fixed. In the usual Standard Model representation with nonzero Yukawa data, the normalized shifted Higgs quaternion runs through the unit quaternions, a three-dimensional gauge orbit, with no radial mode. Degenerate data can have a smaller orbit. A fixed compression can show changes in singular values at second order, so the window has to be named.

Proof. Theorem 22.5: D_F and pi(w)* D_F pi(w) are unitarily equivalent. The same holds with U_w. ■

Known escapes.

These lie outside the hypotheses, and each needs its own construction.

- The coupling X as a field on the wall, with its own kinetic term and potential.

- Connections odd for the grading, as in Corollary 22.2 and lemma 23: the fluctuation K -> K + delta K changes the singular values.
- Bulk masses that do not commute with the gauge action.
- Two or more internal dimensions, where a connection has curvature, although curvature alone does not prove a radial mode.
- An additional composite or other degree of freedom; a quantum effective action on the same configuration space does not by itself add a radial coordinate.
- A fluctuation of the finite data alone, as with the radial parameter of lemma 19, which is not the moving window.

Direction. On the interval, the even connection is pure gauge up to its boundary effect, so the finite orbit is all it can reach. The physical Higgs modulus must then be sought at the wall, in odd directions, or in more dimensions. At the 10 -> 9 endpoint this agrees with T10, where the Higgs is frozen.

The gauge sector and the 10 -> 9 family

Proposition 22.8. Consider the product of M^4 with this parent, with algebra of functions f(y, x) valued in A_F and f(y, l) in C_F. Then the gauge one-forms sum f_1 d_mu f_2 take values in C_F at x = l. Assume, in addition, a flat product, a positive five-dimensional Yang-Mills kinetic form, the trivial background, unrestricted endpoint variation in the allowed directions, and no additional boundary kinetic or mass terms. Then the directions in the Lie algebra of C_F have masses n pi/l, n >= 0, and the orthogonal directions have masses (n - 1/2) pi/l, n >= 1. The physical gauge algebra comes from the skew-adjoint elements through the real gauge representation, with its kernel and any unimodularity condition.

Proof. The first statement is direct evaluation at x = l. The rest is the interval spectrum in the gauge A_5 = 0, with Neumann conditions at 0, and with Neumann or Dirichlet conditions at l. ■

In the Standard Model data with nonzero Yukawas, M_W = M_Z = pi/(2l). The Lie algebra of C_F has dimension 10 before unimodularity; with the unimodularity

condition of Chamseddine-Connes-Marcolli (arXiv:hep-th/0610241, section 2.5), the massless modes are the photon and the gluons. These are the boundary conditions of Higgsless models (Csaki, Grojean, Murayama, Pilo and Terning, Phys. Rev. D 69 (2004) 055006), and the fermion formula tan(m L) = M_D L is equation (5.6) of Csaki, Grojean, Hubisz, Shirman and Terning (Phys. Rev. D 70 (2004) 015012); LITERATURE.md, section C. Boundary spectral geometry for the Standard Model has been studied by Chamseddine and Connes (arXiv:1008.3980, section 5). What is added here, as far as our searches went, is the combination: the real structure, the finite triple as an exact window, the wall algebra C_F, the compression formula and the fluctuations. A wider claim of novelty would need a fuller comparison.

Corollary 22.9 (the 10 -> 9 family). Fix a block M_0 for the light fermions and a block Theta/l on an orthogonal channel, compatible with J_F and the grading, with 0 < Theta < pi/2, for the top and if wanted for the Majorana block. Let D_F(l) have M(l) = M_0 (+) Theta/l, and let l -> 0. Then:

1. X(l) has the light block tan(l |M_0|) V_0* = l M_0* + O(l^3) and a heavy block with singular values tan(Theta): the light couplings vanish as m l at fixed mass m.
2. The heavy channel has mass Theta/l, the broken gauge directions have mass pi/(2l), and the other levels diverge.
3. The massless gauge sector is C_F. Let P_L be the projection reducing D_F(l) to the fixed light block D_L. For constant a, P_L Q_l(a) P_L = sinc(l ad_{D_L})(P_L pi(a) P_L), with ||P_L Q_l(a) P_L - P_L pi(a) P_L|| <= (2/3) l^2 ||D_L||^2 ||pi(a)||, also when pi(a) mixes light and heavy channels. The limiting light compression need not be a representation, since products can pass through the heavy subspace. The full estimate of Theorem 22.4 does not apply, because l ||D_F(l)|| tends to Theta. For example, with D_F(l) = sigma_1 (x) diag(m, Theta/l) and pi(a) = 1 (x) sigma_1, one has Q_l(a) = sinc(Theta - l m) pi(a), which tends to sinc(Theta) pi(a).
4. With l = pi/(2 M_W), we get X_q ~ (pi/2) m_q/M_W, which has the form of y_q = sqrt(2)

m_q/v -> 0 at fixed mass, as the coupling vanishes while the mass stays. Identifying X_q with the four-dimensional Yukawa coupling is a hypothesis.

5. The ratio is m_t/M_W = 2 Theta/pi < 1, and M_W/M_Z = 1.

So the 10 -> 9 proposal is realized at this level as a limit l -> 0, with W, Z and top at the tower scale and the light fermions keeping explicit masses. Item 5 does not reach m_t/M_W = sqrt(8/3), which belongs to the interior. This is a statement in the stated family: lowest level, no bulk mass, strict wall algebra. Bulk masses for the top, and a changed gauge wall condition, are directions beyond it.

A numerical comparison for the interior (Python). A separate model, with a finite wall scalar and Neumann conditions at 0, lies outside Proposition 22.8 and No-go theorem 22.7. In it, the mass of a broken direction satisfies t tan t = s, with t = M l and s = g^2 v^2 l^2/4 (for the Z, s/cos^2 theta_0), which has the form of M_k tan(M_k pi R) = -V in Csaki et al. (arXiv:hep-ph/0305237, section on mixed boundary conditions, read in the LaTeX source), with the sign convention of V there. If g'^2/g^2 = 3/5 is taken at all scales, M_W^2/M_Z^2 goes from 5/8 as s -> 0 to 1 as s -> infinity. It equals the measured 0.777 at M_W l = 1.03, with 1/l = 78 GeV and a next W level at 280 GeV; whether collider searches allow this would need the couplings, branching fractions and an applicable limit, which we have not analyzed. The top needs M_W l < 0.96 for m_t/M_W = sqrt(8/3), or M_W l < 0.73 for the measured ratio, in the lowest level with no bulk mass. So in the flat tree-level toy without bulk masses or running, the top and gauge requirements are incompatible. It does show the interior interpolating between the spectral-action value 5/8 and the endpoint value 1. The directions are bulk masses, warping and running.

What is input and what is derived

The finite data (A_F, H_F, D_F, J_F, Gamma_F) are all inputs, so the existence of the window is close to a reparametrization: X = tan(l D_F). The derived state-

ments are the wall algebra C_F for that input, the order conditions of the parent, the compression formula with its rate, the form of the fluctuations, and the gauge spectrum under the stated assumption. The window identifies D_F, J_F and Gamma_F exactly, while the represented algebra is obtained through compression and is not an exact representation of A_F at l > 0. In particular, this does not select A_F. T0 asks for a commutative parent: a natural next step is a two-stage parent, compressing functions on S^4 and CP^3 at level one to get A_F-valued functions on the interval, and then taking this window, where the difficulty is the same interface problem as after Result 7 of lemma 20.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^5, l ||M|| = 1.3: D u_+(a) = u_-(M a) and D u_-(b) = u_+(M* b) by finite differences, to about 10^-9; boundary conditions and J^ to about 10^-15; norm of W to 10^-13.
- Q_l(a) = sinc(l ad)(a), by quadrature, to 2 x 10^-8. The formula for Q_l(E T), with a nonconstant T, to 10^-14.
- One-generation Standard Model data with Yukawas and the rank-one Majorana block, with signs, order zero and order one exact on B = C + H + M_3(C), and commutant of D in B of real dimension 20. At l = 0.1 and 0.03, ||Q_l(a) - a + (l^2/6)[D, [D, a]]|| is 10^-5 and 4 x 10^-8, and the order-one defect of Q_l is 1.5 x 10^-4 and 1.2 x 10^-5, which scales as l^2.
- The dimensions of the generated algebras quoted above, 15 and 30, from a numerical closure with tolerance 10^-7, not retained and not certified; at smaller l the numerical rank depends on the tolerance.
- The roots of t tan t = s, and the values quoted in the comparison for the interior.
- On the two-point data with J = sigma_1 K: D + A_(1) + A~*(1)* + *A*(2) = U D U* to 4 x 10^-16, and the link of U_w* D U_w equal to m e^{-2 i psi}.

- The bounds of Theorem 22.4, items 2 and 4, on 30 random samples of the one-generation data: largest ratios to the bound 0.33, 10^-4 and 10^-3.

Part 23

Two gradings on an interval: masses as principal angles, the Higgs modulus, and exact W and Z masses

For STRATEGY T0b, T5b, T10 and T13, following lemma 22. Proved by Claude Opus 5.5, with the identities checked in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): four errors and one gap as first written, and amendments, all incorporated below. The simpler form of the mass formulas, and the proof of item 3 of Theorem 23.6, are by the referee.

Outcome. A self-adjoint boundary condition for the derivative on an interval is a grading at each wall. The whole finite Dirac operator is then the relative position of two gradings: its singular values are the principal angles between them, divided by l, and

```
D_F = (1/2l) Gamma_0 log(Gamma_1 Gamma_0).
```

With a bulk prealgebra, this gives four results.

1. Aligned walls compress a prealgebra stable under the grading to its even part, with the grading inherited, which is the step that Example 16.9 had to add by hand.
2. Odd scalar fluctuations move the second grading, so they change the singular values and supply the radial mode that No-go theorem 22.7 excludes for even fluctuations. For M_2(H) the field space is S^4, and the Higgs modulus is half the geodesic distance between the points chosen by the two walls.
3. With a wall connector on one channel and a five-dimensional Yang-Mills action with common normalization, the masses are exact: sin^2(M_W l) = sin^2(2 theta)/2 and sin^2(M_Z l) = (4/5) sin^2(2 theta), for universal fermions of mass theta/l. At small theta this gives M_W^2 = 2 m^2 and M_Z^2 = (16/5) m^2, the constants in the lower

bounds of the fully massive family of lemma 18, with M_W^2/M_Z^2 tending to 5/8 at both ends.

4. The heat-cutoff spectral action on the interval is a Jacobi theta function of theta, decreasing on [0, pi/2]. This family lies outside the hypotheses of the radial no-go of lemma 18: the radial direction is compact, with a stationary point at m = pi/(2l), the Connes-Lott value, where M_W = M_Z = 0.

Everything finite is still an input, and the parent algebra is matrix-valued. For a commutative parent, lemma 24 finds the window exactly, with spectral walls, while direct compression of scalar functions gives only R + R on the S^4 factor. The formulas for gauge masses depend on the assumed action. The relation to measured values fails in the flat toy, as stated at the end.

Two gradings

Let H be a finite-dimensional Hilbert space, and let Gamma_0, Gamma_1 be self-adjoint involutions on H. On L^2([0, l]; C^2) (x) H, with psi = (psi_L, psi_R), let D = E d/dx, where (E psi)_L = psi_R and (E psi)_R = -psi_L, on the domain

```
Gamma_0 psi_L(0) = psi_L(0),   Gamma_0 psi_R(0) =
 ↪  -psi_R(0),   Gamma_1 psi_L(l) = psi_L(l),
 ↪  Gamma_1 psi_R(l) = -psi_R(l).
```

Call it D(Gamma_0, Gamma_1), and let Gamma^ act as +1 on psi_L and -1 on psi_R.

Theorem 23.1.

1. D(Gamma_0, Gamma_1) is self-adjoint with compact resolvent, and it anticommutes with Gamma^.
2. Suppose Gamma_0 and Gamma_1 have eigenspaces of equal dimensions, and that -1 is not an eigenvalue of Gamma_1 Gamma_0. Let theta_k in [0, pi/2) be the principal angles between the +1 eigenspaces. Then K = (1/2l) log(Gamma_1 Gamma_0), with the principal branch, is anti-Hermitian, Gamma_0 K Gamma_0 = -K, and D_F = Gamma_0 K is self-adjoint, odd for Gamma_0, with singular values theta_k/l on the +1 eigenspace. Moreover Gamma_1 = e^{lK} Gamma_0 e^{-lK}.

3. With H_+, H_- the eigenspaces of Gamma_0, and X the H_- -> H_+ block of tan(l D_F), as in lemma 22, D(Gamma_0, Gamma_1) = D_X, and the spectral part of Theorem 22.1 gives the spectrum and the window, with no real structure needed. Conversely, every D_F odd for Gamma_0 with l ||D_F|| < pi/2 arises from exactly one such Gamma_1.
4. Let an antiunitary J_1 satisfy J_1^2 = -1, J_1 Gamma_i J_1^-1 = -Gamma_i for i = 0, 1. Then (J^ psi)_L = J_1 psi_R, (J^ psi)_R = -J_1 psi_L defines a real structure of signs (+, +, -) for D(Gamma_0, Gamma_1). With J_F = Gamma_0 J_1 up to sign, J_F^2 = 1, J_F Gamma_0 = -Gamma_0 J_F and J_F D_F = D_F J_F, which are the signs of lemma 22.

So J_F itself does not anticommute with Gamma_1 in general: J_F Gamma_1 J_F^-1 = -Gamma_0 Gamma_1 Gamma_0, the reflection of Gamma_1 through Gamma_0.

Proof. Item 1: the boundary form is [<psi_R, phi_L> - <psi_L, phi_R>] evaluated between 0 and l, which vanishes on the domain because L-values and R-values lie in orthogonal eigenspaces at each wall, and these are half of the boundary data.

Item 2: U = Gamma_1 Gamma_0 is unitary and Gamma_0 U Gamma_0 = Gamma_0 Gamma_1 = U^-1, so K is anti-Hermitian and anticommutes with Gamma_0. The eigenvalues of U are e^{+-2 i theta_k}, by the standard two-projection decomposition, and K then has eigenvalues +-i theta_k/l. Since e^{lK} Gamma_0 e^{-lK} = e^{2lK} Gamma_0 = U Gamma_0 = Gamma_1, the last formula follows.

Item 3: in the notation of Theorem 22.1, R(l) = e^{lK} carries H_+ to the graph {(C a, -S a)} = graph(-X*), and H_- to graph(X). These are the +1 and -1 eigenspaces of Gamma_1, which is the condition at the wall. The converse uses the same formula, with uniqueness from injectivity of the exponential on the stated range.

Item 4: J^ commutes with E by direct computation, and J^^2 = -J_1^2 = 1. The domain is preserved since J_1 exchanges the eigenspaces of each Gamma_i. For the last

statement, J_1 U J_1^-1 = U, so as J_1 is antilinear, J_1 K J_1^-1 = K in the sense of the principal branch, and J_F = Gamma_0 J_1 then satisfies J_F K J_F^-1 = -K and commutes with D_F = Gamma_0 K. ■

When theta_k = pi/2 the channel has type (c) of lemma 09, with no zero mode and lowest level pi/(2l), the Connes-Lott mass of Theorem A in the paper. There the principal logarithm and the isolated window cease to be unique.

The bulk prealgebra

Let P be a *-subalgebra of End(H), which need not commute with Gamma_0. Let

```
A(Gamma_0, Gamma_1) = {f in C^infinity([0, 1], P) :
 ↪  [f(0), Gamma_0] = 0, [f(1), Gamma_1] = 0},
```

acting pointwise and identically on both spin components.

Theorem 23.2.

1. A function f in C^infinity([0, 1], P) preserves the domain exactly when it lies in A(Gamma_0, Gamma_1). The commutators are [D, f] = E f', and Gamma^ commutes with every f, including those with values odd for Gamma_0.
2. Under the hypotheses of Theorem 23.1, item 2, with W the isometry of Theorem 22.1, Q(T) = l^-1 int_0^l E_0(R(x)* T(x) R(x)) dx, where E_0(Z) = (Z + Gamma_0 Z Gamma_0)/2 is the conditional expectation onto the even part.
3. *Aligned walls.* If Gamma_1 = Gamma_0, then D_F = 0, the window consists of constant L-values in H_+ and constant R-values in H_-, and Gamma^ restricts on it to Gamma_0. For an ambient constant multiplier a in P, Q(a) = E_0(a), and the image of the boundary algebra A(Gamma_0, Gamma_0) is E_0(P), by compactly supported profiles. If Gamma_0 P Gamma_0 = P, then E_0(P) is the even part of P, so the compressions of the prealgebra are exactly its even part, with the grading inherited from the even parent operator Gamma^. Without that hypothesis this can fail: for H = C^2, Gamma_0 = sigma_z and P generated by sigma_x + sigma_z, E_0(P) = span{1,

sigma_z}, while the even part of P is C.

4. *Order conditions.* Let J_1 be as in Theorem 23.1 and let P° = {J_1 b* J_1^-1 : b in P} commute with P. Then the parent has order zero and order one for A(Gamma_0, Gamma_1).

Proof. Item 1: as in Theorem 22.3, using that the conditions at a wall say that L- and R-values lie in the eigenspaces of Gamma_i. Item 2: as in the proof of Theorem 22.4. Item 3 is the case R = 1. Item 4: [D, f] = E f', and pointwise commutation gives both conditions. ■

Item 3 evades the inherited-grading obstruction of Result 11 of lemma 20, because hypothesis 4 there fails: for odd a, here Q(a) = 0, while there the coordinates are retained as Q(x_a) = Gamma_a/5. So at the level of a matrix-valued parent this addresses the question left in step 3 of the framework note: the prealgebra is the bulk algebra, the even algebra is what the window sees, and the grading is a boundary condition. The commutative-parent and retained-coordinate questions remain open, and Theorem 4 of lemma 24 shows a real difference: scalar functions compressed directly give R + R on the S^4 factor, and H + H needs the matrix algebra generated at the intermediate stage.

Remark 23.3 (the twist). On H = E (+) E*, with J(x, y) = (y*, x*), the operator J_1 = Gamma_0 J is the one that anticommutes with both gradings. For an odd anti-Hermitian h in P, J_1 h J_1^-1 is y -> +y h on E*, where J alone gives y -> -y h. So the opposite representation b° = J_1 b* J_1^-1 incorporates on the W-factor the grading automorphism sigma(b) = gamma_W b gamma_W. In this parent the twist comes from the requirement that J_1 preserve both boundary conditions. Twisted spectral triples for the grand symmetry (Devastato, Lizzi and Martinetti, JHEP 01 (2014) 042, arXiv:1304.0415; Devastato and Martinetti, Math. Phys. Anal. Geom. 20 (2017) 2, arXiv:1411.1320, Eqs. (1.8)-(1.11)) use a different automorphism: their rho exchanges left and right spinorial indices and algebra summands, while conjugation by gamma_W is inner on M_2(H) and fixes its centre. So this is a structural analogy between involutive automorphisms, and an identification needs a comparison of the representations, opposites

and first-order conditions.

Odd fluctuations and the radial mode

Theorem 23.4. Let h be smooth with values in the anti-Hermitian elements of P, and let u' = -h u, u(0) = 1, w = u(l). Then

```
D(Gamma_0, Gamma_1) + E h = u D(Gamma_0, w\* Gamma_1
↪  w) u\*.
```

So scalar fluctuations move the second grading along its orbit under the unitary group of P. If h has an odd part, the principal angles between Gamma_0 and w* Gamma_1 w change in general.

Proof. u* D u = D + E u* u', and u* E h u = E u* h u, so the connection cancels. Since u(0) = 1, the wall at 0 is unchanged. At l, psi = u phi gives the stated grading for phi. ■

This lies outside No-go theorem 22.7, whose hypotheses include the even algebra: an even connection gives a w commuting with Gamma_0, and the angles are kept, while an odd connection changes them. For a real fluctuation, add the conjugate one-form: D + E h + J^(E h)J^^-1, with generator of the form (L_F, R_F) in Example 23.5. The invariant is a double coset. With H_i = Stab(Gamma_i) in the connected unitary group G of P, a gauge transformation removing h has the form u(x) c, and it preserves both endpoints exactly when c is in H_0 and w c in H_1, that is, when w is in H_1 H_0. So the scalar moduli of the parent are the double cosets H_1 G / H_0, the relative position of the two gradings. This is with the moving window, as in lemma 22.

Example 23.5 (M_2(H) and S^4). Let W = H^2 = C^4, with the Clifford generators Gamma_a of lemma 16, and for a unit vector p in R^5 put Gamma_p = p_a Gamma_a. Each grading of W with one-dimensional quaternionic +1 and -1 eigenspaces is such a Gamma_p, so these gradings form S^4 = Sp(2)/(Sp(1) x Sp(1)), the quaternionic projective line. For two points, Gamma_q Gamma_p = p . q + (p wedge q), with eigenvalues e^{+- i d}, where d is the geodesic distance; so all the principal angles equal theta = d/2.

Take H = Hom(V, W) (+) Hom(W, V), J(x, y) = (y*,

x*), Gamma_0(x, y) = (gamma x, -y gamma) with gamma = Gamma_{p_0}, and K(x, y) = (theta/l)(F x, y F), with F an odd anti-Hermitian element of M_2(H), F^2 = -1. Then:

- D_F = Gamma_0 K has all singular values equal to theta/l: a universal Dirac mass, for all colours.
- The second grading is Gamma_1(x, y) = (Gamma_p x, -y Gamma_{p'}), where p is at distance 2 theta from p_0 and p' is its mirror image through p_0. J_1 = Gamma_0 J satisfies Theorem 23.1, item 4, while J alone does not.
- The coordinates (theta, q) on S^4, with F determined by the unit quaternion q, are the polar coordinates of a Higgs quaternion: theta is the radial coordinate, bounded by pi/2, and q gives three angular directions, undefined at the poles. At theta = pi/2 the finite logarithm and the lowest window are degenerate. Quotienting by the stabilizer leaves the radial interval. A canonically normalized field needs the scalar kinetic term, which we have not computed.

With bulk masses, by Theorem 12.6, the light masses are about (tan(theta)/l) Phi_Q Phi_U when small compared with the tower thresholds, with a universal matrix before flavour data; general finite-angle eigenvalues need the secular equation. This field space is the coset of the minimal composite Higgs model (Agashe, Contino and Pomarol, Nucl. Phys. B 719 (2005) 165, sections 2-3, with their h/f_pi corresponding to 2 theta). Their construction uses a warped interval and additional fermion and boundary data, so the shared coset does not identify the actions or potentials. The appearance here of the S^4 of the twistor window is an observation, and no identification between the two roles of S^4 is made.

The weak bosons

Take P = M_2(H) (+) M_4(C), acting on E = Hom(V, W) on the left and on the right. Assume the following.

- A five-dimensional Yang-Mills field valued in sp(2) (+) su(4), with the trace form Tr_W + Tr_V, so with one coupling (the normalization of the spectral action, g_3 = g_2 = sqrt(5/3) g_Y).

- At x = 0: Neumann for the stabilizer of p_0, Dirichlet for its complement. At x = l: Neumann for the stabilizer of p_l and of a connector, that is, the a with [a_W, Gamma_p] = 0, a_W w = lambda w and a_V v = lambda v, as in Results 2 and 5 of lemma 20, for a line w in the +1 eigenspace of Gamma_p and a line v in V; Dirichlet for the complement.
- The connector is prescribed gauge-breaking wall data here. If it is instead a term in the fermion boundary operator, as for a unit vector e in the +1 eigenspace of Gamma_1, f = J_1 e, K_c = |f><e| - |e><f| and Gamma_{1,c} = e^{eta K_c} Gamma_1 e^{-eta K_c}, then J_1 still anticommutes with Gamma_{1,c}, the parent order conditions are kept, and the affected fermion tower changes, so the universal fermion formula must be recalculated for that channel.

Theorem 23.6. With theta the angle above, the modes that are massless for all theta are exactly nine, the photon and the eight gluons, with three additional electroweak zero modes at theta = 0 and theta = pi/2. The lowest charged and neutral massive modes satisfy

```
sin^2(M_W l) = sin^2(2 theta)/2,
sin^2(M_Z l) = (4/5) sin^2(2 theta),
```

equivalently tan^2(M_W l) = sin^2(2 theta)/(1 + cos^2(2 theta)) and tan^2(M_Z l) = 4 sin^2(2 theta)/(1 + 4 cos^2(2 theta)).

The photon has 1/e^2 = (8/3)/g^2. Moreover:

1. As theta -> 0, with m = theta/l: M_W^2 -> 2 m^2 and M_Z^2 -> (16/5) m^2, so M_W^2/M_Z^2 -> 5/8.
2. As theta -> pi/2: M_W, M_Z -> 0, with M_W^2/M_Z^2 -> 5/8, while m -> pi/(2l).
3. At theta = pi/4: M_W l = pi/4, tan(M_Z l) = 2, and M_W^2/M_Z^2 = ((pi/4)/arctan 2)^2 = 0.5032..., the global minimum. So M_W^2/M_Z^2 stays in [0.5032, 0.625), with 5/8 as a limit at the ends, where the ratio itself is undefined.
4. m/M_W goes from 1/sqrt(2) at theta -> 0 to infinity at theta -> pi/2. It equals sqrt(8/3) at theta = 1.0589.

Proof. In the gauge A_5 = 0 the field is free, with cos(k x) on the Neumann directions at 0 and sin(k x) on the Dirichlet ones, and the condition at l holds in the frame rotated by Ad exp(theta F). The generators split according to the frequency of Ad F. Write L, R for su(2) on W_- and W_+, S = (L + R)/sqrt(2), A = (L - R)/sqrt(2), and O for the odd generators; Ad F has frequency 0 on S, and frequency 2 on the plane (A, O). So S = s cos kx, A = alpha cos kx, O = beta sin kx.

At l, in the rotated frame, A' = cos(2 theta) A + sin(2 theta) O and O' = -sin(2 theta) A + cos(2 theta) O, with O' = 0 at l, which gives beta = alpha tan(2 theta) cot(k l), and so A'(l) = alpha cos(k l)/cos(2 theta).

Charged directions. The connector forces R^+- to vanish at l, with L^+- Neumann. So S(l) = A'(l), and d/dx (S + A') = 0 at l. Substituting gives sin^2(kl) (1 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Neutral directions. Add B = b cos kx for the generator T_15 = diag(1, 1, 1, -3)/(2 sqrt 6) of su(4). With Y = sqrt(3/5) R + sqrt(2/5) B and Z' = -sqrt(2/5) R + sqrt(3/5) B, the connector forces Z' to vanish at l, with L and Y Neumann. The Neumann conditions give b = -sqrt(3) s, and Z' = 0 gives s = alpha/(4 cos(2 theta)). Substituting in the condition for L gives sin^2(kl) (1/4 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Massless modes and limits. At k = 0, a zero mode is a constant in the intersection of the two stabilizers. This consists of a_W = diag(q, q) on W_+ and W_-, identified by F, with q preserving the line of w, together with a_V = lambda (+) m, and with the trace condition this has dimension 1 + 8. In it, L = R and B/L = sqrt(2/3), which gives 1/e^2 = (1 + 1 + 2/3)/g^2. The limits follow by expanding. A second derivation, from the referee, uses the principal angles between the Neumann subspaces at the two walls: for each charged direction the squared sine of the angle is sin^2(2 theta)/2, and in the neutral sector the single nonzero value is (4/5) sin^2(2 theta); it also covers cos(2 theta) = 0, where the divisions above need a limit. For item 3, with a = sin(2 theta), the ratio is [arcsin(a/sqrt 2)/arcsin(2a/sqrt 5)]^2, and this decreases with a on (0, 1], since x/(sqrt(1 - x^2) arcsin x) increases. ■

The constants in item 1 are the values ||[D_F, a]||^2/||a||^2 in the Hilbert-Schmidt norm, for universal mass m, for a = L^+- and for the direction orthogonal to the photon and Z', which has A-component of squared length 4/5. These are the coefficients of the lower bounds for the poles of the fully massive family in Proposition 14 of lemma 18 (2 x < z_W, (16/5) x < z_Z); the small-x pole coefficients there are 3 and 24/5. The limit 5/8 is common to both, with different behaviour: the finite-action poles grow in the heavy limit of lemma 18, while the interval masses here tend to zero at the antipodal point. The spectral-action Taylor mass for universal m is M_W^2 = 3 m^2, which exceeds the value here by the factor 3/2 of Proposition F in the paper: there the mass matrix carries the relative heat coefficients of the action, and here we have assumed a five-dimensional Yang-Mills action, which gives the commutator norm alone. A direct matching claim needs the full parent spectral action, with its tower and boundary conditions.

Item 2 gives a geometric form of the behaviour that lemma 18 found: heavy fermions with light vectors. At antipodal points Gamma_p = -Gamma_{p_0}, so both walls have the same stabilizer in sp(2) again, and the fermion channels have type (c) with mass pi/(2l).

The spectral action on the interval

Proposition 23.7. For the heat cutoff and universal theta, with s = l Lambda, the trace over a channel is

```
Tr_channel exp(-D^2/Lambda^2) = 2 sum_{n in Z}
 ↪  exp(-(n pi + theta)^2/s^2).
```

This is proportional to the Jacobi function theta_3(theta, exp(-s^2)). It is strictly decreasing on [0, pi/2], with a maximum at theta = 0 and a minimum at theta = pi/2, where it is stationary, and it is symmetric under theta -> pi - theta. In the towerless regime s -> 0 at fixed x = theta^2/s^2, it tends to 2 e^{-x}, the form of the vacuum density in lemma 18 and Proposition 21.2.

Proof. The eigenvalues in a channel are +-(n pi + theta)/l, by Theorem 22.1, and the sum is over Z after combining the signs. By Poisson summation the sum equals (s/sqrt(pi)) sum_{k in Z} exp(-k^2 s^2) e^{2 i k theta}, and the Jacobi triple product writes this as a

constant times the product over n >= 1 of (1 + 2 q^{2n-1} cos(2 theta) + q^{4n-2}), with q = exp(-s^2). Each factor is positive and strictly decreasing in theta on [0, pi/2]. ■

In the form of the Jacobi function, the trace is (2 s/sqrt(pi)) theta_3(theta, exp(-s^2)). This family lies outside the hypotheses of No-go theorem 9 of lemma 18, which concerns finite internal data and a free ray V = c V_*: here the operator family is different, the radial direction is a compact orbit of relative gradings, and the tower, which the finite description omits, supplies a second branch with a stationary point at theta = pi/2. In the present family the literal action has its minimum at antipodal gradings, m = pi/(2l), with M_W = M_Z = 0 by Theorem 23.6, where the lowest eigenspace is degenerate. This is a vacuum with massless vectors, so it is not the physical point. The sign of the term agrees with the fermion loop in the Hosotani mechanism, where gauge-boson loops enter with the opposite sign. So an interior minimum could arise from that competition, which the classical spectral action does not contain. We have not computed it.

The one-loop potential in the same family

From the exact spectra, the products over the towers are, relative to theta = 0 and in Euclidean momentum x = p l,

```
fermion channel:  1 + sin^2(theta)/sinh^2(x),
charged vectors:  1 + sin^2(2 theta)/(2 sinh^2(x)),
neutral vector:   1 + (4/5) sin^2(2
 ↪  theta)/sinh^2(x).
```

So in the gauge-boson and fermion-loop form of the Hosotani mechanism, assuming the usual one-loop formula with three polarizations per vector, and with n_f = 2 x (number of physical Dirac species),

```
8 pi^2 l^4 V(theta) = int_0^infinity x^3 dx [ 3
 ↪  log(1 + s^2/(2 sinh^2 x)) + (3/2) log(1 + (4/5)
 ↪  s^2/sinh^2 x) - n_f log(1 + sin^2(theta)/sinh^2
 ↪  x) ],
```

with s = sin(2 theta). A Dirac top with three colours has n_f = 6, which needs a separately specified fermion sector, since the universal connection gives masses to all:

one generation has n_f = 16 and three have n_f = 48. With I = int x^3/sinh^2 x dx = (3/2) zeta(3) and I' = int x^3/cosh^2 x dx = (9/8) zeta(3), near theta = 0 the bracket is (10.8 - n_f) I theta^2 + o(theta^2), and with epsilon = pi/2 - theta, near theta = pi/2 it is V(pi/2) + (10.8 I + n_f I') epsilon^2 + o(epsilon^2), where we have written V for the left side.

Proposition 23.8 (referee). In the massless flat family with this content, for every n_f > 0, theta = pi/2 is the unique global minimum on [0, pi/2], since the gauge logarithms are nonnegative and vanish there, and the negative fermion logarithm is minimized at sin^2(theta) = 1. For n_f >= 10.8, V is strictly decreasing on the interior. Indeed, with t = sin^2(theta) and J(z) = int_0^infinity x^3 dx/(sinh^2 x + z), the derivative with respect to t is, up to a positive factor,

```
(1 - 2t) [6 J(2t(1 - t)) + (24/5) J((16/5) t (1 -
↪  t))] - n_f J(t).
```

For t <= 1/2, both arguments in the bracket are at least t, and J decreases; for t > 1/2 both contributions have the required sign.

For 0 < n_f < 10.8, on grids of theta (Python), for n_f in {1, 2, 3, 4, 6, 8, 10}, V has one interior maximum, at about 0.33 for n_f = 8, and no interior minimum. Theorem 9 of lemma 24, by Astra 6, proves this for all n_f in that range: exactly one interior stationary point, a strict maximum, using a log-concavity property of the integral.

These are negative results in an exact scope: one interval, no bulk masses, fermions of frequency 1 under F, this gauge content, one loop, this condition at the connector wall. As a formal leading approximation, not uniform at x = 0, V has the form B sin^2(2 theta) - A sin^2(theta) with A, B > 0, whose interior stationary point cos(2 theta) = A/(4B) is a maximum. *Bulk masses.* With bulk masses, the fermion product becomes cos^2(theta) + sin^2(theta) G(x), with G = (w_Q coth w_Q - c_Q)(w_U coth w_U - c_U)/x^2 and w = (c^2 + x^2)^{1/2} in the secular form of lemma 12. For a = c_Q + c_U != 0, G = 1 - a/x + O(x^-2), so the potential has angle-dependent ultraviolet divergences, and a finite value depends on a regulator and subtraction: as the ref-

eree showed, an earlier scan of general pairs here did not establish a potential, and it is withdrawn. The exceptional case c_U = -c_Q = c has a common bulk mass on the two mixed components and G = 1 + w^2/(x^2 sinh^2 w), which decays exponentially. Keeping the flat gauge products as a separate tree-level assumption, a scan for c in {0, 1/2, 1, 3/2, 2, 3, 5}, n_f in {1, 3, 6, 8, 10, 16, 48}, on a grid of 721 values of theta, finds no interior minimum (Python). General bulk masses need a gauge-invariant completion, a regulator and wall counterterms.

So an interior minimum needs another change: a negative coefficient of sin^2(2 theta), as from fermions of frequency 2 under F, which the bimodule E = Hom(V, W) does not contain; general bulk masses or other gauge form factors, as with warping; a wall scalar with its own potential; or tree-level terms at the walls. With h/f_pi = 2 theta, the leading flat form has the angular form alpha cos(h/f_pi) - beta sin^2(h/f_pi) of Agashe, Contino and Pomarol (section 2.1, Eqs. (19)-(23)) with spinorial SO(5) fermions, with alpha = A/2 and beta = -B, up to a constant. Their form factors can give the sign and competition needed for an interior minimum, so the flat results here do not exclude that construction. This points away from a pure flat Wilson line, toward those changed form factors or a wall scalar with its own potential, for the physical interior.

Comparison with measured values in the flat toy

These are numerical comparisons (Python), with no running and no bulk masses, in lowest levels, using the Particle Data Group 2026 values quoted by the referee, m_t = 172.60 GeV as a direct-measurement mass, M_W = 80.3625 GeV and M_Z = 91.1879 GeV, so m_t/M_W = 2.1478 and M_W^2/M_Z^2 = 0.7767. The ratio m_t/M_W reaches sqrt(8/3) at theta = 1.0589 and the measured value at theta = 1.1626, where M_W l = 0.541, so 1/l is about 148.5 GeV. There, M_W^2/M_Z^2 = 0.581. The wall-scalar model of lemma 22 covers [0.625, 1], and this family covers [0.5032, 0.625); in the wall-scalar model, fitting the measured vector ratio gives M_W l = 1.028, where the lowest no-bulk-mass fermion channel has m/M_W < 1.528. So neither flat family fits both ratios on its own, and the flat one-loop potential above does not select an interior point. Whether a tower

at 1/l of this size is allowed would need the couplings and an applicable experimental limit, which we have not analyzed. The directions are bulk masses for the light fermions, warping, running of the couplings, and families combining a Wilson line with a wall scalar. These negative statements hold in the flat tree-level toy only.

Open questions

1. The interior vacuum: the effective potential with a wall scalar, or with warping, or with general bulk masses, with a gauge-invariant completion, a regulator and wall counterterms. The flat cases above have none on the grids.
2. The relation between the two roles of S^4: the base of the twistor window, and the field space of the Higgs quaternion.
3. A commutative parent: functions on S^4 x I and CP^3 x I, with the gradings as local boundary conditions.
4. The factor 3/2 between the five-dimensional normalization and the spectral action: whether the spectral action on the parent itself, with its tower, reproduces the formula of Theorem 23.6.
5. The comparison with twisted spectral triples, Remark 23.3.
6. The contribution of a real connector to the fermion tower, the degenerate antipodal window, and the scalar kinetic normalization needed for a canonically normalized Higgs field.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^4 with l ||M|| = 1.25: Gamma_1 is an involution, Gamma_1 = exp(2 l K) Gamma_0; D_F = (1/2l) Gamma_0 log(Gamma_1 Gamma_0) to 3 x 10^-15; the principal angles over l equal the singular values of M to five digits.
- Sp(2) x SU(4), of dimension 25, with the Neumann and Dirichlet subspaces built from the stated stabilizers: the two-point boundary problem was solved as a linear system on a grid in k l. For theta = 0.2,

pi/4, 1.06 and 1.3, the two lowest nonzero roots are (0.279, 0.356), (0.785, 1.107), (0.647, 0.868), (0.373, 0.479), equal to the formulas of Theorem 23.6 to three digits, with other roots at pi/2 and above. The intersection of the two stabilizers has dimension 9 at each theta tested.

- Small theta: M_W^2/m^2 = 2.000 and M_Z^2/m^2 = 3.200.
- The universal connection on E (+) E*: J D = D J, Gamma_0 D = -D Gamma_0, all |eigenvalues| equal to theta/l, Gamma_1 = (left Gamma_p, right mirror), J_1^2 = -1, J_1 Gamma_1 = -Gamma_1 J_1, while J alone fails; J_1 h J_1^-1 is right multiplication by +h on E*, for odd h.
- The theta function: monotone on a grid for s = 0.3, 1, 2, with the towerless value at s = 0.3 equal to e^{-x} to five digits; Poisson summation and the triple product to 10^-16.
- The ratio M_W^2/M_Z^2 on a grid: range [0.5032, 0.6250], symmetric under theta -> pi/2 - theta to 10^-15, minimum at pi/4. At theta = 1.1625, M_W l = 0.541, and at theta = 1.0589, M_W l = 0.648.
- The one-loop integrals by adaptive quadrature (scipy), and by Gauss-Legendre quadrature on 400 nodes for the scan with c_U = -c_Q. The referee recomputed the derivative with scipy, found one sign change for each n_f in {1, 2, 3, 4, 6, 8}, with the stationary angle about 0.3382 at n_f = 8, and checked I = (3/2) zeta(3).

Part 24

Commutative cylinders, spectral walls, and the two-stage compression

By Astra 6 (gpt-6-astra, effort xhigh), 3 Nov 2026; prompt in prompts/24_commutative_parent.md. Reproduced as delivered, with links made relative. The section "Numerical check" at the end is by Claude Opus 5.5. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): no error in the displayed equations, and one gap in the general statement on compression maps. The record above the section "Referee amendments" stays as deliv-

ered, and those amendments apply to it.

A commutative cylinder admits an exact realization of the **operator window** of lemmas 22 and 23 when the two finite gradings are prescribed through nonlocal spectral boundary conditions. The higher K-modes then decouple as the radius of K tends to zero. Two qualifications remain essential: these boundary conditions restrict scalar functions at the walls, and compression of scalar functions through both stages differs from generating a matrix algebra between the stages. In particular, aligned walls give $\mathbb{R} \oplus \mathbb{R}$ on the quaternionic factor under direct scalar compression, whereas the intervening matrix-algebra construction gives $\mathbb{H} \oplus \mathbb{H}$. Local boundary gradings preserve all scalar functions, but their reduction to the unchanged four-section window has a precise restriction proved below. The radial no-go of lemma 22 survives rechecking in its stated scope. For the massless one-loop potential of lemma 23, an analytic argument strengthens the numerical statement to all $n_f > 0$.

1. Theorem — An exact spectral boundary construction.

Hypotheses and conventions. Let

$$K = \mathbb{CP}^3, \qquad \mathcal{H}_K = L^2(K, \Lambda^{0,*}T^*K \otimes \mathcal{O}(1)),$$

and let

$$A = \sqrt{2}(\bar\partial + \bar\partial^*), \qquad \gamma_K = (-1)^{\deg}.$$

Use the Fubini–Study metric at a fixed reference scale. Write

$$V = \ker A \simeq \mathbb{C}^4, \qquad \Pi = P_K,$$
$$\Delta_K = \inf(\operatorname{spec}|A| \setminus \{0\}) > 0.$$

Here $\gamma_K|_V = 1$. The radius parameter $r > 0$ replaces A by A/r, after the usual unitary identification of the L^2-spaces.

These are the analytic data of Example 16.9. Its S^4 option uses functions pulled back to this same twistor space. An actual operator on S^4 with the required kernel and compression map would need to be specified separately.

Let

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

On $L^2(I_\ell; \mathbb{C}^2 \otimes \mathcal{H}_K)$, set

$$\mathcal{D}_r = E\partial_x + Z\,A/r, \qquad \hat{\gamma} = Z\gamma_K.$$

All unspecified tensor factors are identities.

Choose self-adjoint involutions Γ_0, Γ_1 on V, each with two-dimensional positive and negative eigenspaces, and assume

$$-1 \notin \operatorname{spec}(\Gamma_1\Gamma_0).$$

Extend Γ_i by zero on $V^\perp$, writing $C_i = \Pi\Gamma_i\Pi$. Define $\operatorname{sgn} A = 0$ on V, and impose

$$\chi_0\psi(0) = \psi(0), \qquad \chi_\ell\psi(\ell) = \psi(\ell),$$

where

$$\chi_0 = ZC_0 - S\operatorname{sgn} A, \qquad \chi_\ell = ZC_1 + S\operatorname{sgn} A. \quad (1)$$

Then:

1. The H^1 realization of $\mathcal{D}_r$ is self-adjoint with compact resolvent and anticommutes with $\hat{\gamma}$.
2. $\Pi \otimes 1$ reduces the operator, including its domain.
3. On this reducing subspace, the operator is exactly $D(\Gamma_0, \Gamma_1)$ of lemma 23.
4. Every eigenvalue on the complementary subspace satisfies
$$|\lambda| \geq \Delta_K/r. \quad (2)$$

Proof. Put $B = SA/r$. Since $ES = Z$,

$$\mathcal{D}_r = E(\partial_x + B).$$

The operators χ_i are self-adjoint involutions and anticommute with E. Their positive eigenspaces are therefore maximal isotropic for the boundary form

$$[\langle\psi, E\phi\rangle]_0^\ell.$$

On $V^\perp$, condition (1) selects the negative spectral subspace of B at 0 and its positive spectral subspace at ℓ. These are APS conditions with the inward-normal sign reversed between the two walls. On $\ker B = \mathbb{C}^2 \otimes V$, condition (1) supplies a Lagrangian subspace:

$$\Gamma_i\psi_L = \psi_L, \qquad \Gamma_i\psi_R = -\psi_R.$$

For q in the complementary domain, integration by parts gives

$$\|\mathcal{D}_r q\|^2 = \|q'\|^2 + \|Aq/r\|^2 + [\langle q, Bq\rangle]_0^\ell \geq \|q'\|^2 + \frac{\Delta_K^2}{r^2}\|q\|^2. \tag{3}$$

The boundary term is nonnegative by the spectral choices.

An eigenbasis of A decomposes the complementary problem into finite-dimensional interval problems with maximal isotropic endpoint conditions. Each is self-adjoint. Estimate (3) identifies the domain of their self-adjoint direct sum with the stated H^1 domain and gives compactness of its resolvent. The finite-dimensional tangential kernel contributes an ordinary self-adjoint interval problem. This proves self-adjointness, compactness, and (2).

Every boundary projector commutes with Π, proving reduction. Finally, $\{\gamma_K, A\} = 0$ implies $\{\hat{\gamma}, \mathcal{D}_r\} = 0$, and $\hat{\gamma}$ preserves both boundary subspaces. ■

Define

$$K_F = \frac{1}{2\ell}\log(\Gamma_1\Gamma_0), \qquad D_F = \Gamma_0 K_F, \qquad R(x) = e^{xK_F},$$

with the principal logarithm. Let $P_\pm = (1 \pm \Gamma_0)/2$. The finite-window embedding is

$$W_\ell v(x) = \ell^{-1/2}\begin{pmatrix} R(x)P_+ v \\ R(x)P_- v \end{pmatrix}. \tag{4}$$

Direct differentiation gives

$$\mathcal{D}_r W_\ell = W_\ell D_F, \qquad \hat{\gamma} W_\ell = W_\ell \Gamma_0.$$

Thus the prescribed finite operator and grading are exact, provided $\Delta_K/r > \|D_F\|$. The next eigenvalue in absolute value is at least

$$\min\left\{\frac{\Delta_K}{r}, \frac{\pi - \ell\|D_F\|}{\ell}\right\}. \tag{5}$$

---

2. Proposition — Local walls and spectral walls have different scalar domain algebras.

Hypotheses. Use the product differential expression of Result 1.

For the local option, let $G_i(y)$ be smooth self-adjoint bundle involutions and impose

$$G_i\psi_L = \psi_L, \qquad G_i\psi_R = -\psi_R \quad \text{at } x = i.$$

Assume

$$[G_i, \gamma_K] = 0, \qquad [G_i, \sigma_A(y,\xi)] = 0 \quad (\xi \in T_y^*K), \tag{6}$$

where σ_A is the tangential principal symbol.

Then these local conditions give a self-adjoint elliptic realization with compact resolvent, and **every** smooth scalar function on $K \times I_\ell$ preserves its domain.

For the spectral option (1), the maximal smooth scalar $*$-algebra preserving the domain is

$$\mathcal{A}_{\mathrm{APS}} = \{f \in C^\infty(K \times I_\ell) : f(\,\cdot\,,0), f(\,\cdot\,,\ell) \text{ are constant on } K\}. \tag{7}$$

The same conclusion holds for the subalgebra pulled back from $S^4 \times I_\ell$.

Proof. For the local option, the boundary involution is ZG_i. It anticommutes with the normal matrix E. Condition (6) makes it anticommute with the principal symbol of the adapted tangential operator SA/r. Hence its positive eigenspace is an elliptic Lagrangian boundary condition. The corresponding local ellipticity and adjoint-domain criteria are described in Bär–Ballmann, Corollary 4.18 and Example 4.20. Primary reference

Scalar multiplication commutes pointwise with G_i, so it preserves the endpoint conditions and the H^1 domain. Moreover,

$$[\mathcal{D}_r, f] = E\,\partial_x f + Z[A, f]/r$$

is bounded for each $r > 0$.

For spectral walls, preservation by a scalar $*$-algebra is equivalent to

$$[M_{f_i}, \chi_i] = 0.$$

Independence of the spin matrices Z, S gives

$$[M_{f_i}, C_i] = 0, \qquad [M_{f_i}, \operatorname{sgn} A] = 0.$$

Since $C_i^2 = \Pi$, the first equation implies

$$[M_{f_i}, \Pi] = 0. \tag{8}$$

Consequently multiplication by f_i preserves $H^0(K, \mathcal{O}(1))$. If s is such a section, then

$$0 = \bar{\partial}(f_i s) = (\bar{\partial} f_i)s.$$

The sections of $\mathcal{O}(1)$ have no common zero, so $\bar{\partial} f_i = 0$ everywhere. A holomorphic function on compact connected $\mathbb{CP}^3$ is constant. Constants plainly preserve (1), proving (7). ■

The hypotheses in (6) are a sufficient local ellipticity condition. An arbitrary bundle grading supplies a pointwise isotropic condition, but self-adjointness on an H^1 domain and compact resolvent still require an ellipticity check. Likewise, prescribing a grading only on V leaves the boundary problem incomplete until a condition on $V^\perp$ is supplied.

---

3. Proposition — Compression of a local grading and the unchanged level-one window.

Hypotheses. Let G be a self-adjoint bundle involution acting by multiplication, and let

$$g = \Pi G \Pi|_V, \qquad Q_K(f) = \Pi M_f \Pi|_V.$$

Keep the level-one space and scalar representation of Example 16.9 unchanged.

Then

$$1_V - g^2 = \big((1-\Pi)G\Pi\big)^*\big((1-\Pi)G\Pi\big). \tag{9}$$

In particular, g is a grading exactly when $[G, \Pi] = 0$. In that event,

$$[g, Q_K(f)] = 0 \quad \text{for every scalar } f. \tag{10}$$

For either the full $\mathbb{CP}^3$ scalar algebra or the S^4 pullback algebra, this forces

$$g = 1_V \quad \text{or} \quad g = -1_V. \tag{11}$$

Proof. Expand $\Pi G^2 \Pi = \Pi$ using $1 = \Pi + (1 - \Pi)$; this gives (9). Vanishing of its positive right-hand side implies $(1 - \Pi)G\Pi = 0$, and self-adjointness gives the other off-diagonal block.

Since scalar multiplication commutes with G, reduction then gives (10). The compressed $\mathbb{CP}^3$ functions generate $M_4(\mathbb{C})$. The compressed real S^4 functions generate $M_2(\mathbb{H})$, whose complex-linear commutant on $\mathbb{C}^4$ is also $\mathbb{C}1$. Thus a self-adjoint involution satisfying (10) is ± 1. ■

For **aligned local walls** with $[G, A] = 0$, the finite zero-mode window is explicitly

$$v \longmapsto \ell^{-1/2} \begin{pmatrix} P_+(g)v \\ P_-(g)v \end{pmatrix}.$$

The induced grading is g, and scalar compression is

$$P_+(g)Q_K(f)P_+(g) + P_-(g)Q_K(f)P_-(g) = Q_K(f).$$

By (11), the unchanged four-section window has trivial grading. Its generated algebra remains $M_2(\mathbb{H})$ or $M_4(\mathbb{C})$, according to the scalar algebra chosen. The globally constant choices $G = \pm 1$ give concrete self-adjoint examples with this window and a separated higher spectrum.

There is also an explicit local field illustrating failure of reduction. Add a trivial auxiliary $\mathbb{C}^4$ bundle and put

$$G(y) = \sum_{a=1}^{5} y_a \Gamma_a^{\mathrm{aux}},$$

where $y \in S^4$ is the twistor projection and the Γ_a satisfy $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$. This is a local involution satisfying (6), but now the tangential kernel has dimension 16, and

$$g = \frac{1}{5} \sum_{a=1}^{5} \Gamma_a^{\mathrm{section}} \otimes \Gamma_a^{\mathrm{aux}}.$$

Using $\mathrm{tr}(\Gamma_a \Gamma_b) = 4\delta_{ab}$,

$$\mathrm{tr}(g^2) = \frac{16}{5} \neq 16.$$

Thus g is not a grading. Equation (9) even gives

$$\|(1 - \Pi)G\Pi\| \geq \frac{2}{\sqrt{5}},$$

independently of r. This is a bound on boundary-trace mixing; it does not identify the actual low spectral window of the coupled local problem.

A local $G(y)$ specifies a **field of gradings**. It does not select a marked point p. In the tautological quaternionic line bundle,

$$\Pi_y = \frac{1 + y_a \Gamma_a}{2},$$

the natural field $2\Pi_y - 1$ acts as $+1$ on the fibre $\operatorname{ran} \Pi_y$. By contrast, the finite grading

$$\Gamma_p = p_a \Gamma_a = 5 Q_K(p \cdot y)$$

uses a separately chosen point p. Its action on global section coefficients is nonlocal.

The restriction proved here concerns the unchanged level-one space, a local multiplication grading, and its compression as an actual involution. A different coupled window, additional fibres, or spectral boundary conditions remain available directions.

---

4. Theorem — The composed compression map and the aligned-wall algebra.

Hypotheses. Use the exact spectral construction of Result 1. For a scalar multiplier $f(y, x)$, let

$$T_f(x) = Q_K(f(\,\cdot\,, x)), \qquad \mathcal{E}_0(T) = \frac{1}{2}(T + \Gamma_0 T \Gamma_0).$$

Use real-valued functions and real algebra generation when discussing $M_2(\mathbb{H})$.

Then the compression directly from the scalar cylinder to the finite window is

$$\boxed{\Phi_\ell(f) = \frac{1}{\ell} \int_0^\ell \mathcal{E}_0\big(R(x)^* T_f(x) R(x)\big)\, dx.} \tag{12}$$

It is independent of r.

For aligned walls $\Gamma_1 = \Gamma_0 = \Gamma_p$, direct scalar compression on the S^4 factor generates

$$\mathbb{R} \oplus \mathbb{R}. \tag{13}$$

If instead one first generates $M_2(\mathbb{H})$ and then applies the interval compression, the result is

$$\mathbb{H} \oplus \mathbb{H}. \tag{14}$$

Proof. Insert (4) on both sides of the multiplication operator. Orthogonality of the two interval spin components gives exactly (12).

For real S^4 functions, Example 16.9 implies

$$Q_K(C^\infty(S^4, \mathbb{R})) = \mathrm{span}_{\mathbb{R}}\{1, \Gamma_1, \dots, \Gamma_5\}. \tag{15}$$

Indeed all compressed functions are quaternionic Hermitian matrices; that space has real dimension six, and the displayed six matrices belong to the image and are independent.

For aligned walls $R = 1$. Clifford multiplication gives

$$\mathcal{E}_p(\Gamma_a) = p_a \Gamma_p.$$

Therefore the image of direct scalar compression is

$$\mathrm{span}_{\mathbb{R}}\{1, \Gamma_p\}.$$

This space is already the algebra $\mathbb{R} \oplus \mathbb{R}$, proving (13).

On the other hand,

$$\mathcal{E}_p(M_2(\mathbb{H})) = \{a \in M_2(\mathbb{H}) : [a, \Gamma_p] = 0\} = \mathbb{H} \oplus \mathbb{H},$$

proving (14). The noncommuting products needed to generate the quaternionic diagonal blocks have been inserted between the two compressions in this second procedure. ■

On the single $\mathbb{CP}^3$ factor, real scalar compression is onto the Hermitian 4×4 matrices. A chosen $2+2$ aligned grading therefore gives all Hermitian block-diagonal matrices and generates

$$M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).$$

For the **chosen two-sheet coarse correspondence** of Example 16.9, with

$$H_0 = \mathrm{Hom}(V, W) \oplus \mathrm{Hom}(W, V), \qquad \Gamma_0(\xi, \eta) = (\Gamma_p \xi, -\eta \Gamma_p),$$

the same calculation gives

$$\boxed{\mathrm{alg}_{\mathbb{R}}(\text{direct aligned scalar compression}) = \mathbb{R} \oplus \mathbb{R} \oplus M_4(\mathbb{C}).} \tag{16}$$

The V-side matrices act on the left of η and commute with its right grading, so that factor survives unchanged.

Endpoint restrictions do not change these finite-dimensional images. If a linear functional vanishes on compressions of functions supported in $0 < x < \ell$, its smooth matrix-entry density vanishes in the interior and hence everywhere. Finite-dimensional separation then gives equality with the unrestricted image, exactly as in Result 7 of lemma 20.

At the intermediate stage, however, spectral walls still give scalar endpoint values. Even after algebra generation, those values remain scalar. Enlarging them to the full stabilizers of Γ_0, Γ_1, as in lemma 23, is another change of the parent algebra. Equality of finite compression images does not identify the parent gauge groups or one-forms.

---

5. Theorem — Radius collapse, interval collapse, and controlled mixing.

Hypotheses. First use the exact construction of Result 1. Identify the varying Hilbert spaces by metric rescaling. For resolvent formulas, fix units in which the nonreal spectral parameter is i.

Then, at fixed ℓ,

$$\left\|(\mathcal{D}_r - i)^{-1} - \iota_\Pi (D(\Gamma_0, \Gamma_1) - i)^{-1} \iota_\Pi^*\right\| \le \frac{r}{\Delta_K}. \tag{17}$$

Thus radius collapse gives the entire matrix interval operator in generalized norm-resolvent sense.

If D_F is fixed, $\ell \to 0$, and

$$\Gamma_1(\ell) = e^{\ell \Gamma_0 D_F} \Gamma_0 e^{-\ell \Gamma_0 D_F},$$

then $r \to 0$ and $\ell \to 0$ decouple both towers and leave the finite operator D_F. No relation between the two rates is required in this exactly reducing construction.

Proof. The resolvent is block diagonal with respect to Π. Estimate (2) bounds its complementary block by r/Δ_K,

proving (17). For the finite-window limit, (5) bounds the complementary resolvent by

$$\max\left\{\frac{r}{\Delta_K}, \frac{\ell}{\pi - \ell\|D_F\|}\right\},$$

which tends to zero. ■

A useful mixing statement requires additional hypotheses. Suppose a transformed problem has a common domain and the form

$$\widetilde{\mathcal{D}}_r = \mathcal{D}_r + V_r, \qquad V_r = V_r^*, \qquad \|V_r\| \le b_r.$$

Write

$$v_r = \|(1-\Pi)V_r\Pi\|.$$

For an eigenvector $\psi = p+q$, with $p = \Pi\psi$, $q = (1-\Pi)\psi$, and $|\lambda| \le L$, assume

$$g_r = \Delta_K/r - b_r - L > 0.$$

Then

$$\boxed{\|q\| \le \frac{v_r}{g_r}\|p\|, \qquad \|\text{Schur correction at } \lambda\| \le \frac{v_r^2}{g_r}.} \tag{18}$$

Proof. The complementary eigenvalue equation gives

$$q = -(\mathcal{D}_{r,Q} + V_{QQ} - \lambda)^{-1} V_{QP} p.$$

The inverse has norm at most g_r^{-1}. Substituting this formula into the p-equation gives the second bound. ■

For bounded b_r, v_r, both errors are $O(r/\Delta_K)$. If a common-domain construction instead produces terms of order $1/\ell$, sufficient conditions include

$$\frac{rv_r}{\Delta_K} \to 0, \qquad \frac{rv_r^2}{\Delta_K} \to 0,$$

together with convergence of the Π-diagonal perturbation.

An arbitrary local $G_i(y)$ generally fails $[G_i, \Pi] = 0$. Although the differential expression commutes with Π, its domain then does not. Such a boundary change is not automatically a bounded perturbation satisfying (18). The

explicit local field in Result 3 has nonvanishing boundary mixing at every radius. No convergence to the prescribed two-gradings operator is established here for that nonreducing local problem.

At the level of observables, radius collapse leaves (12) unchanged. For an x-independent $T = Q_K(f)$,

$$\|\Phi_\ell(f) - \mathcal{E}_0(T)\| \le \ell \|D_F\| \, \|T\|. \tag{19}$$

If T is even for Γ_0, the stronger formula of Theorem 22.4 applies:

$$\Phi_\ell(f) = \operatorname{sinc}(\ell \operatorname{ad}_{D_F})(T),$$

$$\|\Phi_\ell(f) - T\| \le \frac{\ell^2}{6} \|[D_F, [D_F, T]]\|. \tag{20}$$

Equation (19) follows by integrating $\|R^* T R - T\| \le 2x\|D_F\|\|T\|$; equation (20) follows by taking the even part of the conjugation series.

Consequently there is convergence of the **compression maps on their scalar operator systems**. The limit is $\mathcal{E}_0 Q_K$. It does not turn that image into every element of $\mathcal{E}_0(P)$; Result 4 computes the difference explicitly. Algebra generation remains a separate operation.

---

6. Proposition — The second sheet, reality, even algebra, and connector remain prescribed data.

Hypotheses. Consider the scalar cylinder construction above and, separately, its extension using the two-sheet coarse correspondence of Example 16.9.

The resulting distinctions are:

Structure	What follows under the stated choices
Second sheet and the two Hom spaces	They are inputs in the coarse correspondence. A single four-section cylinder window does not supply them.
$J_1^2 = -1$, $J_1 \Gamma_i = -\Gamma_i J_1$	These are additional compatibility conditions on the prescribed gradings and antiunitary.

Structure	What follows under the stated choices
$\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C})$	This is the derived even part after choosing the quaternionic real form, the grading, and intermediate algebra generation. Direct aligned scalar compression gives (16).
Connector and $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$	The connector's lines and coupling are chosen. Their intertwining algebra is then derived.

Proof. The one-factor construction starts with $V = \mathbb{C}^4$, and (4) has domain V. The two-sheet construction instead starts with

$$(W \otimes \overline{V}) \oplus (V \otimes \overline{W}).$$

Geometrically, its analytic realization uses product factors and a restricted scalar algebra, as Theorem 16.7 specifies. It is additional correspondence data over the scalar spaces.

For reality, suppose an antiunitary $\mathcal{J}_1$ on the tangential Hilbert space satisfies

$$\mathcal{J}_1^2 = -1, \quad \mathcal{J}_1 A = -A\mathcal{J}_1, \quad [\mathcal{J}_1, \gamma_K] = 0,$$

and its restriction J_1 to V anticommutes with both Γ_i. Then

$$\widehat{J} = E\mathcal{J}_1$$

preserves (1) and obeys

$$\widehat{J}^2 = 1, \qquad [\widehat{J}, \mathcal{D}_r] = 0, \qquad \widehat{J}\hat{\gamma} = -\hat{\gamma}\widehat{J}.$$

On the window,

$$J_F = \Gamma_0 J_1.$$

These identities follow directly from the stated commutation relations. They specify sufficient conditions; the natural $\mathcal{O}(1)$ cylinder has not supplied such an extension or its order-zero property here.

In particular, the quaternionic antiunitary j used to recognize $M_2(\mathbb{H})$ commutes with the Γ_a. It does not itself satisfy the required anticommutation with Γ_p.

The even-algebra assertion is the centralizer calculation in Result 4 after intermediate generation.

Finally, choose

$$w \in W_+, \quad v \in V, \quad \|w\| = \|v\| = 1, \quad \kappa \neq 0,$$

and prescribe

$$T_\kappa \xi = \kappa(w^* \xi v)\, v w^*.$$

Its intertwining conditions require the lines $\mathbb{C}w, \mathbb{C}v$ to be reducing, with the same scalar action. A quaternion preserving $\mathbb{C}w$ lies in its chosen copy of $\mathbb{C}$; $v^\perp$ carries arbitrary $M_3(\mathbb{C})$, and W_- carries arbitrary $\mathbb{H}$. Hence

$$\mathcal{A}(T_\kappa) = \{(\lambda \oplus q, \lambda \oplus m)\} \simeq \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}),$$

as in Result 2 of lemma 20. ■

A marked $p \in S^4$ fixes a quaternionic line, but choosing w also requires a complex line within it. The second line v, the coupling, and an interface realizing the connector remain necessary. Scalar agreement at marked points has unchanged compression image by Result 7 of lemma 20, so it supplies none of this selection by itself.

---

7. No-go theorem — Recheck of the radial statement in lemma 22.

Hypotheses. Let the parent be exactly the one-interval model of lemma 22: fixed wall coupling X, zero bulk mass, the even algebra $\mathcal{A}_\ell$, scalar inner fluctuations Eh with $h^* = -h$, and the **moving spectral window**. For real fluctuations, include the conjugate one-form and assume the stated order-zero representation.

Then every fluctuated finite operator has the same singular values as D_F. This is an exact finite-amplitude spectral statement; it uses no action or loop expansion.

Proof. Solve

$$u' = -hu, \qquad u(0) = 1, \qquad w = u(\ell),$$

and set $v(x) = u(x)w^*$. Then

$$v(\ell) = 1, \qquad v(0) = w^*.$$

Since h is even, w preserves the boundary splitting at 0; the identity at ℓ preserves the fixed coupled wall. Thus v preserves the original domain, and

$$vD_X v^* = D_X + Eh.$$

For the real fluctuation, the analogous implementing unitary is

$$vJ_F vJ_F^{-1};$$

order zero separates the two factors in its derivative.

The full operators are unitarily equivalent, so their moving spectral windows are unitarily equivalent. In the finite identifications of lemma 22, the operators are $w^* D_F w$, or $U_w^* D_F U_w$ in the real case. Their singular values coincide. ■

The moving-window hypothesis matters. For example, with $D_F = m\sigma_1$ and $h = i\sigma_3$, fixed compression gives

$$Q_\ell(Eh) = -\frac{1 - \cos(2m\ell)}{2m\ell}\sigma_2.$$

Adding a multiple of this operator to the fixed $m\sigma_1$ changes its singular values at second order.

The verdict agrees with the author's amended theorem and Terra's independent endpoint-preserving proof. In the usual nondegenerate Standard Model representation, the normalized shifted Higgs quaternion lies on the unit-quaternion orbit; its norm is fixed.

The known escapes are a variable wall coupling, odd connection directions, suitable bulk masses, additional internal directions with nontrivial connection data, or additional elementary or composite fields. A quantum effective action on the same configuration space does not add a radial coordinate. The proposed $K \times I_\ell$ parent has tangential directions and tangential one-forms, so this theorem does not decide its full fluctuation space.

---

8. Lemma — A logarithmic concavity property of the one-loop integral.

Hypotheses. Let

$$\mathcal{J}(z) = \int_0^\infty \frac{x^3}{\sinh^2 x + z}\, dx, \qquad z > 0,$$

and define

$$e(z) = -\frac{z\mathcal{J}'(z)}{\mathcal{J}(z)}.$$

Then $0 < e(z) < 1$, and e is nondecreasing.

Proof. Positivity and

$$0 < \frac{z}{\sinh^2 x + z} < 1$$

show that $e(z)$, expressed as a weighted average of this fraction, lies strictly between zero and one.

Put $x = e^v$, $z = e^u$. Then

$$\mathcal{J}(e^u) = \int_{\mathbb{R}} \exp\left(4v - \log(\sinh^2(e^v) + e^u)\right)\, dv. \tag{21}$$

The function

$$b(v) = 2\log\sinh(e^v)$$

is convex, because, with $x = e^v$,

$$b''(v) = 2x(\coth x - x\ \operatorname{csch}^2 x) > 0.$$

The last inequality is equivalent to $\sinh x \cosh x > x$.

The function $(u, v) \mapsto \log(e^{b(v)} + e^u)$ is therefore convex. The integrand in (21) is jointly log-concave. Marginal integration preserves log-concavity by Prékopa's theorem, so $u \mapsto \log \mathcal{J}(e^u)$ is concave. We use precisely Theorem 6, p. 342, of Prékopa's original paper.

Finally,

$$\frac{d}{du}\log \mathcal{J}(e^u) = -e(e^u).$$

Concavity makes this derivative nonincreasing, proving the assertion. ■

---

9. Theorem — The massless one-loop potential has no interior local minimum.

Hypotheses. Let the action be exactly the one-loop determinant displayed in lemma 23: a flat interval, zero

bulk masses, frequency-one fermions, the stated charged and neutral gauge towers and multiplicities, and the prescribed connector wall condition. Assume that no fermion connector modifies the displayed fermion tower, and add no wall potential or other angle-dependent term.

Let $n_f > 0$, set $t = \sin^2\theta$, and write

$$\mathcal{V}(t) = 8\pi^2\ell^4 V(\theta).$$

Then:

1. $t = 1$, equivalently $\theta = \pi/2$, is the unique global minimum.
2. If $n_f \geq 54/5$, the potential is strictly decreasing on the interior.
3. If $0 < n_f < 54/5$, there is exactly one interior stationary point, and it is a strict maximum.

Proof. Define

$$F(z) = \int_0^\infty x^3 \log\left(1 + \frac{z}{\sinh^2 x}\right) dx, \qquad F'(z) = \mathcal{J}(z).$$

The exact potential is

$$\mathcal{V}(t) = 3F(2t(1-t)) + \frac{3}{2}F\Big(\frac{16}{5}t(1-t)\Big) - n_f F(t). \quad (22)$$

Its gauge terms are nonnegative and vanish at $t = 1$. Since F is strictly increasing, the fermion term is uniquely minimized there. This proves item 1.

Differentiate:

$$\mathcal{V}'(t) = (1-2t)\left[6\mathcal{J}(2t(1-t)) + \frac{24}{5}\mathcal{J}\Big(\frac{16}{5}t(1-t)\Big)\right] - n_f\mathcal{J}(t). \quad (23)$$

For $t \geq 1/2$, this is strictly negative.

For $0 < t < 1/2$ and $c \geq 2$, put

$$R_c(t) = (1-2t)\frac{\mathcal{J}(ct(1-t))}{\mathcal{J}(t)}.$$

Let $a = ct(1-t)$. Then $a \geq t$, and Result 8 gives $e(a) \geq$

$e(t)$. Consequently,

$$\begin{aligned}\frac{d}{dt}\log R_c(t) &= -\frac{2}{1-2t} - \frac{1-2t}{t(1-t)}e(a) + \frac{e(t)}{t} \\ &\le -\frac{2}{1-2t} + \frac{e(t)}{1-t} \\ &< -\frac{2}{1-2t} + \frac{1}{1-t} < 0.\end{aligned}$$

Thus every R_c is strictly decreasing. Since $\mathcal{J}(0)$ is finite and positive,

$$R_c(0+) = 1, \qquad R_c(1/2-) = 0.$$

Therefore

$$H(t) = 6R_2(t) + \frac{24}{5}R_{16/5}(t)$$

decreases strictly from 54/5 to zero. Equation (23) becomes

$$\mathcal{V}'(t) = \mathcal{J}(t)\,[H(t) - n_f] \quad (0 < t < 1/2).$$

For $n_f \ge 54/5$, its sign is negative. For $0 < n_f < 54/5$, it changes sign exactly once, from positive to negative. Since $t = \sin^2\theta$ is strictly increasing on the interior, the same stationary-point classification holds in θ. ■

This proves the massless statement for the entire parameter range, including the previously numerical range. No numerical computation is used in this proof.

The recorded author and Terra referee calculations agree on the sampled massless cases. The logarithmic-concavity proof above is an additional single-agent derivation; it has not received a fresh independent model review in this turn. The recorded agreement therefore should not be described as a multi-model review of this new proof.

The escapes remain changes to the stated determinant: frequency-two fermions, modified form factors through warping or other geometry, suitable bulk-mass sectors, wall potentials, and wall kinetic or interaction terms. The full effective action of the proposed $K \times I_\ell$ parent, including its K-tower, has not been computed here.

---

10. Proposition — Scope of the bulk-mass scans.

Hypotheses. Retain the flat gauge terms of (22) as an independent assumption. Replace the fermion factor by the expression in lemma 23:

$$1 + t(G(x) - 1),$$
$$G(x) = \frac{(w_Q \coth w_Q - c_Q)(w_U \coth w_U - c_U)}{x^2},$$
$$w_i = \sqrt{x^2 + c_i^2}.$$

If $c_U = -c_Q = c$, then the potential is finite and $t = 1$ is its unique global minimum for every $n_f > 0$. The reported absence of additional interior **local** minima at nonzero c remains a numerical statement here.

If $a = c_Q + c_U \neq 0$, the displayed unrenormalized integral has angle-dependent ultraviolet divergences.

Proof. For opposite masses,

$$G(x) = 1 + \frac{w^2}{x^2 \sinh^2 w}, \qquad w = \sqrt{x^2 + c^2}.$$

The added fraction is positive and decays exponentially at infinity. The negative fermion logarithm is strictly decreasing in t. The gauge logarithms are nonnegative and vanish at $t = 1$, proving the global-minimum assertion.

For general masses,

$$G(x) = 1 - \frac{a}{x} + \frac{a^2}{2x^2} + O(x^{-3}),$$

and hence

$$\log(1 + t(G - 1)) = -\frac{ta}{x} + \frac{t(1-t)a^2}{2x^2} + O(x^{-3}).$$

Multiplication by the integration measure $x^3 dx$ produces angle-dependent divergences. A finite potential therefore requires a specified completion, regulator, and counterterms. ■

The opposite-mass scan in lemma 23 samples finitely many masses, multiplicities, and angles. It cannot exclude unsampled local minima. General bulk masses and altered gauge form factors remain directions for constructing an interior vacuum; these conclusions leave that question open.

Consequence for T0 and T0b

For T0, a commutative scalar representation with prescribed spectral walls gives an exact interval operator window and controlled tower decoupling, while local walls preserve the full scalar algebra under the ellipticity hypotheses above. The derivation of a nontrivial local grading and the desired algebra from one coupled scalar window remains open, and intermediate algebra generation must be stated explicitly. For T0b, the second sheet, compatible reality, quaternionic grading choice, and connector remain inputs; their algebraic consequences can be derived once those choices are supplied. Neither the radial theorem nor the specified one-loop potential closes the commutative-parent problem.

Referee amendments

From Terra 6.1. These apply to the text above.

- **Result 2.** Write G_0 at x = 0 and G_l at x = l. Equation (6) supplies the principal-symbol condition of Bar-Ballmann, Corollary 4.18 and Theorem 4.11; their Example 4.20 assumes more. On the unaugmented Dolbeault bundle, the Clifford representation is irreducible, so a fibre endomorphism commuting with every sigma_A is scalar, and a smooth involution of this form is +1 or -1. Additional fibres and other local elliptic projectors lie outside this sufficient criterion.
- **Result 3.** The statement on marked points concerns the natural field displayed there. A local field can encode a chosen point: on the auxiliary bundle, G_p(y) = p_a Gamma_a^aux is a local constant field whose coefficients contain p.
- **Result 4.** Equation (16) has the two Hom spaces and the two scalar algebras of Example 16.9 as inputs. The endpoint argument proves equality of the finite-dimensional linear images, and does not identify the parent algebras, norms, gauge groups or one-forms.
- **Result 5, gap in the final assertion.** Convergence of compression maps needs a common domain and a collapse map. With aligned walls, D_F = 0 and f_l(y, x) = 6 (x/l)(1 - x/l), both endpoint values vanish while Phi_l(f_l) = 1. A cor-

rect form: for F(y, s) fixed on K x [0, 1] and f_l(y, x) = F(y, x/l), Phi_l(f_l) tends to E_0 Q_K(int_0^1 F(., s) ds). For x-independent ambient multipliers the limit is E_0 Q_K, by (19). The rate discussion after (18) also needs control of the denominator: a sufficient addition is r (b_r + L) <= (1 - epsilon) Delta_K. The case b_r = O(1/l) follows from the stronger estimate |D_{r,Q}| >= (Delta_K^2/r^2 + pi^2/(4 l^2))^{1/2}, from the one-ended Poincare inequality on the positive and negative B-spectral components.

- **Result 6.** At the finite level, if the gradings are Gamma_p, Gamma_q, a unit n orthogonal to p and q gives J_1 = Gamma_n j, with J_1^2 = -1 and the required anticommutation. Extending this choice to the parent remains a separate task.
- **Lemma 8 and Theorem 9.** An independent check of the key monotonicity, without log-concavity or Prekopa's theorem: with rho(s) = (arsinh sqrt(s))^3/(2 sqrt(s (1 + s))), J(z) = int_0^infinity rho(s)/(s + z) ds; with L(v) = log rho(e^v), L'(v) = 3 tanh(x)/(2x) - (1 + tanh^2 x)/2 for v = 2 log sinh x, which is strictly decreasing. For z = e^u, with the probability density in y proportional to e^y rho(e^{u+y})/(1 + e^y), e(e^u) is the mean of 1/(1 + e^y), and its derivative in u is Cov(1/(1 + e^y), L'(u + y)) > 0, as both are strictly decreasing. Theorem 9 is for theta in [0, pi/2]. At theta = pi/2 the determinant has its continuous value, while the principal-logarithm window is degenerate. Its hypotheses are the flat massless one-loop determinant, its gauge multiplicities and connector wall condition, frequency-one fermions, no fermion-connector modification, and no other angle-dependent term.
- **Result 10.** The divergence is stated for nonzero fermion multiplicity, with no cancellation by additional sectors. The opposite-mass scan in lemma 23 is a numerical result on its sampled parameters.

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026, in a temporary directory, not kept: E_p(Gamma_a) = p_a Gamma_p to 10^-15, which

gives equation (13); tr(g^2) = 16/5 for the auxiliary field of Result 3; on grids, e(z) lies in (0, 1) and is nondecreasing for z in [10^-3, 50]; H(t) is strictly decreasing on (0, 1/2), with H near 54/5 at small t and near 0 at t = 1/2; for n_f in {1, 3, 6, 8, 10}, H - n_f has one sign change, with the stationary angle 0.34 at n_f = 8, which agrees with the referee of lemma 23. Results 1, 2, 5 and 6 were read and not separately computed.

Part 25

The weak-boson sum rule and the effective dimension of the interval

For STRATEGY T5b, T13 and T1, following lemma 18, lemma 22 and lemma 23. Proved by Claude Opus 5.5, with checks in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors and two gaps as first written, and amendments, all incorporated below.

Outcome. Unfolded, the interval parent is a circle of length 2l, and the parent operator is -i d/dx (x) 1 + 1 (x) D_F: the tower is D_F shifted by the momenta k_n = n pi/l. So the exact quadratic kernel of lemma 18 applies termwise over the tower. For small masses, at leading order, for the heat cutoff, the Taylor mass of a vector with constant profile in the transported frame is

```
M^2 = ((3 - d_eff)/2) ||[D_F, T]||^2 / ||T||^2,
 ↪  d_eff(s) = d log S / d log s,      S(s) = sum_{n
 ↪  in Z} exp(-n^2 pi^2/s^2),   s = l Lambda,
```

in Hilbert-Schmidt norms, where d_eff is the spectral dimension at the scale of the cutoff of the unfolded circle, equivalently of the paired Neumann and Dirichlet interval: 0 for l Lambda « 1 and 1 for l Lambda » 1. For the Standard Model representation, at the same order,

```
M_W^2 = ((3 - d_eff)/24) sum_f N_f m_f^2,
 ↪  m_t/M_W = (8/(3 - d_eff))^{1/2} for the top
 ↪  alone,
```

with M_W^2/M_Z^2 = 5/8 for all s. So the relation 8 M_W^2 = sum N_f m_f^2 of the asymptotic spectral action is the zero-dimensional value, with m_t/M_W = sqrt(8/3). A resolved interval gives 12 M_W^2 = sum

N_f m_f^2, with m_t = 2 M_W, as five-dimensional covariance requires for this operator on the circle. The endpoint m_t = 2 M_W is known in flat gauge-Higgs models with suitable representations (references below); in the inspected records the referee did not find the coefficient (3 - d_eff)/24 or its crossover, and priority remains open pending a wider search. The factor 3/2 in Proposition F of the paper, the finite-action coefficient relative to the Yang-Mills commutator quotient, is the ratio of these two endpoints. The crossover is at l Lambda about 2.

Unfolding

Let D_X = E d/dx be the parent of lemma 22, and let D_0 + E Gamma_F D_F be its form after the transport of Corollary 22.2, where D_0 is the operator with X = 0.

Proposition 25.1. On the circle x in [-l, l] with periodic conditions, for psi in the domain of D_0, put

```
Psi_Q(x) = Q_L(|x|) + i sgn(x) Q_R(|x|),
 ↪  Psi_U(x) = U_R(|x|) - i sgn(x) U_L(|x|).
```

After multiplying by 1/sqrt(2), this is a unitary from the parent Hilbert space onto L^2(circle) (x) (H_+ (+) H_-), which maps the H^1 domain of D_0 onto the periodic H^1 domain, and it carries D_0 + E Gamma_F D_F to

```
-i d/dx (x) 1 + 1 (x) D_F.
```

So the signed spectrum is k_n + mu_i, with k_n = n pi/l for n in Z and mu_i the eigenvalues of D_F. An even operator T = T_Q (+) T_U on H_F, constant in x in this transported frame and acting identically on both spin components, is carried to 1 (x) T.

Proof. Since Q_R(0) = Q_R(l) = 0 and U_L(0) = U_L(l) = 0, the functions Psi are continuous and periodic, and every smooth function on the circle arises from its even and odd parts. For x > 0, -i d/dx Psi_Q = Q_R' - i Q_L' = (D psi)_L + i (D psi)_R on Q, and -i d/dx Psi_U = -i U_R' - U_L' = (D psi)_R - i (D psi)_L on U, which are the transforms of D_0 psi; x < 0 is the same with the odd parts. For the connection term, with M* the block of D_F, the term E Gamma_F D_F sends Psi_Q to M* Psi_U and Psi_U to M Psi_Q. ■

In this form D_F is a constant connection on the circle, with holonomy exp(2 i l D_F) up to the convention for parallel transport, and the two walls are the fixed points of the reflection that carries D_F to -D_F. With U = exp(-i pi Gamma_0/4), the identity of lemma 23 reads Gamma_1 Gamma_0 = exp(2 l Gamma_0 D_F) = U exp(2 i l D_F) U*. This form of the interval as a circle with a reflection is the usual orbifold description. It also gives a one-line proof of the spectrum in Theorem 22.1.

The frame. By Corollary 22.2, D_X = R (D_0 + E Gamma_F D_F) R*, with R(x) = exp(x Gamma_F D_F). A generator T constant after transport is R(x) T R(x)* in the original frame of D_X, which generally varies with x and has an odd bulk component. Conversely, a constant generator in the original frame need not preserve the coupled wall, if it is broken. So the field used here requires an admissible enlarged gauge algebra, as in the prealgebra P of lemma 23, and it is generally not a field of the even algebra of lemma 22.

The kernel over the tower

Let A_mu(y) = a_mu(y) T on M^4 = R^4, with T even and constant along x, and consider D = -i gamma^mu (d_mu + i A_mu) + gamma_5 V with V = -i d/dx (x) 1 + 1 (x) D_F. Theorem 1 of lemma 18 has this form, with V_0 = V, whose eigenvalues are k_n + mu_i, and T_{n i, n' j} = delta_{n n'} T_{ij}.

Theorem 25.2. With d_ij = mu_i - mu_j, for the heat cutoff t = 1/Lambda^2, the kernel is

```
C_t(q^2) = (1/(8 pi^2)) sum_{n in Z} sum_{ij}
 ↪  |T_ij|^2 [ (2/t + d_ij^2 + q^2) I_t((k_n +
 ↪  mu_i)^2, (k_n + mu_j)^2; q^2) - (e^{-t (k_n +
 ↪  mu_i)^2} + e^{-t (k_n + mu_j)^2})/t ],
```

with I_t as in lemma 18. The sum converges, and it holds without expansion in D_F. Expanding in D_F, with C_t = C_0 + q^2 C_1 + ..., and with t = 1, so with k_n = n pi/(l Lambda) dimensionless,

```
C_0 = (1/(8 pi^2)) ||[D_F, T]||^2 sum_n e^{-k_n^2}
 ↪  (1 - (2/3) k_n^2) + O(||D_F||^4),
C_1 = (1/(8 pi^2)) (2/3) ||T||^2 sum_n e^{-k_n^2} +
 ↪  O(||D_F||^2).
```

So, with <k^2> = sum k_n^2 e^{-k_n^2} / sum e^{-k_n^2}, at fixed s and for fixed nonzero T,

```
M^2 = C_0/C_1 = (3/2 - <k^2>) ||[D_F, T]||^2/||T||^2
 ↪  + O(||D_F||^4 / Lambda^2).
```

Proof. V and T split into finite-dimensional blocks indexed by n, so equation (1) of lemma 18 applies to symmetric finite sets of blocks, and the limit is justified by Gaussian bounds, uniform for D_F in a bounded neighborhood of zero. The same bounds justify the Taylor expansion under the sum. Terms odd in k cancel between n and -n: exactly, the bracket at q = 0 is d_ij^2 int_0^1 E(u) [1 - u(1-u)(2k + mu_i + mu_j)^2] du, with E(u) = exp(-(1-u)(k + mu_i)^2 - u (k + mu_j)^2), whose next term is d_ij^2 e^{-k^2} ((2k^3 - 5k)/3)(mu_i + mu_j), and the bracket at order q^2 begins e^{-k^2} (2/3 - (2k/3)(mu_i + mu_j)). For the leading term of the expansion, put a_i = (k + mu_i)^2 = k^2 + p mu_i + mu_i^2 with p = 2k. To second order in mu,

```
I(a_i, a_j) = e^{-k^2} [1 - (p/2)(mu_i + mu_j) -
 ↪  (mu_i^2 + mu_j^2)/2 + (p^2/6)(mu_i^2 + mu_j^2 +
 ↪  mu_i mu_j)],
```

and e^{-a_i} = e^{-k^2} [1 - p mu_i - mu_i^2 + p^2 mu_i^2/2]. Therefore

```
2 I(a_i, a_j) - e^{-a_i} - e^{-a_j} = -(p^2/6)
 ↪  d_ij^2 e^{-k^2},
```

and adding d_ij^2 I gives d_ij^2 e^{-k^2} (1 - p^2/6) = d_ij^2 e^{-k^2} (1 - (2/3) k^2). Finally, sum_ij |T_ij|^2 d_ij^2 = ||[D_F, T]||^2. For C_1, at D_F = 0 the bracket in equation (3) of lemma 18 has d_ij = 0 and integrates to 2/3. ■

Differentiating directly, <k^2> = (1/2) d log S/d log s = (s/2) d log S/ds, with S(s) = sum_n exp(-n^2 pi^2/s^2). This S is the heat trace of the Dirac operator of the unshifted circle of length 2l at t = 1/Lambda^2, equivalently the paired Neumann and Dirichlet trace on the interval, with S_N = (S + 1)/2 and S_D = (S - 1)/2 separately. So d_eff = d log S/d log s = -2 d log Tr e^{-t D^2}/d log t is its spectral dimension at that scale. For nonzero D_F, the parent trace has e^{-t (k_n + mu_i)^2}, and S is the unperturbed trace used in the leading expansion. Hence, at leading order,

```
M^2 = ((3 - d_eff)/2) ||[D_F, T]||^2/||T||^2.
```

The two limits. We first take the coefficient at small D_F/Lambda at fixed s, and then the limit in s. Explicitly, d_eff ~ (4 pi^2/s^2) exp(-pi^2/s^2) as s -> 0, and d_eff = 1 - 4 s^2 exp(-s^2) + O(s^2 exp(-2 s^2)) as s -> infinity.

1. For s « 1, only n = 0 contributes. Then d_eff = 0, the factor is 3/2, and this is the towerless regime of lemma 18 and Proposition F of the paper.
2. For s » 1, the sum is a Gaussian integral, with <k^2> = 1/2. Then d_eff = 1, and the factor is 1. This is forced by covariance on the unfolded circle: V is the fifth component of a five-dimensional Dirac operator with constant A_5 = D_F, and the local five-dimensional heat coefficient contains F_{MN} F^{MN}, with F_{5 mu} = i [D_F, A_mu] up to convention and with the same coefficient as F_{mu nu}^2. So the mass is the Hilbert-Schmidt commutator norm alone, as at small angle for the Yang-Mills parent of Theorem 23.6. On a general space with boundary, the bulk coefficient alone does not exclude boundary contributions; here the operator with these transported fields is unitarily equivalent to the circle problem, which is the reason they are absent.

In table form, at small masses:

s = l Lambda	0.3	1	1.5	2	2.5	3	5
(3 - d_eff)/2	1.5000	1.4990	1.3935	1.1414	1.0240	1.0022	1.0000

Corollary 25.3 (the sum rule). For the three-generation Standard Model representation, with the literal particle and antiparticle trace, with Dirac masses small compared with Lambda and with 1/l, and with T a charged generator of SU(2)_L, at leading order,

```
M_W^2 = ((3 - d_eff)/24) sum_f N_f m_f^2,
```

which at d_eff = 0 is 8 M_W^2 = sum N_f m_f^2, and at d_eff = 1 is 12 M_W^2 = sum N_f m_f^2. For the top alone,

```
m_t/M_W = (8/(3 - d_eff))^{1/2},
```

from sqrt(8/3) = 1.633 at d_eff = 0 to 2 at d_eff = 1. The neutral direction scales with the same factor, so M_W^2/M_Z^2 = 5/8 at all s to this order, and the massless directions keep zero mass coefficient, while their kinetic coefficients can change. These are leading-order statements: already in the towerless top-only family, M_W^2 = (3/8) m_t^2 - (23/128) m_t^4/Lambda^2 + … (equation (19) of lemma 18).

Proof. The trace gives ||T||^2 proportional to Q_11 = 12 and ||[D_F, T]||^2 proportional to K_11 = sum N_f m_f^2, which gives the coefficient. At d_eff = 0 this is the CCM relation (hep-th/0610241, Eq. (4.16)), as recovered in Proposition 4 and Remark 16 of lemma 18 at small x. The factor is common to all generators in Theorem 25.2. ■

In terms of Yukawa couplings, with m_t = y_t v/sqrt(2) and M_W = g v/2, this reads y_t^2 = 4 g^2/(3 - d_eff), so y_t = 2 g/sqrt(3) at d_eff = 0, the top-dominance value of CCM (Eq. (5.27) there), and y_t = sqrt(2) g at d_eff = 1.

Remark (positive Laplace cutoffs). For f(z) = int e^{-tz} dnu(t) as in equation (4) of lemma 18, with t measured in units of 1/Lambda^2, the same computation gives at leading order

```
M^2 = ((3 - d_nu)/2) ||[D_F, T]||^2/||T||^2,
 ↪  S_nu(s) = int S(s/sqrt(t)) dnu(t),     d_nu = d
 ↪  log S_nu/d log s = 2 <t k^2>_nu,
```

where the average has weights dnu(t) e^{-t k_n^2}. Under the moment hypotheses there, dominated convergence gives the same endpoints, and the crossover and its corrections depend on nu. A general smooth cutoff, such as one flat near zero, need not belong to this class.

Scope

- This is the Taylor quotient of a prescribed constant-profile vector, at leading order in D_F, about a prescribed background and not a demonstrated vacuum. Taylor and pole masses differ in general. Calling it the leading term of a lowest-mode mass also needs an admissible gauge field, a positive kinetic

form and an isolated lowest branch; on the circle, translation invariance separates the zero Fourier sector, within which the generator matrix must still be diagonalized. For finite angles in the Yang-Mills parent, the exact masses are in Theorem 23.6, which expands as M_W^2 = 2 m^2 - (4/3) m^4 l^2 + … and M_Z^2 = (16/5) m^2 - (64/75) m^4 l^2 + …, in agreement with d_eff = 1 at leading order. That theorem is for its own Yang-Mills boundary problem, and it does not establish the full finite-cutoff gauge spectrum of the spectral action.

- T must be even and constant in the transported frame, so it preserves both walls there, and it needs the enlarged gauge algebra noted above. Vectors that are Dirichlet at a strict wall, as in Proposition 22.8, are a different boundary problem.
- All fields have flat profiles on one interval, with no bulk masses. In particular, for the top-only statement, the kinetic term counts all twelve doublets of three generations on the same interval, with the same S. With bulk masses to localize light fields, the spectra and overlaps change, and the kinetic coefficient must be recomputed from all eigenstates and overlaps.
- The spectral action is taken as a function of the fermion operator at a matching scale, with no running.
- For a boundary with reflection, the walls contribute heat-kernel boundary terms in general. Here the unfolding removes them for this operator and these generators.
- Other priority: the endpoint m_t = 2 M_W appears at tree level in Cacciapaglia, Csaki and Park, Fully radiative electroweak symmetry breaking, arXiv:hep-ph/0510366, section 5, for a suitable rank-four representation, and a limit approaching 2 M_W in Adachi and Maru, arXiv:1804.06012, Eq. (3.23). These are representation-dependent mechanisms, and they do not give the spectral-action crossover. Universal propagation alone does not impose it in ordinary universal extra dimensions (Cordero-Cid et al., arXiv:1108.2926, section 2).

Reading

This connects three items of the project.

1. *The emergence of a dimension.* The heat dimension of the interval, which distinguishes the labels 10 and 11, enters the mass relation directly. As the cutoff resolves the interval, the coefficient moves from the finite-space value to the five-dimensional value.
2. *Proposition F and lemma 13.* The factor 3/2 is the finite-action coefficient relative to the Yang-Mills commutator quotient, and in this constant-profile calculation the two are the two ends of one function. Proposition F also contains a bounded-fermion against tower-scale obstruction and overlap conditions, which keep their separate hypotheses.
3. *The measured ratio.* With the Particle Data Group 2026 listings, m_t = 172.60 +- 0.27 GeV as a direct-measurement Monte Carlo mass, and M_W = 80.3625 +- 0.0077 GeV, the ratio is 2.148, which would formally correspond to d_eff = 1.27, outside the range. With a running mass of about 163 GeV (Yoon and Peskin, arXiv:1810.12352, section 3.3 and Eq. (23), for a pole mass of 173 GeV, with about 147 GeV at 2 TeV), it is 2.03, which corresponds to d_eff = 1.06. Dividing a running mass by an on-shell W mass is only a formal orientation and not a matching test at a common scale, and no running from the matching scale has been computed, so we draw no conclusion from this comparison. The known spectral-action value y_t = 2 g/sqrt(3) is the d_eff = 0 boundary condition. A tower near the weak scale, as in the separate unrefereed estimate of the steer note, item 9, would use the d_eff = 1 boundary condition y_t = sqrt(2) g, at a matching scale of that order, in a specified effective theory with thresholds and a renormalization prescription. In a one-loop orientation with universal bulk fields (the steer note, item 11, no referee), the running approaches an infrared quasi-fixed point near 211 to 232 GeV, nearly independently of the boundary value, so that family does not support the closeness of 2 M_W to the running top mass.

Numerical check

By Claude Opus 5.5, in Python (numpy), 3 Nov 2026, in a temporary directory, not kept. The kernel of lemma 18, with quadrature in u, was summed over the tower, with explicit 4 x 4 level blocks for the operator on the interval, independently of the unfolding.

- Random complex K_1 and Hermitian T_Q, T_U on C^3, with ||K|| = 10^-3 min(s, 1): the ratio of C_0/C_1 to the Hilbert-Schmidt quotient agrees with 3/2 - <k^2> to six digits, for s in {0.3, 1, 1.5, 2, 2.5, 3, 5, 10}.
- Three generations, top only, with T = sigma_x/2 on the U-fields: m_t/M_W agrees with (8/(3 - d_eff))^{1/2} to six digits on the same grid, from 1.63299 at s = 0.3 to 2.00000 at s = 5.
- The neutral direction, with weights for lepton and colour: M_Z^2/m^2 goes from 4.801 to 3.200 for universal mass, and M_W^2/M_Z^2 stays at 0.625 to four digits.
- Finite mass in the top-only family: M^2/m^2 at s = 5 is 0.2500, 0.2498, 0.2477, 0.2409, 0.2241, 0.1478 for m/Lambda = 0.02, 0.1, 0.3, 0.6, 1, 2, and at s = 0.5 it is 0.3749, 0.3732, 0.3591, 0.3145, 0.2277, 0.0555, the latter matching lemma 18.

Part 28

Two sheets as the two ends of one connected commutative parent

For STRATEGY T0 and T0b, following lemma 16 (Example 16.9), lemma 24 (Proposition 6) and lemma 09. Propositions 28.1 to 28.4 and 28.13 are by Claude Opus 5.5. Results 28.8 to 28.12 and 28.14 to 28.21 are by Astra 6, job 28 (report, with the full proofs; prompt), checked by Claude Opus 5.5 by hand and in Python. Proposition 28.13 and the numerics are refereed by Terra 6.1 (report), with its corrections applied. The window is a conjecture with numerical support; the numbers come from a radial solve in Python (numpy and scipy), described in the section Numerical check; the script is not kept.

Correction, 3 Nov 2026. The first version of Result 28.5 and of the numbers in Remark 28.7 (commits

dd1b14c, 530fa06) used the wrong sign of the spin connection in the Python slice operator. That changed the singular values and the boundary fibre parts, so those numbers are withdrawn, including the next singular value 1.04, the value 1.35 on the representation 16, and the boundary fibre parts -0.011 to -0.853. The exponents at the origin and the multiplicities are unaffected. Astra 6 did not reproduce the old boundary values and its proofs do not use them. We found the error when that operator gave a core zero mode in the representation 20, against the exact count of Proposition 28.11. The singular values, fibre parts and densities below are recomputed, and the script tests the operator against exact statements: {A_0, A_1} = 0, A_0^2 equal to the Casimir formula on S^4, D^2 on (0, 0)-forms with values in O(1) at the Fubini-Study radius equal to j(j + 4), and A_2 equal to its curvature contraction (the last test proposed by Terra 6.1).

Update, 4 Nov 2026. Lemma 29 proves Conjecture 28.6 in case (b) for the metric class of its Setting 29.0, and in case (a) at the Fubini-Study radius, refereed by Terra 6.1. Item 3 below and the Status of Proposition 28.13 describe the position before that lemma.

Outcome.

1. *The mechanism is proved at the level of compression maps.* A smooth function on the disc bundle X has independent values on the zero section S^4 and on the boundary CP^3, and at the zero section it is a function on S^4 alone (Proposition 28.8). For prescribed windows over this X, real functions generate exactly M_2(H) + M_4(C) at every localization scale (Theorem 28.21). In Example 16.9 the second sheet and the restriction to S^4 were inputs, so this is a connected source for them.
2. *The two ends of an actual local operator are understood.* There is a self-adjoint operator with a local wall (Theorem 28.9). The core carries exactly four states in the representation 4 (Proposition 28.11). At the wall, large mass keeps a tangential mixing (Lemma 28.12). For the pullback connection the boundary mode then has the fibre part -w(rho) in closed form, with w(rho) = 2(rho^2+2)^2/(9+2(rho^2+2)^2), which reaches

the level-one value of Example 16.9 only as rho -> infinity (Proposition 28.13). This is a statement on the finite wall model, and its convergence from the full operator is numerical. A connection preserving the line summands in a collar, with the Fubini-Study metric at the wall, gives the level-one value in that model.

3. *The isolated window of the whole operator stays a conjecture* (Conjecture 28.6): what is missing is the normal reductions with their errors, the matching, and a coercive estimate uniform in the mass and in all representations of Sp(2) (Lemma 28.14).
4. *The algebra M_2(H) + M_4(C) is the algebra of the limit map.* Generated algebras need not converge when compression maps do (Counterexample 28.17). At the sample points resolved numerically, the core mode has a nonzero density breaking the quaternionic structure, empirically of order lambda^2; if it is nonzero in the continuum, Proposition 28.16 gives M_4(C) for the core block at those masses. This is numerical evidence and does not establish the exact algebra at any finite mass.

Setting

Let G = Sp(2), K = Sp(1)_a x Sp(1)_b and H = Sp(1)_a x U(1)_b, so that G/K = S^4, K/H = S^2 and G/H = CP^3. Let S be the bundle G x_K (1, 2), so that CP^3 = P(S). The normal bundle of the zero section is G x_K (1, 3), and away from it X = (0, 1] x CP^3, with radial coordinate x and S^2 fibres of radius rho(x), rho(x) ~ x at x = 0. Charges of U(1)_b are normalized so that the doublet of Sp(1)_b has +-1. The isotropy representation is (2)_{+-1} (+) (1)_{+-2}. The spinors of X restrict on a slice to Sigma_6 = Sigma^+ (+) Sigma^-, with c(d_x) = Z the chirality of Sigma_6, where

```
Sigma^+ = (1)_2 (+) (1)_0 (+) (2)_{-1},   Sigma^- =
↪  (2)_1 (+) (1)_0 (+) (1)_{-2}.
```

The representation 4 of G restricts to (2)_0 (+) (1)_1 (+) (1)_{-1}, that is (2, 1) (+) (1, 2) of K. Let Gamma_a be the Clifford generators on 4, with Gamma_5 commuting with K and equal to +-1 on the two summands, and let x_a(g) be defined by lambda(g) Gamma_5

lambda(g)^-1 = x_a(g) Gamma_a.

Proposition 28.1 (the twist).

1. For k != 0, O(k) on CP^3 does not extend to a line bundle on X, since H^2(X) = H^2(S^4) = 0 while O(k) has degree k on the fibre S^2.
2. The pullback of S to X restricts on CP^3 to O(1) (+) O(-1), as the tautological splitting. So E = S^3(S), which is pulled back from S^4 and has trivial connection along the fibres, restricts to O(3) (+) O(1) (+) O(-1) (+) O(-3). The summand O(3) gives the Dolbeault operator on O(1) of Example 16.9, since Sigma (x) O(3) = Lambda^{0,*} (x) O(1).
3. On E the section Phi(v) = phi(|v|) (v-hat . T), with T the spin-3/2 generators and phi odd and smooth, is smooth and G-equivariant, with zero set exactly S^4. In the frame of the slice it is phi(x) tau, with tau diagonal in the summands, with eigenvalues +-3/2, +-1/2.

So we take the Callias-type operator T = D_X(E) + i Phi on spinors of X tensored with E, and the self-adjoint operator D-hat = [[0, T^*], [T, 0]], with the grading Gamma-hat = diag(1, -1). At x = l we take the local condition Gamma-hat Z = -1 on the doubled spinor, that is u_even(l) = 0 for T and v_odd(l) = 0 for T^*. It is a chirality condition, elliptic and local, and it preserves the domain under all smooth functions on X.

Proposition 28.2 (multiplicities). On a slice, Hom_H(4, Sigma^+ (x) E) and Hom_H(4, Sigma^- (x) E) each have dimension 5. For the summand O(3), the dimensions are 1 and 0. So on 4, the operator on the summand O(3) with its U(1) connection vanishes for every G-invariant metric on the slice, and its kernel contains H^0(CP^3, O(1)) = 4 for every fibre radius. This concerns the separate line connection. The pull-back connection of E does not preserve the summands, and its slice operator is invertible for small fibres (Proposition 28.10).

Proof. Tensoring the lists above with the charges of E gives the multiplicities. The operator is odd under Z and maps the multiplicity space of 4 into itself. ■

Proposition 28.3 (the zero section). The fibre operator c_N . nabla + i Phi on R^3 with values in E has

zero modes (1, 3) (+) (1, 1) of K, by the Callias count for the positive weights 3/2 and 1/2, with one sign of Phi. On S^4 this gives the Dirac operator twisted by (1, 3) (+) (1, 1). Then Hom_K(4, S^- (x) (1, 3)) has dimension 1, through (1, 2), and Hom_K(4, S^+ (x) (1, 3)) = 0, so the twisted operator vanishes on that copy of 4, in agreement with its index 4. The untwisted part (1, 1) has eigenvalue 2 on the representation 4. Every map in Hom_K(4, S^- (x) (1, 3)) annihilates (2, 1) in 4. These are the twistor spinors of S^4, as sections of S.

Proof. The representation count is as in Proposition 28.2, and the Callias count is a reference, with the exact count in Proposition 28.11. For the last sentence, the first-order operator from (1, 2) to (2, 3) in the slice limit is the twistor operator, and its kernel is the 4. ■

On a representation V the twisted operator has D^2 = Cas_G(V) + 3/2 - Cas_K(tau), for the twist tau, with Cas_G(4) = 5/2, Cas_G(16) = 15/2, Cas_G(20) = 21/2, Cas_G(40) = 29/2, Cas_K(j_a, j_b) = 2 j_a(j_a+1) + 2 j_b(j_b+1). With the twist (1, 3) this gives 0, 5, 8, 12, and with the twist (1, 1) on the representation 4 it gives 4. The slice operator reproduces these (Result 28.5).

Proposition 28.4 (the compression). Let s_v(g) = F lambda(g)^-1 v be a section in the multiplicity space of 4 on a slice, with B = F^* F, which commutes with lambda(H). Write t-hat = 2 i t_3 of Sp(1)_b. Then, with normalization by the norm of s_v,

```
Q(x_a) = (tr(B Gamma_5)/tr B) Gamma_a/5,
```

and for the functions y_T(g) = tr(T Ad(g) t-hat) with T in the Lie algebra sp(2), Q(y_T) is a multiple of T, proportional to tr(B t-hat)/tr B.

Proof. End_H(4) is spanned by 1, Gamma_5 and t-hat. Since int lambda(g) Gamma_5 lambda(g)^-1 x_a = int x_a x_b Gamma_b = Gamma_a/5, the part with Gamma_5 gives the first formula. Equivariant maps from 5 to End(4) are multiples of Gamma_a, and tr(Gamma_a Ad(g) t-hat) = 0, so 1 and t-hat contribute nothing. The second formula is Schur's lemma on the representation 10. ■

Consequently, a mode with F supported on (1, 2) and K-

symmetric has tr(B Gamma_5)/tr B = +-1 and tr(B t-hat) = 0. That is the zero section: Q(x_a) = +-Gamma_a/5, and the part of a function that depends on the fibre gives zero. On the boundary with the pure level-one mode, B is the projection on (1)_{-1}, so that tr(B t-hat)/tr B = +-1. Example 16.9 has these values.

The operator and its two ends

Proposition 28.8 (endpoint restrictions). Restriction C^infinity(X, R) -> C^infinity(S^4, R) + C^infinity(CP^3, R) is surjective, and in normal coordinates a(y, v) = a(y, 0) + O(|v|), uniformly on bounded subsets of C^1(X). (Report, Proposition 1.)

Theorem 28.9 (a self-adjoint realization). Take the connection metric g = dr^2 + h(r)^2 pi^* g_{S^4} + rho(r)^2 g_{S^2} with h smooth, even and positive at 0, rho odd and smooth with rho(r) = r + O(r^3), and a product near r = l. Let Phi_M = M b(|v|^2) v . J with b smooth and positive, T_M = D_X^E + i Phi_M, and D-hat_M the doubled operator. At r = l set Z = -i c(dr), chi = Gamma-hat Z, and impose chi psi = -psi. Then D-hat_M on this H^1 domain is self-adjoint, elliptic and odd, with compact resolvent, and every smooth scalar function preserves the domain. No boundary condition enters at the zero section: after the outer condition is fixed, removing S^4 creates no further self-adjoint extension. (Report, Theorem 2, using Bär-Ballmann, Corollary 4.18.)

Proposition 28.10 (three slice operators). For the standard complex structure on CP^3: (1) the kernel of the Dolbeault-Dirac operator on O(1) is C^4 for every Hermitian metric; (2) the Levi-Civita Dirac operator twisted by the separate line bundle O(3) with its invariant connection vanishes on the 4 for every invariant squashing, which gives a kernel of dimension at least four, with any further kernel in other representations at a locally discrete set of squashings; (3) the Levi-Civita Dirac operator twisted by the full pull-back connection on E is invertible for small fibres, since (D^E_rho)^2 >= 1/(2 rho^2) - C_0 - C_1 rho^2. (Report, Proposition 3.) In Python, on the representation 4, the smallest |eigenvalue| of that operator is 3.235, 1.842, 1.200, 0.722, 0.148 at rho = 0.3, 0.5, 0.707, 1, 2. So the window comes from the

competition between the normal operator and the mass.

Proposition 28.11 (normal model). Let Q = sigma . d_y + y . J on L^2(R^3; C^2 (x) C^4), with J Hermitian of spin 3/2. Then ker Q = 0 and ker Q^* = 3 (+) 1 under the normal Sp(1), with a positive gap above the zero modes; changing the sign of the mass exchanges the two kernels. The proof uses Q^* Q = -Laplacian + (y . J)^2 - sigma . J >= 0, with equality excluded since sum_i J_i^2 = 15/4, and the Callias index for the positive eigenbundles of n . J, of degrees 3 and 1. The induced bundle over S^4 is S^2(S) (+) 1, the operator on Sigma (x) S^2(S) has exactly four zero modes, in one chirality, and on the parallel inclusion S -> S (x) S^2(S) the twisted equation is a multiple of the twistor equation. So the kernel consists of harmonic spinors of the twisted operator, which are twistor spinors as sections of S. Rescaling gives the localization scale r = O(M^{-1/2}) at the core and O(M^{-1}) at the wall. (Report, Proposition 4.) In Python, in an oscillator basis: Q^* Q starts at 1.02, QQ^* has four near-zero values, with total spin 0 and three values near spin 1, then 1.02.

Lemma 28.12 (tangential mixing). On a product collar, normalized stable profiles for mass eigenvalues M q_i overlap by 2 sqrt(q_i q_j)/(q_i + q_j), independently of M, which is sqrt(3)/2 for q = 1/2, 3/2. So large mass does not remove the coupling between the stable channels O(3) and O(1), which in the pull-back connection has order 1/rho. (Report, Lemma 5.)

Proposition 28.13 (the boundary operator on the representation 4). Take the sign of Phi for which the right singular vector of T is at the zero section. Its left singular vector v is at the wall, in the domain Z v = +v of Theorem 28.9 for T^*. A wall profile exp(q(x - l)), q > 0, for T^* = i Z d/dx + D_6 + i phi tau, with phi > 0, has q = -phi Z tau at leading order. So the stable channels of v are (Z, tau) = (+, -3/2) and (+, -1/2) twice, and they map to the channels (-, -1/2) twice. In unitary bases of these, the compression of D_6(rho) with the overlaps of Lemma 28.12 is

```
A(rho) = ( 3/(2 rho)   1/rho   rho + 1/rho )
         (     0       sqrt 2     -sqrt 2  )
```

with the matrix elements computed in Python, and re-

produced by Terra 6.1 in exact symbolic arithmetic from the bracket and Clifford data. Its kernel is spanned by c = (-(2/3)(rho^2 + 2), 1, 1). The channel tau = -3/2 and one channel tau = -1/2 are supported on (1)_{-1} of the 4; the other channel tau = -1/2 is supported on (1)_{+1}. All three lie in Gamma_5 = -1, and their target spaces are mutually orthogonal, so the cross terms vanish in the traces of Proposition 28.4. Hence for the mode c

```
tr(B Gamma_5)/tr B = -1,    tr(B t-hat)/tr B =
↪  -w(rho),    w(rho) = 2(rho^2 + 2)^2 / (9 +
↪  2(rho^2 + 2)^2),
```

for every rho, with horizontal scale 1; for horizontal scale h, replace rho^2 by rho^2/h^2. So w = 8/17 as rho -> 0, w = 2/3 at the Fubini-Study radius rho = 1 (the nearly Kähler radius is 1/sqrt 2), and w -> 1 as rho -> infinity, the level-one value. The nonzero singular values of A are the next wall values on the 4: 1.863 and 2.789 at rho = 1, with the core value 2 in between. With the opposite sign of Phi, the right singular vector is at the wall, in the channels Z = -1, tau > 0, with fibre part +w(rho), and its left singular vector at the zero section.

For the connection preserving the line summands, the first row is replaced by (0, 1/rho, rho + 1/rho), the kernel is (1, 0, 0), so w = 1 for every rho, and the block on tau = -1/2 has singular values with sum of squares rho^2 + 2/rho^2 + 6 and product sqrt 2 (rho^2 + 2)/rho, so it is invertible for every rho > 0, with D^2 = 3, 6 at rho = 1. This is the collar repair of the report: an invariant connection preserving the four line summands in a collar extends into X with a cutoff, with the pull-back connection near the zero section. At the Fubini-Study metric the stable boundary model is then a sum of line-twisted Dirac operators, with four zero modes on O(3) and none on O(1), since H^q(CP^3, O(-1)) = 0 for all q.

Status. The kernel and the traces are exact for this finite block, which is the leading formal wall operator; the next correction is O(1/M) in the fixed-block expansion and does not change the limiting kernel, since the kernel is separated from the other singular values. That the boundary mode of T_M converges to it as M -> infinity is a numerical statement here (Result 28.5), since a uniform global reduction is unproved: the difference falls

as phi_0 grows, from 0.037 to 0.009 at wall radius 0.5 and from 0.008 to 0.002 at wall radius near 1, between phi_0 = 16 and 32. The value 2/3 tests the factor sqrt 3/2, since without it w(1) would be 3/5.

Lemma 28.14 (a sufficient spectral criterion). Let U_M : C^8 -> Dom D-hat_M be an isometry with ||D-hat_M U_M|| <= eps_M -> 0 and ||D-hat_M psi|| >= delta ||psi|| for psi orthogonal to the range, with delta > 0 independent of M. Then for large M, D-hat_M has exactly eight eigenvalues in (-delta, delta), and the spectral space is within O(eps_M/delta) of the range. (Report, Lemma 6.) The missing steps for Conjecture 28.6 are the normal reduction with its errors, and the coercive estimate on every representation of Sp(2).

Conjecture 28.6 (the window). Take Theorem 28.9 with the sign of Proposition 28.13 and with (a) the pull-back connection and wall radius rho, or (b) the collar connection of Proposition 28.13 and the Fubini-Study metric at the wall. For large M the hypotheses of Lemma 28.14 hold, with the eight states being one copy of the 4 at the zero section in one Gamma-hat degree and one at the boundary in the other. For a family of metrics and mass profiles with a uniform positive decay rate, the off-diagonal block is exponentially small in l. The compression of a smooth function tends to (Q_B(a|*{S^4}), Q_Y(a*/{CP^3})), with Q_B and Q_Y(x_a) = -Gamma_a/5, and Q_Y(y_T) a multiple of T, equal to the level-one value of Example 16.9 times w(rho) in case (a), and equal to it in case (b). This concerns compression maps, in line with Counterexample 28.17.

Result 28.5 (Python, the radial system). We solve T on the multiplicity spaces, with Hermitian Clifford generators, so that c(dx) = i Z in the skew convention of Theorem 28.9. For horizontal scale h, after absorbing the half-density h^2 rho, T = i Z d/dx + A_0/h + A_1/rho + (rho/h^2) A_2 + i phi tau, with -s(x) A_E/rho added for a collar connection. We take h = 1, so that T = i Z d/dx + D_6(rho(x)) + i phi(x) tau, with D_6 built from the structure constants of sp(2), rho(x) = rho_inf tanh(x/rho_inf), phi(x) = phi_0 tanh(x/w_0), w_0 = 0.3, in units with S^4 of radius 1, by the quadratic form of T^* T with P1 elements on a graded grid. This rho

is not exactly a product near the wall. Rayleigh quotients of the exact form bound the values from above, and with three-point quadrature they approximate those bounds. Small entries have a mesh floor that depends on the parameters: at (phi_0, l) = (6, 4) the lowest value is 0.00548, 0.00401, 0.00355 for 2500, 5000, 10000 elements, so such entries do not establish a zero continuum value. At x = 0 the exponents are the values for R^3: +-1, +-2 (three times), +-3 on the 4.

phi_0	l = 2	l = 3	l = 4	next singular value, l = 4
3	0.406	0.159	0.066	1.293
4	0.288	0.088	0.028	1.452
5	0.200	0.045	0.011	1.665
6	0.132	0.021	floor	1.818
8	0.049	floor	floor	1.852

(rho_inf = 1, representation 4; the next value at larger phi_0 is 1.86 to 1.87.) The lowest value falls with the exponent of the constant-coefficient system, the smallest |Re kappa| of i Z (D_6(rho) - i phi tau): measured between l = 3 and 4, 0.877, 1.155, 1.464 at phi_0 = 3, 4, 5, against 0.856, 1.151, 1.476, as numerical evidence for these profiles, without a general tunnelling theorem. At large phi, kappa = phi/2 - 2/rho + O(1/phi) (phi = 64: 30.07 at rho = 1, 31.07 at rho = 2). The formula 2 mu exp(-mu l) of lemma 09 belongs to a separable interval and does not carry over.

On the two modes, at rho_inf = 1, l = 3, with the sign of Proposition 28.13, the boundary mode being the left singular vector; the core fibre parts at phi_0 = 16, 40 are below what the computation resolves:

phi_0	zero section: position, Q(x_a)/(Gamma_a/5), tr(B t-hat)/tr B	boundary: position, Q(x_a)/(Gamma_a/5), tr(B t-hat)/tr B, Z, tau
10	0.39, -1.0000, -3e-7	2.93, -1.0000, -0.645, 0.944, -1.117
16	0.28, -1.0000, 1e-11	2.96, -1.0000, -0.658, 0.976, -1.146
40	0.16, -1.0000, 1e-11	2.99, -1.0000, -0.663, 0.996, -1.162

Against Proposition 28.13, at l = 3:

rho_inf	wall radius	-w(rho)	phi_0 = 16	phi_0 = 32	next value, phi_0 = 16	second singular value of A
0.5	0.500	-0.529	-0.493	-0.521	1.992	1.992
0.7	0.700	-0.579	-0.562	-0.575	1.966	1.966
1.0	0.995	-0.665	-0.658	-0.663	1.864	1.865
1.5	1.446	-0.788	-0.785	-0.787	1.660	1.661
2.0	1.810	-0.861	-0.858	-0.860	1.557	1.556
3.0	2.285	-0.921	-0.918	-0.919	1.487	1.486

Collar connection. With rho(l) = 1 and s(x) = (1 + tanh((x - l + 1)/0.15))/2 multiplying the vertical connection term to be removed, at phi_0 = 8, 16, 32: the boundary mode has tr(B t-hat)/tr B = -1.0000 and Q(x_a)/(Gamma_a/5) = -1.0000, the next value is 1.733, that is sqrt 3, and the core mode is unchanged. The switch is nonzero, though tiny, at x = 0, whereas the smooth extension needs a cutoff vanishing near the zero section, so these rounded values support the limiting model and do not establish exact finite-mass equality.

Other representations. The multiplicity is zero on 1, 5, 10, 14, 35. At phi_0 = 16, rho_inf = 1, l = 3, the lowest singular values are 2.188 (wall) and 2.235 (core) on the 16, 1.920 (wall) and 2.828 (core) on the 20, and 3.381 (wall) and 3.463 (core) on the 40, the core values being sqrt 5, sqrt 8 and sqrt 12 up to the mass. The multiplicity zero follows from the central character. The finite wall operator A_V(rho) has no zero at 1500 sampled radii in [0.1, 20] on the 16, 20, 40, 64, a sampled statement; its smallest singular value tends to the core value as rho -> 0 and falls slowly at large rho on the 16 and the 40 (0.33 and 0.67 at rho = 20). Higher representations were not computed, and no estimate in the representation is proved.

Core block at finite mass. The density of Proposition 28.16 is nonzero at these points, stable in the mesh to five digits:

phi_0, l	lambda	tr(B t-hat)/tr B, integrated	max of abs tr(B t-hat) over max of tr B	sup of the pointwise ratio	integrated value over lambda^2
4, 3	0.0882	-5.71e-3	6.48e-3	0.849	-0.73
5, 3	0.0453	-1.84e-3	1.97e-3	0.856	-0.89
6, 3	0.0212	-4.55e-4	4.73e-4	0.858	-1.01

The pointwise ratio reaches its maximum at the outer wall, where the core density is small, so the breaking comes from the tail of the core mode at the wall. Over phi_0 = 3 to 6, l = 2 to 4, the integrated value over lambda^2 lies between -0.4 and -1.0, an empirical observation with no O(lambda^2) asymptotic proved. These values support the hypothesis of Proposition 28.16 at the sample points, conditional on the density being nonzero in the continuum.

The compression and the algebra

Theorem 28.15 (compression, conditional on localization). Assume an eight-dimensional spectral window splitting under Gamma-hat as W_M (+) V_M with dimensions 4, 4. Then scalar compression is exactly block diagonal. If the isometries differ by at most eta_M from model embeddings with angular compressions Q_B, Q_Y and normal measures with first moments at most C M^{-1/2} at the core and C M^{-1} at the wall, then

```
|| Q_M(a) - (Q_B(a|_B) (+) Q_Y(a|_Y)) || <= C
 ↪  ||a||_{C^1} M^{-1/2} + 4 ||a||_infinity eta_M.
```

(Report, Theorem 7.) For a spectral wall as in lemma 24, the domain has constant boundary values, and if a$|_Y$ = *c then any boundary-localized family has* //Q{Y,M}(a) - c|| <= C M^{-1} ||d_r a||_infinity. So localization just inside a spectral wall does not recover arbitrary boundary functions for a fixed smooth a, and the limit needs a local wall or an explicitly different comparison algebra.

Proposition 28.16 (the exact symmetry test). For an equivariant copy of 4, B(r) = F(r)^* F(r) = diag(b_a(r) I_2, b_+(r), b_-(r)). All real scalar functions compress to operators commuting with the quaternionic antiunitary j if and only if b_+(r) = b_-(r) for

almost every r. If in addition the 5 component is nonzero, the block generates M_2(H). If both the 5 and the 10 components are nonzero, it generates M_4(C). A vanishing integral of tr(B(r) t-hat) alone does not decide this. If a radial h has Q_M(h) = alpha I_W + beta I_V with alpha != beta, then the block projections lie in the generated algebra. (Report, Proposition 8.) At the limit, the real linear image is Herm_2(H) + Herm_4(C), of real dimension 6 + 16 = 22, and the generated real algebra has dimension 16 + 32 = 48.

Counterexample 28.17 (generation is discontinuous under convergence of compression maps). Let R_0 = Q_B E_fib, with E_fib the fibre average on CP^3, R_1 = Q_Y, and R_eps = (1 - eps) R_0 + eps R_1. These are unital completely positive, ||R_eps - R_0|| <= 2 eps, and real functions generate M_4(C) under R_eps for every eps > 0, and M_2(H) at eps = 0. So the correct limit statement concerns specified compression maps, and it gives no convergence of the generated algebras. (Report, Counterexample 9.)

Theorem 28.21 (prescribed windows on the same X). Let P_y = (1 + y_a Gamma_a)/2 and p_z <= P_{pi(z)} the tautological rank-one projector. With normalized radial profiles chi_0, chi_l supported in disjoint eps-neighbourhoods of the two ends, U_0 v = sqrt 2 chi_0 P_y v and U_l v = 2 chi_l p_z v are isometries into L^2(X; C^4) (+) L^2(X; C^4). Real functions compressed by them generate M_2(H) + M_4(C) at every eps, and the compression differs from Q_B (+) Q_Y by at most C eps ||a||_{C^1}. (Report, Theorem 13.) This proves the connected-space mechanism without Conjecture 28.6, with the window as input. Allowing nonlocal smoothing perturbations, (1 - P) D (1 - P) + P d P makes such a window reducing, which enlarges the operator class and keeps the window as input.

The finite data

Proposition 28.18 (charge conjugation and the wall). With Hermitian generators J_i, C = C_7 (x) j_E has C^2 = -1 and commutes with T_M, D-hat_M and Gamma-hat as a differential expression, and CZ = -ZC. So it exchanges the two local wall conditions and does not preserve the chosen domain. If a pointwise par-

ent antiunitary preserves a window and commutes with real multiplication, every real compressed multiplier commutes with its restriction, which cannot hold on a generated full M_4(C). (Report, Proposition 10.) The complex boundary block therefore requires this loss of the inherited symmetry, or a different real correspondence.

Remark 28.7 (why one block is quaternionic and the other complex). Proposition 28.18 proves the signs argument of the first version, with one correction of wording: j_E commutes with the anti-Hermitian Lie algebra action and reverses the Hermitian generators, so that C commutes with i Phi. Numerically (Python, phi_0 = 16, l = 3, rho_inf = 1), exchanging the wall condition exchanges (Z, tau, tr(B t-hat)/tr B) = (0.976, -1.146, -0.658) and (-0.976, 1.146, 0.658) on the boundary mode, with the same singular values, and leaves the zero-section mode unchanged. The formal parent symmetry does not by itself prove that the finite-mass core space is preserved, since C changes the wall domain, and the breaking computed in Result 28.5 shows the size of the effect at the sample points.

Proposition 28.19 (tunnelling and the grading). If the window consists of one copy of the 4 in each Gamma-hat degree, then d_M = [[0, conj(lambda_M) I_4], [lambda_M I_4, 0]] after equivariant unitary identifications, by Schur's lemma, and lambda_M = (1/4) Tr(U_{Y,M}^* T_M U_{B,M}). The inherited grading is I_W (+) (-I_V), the auxiliary grading of the doubling, which labels the two ends under localization. The radial Clifford involution Z does not extend across the zero section, so it does not serve as an interval chirality. This grading leaves all four core states in one degree, so W = H (+) H, for the even algebra H + H + M_4(C), still needs a chosen quaternionic grading, such as a marked Gamma_p. (Report, Proposition 11.)

Proposition 28.20 (the doubled bimodule). With H = W (+) V, Theorem 16.2 gives K = H (x) conj(H), of dimension 64, with DD = d (x) 1 + Gamma-hat (x) d^c. Its odd part is exactly K^- = Hom(V, W) (+) Hom(W, V), with the desired bimodule actions, and order zero and order one hold exactly on the full K, for its actual even algebra. If lambda != 0, then DD maps K^- into K^+, P_- DD P_- = 0 and DD^2 = 2 lambda^2. So K^- is

not a reducing subspace. On K^- the inherited tensor grading is constant; the sheet-sign grading of Theorem 16.7 gives J(x, y) = (y^*, x^*), J^2 = 1, J Gamma_6 = -Gamma_6 J, with zero Dirac operator. The tunnelling map W -> V does not provide the map Hom(V, W) -> Hom(W, V) of lemma 20. (Report, Proposition 12.)

Balance against Proposition 6 of lemma 24

(Report, Proposition 14, with the numerical statements updated.)

Structure	What follows
Connected scalar parent	Supplied by the chosen disc bundle X.
Independent endpoint functions	Derived, by restriction and extension.
Base-function restriction at the collapsed end	Derived geometrically; quantitative for localized states.
Four core states	Proved for the linear normal model and the resulting twisted operator on S^4.
Four boundary states	Proved for the separated collar model; for the pull-back connection, the finite block of Proposition 28.13, with the convergence numerical.
Isolated 4 + 4 window	Conjecture 28.6.
Limiting quaternionic core compression	Derived once convergence to the core model is established.
Exact finite-mass quaternionic block	Requires b_+ = b_-; numerically not satisfied at the sample points resolved.
Exact level-one boundary compression	Requires rho -> infinity or the separated collar connection.
End grading	Inherited from the chosen self-adjoint doubling.
Quaternionic grading axis	Input.

Structure	What follows
The two Hom spaces	The odd part of the coarse correspondence; the selection is additional.
KO-6 reality and order conditions	Exact after the specified correspondence and grading choices.
Nonzero connector of lemma 20	Input.
Mass value and dependence on l	Radial matching; equivariance gives only lambda I_4; the exponent is computed numerically in Result 28.5.

Choices that stay inputs: X, E = S^3(S), the hedgehog, the wall condition, the collar connection in case (b), and the doubling of Theorem 16.2. A double with two identical collapsed zero sections removes the outer boundary condition, but the same local core construction gives two quaternionic endpoint maps, so obtaining a complex block needs different structure at one end. T0 stays open, and this is a candidate for its first half.

Direction. The next analytic step is the coercive estimate of Lemma 28.14, in case (b) first, where both end models have exact kernels. Since Phi commutes with the Clifford symbol, D + iM Phi is a concentrating pair in the sense of Maridakis (arXiv:1510.07004, Definition 2.1 and Lemma 2.2), with [D, Phi] of order zero; the symbol condition goes back to Prokhorenkov and Richardson (arXiv:math/0307251). Two things are missing there. The spectral separation theorem of arXiv:2110.11654 (section 2, Assumption 3) needs a leading endomorphism independent of the normal direction, and this fails here, since (n . J)^2 depends on n for spin 3/2; so a separate core analysis or an anisotropic extension is needed. And the wall adds a boundary layer: on a half-line ||(d_t + M) f||^2 = ||f'||^2 + M^2 ||f||^2 - M |f(0)|^2, and exp(-Mt) is a zero mode away from the zero set of the mass. (Read by Terra 6.1 in arXiv HTML; the LaTeX sources were not available to it.)

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026, in a script that ran the tests below as assertions and printed the tables of Result 28.5; by the rule of the project the script is not kept (removed on 4 Nov 2026; Terra 6.1 had reproduced its matrix of Proposition 28.13 in exact arithmetic). The slice operator on each multiplicity space is Hermitian, odd under Z, and closes on Hom_H. The structure constants reproduce [t_1, t_2] = t_3 in Sp(1)_a and Sp(1)_b. The decomposition D_6(rho) = A_0 + A_1/rho + A_2 rho holds. The form uses three-point Gauss quadrature on each element. The exact tests are {A_0, A_1} = 0, A_0^2 = 0, 4 on the 4, 5, 9 on the 16, 0, 8 on the 20, 12, 16 on the 40 (the value 0 on the 20 has the twist (1, 5) and gives no core zero mode), A_2 = -(i/8) sum C_ab^alpha c_a c_b c_alpha, and at rho = 1 with the line connection the values D^2 = 0, 5, 5, 12 on the summand O(3) of the 4, 16, 20, 40, against j(j + 4) for (0, 0)-forms in O(1); the whole summand also has the value 10 on the 20. The closed form of Proposition 28.13 is tested to ten digits, and the core mode density in the mesh.

Part 29

The eight-state window on the disc bundle

Written by Claude Opus 5.5 on 4 November 2026, for STRATEGY T0, without an Astra job. It continues lemma 28 and answers the prompt 29_window_coercive_estimate.md. Refereed by Terra 6.1 (report): case (b) stands with amendments, and the two errors it found in case (a), the definition of A_eff and the sign of the fibre part, are corrected below. The referee ran no computation. A fresh reading of the corrected case (a) in the referee report on lemma 30 found one more sign and one missing step in Proposition 29.10; these are corrected, and the corrections have not been read again.

Outcome

1. **Conjecture 28.6 holds in case (b)** for the class of metrics and mass profiles of Setting 29.0: for large M, D-hat_M has exactly eight eigenvalues in

(-delta, delta), for every delta < sqrt 3 (Theorem 29.7).

2. **Conjecture 28.6 holds in case (a) at the Fubini-Study radius** rho = 1, with fibre part -2/3 (Theorem 29.11). At another wall radius, case (a) follows from one kernel condition K(rho), which is a finite check at each radius.
3. **The off-diagonal block is exponentially small**, |lambda_M| <= C exp(-(1 - eta)(M/2) int f dr), by an exact tunnelling identity and Agmon estimates (Proposition 29.8).
4. **The compression converges** to (Q_B(a|*{S^4}), *Q_Y(a/*{CP^3})) with error O(M^{-1/2}) (Theorem 29.9). The statement concerns compression maps, in line with Counterexample 28.17.

The proof is short because Sp(2)-equivariance gives exact sums of squares at the core and at the wall. Each estimate holds on all of L^2 of a fixed manifold, with the boundary operators elliptic on CP^3 or S^4, so uniformity over the representations of Sp(2) needs no separate argument. The spectral separation hypothesis of Maridakis, which fails for this mass (reviews/2026-11-03_terra_check_28.md, item 6), is never used. General forms of two identities are in the literature: the square with its zero-order cross term in the bulk (Maridakis, arXiv:1510.07004, Lemma 2.2), and a horizontal and vertical factorization with a curvature contraction (Kaad and van Suijlekom, arXiv:1710.03182, Proposition 18, under other hypotheses).

Scope. These are spectral statements for a chosen metric family, with flat fibres near the zero section. They give the finite window from the operator, for the sign of Proposition 28.13. They leave open: other metrics in the class of Theorem 28.9, including the Bryant-Salamon metric; the connector of Proposition 28.20; the quaternionic grading of Proposition 28.19; any dynamics selecting M, l or rho. Finite-mass generated algebras stay outside the statement (Counterexample 28.17).

Setting

Setting 29.0. Let X be the disc bundle of lemma 28, G = Sp(2), K = Sp(1)_a x Sp(1)_b, H = Sp(1)_a x U(1)_b, E = S^3(S), Phi_M = eps M f(r) n . J, and

T_M = D_X^E + i Phi_M, with D-hat_M the doubled operator and the wall condition of Theorem 28.9: Z u = -u for T_M, and Z v = +v for T_M^*. Assume:

- (H1) g = dr^2 + pi^* g_{S^4} + rho(r)^2 g_{S^2}, with S^4 of radius 1, rho(r) = r on [0, 2e], and rho constant, equal to rho_l, on the collar [l - 2a, l]. Case (b) and Theorem 29.11 take rho_l = 1, the Fubini-Study metric.
- (H2) f(r) = r on [0, 2e], f = f_0 > 0 on the collar, and f > 0 on (0, l]. This is M b(|v|^2) v . J with b positive.
- (H3) The connection on E is the pull-back connection on r <= 2e. In case (b) it preserves the four line summands on the collar, with any invariant interpolation in between. In case (a) it is the pull-back connection throughout. We use radial parallel gauge: nabla_{d/dr} = d/dr. An invariant dr-component a(r) q could otherwise be added to a line-preserving connection; it is removed by radial parallel transport, which changes the displayed formulas by a phase on each line.
- (H4) eps = -gamma_b, where gamma_b is the chirality of Sigma_4 whose K-type is (1, 2). This is the sign of Proposition 28.13: its right singular vector is at the zero section. The opposite sign exchanges T_M and T_M^* throughout. In the convention of the radial computation of lemma 28, gamma_b = +1.

On the collar put q = eps n . J, with eigenvalues +-3/2, +-1/2 on the line summands, and m = M f_0. In general q = -gamma_b n . J. In the notation of Proposition 28.13 and of the radial computation of lemma 28, q = -tau, and T_M = i Z d/dx + A + i m q on the collar, with x increasing towards the wall. The line q = +3/2 is the summand O(3), and q > 0 are the stable channels of Proposition 28.13.

Let C_R = ||A_2||, the norm on Sigma_6 (x) E of the curvature contraction A_2 = -(i/8) sum C_ab^alpha c_a c_b c_alpha; it does not depend on V, and C_R = 1 in Python. Let mu_1 > 0 be the smallest positive eigenvalue of Q^* Q for Q = sigma . d_y + y . J of Proposition 28.11. It is positive since the resolvent is compact, because (y . J)^2 >= |y|^2 / 4, and ker Q =

0. Python gives 1.02.

The zero section

Proposition 29.1 (exact splitting). Identify sections over the tube r <= 2e with K-equivariant maps s : G -> F, F = Sigma_4 (x) L^2(R^3; C^2 (x) C^4), with c(e_a) = gamma_a (x) 1 and c(f_alpha) = gamma_5 (x) sigma_alpha. Let D_H = i sum gamma_a e_a, with e_a acting as left-invariant derivatives on G, and Q^ = gamma_5 sigma . d_y + eps M y . J. Then on this tube T_M = T_0 + R, with T_0 = D_H + i Q^ and R Hermitian of order zero with |R| <= C_R |y|, and

```
T_0^* T_0 = D_H^2 + Q^^* Q^,    T_0 T_0^* = D_H^2 +
 ↪  Q^ Q^^*,
```

with D_H^2 commuting with Q^^* Q^ and Q^ Q^^*.

Proof. The fibres are flat and totally geodesic by (H1), so the Levi-Civita connection of X differs from the connection associated with the principal K-bundle G -> S^4 only by the O'Neill tensor. Its component is the curvature of N evaluated at v. This gives the zero-order term R, which equals r A_2 in the radial form D_6 = A_0 + A_1/rho + A_2 rho, of norm at most C_R |y|. For the pull-back connection, the remaining operator is D_H + i gamma_5 sigma . d_y + i Phi_M. The operator Q^ acts on the values of s, and it is K-equivariant, since sigma . d_y and y . J are invariant under the simultaneous rotation of y, of C^2 and of C^4. So Q^ maps equivariant maps to equivariant maps, and it commutes with each derivative e_a, because it does not depend on the point of G. Since gamma_a anticommutes with gamma_5 and commutes with y . J, D_H Q^ = Q^^* D_H, where Q^^* = -gamma_5 sigma . d_y + eps M y . J. Therefore

```
T_0^* T_0 = (D_H - i Q^^*)(D_H + i Q^) = D_H^2 +
 ↪  Q^^* Q^ + i (D_H Q^ - Q^^* D_H) = D_H^2 + Q^^*
 ↪  Q^,
```

and likewise for T_0 T_0^*. Finally D_H Q^^* Q^ = Q^ Q^^* D_H, so D_H^2 commutes with both. ■

In radial form. T_0 = i Z d/dx + A_0 + A_1/rho + i phi q, and the cross terms vanish by {A_0, Z} = 0, [A_0, tau] = 0 and {A_0, A_1} = 0. these three

identities were asserted on the representations 4, 16, 20, 40 (Python, script not kept).

Proposition 29.2 (the model spectrum). On the total space of N, with f = r and T_0 as above, let Pi be the projection onto ker T_0. Then:

1. ker T_0 is one copy of the 4 of G; ker T_0^* = 0.
2. ||T_0 u||^2 >= min(4, mu_1 M) ||u||^2 for u orthogonal to ker T_0, and ||T_0^* v||^2 >= min(4, mu_1 M) ||v||^2 for all v in the domain.

Proof. Write Q^_gamma for Q^ on chirality gamma. By Proposition 28.11, Q^_gamma has kernel exactly when gamma = -eps; this is the rank-four K-representation W_0 = (1, 3) + (1, 1), and on the orthogonal complement Q^^* Q^ >= mu_1 M, after the rescaling y -> M^{-1/2} y. The same bound holds on the other chirality, and for Q^ Q^^* with the chiralities exchanged, since Q^* Q and Q Q^* have the same nonzero spectrum. The fibrewise projection P_W onto W_0, on both chiralities of Sigma_4, commutes with D_H, by Proposition 29.1. (The projection onto the fibre kernel of Q^ also selects gamma = -eps, so it does not commute with D_H, which is odd; it commutes with D_H^2.) On chirality -eps, Q^ vanishes on the range of P_W and is bounded below by sqrt(mu_1 M) on its complement; on chirality +eps, it is bounded below by sqrt(mu_1 M) throughout. On Sigma^gamma (x) W_0, D_H is the Dirac operator of S^4 twisted by the homogeneous bundle of W_0, so by Parthasarathy's formula D_H^2 = Cas_G(V) + 3/2 - Cas_K(tau), with Cas_K = 2 j (j + 1) on Sp(1)_b, that is 0 and 4 on W_0. Only V odd under -1 occur, with Cas_G(V) >= 5/2 and Cas_G(V) >= 15/2 for V not the 4. So D_H^2 >= 4 except for D_H^2 = 0 at V = 4, twist j = 1. This kernel needs Hom_K(4, Sigma^gamma (x) (1, 3)) != 0, which holds exactly for Sigma^gamma = (1, 2), where (1, 2) (x) (1, 3) contains (1, 2) once. By (H4), gamma_b = -eps, so T_0 sees this chirality and T_0^* the other. Decomposing u by chirality and by P_W, and using the sum of squares, gives both bounds. The computation holds on compactly supported smooth sections, and extends by closure, since the doubled model is essentially self-adjoint on that core; no boundary condition enters at infinity. ∎

Lemma 29.3 (core estimate). Let u be in H^1 with

support in r < 2e. Then

```
|| |y| u || <= (2/M) ||T_0 u|| + sqrt(10/M) ||u||,
```

and the same holds with T_0^*. So ||T_M u|| >= (1 - 2 C_R/M) ||T_0 u|| - C_R sqrt(10/M) ||u||, and likewise for T_M^* with T_0^*. In particular, for such u with Pi u = 0, and for all such v,

```
||T_M u|| >= (2 - o(1)) ||u||,    ||T_M^* v|| >= (2
↪  - o(1)) ||v||,    M -> infinity.
```

Proof. Q^^* Q^ = -Laplacian_y + M^2 (y . J)^2 - eps M gamma_5 sigma . J. The eigenvalues of sigma . J are 3/2 and -5/2, and (y . J)^2 >= |y|^2/4. Hence M^2 || |y| u ||^2 / 4 <= ||Q^ u||^2 + (5/2) M ||u||^2 <= ||T_0 u||^2 + (5/2) M ||u||^2, by Proposition 29.1. The rest is the triangle inequality with |R| <= C_R |y|, and Proposition 29.2. ■

The bulk

Lemma 29.4 (bulk estimate). Let w be in H^1 with support in a compact subset of 0 < r < l, and let f_1 be the minimum of f on that support. Then

```
||T_M w||^2, ||T_M^* w||^2 >= (M^2 f_1^2 / 4 - C_Phi
↪  M) ||w||^2,
```

with C_Phi the supremum of the operator norm of the Clifford contraction sum_i c(e_i) nabla_{e_i} (f n . J). With 0 < r_1 < r_2 < l and a radial cutoff psi equal to 1 on [r_1, r_2], with compact support in 0 < r < l, we have for u in Dom T_M,

```
||u||^2_{L^2(r_1 <= r <= r_2)} <= C M^{-2} (||T_M
↪  u||^2 + ||u||^2).
```

Proof. ||(D + i Phi) w||^2 = ||D w||^2 + ||Phi w||^2 + <w, i [D, Phi] w>, with [D, Phi] of order zero and pointwise bounded independently of V, and Phi^2 >= M^2 f^2 / 4. For the second statement, T_M (psi u) = psi T_M u + [D, psi] u. ■

The interpolation of the connection in (H3) takes place in this region, where it enters only through C_Phi.

The wall

Lemma 29.5 (line-twisted operators at the Fubini-Study metric). Let D_k be the Dirac operator on

CP^3 twisted by O(k) with its invariant connection, k odd, at rho = 1. Then D_k^2 = C_2(L) + 3/2 - (3/8) k^2 on the SU(4)-type L, with C_2 normalized by C_2(4) = 15/8. Consequently:

1. On O(3): ker D_3 is one copy of the 4 of Sp(2), in Z = +1; and D_3^2 >= 5 on the V-isotypic part for V not the 4.
2. On O(1): ker D_1 = 0; D_1^2 >= 3 on the 4 and D_1^2 >= 6 on the V-isotypic part for V not the 4.

So the smallest nonzero |eigenvalue| on O(3) + O(1), over all V, is kappa_w = sqrt 3, attained on the 4; the lowest wall value on the 16 is sqrt 5.

Proof. The form of D_k^2 is Parthasarathy's formula for SU(4)/U(3), since the twist is a character of U(3). The Sp(2)-invariant connection on a line summand is unique, because the isotropy representation has no trivial summand, so it is the Chern connection. The normalization: j(j + m + 3) is the eigenvalue of 2 box on (0,0)-forms in O(m) on the type [m + j, 0, j], with C_2([m + j, 0, j]) - C_2([m, 0, 0]) = j (j + m + 3), which fixes the coefficient of C_2; scalar curvature 12 gives the constant 3/2; and the term in k^2 is fixed by the kernel of D_3 on the 4. Two checks: D_0^2 = 4 on the type 6, the Kirchberg limit R/3 at R = 12 (Ann. Global Anal. Geom. 4 (1986) 291-325); and the values in the Python table below. The kernel statements are H^q(CP^3, O(1)) = C^4 at q = 0 only, and H^q(CP^3, O(-1)) = 0, with K^{1/2} = O(-2). The types of odd 4-ality with C_2 < 55/8 are the 4, its conjugate, the 20 and its conjugate (enumerated in Python). The 4 occurs on O(3) once, by the multiplicity on the slice, and this copy is the kernel; since each 20 restricts to Sp(2) as 4 + 16, neither 20 occurs on O(3). So V != 4 has C_2 >= 55/8 on O(3), giving D_3^2 >= 5. On O(1), the 4 and its conjugate restrict only to the 4 of Sp(2), so V != 4 has C_2 >= 39/8, giving D_1^2 >= 6; and the slice block on the 4 has D_1^2 = 3, 6. The chirality of ker D_3 is that of (0,0)-forms, and the slice block fixes it as Z = +1 for q = +3/2 (Python). ■

Lemma 29.6 (wall estimate in case (b)). Let w be in H^1 with support in the collar, vanishing near its inner end, and p(x) = sqrt(3 m) exp((3/2) m (x - l)), so that the kernel of T_M^* on the half-cylinder is p (x) ker

D_3. Then:

1. If Z w = -w at r = l, then ||T_M w||^2 >= min(m^2/4, 3) ||w||^2.
2. If Z w = +w at r = l, and w is orthogonal to p (x) ker D_3, then ||T_M^* w||^2 >= min(m^2/4, 3) ||w||^2.

Proof. On the collar, A commutes with q and with d/dx, and anticommutes with Z. Writing w = w_+ + w_- by Z, integration by parts gives, for T_M,

```
||T_M w||^2 = ||(d + m q) w_+||^2 + ||A w_+||^2 +
↪  ||(d - m q) w_-||^2 + ||A w_-||^2,
```

where the cross terms are purely imaginary, using w_+(l) = 0. For T_M^* the same holds with q replaced by -q and with w_-(l) = 0. There are three one-dimensional cases on (-infinity, l]. (i) If g(l) = 0, then ||(d +- m q) g||^2 = ||g'||^2 + m^2 q^2 ||g||^2. (ii) With g(l) free, ||(d + m |q|) g||^2 = ||g'||^2 + m^2 q^2 ||g||^2 + m |q| |g(l)|^2. (iii) With g(l) free, ||(d - m |q|) g||^2 is the form of -d^2 + m^2 q^2 with a Robin condition at l, whose spectrum consists of 0, on the profile sqrt(2 m |q|) exp(m |q| (x - l)), and of [m^2 q^2, infinity). Cases (i) and (ii) give the lower bound m^2/4. Case (iii) arises exactly in the channels (Z, q) = (-, q > 0) for T_M and (+, q > 0) for T_M^*. On those, expand in eigenvectors of A^2: the term ||A w||^2 is at least 3 ||w||^2, except on ker D_3 in Z = +1, which by Lemma 29.5 only arises for T_M^* at q = 3/2, where the Robin gap (9/4) m^2 applies. ■

The window

Theorem 29.7 (the window, case (b)). Under Setting 29.0 with rho_l = 1 and connection (b), let delta < sqrt 3. For M large, D-hat_M has exactly eight eigenvalues in (-delta, delta). The spectral subspace splits under Gamma-hat as W_M (+) V_M, each a copy of the 4 of Sp(2), with W_M within O(M^{-1/2}) of the cut-off core modes in the domain of T_M, and V_M within O(M^{-1/2}) of the cut-off wall modes p (x) ker D_3 in the domain of T_M^*.

Proof. We verify Lemma 28.14. Take radial functions chi_c, chi_b, chi_w with chi_c^2 + chi_b^2 + chi_w^2 = 1, chi_c = 1 on r <= e with support in r < 2e, chi_w = 1 on [l - a, l] with support in the collar. Let U_M span

(chi_c u_i, 0) and (0, chi_w v_k), where u_i is a basis of ker T_0 and v_k of p (x) ker D_3, with Gram-Schmidt. Radial functions preserve the wall condition.

Trial states. T_M (chi_c u_i) = [D, chi_c] u_i + R chi_c u_i, of norm O(M^{-1/2}), since u_i has Gaussian decay at the scale M^{-1/2}; and T_M^* (chi_w v_k) = [D, chi_w] v_k, which is O(exp(-(3/2) m a)). So eps_M = O(M^{-1/2}).

Coercivity. Since [D, chi] = c(d chi) with c(d chi)^2 = |d chi|^2, and sum chi_i d chi_i = 0,

```
||T_M u||^2 = sum_i ||T_M chi_i u||^2 - sum_i || |d
 ↪  chi_i| u ||^2,
```

and likewise for T_M^*. The functions d chi_i vanish outside the bulk region e <= r <= l - a. So by Lemma 29.4,

```
(1 + C M^{-2}) ||T_M u||^2 >= ||T_M chi_c u||^2 +
 ↪  ||T_M chi_w u||^2 - C M^{-2} ||u||^2,    ||chi_b
 ↪  u||^2 <= C M^{-2} (||T_M u||^2 + ||u||^2).
```

If u is orthogonal to every chi_c u_i, then Pi (chi_c u) = 0 exactly, since <u_i, chi_c u> = <chi_c u_i, u>, so Lemmas 29.3 and 29.6 (1) give ||T_M u||^2 >= (3 - o(1)) ||u||^2. If v is orthogonal to every chi_w v_k, then chi_w v is orthogonal to p (x) ker D_3, so Lemmas 29.3 and 29.6 (2) give the same for T_M^*. Since ||D-hat (u, v)||^2 = ||T_M u||^2 + ||T_M^* v||^2, Lemma 28.14 gives the count and the distance to the range. The window is G-invariant and close to the range of U_M, so as representations W_M and V_M are copies of the 4. ■

Remark. The gap is set by the wall at sqrt 3, with the core next at 2. On the 16, 20 and 40 the lowest values are sqrt 5, sqrt 5, sqrt 12.

Proposition 29.8 (tunnelling). Let T_M u = lambda v and T_M^* v = lambda u be a singular pair of the window, with unit vectors. By Proposition 28.19, the four singular values coincide. For a radial chi equal to 1 near r = 0 and 0 near r = l,

```
lambda (<u, chi u> - <v, chi v>) = <v, c(d chi) u>.
```

For every eta in (0, 1) there are M_eta and C_eta with

```
|lambda_M| <= C_eta exp(-(1 - eta)(M/2) int_e^{l -
↪  a} f dr),    M >= M_eta.
```

Proof. The function chi u lies in Dom T_M, since chi vanishes near the wall. So <T_M^* v, chi u> = <v, T_M chi u>, and T_M chi u = chi lambda v + c(d chi) u. This gives the identity.

For the decay of u, let xi = psi exp(g) be radial, with psi = 0 on r <= e/2 and psi = 1 on r >= e, with g = 0 on r <= e, g' = (1 - eta)(M/2) f on [e, r_*], and g constant on r >= r_*. Since xi^2 u lies in Dom T_M,

```
||T_M (xi u)||^2 = lambda^2 ||xi u||^2 + || |d xi| u
↪  ||^2.
```

As in the proof of Theorem 29.7, with no core piece, the bulk bound of Lemma 29.4 and Lemma 29.6 (1), or Proposition 29.10 in case (a), bound the left side from below by the integral of (M^2 f^2/4 - C M) |xi u|^2 on the bulk, plus (3 - o(1)) ||xi u||^2 on the collar, minus C M^{-2} ||xi u||^2. On the bulk the excess over (g')^2 is at least (eta/4)(M^2 f^2 / 4) for M large, with f >= f_1 > 0, so ||exp(g) u||*{L^2(r >= e)}* <= *C_eta. For v we use the same argument from the wall inward, with g-tilde = 0 on [l - a, l], with -g-tilde' = (1 - eta)(M/2) f, and with Lemma 29.3 replacing Lemma 29.6 on the core, since T_M^* has no low states there. Take chi with |d chi| <= 2/s, supported on [r_, r_ + s] in the bulk. By Theorem 29.7,* <u, chi u> -> 1 *and* <v, chi v> -> 0, *so the identity gives |lambda| <= (4/s)* ||u||*{[r_, r_ + s]}* ||v||*{[r_, r_ + s]}. Run the weights with eta' = eta/2, and take s so small that (1 - eta')(I - int*{r_}^{r_ + s} f) >= (1 - eta) I, with I = int_e^{l - a} f. This gives the stated exponent. The absorption uses lambda -> 0, supplied by the trial states. The factor 1/2 holds on every channel, since (n . J)^2 >= 1/4. ■

For a family with l -> infinity, M >= M_eta, with uniform bounds on the end geometry, the cutoffs, the connection and the positive bulk mass, the constants are uniform, so lambda is exponentially small in l at the rate (1 - eta)(M/2) inf f. This is the uniform decay rate required by Conjecture 28.6, as amended by the referee.

Theorem 29.9 (compression). Under Theorem 29.7 or Theorem 29.11, for a in C^1(X),

```
|| Q_M(a) - (Q_B(a|_{S^4}) (+) Q_Y(a|_{CP^3})) || <=
 ↪  C ||a||_{C^1} M^{-1/2},
```

where Q_B(a) = int a(x) 2 P_x d mu(x), with P_x = (1 - x . Gamma)/2, and Q_Y(a) = int a(z) B(z) d mu(z). In case (b), B(z) = 4 p_z, the level-one map of Example 16.9. In case (a) at radius rho, B(z) = 2((1 + w) p_z + (1 - w) p_{jz}), with p_{jz} the quaternionic partner of p_z and w = w(rho); so Q_Y(x_a) = -Gamma_a/5, Q_Y(y_T) is w(rho) times the level-one value, and the signed fibre part is -w(rho). Restriction to the two ends is surjective (Proposition 28.8), so real functions on X generate M_2(H) + M_4(C) under the limit map, by lemma 10 and Example 16.9, for every w(rho) > 0.

Proof. Theorem 28.15, with eta_M = O(M^{-1/2}) from Theorems 29.7 and 29.11, after choosing equivariant unitary identifications of the window spaces with the model copies of the 4. The normal first moments are O(M^{-1/2}) at the core, by the Gaussian decay of the model, and O(M^{-1}) at the wall, by the profiles. At the core, F in Hom_K(4, (1, 2) (x) (1, 3)) kills (2, 1), so B = 2 P_x in the normalization Q_B(1) = I. The sign in P_x is the value tr(B Gamma_5)/tr B = -1 of Proposition 28.13. At the wall, the pointwise inner product of different line summands vanishes, so the density is a sum over channels with weights from Proposition 28.13. ■

Case (a)

On the collar in case (a), A = A_d + A_o, where A_d = D_line preserves the lines and A_o = A_E/rho has order zero, with [A_o, q] != 0. In Python, the block of A_E between q = 3/2 and q = 1/2 has norm sqrt 3 and ||A_E|| = 2, on the full fibre, hence for all V. Let P be the projection onto {sum_{q > 0} p_q g_q : Z g_q = -g_q}, with p_q = sqrt(2 m q) exp(m q (x - l)), and Y the same with Z = +1, with Pi_Y the projection onto the second space. Let F = Pi_Y (A_d + A_o) P, from the range of P to Y, and let A_eff be the operator on the sum of both ranges with blocks F and F^*. Identified with sections over CP^3 of the lines q > 0, it is, by Lemma 28.12,

```
A_eff = D_3(rho) (+) D_1(rho) + (sqrt 3 / 2) B(rho),
```

with B(rho) the block of A_o between the stable chan-

nels, of norm sqrt 3 / rho. This is self-adjoint and elliptic on the fixed manifold CP^3, so its nonzero spectrum has a gap kappa_a(rho) > 0, uniformly in V. On the 4, F^* is the matrix A(rho) of Proposition 28.13, from Y to the range of P.

Condition K(rho). The kernel of A_eff(rho) is exactly one copy of the 4, in Z = +1, so F is injective and ker F^* is that copy.

Proposition 29.10 (wall estimate in case (a)). Assume K(rho). For w as in Lemma 29.6, with the kernel p (x) ker D_3 replaced by the kernel of the half-cylinder operator T_M^*, the two statements of Lemma 29.6 hold with the constant 3 replaced by kappa_a(rho)^2 - O(1/m). The half-cylinder kernel is exactly one copy of the 4 for m large, with fibre part tending to -w(rho) at rate O(1/m).

Proof. T_d = i Z d/dx + A_d + i m q preserves the block structure: T_d P = Pi_Y T_d P, since T_d (p_q g) = p_q A_d g. Also T_d maps the complement of P to the complement of Y, because for y in Y the function T_d^* y = A_d y lies in P. On the complement of P, ||T_d w_2|| >= (m/2) ||w_2||, by the sum of squares of Lemma 29.6, where the channels with Robin kernel are removed. So with w = w_1 + w_2 and w_1 = P w,

```
||T_M w|| >= ||(1 - Pi_Y) T_M w|| >= (m/2) ||w_2|| -
 ↪ ||A_o|| ||w||,
||T_M w|| >= ||Pi_Y T_M w|| >= ||F w_1|| - ||A_o||
 ↪ ||w_2||.
```

Eliminating ||w_2||, one gets (1 + 2 ||A_o|| / m) ||T_M w|| >= ||F w_1|| - (2 ||A_o||^2 / m) ||w||. Under K(rho), F is bounded below by kappa_a(rho), and F^* is bounded below on the complement of its kernel. The same elimination holds for T_M^*, with P and Y and F and F^* exchanged. On the half-cylinder T_M is thus injective with closed range, so T_M^* is surjective; its kernel is the kernel of the Schur complement of the block T_d + (1 - Pi_Y) A_o (1 - P), whose inverse has norm at most (m/2 - ||A_o||)^-1. So this Schur complement differs from F^* by at most ||A_o||^2 / (m/2 - ||A_o||), and the graph of the Schur solution gives ||P_{ker T_M^*} - P_{ker F^*}|| = O(1/m). This projection estimate turns orthogonality to the exact half-cylinder kernel into orthogonality to ker F^* up to O(1/m), which gives the

adjoint bound. Off the 4, this is invertible. On the 4 the block is a 2 x 3 matrix of rank 2, so its kernel has dimension exactly 1, within O(1/m) of (-(2/3)(rho^2 + 2), 1, 1). ■

Theorem 29.11 (the window in case (a)). Under Setting 29.0 with the pull-back connection, K(1) holds, and for delta < min(2, kappa_a(1)) and M large the conclusion of Theorem 29.7 holds at the Fubini-Study radius, with the wall modes replaced by the kernel of Proposition 29.10, and with fibre part tending to -w(1) = -2/3. Here kappa_a(1) >= sqrt 5 - 3/2 off the 4, and it equals 1.863 on the 4, the squared values there being (45 +- 3 sqrt 33)/8. At another wall radius the same conclusion holds under K(rho).

Proof. By Lemma 29.5, on V != 4 we have |D_3 (+) D_1| >= sqrt 5, while ||(sqrt 3/2) B(1)|| = 3/2. So A_eff is invertible there, with lower bound sqrt 5 - 3/2 > 0. On the 4, F^* is A(1), with kernel of dimension one in Z = +1, the vector (-2, 1, 1), and nonzero singular values 1.863 and 2.789, by Proposition 28.13. So K(1) holds. The proof of Theorem 29.7 then applies with Proposition 29.10 in place of Lemma 29.6. The trial wall states are cut-offs of the exact half-cylinder kernel, so their defect is exponentially small. ■

Other radii. For K(rho), only finitely many V can carry a kernel, those with |D_3(rho) (+) D_1(rho)| <= 3/(2 rho), by ellipticity. In Python, on the 16, 20, 40 and for rho in [0.2, 4], the compressed block has smallest singular value at least 1.30. A check of K(rho) on all relevant V at a given radius needs a lower bound for D_line(rho) on all V, which we have not written down away from rho = 1.

Remarks

Remark 29.12 (the index). The summands of E on CP^3 have indices 4, 0, 0, -4, of total zero, as equivariant cobordism invariance requires: E extends over X (Witten 1983, preprint pp. 43-44). The window has one copy of the 4 in each Gamma-hat degree, so its Sp(2)-index is 4 - 4 = 0. So index theory allows lambda != 0, and the eight states come from the estimates above. The sign in (H4) is consistent with this: with the opposite sign, the end-model modes at the core belong to T_M^* and those on

O(-3) at the wall to T_M (Proposition 28.13), so the two ends never lie in the same Gamma-hat degree. At finite mass the full operator can have nonzero lambda, so these states need not lie in exact kernels. The relative sign of the two ends enters through the chirality of ker D_3. It follows from the slice weights, which contain the 4 once in Z = +1 and not in Z = -1, and the sign of the core is the chirality statement behind (H4). If the O(3) zero mode lay in Z = -1 with q > 0, it would be a wall zero mode of T_M, in conflict with Lemma 29.6 (1). As a check in Python, the mean position of the modes is 0.51 at the core for T_M and 2.96 at the wall for the adjoint problem, with l = 3.

Remark 29.13 (physical units). All statements are in shape units. Under the dilation g -> R^2 g, with D -> D/R and the potential divided by R, the operator is unitarily equivalent to D-hat_M / R. So the light mass is lambda_M / R, the gap delta / R, and their ratio is unchanged, while the compression maps stay the same; this observation is from the audit of 3 November 2026 by Astra 6, section 5.2, conditional there on the estimates proved here. A family with R -> infinity, with lambda_M / delta small, closes the physical gap and keeps the eight states separated. Whether a dynamics selects such a family is open. The limiting geometry of such a family is a separate question: nothing here identifies it with a complete Bryant-Salamon manifold.

Remark 29.14 (other metrics). In the class of Theorem 28.9, T_M - T_0 also contains first-order terms with coefficients O(r^2), and M (b(r^2) - b(0)) y . J. One route shrinks the core piece to r < M^{-1/2 + s}, where Lemma 29.4 controls the complement since M^2 r^2 dominates. We have not carried this out, and the horizontal first-order terms need a bound in terms of D_H that takes account of the unbounded curvature terms.

Numerical check

Python (numpy, scipy), scripts not kept by the rule of the project. The exact inputs were asserted: [A, q] = 0 and {A, Z} = 0 at rho = 1 with the line-preserving connection, the values of Lemma 29.5 against the formula, the norms used above, and the identities of Proposition 29.1. The direct solve is a test of the prediction: the

limit values below were fixed by Proposition 29.2 and Lemma 29.5 before the run, and a failure would have refuted the argument. Setting: rho(x) = x for x <= 0.4, rho(x) = 1 for x >= 1.2, phi = M rho, l = 3, with the connection switched on [0.6, 1.2], and P1 elements at 2500 cells. Lowest singular values of T_M:

V	M = 8	M = 16	M = 32	predicted limit
4	small, 1.7321, 1.9861, 2.4495	small, 1.7321, 1.9963, 2.4495	small, 1.7323, 1.9991, 2.4497	0, sqrt 3, 2, sqrt 6
16	2.2245, 2.2361, 2.4495	2.2324, 2.2367, 2.4495	2.2351, 2.2409, 2.4497	sqrt 5 (core), sqrt 5 (wall), sqrt 6
20	2.2361, 2.8148, 2.8284	2.2367, 2.8248, 2.8284	2.2409, 2.8275, 2.8286	sqrt 5 (wall), sqrt 8 (core), sqrt 8 (wall)
40	3.4461, 3.4641, 3.4641	3.4585, 3.4641, 3.4645	3.4626, 3.4642, 3.4672	sqrt 12 three times

The wall values have a mesh error growing with M, of size 0.005 at M = 32 on the 16 and 20. The small value on the 4 is uninformative in this table: it halves when the mesh doubles, so it is a mesh floor, and Proposition 29.8 is proved without it. The wall mode of the adjoint problem has fibre part -1.00000 and Gamma_5 part -1.00000 at M = 8, 16, 32, as Theorem 29.9 says in case (b).

Part 30

The massless limit as a joint limit: a finite window and an opening tower

Written by Claude Opus 5.5 on 4 November 2026, for STRATEGY T14. Theorem 30.1, Corollary 30.2, Proposition 30.3, Corollary 30.4 and Proposition 30.6 were proposed in an audit of commit b2ddf98 by Astra 6 (assessment in the review); the proofs below were redone

against the source lemmas. Theorem 30.5 and Proposition 30.7 are new here and rest on lemma 29, whose referee is queued. Refereed by Terra 6.1, together with the note (report); its corrections are applied and have not been read again, since the Codex quota is spent.

Outcome

The claim that W and Z become massless as a dimension opens needs a family. Along a general path to M_W = M_Z = 0 nothing forces a tower to open: lemma 23 has theta -> 0 at fixed l, with the tower spacing fixed. The families below have the hierarchy

```
m_weak << E_eps << Delta_KK -> 0,
```

so a moving low-energy window keeps a fixed finite number of states. The hierarchy alone does not make the count at fixed energy diverge: the signed spectrum 0, +-eps^alpha, +-1, +-2, … has the hierarchy with a count bounded at fixed energy below 1. So the divergence at fixed positive energy is a separate property, proved below for each family.

1. On the exact interval this holds with an internal dimension of one (Theorem 30.1), also with localized end modes (Proposition 30.3).
2. On the cylinder of lemma 24 and on the disc bundle of lemma 29 it holds with seven internal dimensions. On the disc bundle the window comes from the operator, with one copy of the 4 at each end, and the count at fixed energy has seven-dimensional Weyl growth (Theorem 30.5).
3. On the disc bundle the core block of the window is quaternionic up to a defect that is O(M^{-1/2}) in general, and exponentially small when the connection preserves the quaternionic structure of E, for all bounded real observables (Propositions 30.6, 30.7).

Scope. These are statements about operators, compression maps and spectra along families given by hand. No dynamics selects them (Remark 30.8). The vector masses of Corollary 30.2 belong to the five-dimensional Yang-Mills toy of lemma 23. On the disc bundle only the fermion window is treated, so the relation between lambda and M_W is unproved there.

The interval

Theorem 30.1 (window and tower on the interval). Let (H_F, D_, *J_F, Gamma_F) satisfy the hypotheses of Theorem 22.1, with the algebra A_F represented by pi as in Theorem 22.4, r = dim H_F, and for 0 < alpha < 1,* l_ > 0 and small eps > 0 let D_F(eps) = eps D_, *l_eps =* l_ eps^{-alpha}, with D_eps the interval operator D_X of Theorem 22.1 for these data. Let Delta_eps be the smallest element of spec |D_eps| outside spec |eps D_*|. Then for eps small:

1. W_eps^* D_eps W_eps = eps D_*, with J_F and Gamma_F exact.
2. Delta_eps = (pi / l_) *eps^alpha - eps* //D_||.
3. ||Q_eps(a) - pi(a)|| <= (2/3) l_^2 *//D_||*^2 eps^{2(1 - alpha)} ||a||.
4. For alpha < beta < 1, E_* > 0 and E_eps = E_* eps^beta, the count N_eps(E_eps) = #{|D_eps| <= E_eps} equals r.
5. For fixed E > 0, N_eps(E) ~ (2 r l_* E / pi) eps^{-alpha}.

Proof. The condition l_eps ||D_F(eps)|| = l_* ||D_// *eps^{1 - alpha} < pi/2 holds for small eps. Theorem 22.1 gives item 1, and the next level is (pi - theta_max)/l = pi/l - //D_F//, which is item 2. Theorem 22.4 (2) gives item 3. For item 4, eps //D_||* « eps^beta « eps^alpha. For item 5, the signed spectrum is (n pi +- theta_k)/l_eps, which is r arithmetic progressions of spacing pi / l_eps, counted with multiplicity also when singular values coincide or vanish, each with (2 E l_eps / pi)(1 + o(1)) points in [-E, E]. ■

Theorem 22.1 starts from supplied finite data and a matrix-valued parent algebra, and the wall data change with eps by hand. A fixed Majorana block does not scale as eps D_*, so applying the theorem to an electroweak Dirac block needs a reducing sector or a treatment of the heavy neutral sector.

Corollary 30.2 (vector poles in the toy of lemma 23). For the universal channel of lemma 23, with m_eps = eps m_* and theta_eps = l_eps m_eps -> 0,

```
M_W = arcsin(sin(2 theta_eps)/sqrt 2) / l_eps,
 ↪  M_Z = arcsin(2 sin(2 theta_eps)/sqrt 5) / l_eps,
```

so M_W^2 = 2 m_eps^2 (1 + O(theta_eps^2)) and M_Z^2 = (16/5) m_eps^2 (1 + O(theta_eps^2)). The vector poles follow the hierarchy. The constants 2 and 16/5 belong to this five-dimensional action. The small-mass pole coefficients of the fully massive finite spectral action are 3 and 24/5 (lemma 18, Proposition 14, and lemma 23), for the family with both members of every doublet at the same Dirac mass and no Majorana mass. Lemma 25 connects the two pairs at the level of the leading coefficient: its M^2 = ((3 - d_eff(s))/2) |||[D_F, T]||^2 / ||T||^2, with s = l Lambda, is the leading Taylor quotient of the heat-cutoff action for a generator constant in the transported frame, so the coefficient is 3 and 24/5 at d_eff = 0, and 2 and 16/5 at d_eff = 1. This does not identify the spectral-action pole problem with the Yang-Mills toy, and a joint-limit use needs a remainder uniform in s, since lemma 25 expands at fixed s. Along Lambda = E_eps we have s = l_* E_* eps^{beta - alpha} -> 0, the finite value, and at fixed Lambda > 0 we have s -> infinity, the interval value. Varying the cutoff of the action is an extra prescription: an observational cutoff alone does not change the action.

Proposition 30.3 (a localized family). Take M = -mu and the mixed conditions of lemma 9, with mu_* > 0, 0 < alpha < 1, mu_eps = mu_* eps^alpha and the length l_eps^loc = c_eps / mu_eps, where c_eps = (1 - alpha) log(1/eps) + c_0. This length differs from the l_eps of Theorem 30.1. Let s = kappa l be the root of s = c tanh s, and m = mu sech s. Then

```
m_eps ~ 2 mu_* exp(-c_0) eps,    Delta_eps ~ mu_*
 ↪  eps^alpha,    (1/kappa) / l_eps = 1/s -> 0.
```

With the left profile L(x) proportional to exp(-kappa x) - r exp(kappa x), r = exp(-2 kappa l), the following identities are exact:

```
r = (m / (mu + kappa))^2,    mu - kappa = m^2 / (mu
 ↪  + kappa).
```

Proof. Lemma 9.5 gives m = 2 mu exp(-c)(1 + o(1)) and kappa/mu -> 1. The first oscillatory root has pi < k_1 l < 3 pi/2 and lambda_1^2 = mu^2 + k_1^2, so Delta/mu -> 1. From mu = kappa coth s and m = kappa / sinh s we get mu + kappa = kappa exp(s) / sinh s, hence m / (mu + kappa) = exp(-s). The second identity is m^2 = mu^2 - kappa^2. ■

The localization width and the length both grow, with a shrinking ratio. The finite two-point distance 1/m and the length l are both divergent, with (1/m)/l ~ exp(c)/(2c), so a metric convergence statement would need its own comparison maps.

Corollary 30.4 (the cylinder of lemma 24). Use the operator D_r = E d_x + Z A / r on the interval times CP^3, with A as in lemma 24, finite data on V = ker A = C^4 with gradings of type (2, 2) as in lemma 24, so that r = 4, the scaling of Theorem 30.1, and r_eps = r_* eps^{-gamma}, 0 < gamma < 1. Other values of r need an enlarged tangential bundle, which we do not construct. Then for max(alpha, gamma) < beta < 1, the cutoff E_* eps^beta isolates the finite window, since the next level is at least min((Delta_K / r_) *eps^gamma, (pi / l_)* eps^alpha - eps ||D_*||), by lemma 24, (5). At fixed E > 0 the count is proportional to l_eps r_eps^6 E^7, the seven-dimensional Weyl growth, by Dirichlet and Neumann bracketing of the squared product operator as l_eps and r_eps grow. For gamma = alpha the geometry expands homothetically in all seven directions.

The window here is prescribed by the spectral walls of lemma 24.

The disc bundle

Theorem 30.5 (joint limit on the disc bundle). Fix a shape as in Setting 29.0, with connection (b), or with connection (a) at rho = 1. Let delta > 0 be as in Theorem 29.7 or 29.11, G_0 = int_e^{1 - a} f dr, 0 < alpha < beta < 1, E_* > 0, eta in (0, 1), and

```
M_eps = (2 (1 - alpha) / ((1 - eta) G_0))
 ↪ log(1/eps),    R_eps = eps^{-alpha},    D_eps =
 ↪ D-hat_{M_eps} / R_eps.
```

By the dilation of Remark 29.13, which is exact under the canonical spinor identification and the L^2 unitary factor R^{-7/2}, D_eps is the operator of the metric R_eps^2 g with potential (M_eps / R_eps) f(s / R_eps) n . J in the physical radial coordinate s = R_eps r, of size M_eps / R_eps -> 0. For eps small:

1. D_eps has exactly eight eigenvalues in (-delta eps^alpha, delta eps^alpha), and they are +- m_eps with multiplicity 4 (or zero with multiplicity

8), where m_eps <= C eps. The next level is at most C log(1/eps) eps^alpha, so it tends to zero.

2. For E_eps = E_* eps^beta the count N_eps(E_eps) = #{|D_eps| <= E_eps} equals 8.
3. The compression converges to the endpoint maps, with error C ||a||_{C^1} (log 1/eps)^{-1/2} in shape coordinates.
4. For fixed E > 0, N_eps(E) ~ C_X E^7 eps^{-7 alpha}, with C_X = 64 omega_7 Vol_g(X) / (2 pi)^7 the Weyl constant of the elliptic operator D-hat_0 on X, where omega_7 is the volume of the unit ball and 64 = 8 x 4 x 2 counts the spinors, E and the doubling.

Proof. Item 1: Theorem 29.7 or 29.11 applies since M_eps -> infinity, Proposition 28.19 gives the multiplicity, and Proposition 29.8 gives lambda <= C exp(-(1 - eta)(M_eps/2) G_0) = C eps^{1 - alpha}, hence m_eps = lambda / R_eps <= C eps. The bound on the next level: the shape-unit value is at most s_9(D-hat_0) + c M_eps, with the comparison below. Item 2: eps « eps^beta « eps^alpha. Item 3: Theorem 29.9, with M_eps^{-1/2} of the stated order. Item 4: D-hat_M = D-hat_0 + M V on the same domain, with V bounded by c = (3/2) sup f, so by min-max N(D-hat_0, Lambda - c M) <= N(D-hat_M, Lambda) <= N(D-hat_0, Lambda + c M). Take Lambda = E R_eps, so that M_eps / Lambda -> 0, and use the Weyl law for the fixed self-adjoint elliptic operator D-hat_0. ■

So the eight finite states stay separated at the moving scale, and at fixed physical energy the spectrum has the growth of a seven-dimensional internal space, for eleven dimensions in M^4 x X. Item 1 is an upper bound on m_eps, since a lower bound on lambda is unproved. A family with l_eps growing at fixed M has the same form, with the rate of Proposition 29.8. The constants M_eta, C_eta of Proposition 29.8 are fixed once the shape and eta are fixed, so no uniformity in a varying shape is used. The limit at fixed energy resolves a neighbourhood whose shape is fixed and whose size grows, so the pointed limit is flat at a fixed interior point and a half-space at a fixed boundary point. Nothing here identifies it with a complete Bryant-Salamon manifold, or with a solution of field equations.

The quaternionic defect

For the 4-copy of the window at the core, let B(r) = F(r)^* F(r) = diag(b_a I_2, b_+, b_-) as in Proposition 28.16, let j be the quaternionic antiunitary commuting with Sp(2), and P_H(A) = (A + j A j^-1)/2. Put

```
eta_t = int |b_+(r) - b_-(r)| dr / int tr B(r) dr.
```

Proposition 30.6 (all bounded real observables). For the normalized compression Q of this mode, ||Q(f) - P_H Q(f)|| <= 2 eta_t for every real f with ||f||_infinity <= 1. If X_1, ..., X_k are such compressions, then ||X_1 ... X_k - P_H(X_1 ... X_k)|| <= 2 k eta_t.

Proof. With angular Haar measure of total mass 1, Schur's lemma gives the angular average (tr B / 4) I, so Q(f) = (4 / int tr B) int int f(r, g) lambda(g)^-1 B(r) lambda(g) dg dr. Since j commutes with lambda(g), Q(f) - P_H Q(f) is the same integral with B - P_H B. Here j exchanges b_+ and b_- and fixes the block b_a, so ||B - P_H B|| = |b_+ - b_-|/2. The triangle inequality gives the first bound. For the second, write X-tilde = j X j^-1, so that X - P_H X = (X - X-tilde)/2 and ||X_i - X-tilde_i|| <= 4 eta_t. Telescoping, X_1 ... X_k - X-tilde_1 ... X-tilde_k is a sum of k terms, each a product of contractions with one factor X_i - X-tilde_i, so its norm is at most 4 k eta_t; dividing by 2 gives 2 k eta_t. ■

So bounded expressions become quaternionic when eta_t -> 0. An unrestricted generated algebra may use coefficients of size 1/eta_t and retain the other directions, as in Counterexample 28.17.

Proposition 30.7 (the size of the defect). Under Theorem 29.7 or 29.11, eta_t = O(M^{-1/2}). Assume moreover that the quaternionic structure j_E of E is parallel on all of X. This holds for the pull-back connection and for the interpolation (1 - s(r)) nabla^pull + s(r) nabla^line with real s, and it can fail for other invariant interpolations allowed by (H3). Then for every eta in (0, 1) there are M_eta and C_eta with

```
eta_t <= C_eta exp(-(1 - eta) M int_{2e}^{l - a} f
 ↪  dr),    M >= M_eta.
```

This is the exponent of the bound on lambda in Proposition 29.8 doubled, with the integral starting at 2e. A

bound in terms of lambda itself would need a lower bound on lambda, which is open.

Proof. For the first bound, fix a normalized equivariant model F_0 at the core with C F_0 = F_0 for the local real structure below, so that j B_0 j^-1 = B_0. Theorem 29.7 gives ||u - chi_c F_0|| = O(M^{-1/2}) in L^2, and |b_+ - b_-| <= (|F| + |C F|) |F - C F| gives the bound, by the Cauchy-Schwarz inequality.

For the second, j_E is parallel and j_E (n . J) j_E^-1 = -n . J, so C = C_7 (x) j_E commutes with D_X^E and with i Phi_M, hence with T_M as a differential expression, with C^2 = -1, and it exchanges the wall conditions (Proposition 28.18). For the line-preserving connection, j_E P_q j_E^-1 = P_{-q} gives j_E nabla^line j_E^-1 = nabla^line, and real affine interpolation keeps this. On Hom_G(4, -) put (C F) = C F j^-1, which is antilinear with square +1. It commutes with D_H, with the full fibre term i Q^ and with R separately, since each part of the Clifford structure transforms in the same way, so it acts on the one-dimensional space Hom_G(4, ker T_0) as a real structure. Fix F_0 there with C F_0 = F_0, so that B_0 is j-invariant. Let u = F be the mode with T_M u = lambda v, T_M^* v = lambda u, with the phase for which <chi_c F_0, u> is real, with chi_c as in Theorem 29.7. Put zeta = u - C u and nu = v - C v. In the interior,

```
T_M zeta = lambda nu,    Pi (chi_c zeta) = 0,
```

where the second holds since <chi_c F_0, C u> is the conjugate of <chi_c F_0, u>. Take the weights of the proof of Proposition 29.8 starting at r = 2e and meeting at r_* = l - a: g = 0 on r <= 2e, g' = (1 - eta)(M/2) f on [2e, l - a], g constant beyond; and g-tilde = G - g on r <= l - a, g-tilde = 0 beyond, with G = g(l - a). So g-tilde is constant on r <= 2e. That proof gives ||exp(g) u|| <= C_eta on X, and ||exp(g-tilde) v|| <= C_eta on r <= l - a/2, and lambda <= C_eta exp(-G). The same bounds hold for C u and C v, since C is pointwise isometric.

Let xi = psi-tilde exp(g-tilde), with psi-tilde = 1 on r <= l - a and 0 on r >= l - a/2. Then ||T_M (xi zeta)|| <= lambda ||xi nu|| + || |d xi| zeta ||. Since xi is constant on the support of chi_c, Pi (chi_c xi zeta) = 0, so Lemmas 29.3 and 29.4 and the partition of Theorem 29.7 bound

||T_M (xi zeta)|| from below by the bulk term with (M/2) f and by (2 - o(1)) on the core. On the bulk |d xi| = (1 - eta)(M/2) f xi, which is absorbed with the factor eta to spare, leaving

```
||xi zeta|| <= C (lambda + ||u||_{L^2(1 - a <= r <=
↪  1 - a/2)}) <= C exp(-G).
```

Finally B - j B j^-1 = F^* F - (C F)^* (C F), so |b_+ - b_-| <= (|F| + |C F|) |zeta| pointwise in r. On r <= 1 - a, the Cauchy-Schwarz inequality with the weight exp(g) gives

```
int (|F| + |C F|) |zeta| <= 2 ||exp(g) u|| ||exp(-g)
↪  zeta|| = 2 ||exp(g) u|| exp(-G) ||xi zeta|| <= C
↪  exp(-2 G),
```

since exp(-g) = exp(g-tilde - G). On r >= 1 - a, both |F|^2 and |C F|^2 integrate to at most C exp(-2 G). ■

The samples in section 6.1 of the audit (Python, in the pull-back connection, rows reproduced by Claude Sonnet 5.5) have eta_t / lambda^2 between 0.5 and 0.9. They vary the length at fixed mass, with profiles outside Setting 29.0, so they are a qualitative comparison and do not test this statement as M -> infinity. Under the added hypothesis the defect comes from the wall, which breaks C: for an operator with the same interior and no wall, on a complete manifold, Hom_G(4, ker T) carries a real structure, and if it is one-dimensional the core block is exactly quaternionic.

Remark 30.8 (what is missing, and the direction). The next steps are, in order:

1. The vector sector on the family of Theorem 30.5: the kinetic matrix and mass Hessian of the vector fluctuations, H_eps u = M^2 K_eps u, and then the full momentum-dependent kernel. The aim is M_W, M_Z ~ c lambda / R with finite nonzero couplings. The tunnelling eigenvalue alone does not establish this.
2. The gauge normalization. For a bulk Yang-Mills field with fixed microscopic coupling and constant profile, g_4^-2 = Vol(X) / g_11^2, so plain dilation gives g_4 proportional to R^{-7/2}, which must be separated from restoration at finite coupling. Gauge fields from isometries have Killing overlap integrals instead.

3. The action in the radius and the electroweak order parameter. A toy V = a r^-p + b eps^2 r^q with a, b, p, q > 0 has its unique minimum at r proportional to eps^{-2/(p+q)}, so alpha = 2/(p+q) lies in (0, 1) if p + q > 2. The repository has not derived such a potential, and adding it by hand would derive nothing.
4. The connector of Proposition 28.20, and the origin of the quaternionic grading of Proposition 28.19.

Numerical check

No statement here needs a computation. The audit tabulates the localized family in Python (mu_* = 1, alpha = 1/2, c_0 = 2), and Claude Sonnet 5.5 recomputed the rows; for eps = 10^-2, c = 4.3026 and m = 2.710 x 10^-3, against the asymptotic 2.707 x 10^-3.

Part 11D, README, referee pending

Eleven dimensions from massless W and Z

Branch `11D`, opened 3 Nov 2026. This folder studies how the limit M_W, M_Z -> 0 of hypothesis H could give an eleven-dimensional theory, with proposed connections to Witten's seven-manifolds and the M2 and M5 readings of lemma 21 and STRATEGY T12. The branch also contains the independent referee report and related ledger and status updates.

Attempted by Claude (Anthropic) on 3 Nov 2026. The Astra 6 audit of the same day (reviews/2026-11-03_astra_audit_d11.md, of snapshot b2ddf98) supplies the joint-limit family and corrections to Proposition 1.4 and Proposal 1.3. GPT-6.1 Terra has addressed the requested items analytically in the referee report, with amendments and explicitly conditional source comparisons. The full EPJC paper and several cited primary sources were unavailable, and the original localized-family numerical table was not recomputed. The general third-charge extension of the referee has been checked by Claude Sonnet 5.5. The original scoped no-go results had two-model support after the Terra report; Claude Sonnet 5.5 has since checked the Terra

sections on files 01 to 04 a second time (second check), so the no-go results of those files have three model identifiers, the ledger's general-consensus threshold, each in its stated scope. Files 05 to 09 are being checked in the same way. Opening statements in the numbered files that say "referee pending" are historical; their adjacent referee notes and the table below give the current status. prompts/terra_check_11d.md is the referee request. The conversation in which the folder was written is recorded at https://claude.ai/code/cshk_01BpscANdX3GG8D45SQ1BxsaHviSe7ufxxCLNiNaDv4fW

The picture

H reads the two sheets of the finite geometry as the minimal truncation of an interval with opposite chiral walls, and v -> 0 as the interval growing. The labels 9, 10 and 11 of H are KO labels; the heat dimension at the 11 endpoint is 5. To make 11 a metric count, the finite KO-6 factor is resolved into a six-dimensional base, B_6 = CP^2 x S^2, which has isometry SU(3) x SU(2) and is the base of Witten's spaces:

```
11 = 4 (spacetime) + 1 (the interval of H) + 6
↪  (B_6).
```

The following table summarizes two proposed completions of 4 + 6 and their brane readings. The M^{111} identification uses the recalled fibration, and the physical dictionaries and the single-family construction remain open:

	electric (M2) completion	magnetic (M5) completion
eleven-dimensional vacuum	AdS4 x Y_{3,2}, Y_{3,2} = M^{111}, a circle bundle over B_6	AdS5 x CP^2 x S^2
the interval of H	the fibre circle, antiperiodic, two walls at the fixed points of a reflection	the radial direction of AdS5, infrared end at z_IR ~ 1/v
SU(3) x SU(2)	isometries	isometries

	electric (M2) completion	magnetic (M5) completion
hypercharge	the vector of C_3 on the harmonic two-form (b_2 = 1)	one combination of the two C_3 vectors (b_2 = 2)
v -> 0	the M2-brane moves to the apex; AdS4 x Y_{3,2} with the Standard Model group unbroken	the infrared end moves to infinity

Results

#	result	status	file
1.1	The chiral interval of length l is unitarily the scalar antiperiodic circle of radius R = l/pi; the grading is reflection, the walls its fixed points, m = 1/(2R)	Proposition independently checked, with counting clarification	01; referee
audit §3, §4	Joint family: D_F = epsilon D_*, l ~ epsilon^{-alpha}, alpha < 1, gives a finite window and an opening tower; a localized variant uses a bulk mass and logarithmic length factor	Window, Weyl constant and localized asymptotics independently checked analytically	01; referee
1.3	The circle of H as the M-theory circle: v -> 0 along the joint family is strong coupling with the Wilson-line angles going to zero, the 9 endpoint has no bulk fermions as in type 0 strings	Proposal	01

#	result	status	file
1.4	The angle-dependent one-loop term of lemma 24 is -93 zeta(5) n_f/(512 pi^2 l^4) at its minimum, relative to theta = 0, where M_W = M_Z = 0 at fixed l	Proposition independently checked in the scope of lemma 24	01; referee
1.5	An isolated stationary angle of a scale-free potential with theta = l m gives l ~ 1/m; flat directions allow other stationary paths. The specified one-loop determinant has a small-angle obstruction when n_f != 54/5	Corrected conditional result; universal claim has a counterexample	01; referee
2.1	The hypercharge quotient of the 5-bar is the circle bundle of class ±(3,2), with effective group (U(3) x U(2))/U(1)_Y and a trivial-canonical cone; its identification with M^{111} uses the recalled fibration	Quotient checked; literature identification conditional, with metric and dual-action amendments	02; referee
2.2	Only the 5 and 5-bar of SU(5) give a seven-dimensional free hypercharge quotient; among the specified circle quotients only (3, 2) has the required trivial canonical character	Proposition independently checked	02; referee
2.3	Y_{a,b}: H^2 = Z (+ torsion), H^4 of order a^2 (Z/9 for (3,2)), spin iff a odd and b even; H^2(B_6) -> H^2(Y_{3,2}) is (x, y) -> 2x - 3y	Proposition independently checked; literature comparison pending	02; referee

#	result	status	file
2.4	The Z_6 is the kernel of the displayed SU(3) x SU(2) action on Y_{3,2}; tensor Kaluza-Klein modes on an invariant background contain no triplet or doublet	Proposition independently checked in the tensor-mode scope	02; referee
2.5	No generation from one abelian line bundle per field on CP^2 x CP^1 with fluxes along Y and B - L; the asserted equal images and unsigned linearity need common ten-dimensional chirality	Scoped no-go independently checked, with amendment	02; referee
2.5, third charge	The original bounded third-charge search is excluded analytically; the proof extends to any real third charge and independent source chiralities	Bounded claim independently checked; the extension to any real third charge checked by Claude Sonnet 5.5 (two model identifiers)	02; referee
2.6	A line-bundle index gives one generation using a universal charge, chi and B - L; the colour-action convention changes the quark bidegrees, and compatible global flux quantization remains to be specified	Conditional index result independently checked, with amendment	02; referee
3.1	Hypercharge is the Betti vector of Y_{3,2}; the N = 1 fivebrane baryons are d^c and L	Proposal	03
3.2, 3.3	One generation is Λ^odd(5-bar); its realization as a Fock space of wrapped fivebranes with 6Y as homology class is conjectural	Standard representation count independently checked; brane interpretation remains a conjecture	03; referee

#	result	status	file
3.4	The Higgs as an M2 position is a proposal; the FI quotient has residual Z_3 or Z_2 stabilizers, and the fixed-N tower remains finite	Coordinate stabilizer independently checked; orbifold and physical-identification amendments	03; referee
3.5	AdS5 x CP^2 x S^2 with the displayed magnetic G_4 solves the classical field equations for scaled Einstein factors iff abs(f_1) < 2 abs(f_2); quantization and stability remain open	Proposition independently checked	03; referee
3.6, 3.7	Two completions of one base; the M2/M5 exchange as the electric-magnetic duality of hypercharge, which does not permute the 96 states	Proposals	03
3.8, 3.9	The 84 of C_3 is not colour-equivariantly either family of 84, through the specified little-group action or as tensor Kaluza-Klein modes on Y_{3,2}	Scoped no-go and corollary independently checked	03; referee
4.1	The ratio-preserving 12-state blocks of whole species are one up-type quark, or the three neutrinos	Proposition independently checked	04; referee
4.2, 4.3	The colour contents of F1 and F2 differ, and Lambda^3 C^9 is neither family for any linear SU(3) action; Lambda^3 C^9 has only 1, 3, 3bar exactly for C^9 = 9 (1), 3 + 6 (1) or its conjugate	Scoped no-go independently checked, with amendments	04; referee

#	result	status	file
4.A, 4.4	Two commuting order-3 gradings of e8 place the left-handed 10 and H_u in the C_3 sector and (d^c, L), nu^c in the other sectors; top and antitop each have chiralities in both sectors	Branching and counts independently checked, with chirality amendment	04; referee
4.5	Up-type grade selection is within the C_3 sector and the other Yukawa triples use one field per sector; the physical top mass needs family-breaking or compactification data	Grade selection independently checked, with family-tensor amendment	04; referee
4.6	The Chevalley involution exchanges conjugate sectors and preserves both families; a grading-preserving E8 automorphism cannot exchange the specified top and neutrino blocks. An antiunitary J needs an extra conjugation; its physical duality identification is a proposal	Scoped no-go independently checked, with amendment	04; referee
4 §7, §8	nu^c as the gravitational state selects t = +1/2; a discrete Z_3 from the M-grading	Proposal; discrete-grade selection independently checked	04; referee
5.1	As configurations of aligned three-family E8 roots the two families of 84 have equal pair counts but are not isometric; no element of W(E8) exchanges them	Scoped no-go independently checked	05; referee

#	result	status	file
5.2	With four families, M = -P in W(E8) transposes colour with family and maps the up-type quark block of family 0 onto the neutrino blocks of families 1, 2, 3	Proposition independently checked, with setwise-invariance clarification	05; referee
5.3	With a unitary colour-family identification, the flip commutes with gamma and J and conjugates the heavy-top and three-heavy-neutrino limits; the displayed algebras are not flip-invariant	Proposition independently checked, with unitarity amendment	05; referee
5.4	No assignment of the four types to three of four family slots, aligned or misaligned, admits the exchange as an isometry of the specified E8 roots; the original universal neighbour counts need correction	Scoped no-go independently checked analytically, with amendment	05; referee
5.5	Families as zero modes of an SU(4) bundle escape the single-root-slot assumptions; the index counts net chirality and a compatible Weyl action remains open	Proposal, independently reviewed with amendments	05; referee
6.1	The SO(32) charged Wilson-line sector has 30 copies of the lemma-22 channel spectrum; its light mass is a/R for 0 <= a <= 1/2	Spectrum independently checked, with multiplicity and range amendments	06; referee
6 §2, §3, §5	The heterotic web and type-I' dictionary as a route to the target family	Proposals reviewed; Wilson-line map, winding duality and slope rule confirmed against the two sources in the second check	06; referee

#	result	status	file
7.1	A channel spectrum has fundamental domain [0, pi/2], with doubly degenerate periodic and antiperiodic endpoint spectra	Proposition independently checked, with endpoint and scope clarifications	07; referee
7.2	The SO(32) endpoint has centralizer S(O(30) x O(2)); the lemma-24 one-loop minimum is a = 1/2 under its stated hypotheses	Proposition independently checked, with disconnected-centralizer amendment; brane interpretation remains a proposal	07; referee
7.3	A doubled flux torus gives KO-6 signs, a 2p-dimensional kernel and a generated M_p(C) corner; adding a scalar summand gives a unital finite datum	Error in the stated unital triple; signs and matrix corner independently checked	07; referee
7 §2, §4, §5, §6	Readings of dualities for finite geometry; the torus flux and M2 wrapping need a constructed identification	Proposals reviewed; duality recalls conditional on unavailable primary sources	07; referee
8.1	The descended antiunitary gives O(n) or USp(n) on each eigenspace of d^2; preservation of the rest of the data requires further conditions	Proposition independently checked, with symmetry-scope clarification	08; referee
8.2	A signed channel eigenvalue has multiplicity k in the interior and 2k at the endpoints; the corresponding abs(D) counts differ	Proposition independently checked, with counting clarification	08; referee
8.3, 8 readings	Wall slopes and 2L/3 placement follow from the recalled rule; E-type enhancements require tuned coupling, and the electroweak identification remains a proposal	Arithmetic independently checked; slope rule and E-series confirmed against the sources in the second check, physical readings conditional	08; referee

#	result	status	file
9.1	The integrated profile and far-wall condition follow from the recalled slope rule; its spectral trace uses sixteen selected channels	Integration independently checked; slope rule confirmed against Polchinski-Witten in the second check, channel dictionary linear only at constant dilaton	09; referee
9.2	Antitriplet counts are (6, 6, 3) in the antisymmetric block and (3, 6, 1) in the symmetric block; the single-line antisymmetric component vanishes	Representation counts independently checked; orientifold projection and physical identification conditional	09; referee
9.3	The stated KO signs force a particle-antiparticle block into S^2 V, excluding nonzero Lambda^2 V blocks; the full symmetric spectrum and interval scalar modes require further construction	Scoped no-go independently checked, with amendment	09; referee
9 §1	SO(32) branching and the SU(16) alternative; the archived abstract also states an extra chiral +4/3 quark per generation	Algebra independently checked; detailed full-paper comparison pending	09; referee
9 §5	E-type wall and del Pezzo correspondence; the two degree-eight surfaces have distinct root lattices and engineering assignments	Error in the E1 surface alternatives; dP_2 comparison conditional	09; referee
9 §5(2)	The E8 Weyl transposition conjugates a Wilson line and acts on the Narain lattice; a physical exchange must preserve the full background	Gauge action independently checked, with background-scope clarification	09; referee

#	result	status	file
5 Readings	Three families as the obstruction; M exchanges the two heavy directions of the algebra chain; 128 = 44 + 84 under Spin(9)	Proposals	05

The author's numerical checks are in checks_11d.py (Python 3, exact fractions). The independent referee arguments are analytic.

Added on master, 4 Nov 2026, by Claude Opus 5.5, with one independent reading by Claude Sonnet 5.5 (review): 10 tests the dilaton as second scale (negative in scope; computes a_0; corrects the dictionary of 06 to 09: nonlinear angle map, T-dual lengths, Romans mass) and 11 computes the SO(30) fluctuations (the antisymmetric block needs M_15(C) at both walls, and the window sees its commutator with the masses).

How the massless limit gives eleven dimensions here

Vanishing W and Z masses open a dimension only along a specified path (audit §2.2): at fixed l, both theta -> 0 and theta -> pi/2 make them massless with the tower fixed. The criterion used here is the joint family of the audit, m_weak « E_epsilon « 1/R -> 0, with finite gauge couplings.

1. In the circle language of Proposition 1.1, the joint family is R -> infinity for an antiperiodic circle with two fixed points, with the Wilson-line angles of D_F going to zero. Read as the M-theory circle (Proposal 1.3), it is the strong-coupling limit, with the weak scale below the momentum tower of the eleventh dimension.
2. What selects the path: an isolated stationary angle of a potential with l and the angles as its only scales gives alpha = 1, the original l ~ 1/M_W of H, on which the finite window does not separate. A flat direction permits other paths without selecting one (referee of Proposition 1.5). A second scale could select a joint family; the localized family of the audit

(bulk mass mu, mu l ~ (1 - alpha) log(1/epsilon)) is the first candidate, and the four-dimensional couplings must stay finite, which on the circle needs g_5^2 ~ R.

3. In the brane language of Proposal 3.4, v -> 0 moves the M2-brane to the apex of the cone over the Witten space with Killing spinors, and the near-horizon geometry is AdS4 x Y_{3,2}: 4 + 7 = 11 with SU(3) x SU(2) from isometries and U(1)_Y from C_3 (Proposal 3.1). This is the family of manifolds with the Standard Model group, singled out by the 5-bar (Theorem 2.1, Proposition 2.2). At fixed N it is the fixed-tower path; a joint family there needs N -> infinity, in tension with the N = 1 matter content.
4. On the magnetic side, AdS5 x CP^2 x S^2 carries the same base, and v -> 0 sends the infrared end of AdS5 to infinity (Proposal 3.6), which is the interval of H growing.

The angle-dependent one-loop term (Proposition 1.4) is minimized at the massless point at fixed l; the direction in which l moves depends on the angle-independent Casimir coefficient, which is not yet computed.

Next computations, ranked

0a. **The main point** (06, PLAN task B): one family realizing the three regimes; first steps listed in 06 §6 (type I' interval with the D8 configuration 15 + 1 as the parent of the lemma-22 window; Chan-Paton algebras against C + H + M_3(C); inner fluctuations against $\Lambda^2(30)$). 0b. **A nine-dimensional spectral triple on the type I' interval** (07 §7): sixteen D8 positions as the eigenvalues of D_F, and its strong-coupling limit with E8 walls. File 09 gives a conditional dilaton profile and representation counts; the physical string dictionary and whether the coupling can select the path remain open. 0. **A potential that selects alpha $<$ 1** (Proposition 1.5, its referee amendment and audit §7.1): the one-loop potential of the localized family, G(l m, l mu), from the exact secular equations of lemma 09, and the angle-independent Casimir coefficient a_0 of the flat family; then the vector kinetic matrix and mass Hessian along the selected path (audit eq. (17)), with g_4 finite. 1. **Remaining referee work** (report, second check): compare the EPJC paper

and cited primary sources when available, check the original localized numerical table if needed, and compare the EPJC paper when available; the second check of files 01 to 09 and of the third-charge extension is complete. 2. **Spinors on Y_{3,2}** (prompts/astra_11d_01.md): the lift of (U(3) x U(2))/U(1)_Y to the spin bundle, whether fermionic Kaluza-Klein modes can carry triality or isospin 1/2, and which spin structures give the antiperiodic fibre momenta (k + 1/2)/R of Proposition 1.1. 3. **W and Z masses from the resolution**: specify the partial orbifold resolution, any further smoothing, and the brane background; compute the broken SU(2) x U(1)_Y vector masses as functions of zeta and N and compare M_W/M_Z with the cos^2 theta = 5/8 of the spectral action. 4. **The 9D reduction of Proposition 2.6**: with SU(2)_R broken by wall data, derive the mass basis of u and d against the charge basis, and the vectorlike limit at the 9 label, where the left-handed interactions are off. 5. **Fermionic zero modes of wrapped fivebranes** (Conjecture 3.3) at N = 1, and three generations. 6. **The magnetic completion**: flux quantization, stability and the two C_3 vectors of AdS5 x CP^2 x S^2 against the fluxed pair (chi, B - L) of Proposition 2.6. 7. **A four-family completion with the fourth family decoupled** (PLAN, steps 4-5): a finite geometry or Hořava-Witten configuration in which the colour-family transposition M of Proposition 5.2 is a symmetry, so that the three-family statement of lemma 21 follows as a limit; Propositions 5.1 and 5.4 (one root per state) show that with three families the exchange is not an isometry.

The Astra 6 audit of 3 Nov 2026

The audit (reviews/2026-11-03_astra_audit_d11.md) reviews the master snapshot b2ddf98 and proposes a route from the massless electroweak limit to eleven dimensions with X = D(Λ^2_- T^*S^4) as principal candidate and M^{111} as comparison. Its use here:

audit item	use in this folder
§2.2, the massless limit must specify a path	adopted as the criterion; Proposal 1.3 revised; Proposition 1.5 shows what a scale-free action selects

audit item	use in this folder
§3, joint family	restated in circle language in 01, checked against Theorems 22.1 and 22.4; the constant 2 r l E/pi of its Weyl count confirmed (Python, section H)
§3.1, vector poles of the lemma-23 toy	quoted in 01: M_W^2/ M_Z^2 -> 5/8 along the family
§4, localized family and tail identity	table of roots and the identity (8) reproduced (Python, section H, bisection), independently of the audit's script
§7.1, modulus target	In the stated flat one-loop family the electroweak term is m^2/l^2 (q = -2); a second scale could select the audit's q > 0 path. General scale-free potentials can have flat directions
§7.2, gauge normalization	test 2 of Proposal 1.3, and the large-N couplings in 03
Appendix A, M^{111}	agrees with Theorem 2.1 and Proposition 2.3 for (3, 2), and with the placement of hypercharge in the charged module (03)
§2.1, §5, §6, X and lemma 28	outside this folder; the 7-dimensional parent X is a third possible completion, M-theory on R^4 x X with the Bryant-Salamon comparison and the circle of solidification §7 as a candidate M-circle, listed as a direction only
§2.4, docs and distribution	outside this branch: refreshing the PDFs, adding lemmas 17 and 28 to paper/ build_supplement.sh and repairing the supplement layout belong to master's build, and LaTeX is not available in this session

References used here

Read in this session (3 Nov 2026), from the arXiv abstract pages or the ar5iv text through a summarizing fetch: D. Fabbri, P. Fré, L. Gualtieri and P. Termonia, *M-theory on AdS4 x M^{111}: the complete Osp(2|4) x SU(3) x SU(2) spectrum from harmonic analysis*, Nucl. Phys. B 560 (1999) 617, hep-th/9903036 (coset, Killing spinors only for p = q, R-symmetry, Betti multiplet, Kaluza-Klein states neutral under it); D. Fabbri, P. Fré, L. Gualtieri, C. Reina, A. Tomasiello, A. Zaffaroni and A. Zampa, *3D superconformal theories from Sasakian seven-manifolds*, hep-th/9907219 (D-term 2|U|^2 = 3|V|^2, U(1) fibration over P^1 x P^2, baryons det U and det V as wrapped M5-branes, E_0 = 4N/9 and N/3, b_2 = b_5 = 1). M. Kreck and S. Stolz, *A diffeomorphism classification of 7-dimensional homogeneous Einstein manifolds with SU(3) x SU(2) x U(1)-symmetry*, Ann. of Math. 127 (1988) 373 (title and scope only).

Recalled and not reread, to be checked before any of them carries an argument: E. Witten, *String theory dynamics in various dimensions*, hep-th/9503124 (D0 states as momentum modes); O. Bergman and M. Gaberdiel, *Dualities of type 0 strings*, hep-th/9906055; M. Fabinger and P. Hořava, *Casimir forces between end-of-the-world branes*, hep-th/0002073; E. Witten, *SL(2, Z) action on three-dimensional conformal field theories with Abelian symmetry*, hep-th/0307041; C. N. Pope and P. van Nieuwenhuizen, *Compactifications of d = 11 supergravity on Kähler manifolds*, Commun. Math. Phys. 122 (1989) 281; J. Gauntlett, D. Martelli, J. Sparks and D. Waldram, *Supersymmetric AdS5 solutions of M-theory*, hep-th/0402153.

Already in the corpus: Witten 1981 and Castellani-D'Auria-Fré (LITERATURE.md §F); Hořava-Witten, Atiyah-Witten and Acharya-Witten-type cones (solidification §6).

Prior work for the observation of Theorem 2.1, that the weights of M^{111} are the hypercharges of the 5-bar: a thin web search of 4 Nov 2026 found no statement of it, as recorded in LITERATURE.md; this does not establish priority.

Files

- 01_interval_and_eleventh_dimension.md: the "1".
- 02_witten_spaces_hypercharge_quotient.md: the "6 + 1" of Witten's spaces, the Z_6, chirality from fluxes.
- 03_branes_hypercharge_and_the_84.md: C_3, M2, M5 and the 84.
- 04_two_families_m2_m5.md: the two 12-state blocks, two gradings of E8, Yukawa grades, and a proposed M2/M5 reading. An antiunitary J requires more than the complex-linear Chevalley involution.
- 05_colour_family_exchange.md: the two families as E8 root configurations, the colour-family transposition with four families.
- 06_main_point_heterotic_route.md: the main point: target, the heterotic web 9/10/11, dictionary, Proposition 6.1.
- 07_dualities_9_11.md: recalled dualities, the proposed D8-position dictionary, the conditional one-loop endpoint, and the doubled flux-torus matrix corner with a unitality amendment.
- 08_orientifold_sign.md: antiunitary commutants, a proposed orientifold interpretation, and conditional wall-enhancement arithmetic.
- 09_dilaton_spectral_and_so32.md: a conditional type I' coupling profile, SO(32) branching and diquark counts, and proposed string and del Pezzo readings; the full EPJC comparison remains pending.
- PLAN.md: the handover plan of the line.
- checks_11d.py: exact checks.
- ledger_11d.pl: ledger facts, included by ../ledger.pl.
- prompts/: the referee request and the first Astra job.
- ../reviews/2026-11-03_astra_audit_d11.md: the Astra 6 audit used in the second version.

Part 11D, 01 interval and eleventh dimension, referee pending

The interval of H as an eleventh dimension

Part of the 11D line of research (README). Propositions 1.1, 1.4 and 1.5 are proved here; Proposal 1.3 is a dictionary to be tested. The joint-limit family of the section of that name is from the Astra 6 audit of 3 Nov 2026 (reviews/2026-11-03_astra_audit_d11.md, §3 and §4), restated here in circle language and checked. Attempted by Claude (Anthropic), 3 Nov 2026; checks in Python, checks_11d.py, sections F, F2 and H. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

Referee note (GPT-6.1 Terra, 4 Nov 2026). Proposition 1.1 and the joint family's Weyl constant are independently checked; the unfolded circle is scalar, and even extensions of arbitrary smooth interval functions need not be smooth at the fixed points. See the referee report. Proposition 1.5 is amended at its statement below.

Referee note on Proposition 1.4 (GPT-6.1 Terra). The endpoint value, minimum and small-angle coefficient are independently checked under the stated one-loop hypotheses; see the report.

Correction (3 Nov 2026). There was an error in the first version of Proposition 1.4: it read the potential of lemma 24 as the whole energy of the walls, but that potential is taken relative to theta = 0 and omits the theta-independent Casimir energy. The corrected statement is below; the reading "the walls attract" is withdrawn. The first version of Proposal 1.3 also placed the W at the bottom of the tower, M_W ~ 1/(2R), along every path to v = 0; after the audit (its §2.2), that holds only on the scale-free path of Proposition 1.5.

Setting

Hypothesis H of STRATEGY.md reads the two sheets of the finite geometry as the minimal truncation of an interval I_l with opposite chiral boundary conditions, with l ~ 1/M_W. Its three regimes carry the labels 9, 10 and 11.

These are KO labels. At the 11 endpoint (v -> 0, l -> infinity) the heat dimension is 4 + 1 = 5, since the finite factor F keeps heat dimension 0 for every v (canonical §2.1). To read 11 as a metric dimension, the KO-6 factor must itself be resolved into a six-dimensional space, which gives the count

```
11 = 4 (spacetime) + 1 (the interval) + 6 (a space
↪  resolving F).
```

The files of this folder take the six-dimensional space to be B_6 = CP^2 x S^2, the base of Witten's spaces (02), and study two ways of completing 4 + 6 to eleven dimensions (03). This file treats the "1", and the path along which v -> 0 opens it.

Proposition 1.1 (the chiral interval is the antiperiodic circle)

Statement. Let D_l act on H_l = L^2([0, l], C^2) as in lemma 05, eq. (8): D_l(L, R) = (R', -L'), with R(0) = 0 and L(l) = 0, grading Gamma = diag(1, -1) and the scalar functions C^infinity([0, l]). Let S_R be the circle of length 2l, that is of radius R = l/pi, with coordinate x in [-l, l] and antiperiodic functions u(x + 2l) = -u(x). The map

```
(W(L, R))(x) = (L(|x|) + i sgn(x) R(|x|)) / sqrt 2
```

is a unitary H_l -> L^2_antiperiodic(S_R) with

1. W D_l W^* = -i d/dx on its antiperiodic domain;
2. W Gamma W^* = P, the reflection (Pu)(x) = u(-x);
3. W f W^* = multiplication by f(|x|), so the algebra goes to the reflection-even functions;
4. the walls x = 0 and x = l go to the two fixed points x = 0 and x = l ~ -l of the reflection.

Hence spec D_l = {(k + 1/2)/R : k in Z}, each eigenvalue simple, which is (9) of lemma 05, and the Connes-Lott mass of the minimal truncation is m = pi/(2l) = 1/(2R).

Proof. The inverse map is L(x) = (u(x) + u(-x))/sqrt 2, R(x) = (u(x) - u(-x))/(i sqrt 2) for x in [0, l], so W is a bijection, and |u(x)|^2 + |u(-x)|^2 = |L(x)|^2 + |R(x)|^2 gives unitarity. For H^1 data, u is continuous at 0 exactly when R(0) = 0, and u(l) + u(-l) = sqrt 2 L(l), so the antiperiodic matching u(-l) = -u(l) is the condition L(l)

= 0. On x > 0, -i (L + iR)' = R' + i(-L'), the image of (R', -L') = D_l(L, R). On x < 0, u(x) = (L(-x) - i R(-x))/sqrt 2 and -i u'(x) = (R'(-x) - i(-L'(-x)))/sqrt 2, which is again the image of (R', -L'). Gamma(L, R) = (L, -R) maps to L(|x|) - i sgn(x) R(|x|) = u(-x). Items 3 and 4 are immediate. The antiperiodic spectrum of -i d/dx on a circle of length 2l is pi(2k + 1)/(2l) = (k + 1/2)/R. ■

So the two sheets of H are the two fixed points of a reflection on a circle whose spinors are antiperiodic, and the grading of the finite geometry is that reflection. With finite data D_F supplied as in lemma 22, the parent unfolds to -i d/dx + D_F on the periodic circle of length 2l (lemma 25, Proposition 25.1), and the angle theta interpolates between that circle (theta = 0) and the antiperiodic circle of Proposition 1.1 (theta = pi/2): D_F is a constant connection, a Wilson line, and the wall angles theta_k = l mu_k (mu_k the eigenvalues of D_F) are its holonomy phases.

The joint family: weak masses below a descending tower

At fixed l, the vector masses of lemma 23 vanish as theta -> 0 and also as theta -> pi/2, while the tower spacing pi/l stays fixed; vanishing masses alone open no dimension (audit §2.2). The audit gives a family in which they do.

Proposition (Astra 6, audit §3; restated). Let D_F(epsilon) = epsilon D_* and l_epsilon = l_* epsilon^{-alpha}, 0 < alpha < 1, with wall data X = tan(l D_F) as in lemma 22. Then the exact window is W^* D W = epsilon D_, *the rest of the spectrum lies at /lambda/ >= pi/l_epsilon - epsilon //D_||*, the compression error is at most (2/3) l_^2 //D_||^2 epsilon^{2(1-alpha)} ||a|| (Theorem 22.4, item 2), and for alpha < beta < 1 a cut-off E_epsilon = E_* epsilon^beta contains exactly the r = dim H_F finite states, while at every fixed E > 0 the number of states N_epsilon(E) grows like l_epsilon E, which is proportional to epsilon^{-alpha}.

In the circle language of lemma 25 (the periodic circle, on which D_F is a Wilson line; the window is its n = 0 mode) this reads: the radius R_epsilon = l_epsilon/pi goes to infinity, and the Wilson-line angles theta_epsilon =

l_epsilon epsilon mu_k go to zero like epsilon^{1-alpha}. The hierarchy is

```
m_weak ~ epsilon  <<  E_epsilon ~ epsilon^beta  <<
 ↪  1/R_epsilon ~ epsilon^alpha  ->  0.
```

In the five-dimensional Yang-Mills toy of lemma 23 the vector poles along this family are M_W^2 = 2 m^2 (1 + O(theta^2)) and M_Z^2 = (16/5) m^2 (1 + O(theta^2)), so M_W^2/M_Z^2 -> 5/8 (audit §3.1, from Theorem 23.6).

Check (Claude). The parent space of lemma 22 is L^2([0, l]; C^2) ⊗ H_F, of fibre dimension 2r. By Theorem 22.1 its signed spectrum is (n pi +- theta_k)/l, n in Z, two progressions of spacing pi/l for each of the r/2 singular values, so N(E) = 2 r l E/pi + O(r), the constant of the audit (the scalar circle of Proposition 1.1 gives 2 E l/pi; with fibre H_F the count is 2 r l E/pi). The window, gap and compression statements follow from Theorems 22.1 and 22.4 as stated. (Python, section H: exact spectra for random singular values.)

The audit also gives a localized family (its §4): a bulk mass mu_epsilon = mu_* epsilon^alpha and mu l = (1 - alpha) log(1/epsilon) + c_0, with light mass m ~ 2 mu_* e^{-c_0} epsilon, gap ~ mu_* epsilon^alpha, and the exact tail identity e^{-2 kappa l} = (m/(mu + kappa))^2. Its table of roots is reproduced in section H (Python, bisection on tanh s = s/c and tan y = y/c): the four light masses and next masses agree with the audit to the digits printed there.

Referee note (GPT-6.1 Terra, 4 Nov 2026). The localized asymptotics and tail identity are independently checked analytically; the length includes a logarithmic factor. The numerical table was not recomputed. See the referee report.

Proposal 1.3 (the circle of H as the M-theory circle)

Take the circle of Proposition 1.1 to be the eleventh direction of M-theory over a ten-dimensional type IIA background, so R = g_s l_s. Then, along the joint family:

H	circle of Proposition 1.1	M-theory / string side (recalled, to check)
v -> 0 along the joint family, label 11	R -> infinity, Wilson-line angles -> 0, m_weak R -> 0	strong coupling: D0-brane bound states of mass n/(g_s l_s) become light and are the momentum modes of the eleventh dimension (Witten 1995); the weak masses lie below this tower
v finite, label 10, two sheets	R finite, fixed points of a reflection	M-theory on an interval with two walls (Hořava-Witten); here with walls of opposite chirality
v -> infinity, label 9, sheets merge	R -> 0, antiperiodic spinors	a circle with antiperiodic fermions shrinking: bulk fermions get masses >= 1/(2R); compare type 0 strings, which have no bulk spacetime fermions (Bergman-Gaberdiel 1999), and the walls of opposite chirality of Fabinger-Hořava 2000

Each row matches a statement already in the corpus: the 11 endpoint needs the tower (canonical §2.1, lemma 05 Part 4), and at the 9 endpoint the bare interval keeps no light fermion, so light fermions must come from another sector (STRATEGY, T1 and T3). On the string side the same happens: the light fermions of a type 0 theory come from open strings on branes. The proposal therefore places the light fermions of the 9 endpoint on branes or walls, and 03 develops a brane origin for them.

Tests.

1. Along the joint family m_weak/M_KK ~ epsilon^{1 - alpha} -> 0, so the weak scale is far below 1/(g_s l_s); the scale-free path of Proposition 1.5, with M_W R fixed, is the other possibility and

keeps no finite window.

2. Gauge couplings. A five-dimensional gauge field with constant profile on the interval has g_4^{-2} = l/g_5^2 = pi R/g_5^2 (audit §7.2 for the general volume formula), so at fixed g_5 the four-dimensional couplings vanish as R -> infinity, and M_W = g v/2 could vanish through g. The limit sought keeps g finite, which needs g_5^2 ~ R, and this must be shown for the vectors in question before the masses are read as restoration of the symmetry.
3. The path must be selected by an action: Proposition 1.5 shows what the flat interval alone selects.

Proposition 1.4 (the angle-dependent one-loop term at the vacuum angle)

Statement. Under the hypotheses of Theorem 9 of lemma 24 (flat interval, zero bulk masses, frequency-one fermions, the stated gauge towers and multiplicities, no wall potential), the angle-dependent part of the one-loop potential, taken relative to theta = 0 as in lemma 23, has its minimum at theta = pi/2 with value

```
V(pi/2) - V(0) = - 93 zeta(5) n_f / (512 pi^2 l^4),
```

and near theta = 0 it is (3 zeta(3)/(16 pi^2)) (54/5 - n_f) theta^2/l^4 + o(theta^2). At theta = pi/2 the exact vector poles of lemma 23 give M_W = M_Z = 0, with the fermion at the tower scale m = pi/(2l).

Proof. In the notation of lemma 24, V = $\mathcal{V}$/(8 pi^2 l^4), with F(0) = 0, and $\mathcal{V}$(1) = -n_f F(1) since the gauge terms vanish at t = 1. With 1 + 1/sinh^2 x = coth^2 x and log coth x = 2 sum over odd k of e^{-2kx}/k,

```
F(1) = 2 int_0^infinity x^3 log coth x dx = 4 sum_{k
 ↪  odd} (1/k) 3!/(2k)^4 = (3/2) sum_{k odd} k^{-5}
 ↪  = (3/2)(31/32) zeta(5) = 93 zeta(5)/64.
```

Near t = 0, $\mathcal{V}$'(0) = (6 + 24/5 - n_f) J(0) with J(0) = int x^3/sinh^2 x dx = (3/2) zeta(3), the coefficient already given in lemma 23. Python: F(1) = 1.506786, J(0) = 1.803085, series against quadrature to 10^-6. ■

Reading. The θ-independent Casimir energy of the towers, a_0/l^4, is not part of this potential and is not computed in lemmas 23 or 24; the direction in which l moves

depends on the sign of a_0 - 93 zeta(5) n_f/(512 pi^2). The minimum over the angle is the point with massless W and Z at fixed l, which opens no dimension (lemma 23 already notes that it is not the physical point).

Proposition 1.5 (a scale-free potential with an isolated stationary angle fixes M_W l)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The derivative identity is correct, and an isolated stationary angle gives $l \propto 1/m$. The universal conclusion has a flat-direction counterexample $G(\theta) = \theta^4$; for the displayed one-loop determinant the small-angle obstruction is recovered when $n_f \neq 54/5$. See the referee report.

Statement. Let V(l, theta) = l^{-4} G(theta) with G of class C^1, the form of every one-loop potential of a flat interval whose only scales are l and the wall angles (lemmas 23 and 24, with a_0 included in G).

1. If the angle is free, V is monotone in l at every fixed theta with G(theta) != 0. At the minimizing angle theta_min the length runs away: to l -> infinity if G(theta_min) > 0, to l -> 0 if G(theta_min) < 0. Stationary points exist only on the set G = G' = 0.
2. If the angle is tied to a fixed mass, theta = l m (as for X = tan(l M) in lemma 22), the stationary lengths satisfy theta G'(theta) = 4 G(theta) at theta = l m. So l = theta_/m *for a root theta_* that does not depend on m, and M_W l, M_Z l (functions of theta alone by Theorem 23.6) are the same at every m.

Hence, when the roots of theta G' - 4 G are isolated, a stationary branch has l = theta_*/m: the path $l \sim 1/M_W$ of H's original statement, alpha = 1 in the scaling of the joint family (Remark 10.7 of 10 for a small fixed ratio). A root interval, as for G = theta^4 near 0, makes every alpha < 1 stationary and selects none (referee amendment). The joint family with alpha < 1 needs a second scale in the action: a bulk mass (the localized family, G(l m, l mu)), the cutoff of the spectral action (lemma 25, G(theta, l Lambda)), curvature, flux or a wall term.

Proof. d/dl [l^{-4} G(l m)] = l^{-5} (theta G'(theta) - 4 G(theta)) at theta = l m. For item 1, d/dl V = -4

l^{-5} G(theta) and d/dtheta V = l^{-4} G'(theta). ■

Amendment (4 Nov 2026). The window can separate on the scale-free path, with a fixed small ratio, if the light mass is exponentially small in a stabilized shape parameter, and a_0 is now computed. See 10, Remark 10.7.

Consequence for the modulus target of the audit (§7.1). The audit proposes V = V_*(a r^{-p} + b epsilon^2 r^q) with a, b, p, q > 0, whose minimum r ~ epsilon^{-2/(p+q)} gives alpha = 2/(p + q) < 1 when p + q > 2. In the flat one-loop family the epsilon-dependent term near theta = 0 is (3 zeta(3)/(16 pi^2)) (54/5 - n_f) m^2/l^2, so q = -2, and with the Casimir term p = 4 the stationary point has l ~ 1/m, alpha = 1, in agreement with item 2. A term growing with l at fixed v, as the ansatz requires, is absent from the flat family and must come from one of the second scales listed above.

What remains open here

- Whether the identification of Proposal 1.3 survives the gauge sector: lemmas 13, 14 and 25 tie M_W to the tower with model-dependent coefficients, which here become a relation between M_W, g_s and l_s, and test 2 above requires g_5^2 ~ R.
- The theta-independent Casimir coefficient a_0 of the flat family, which decides the runaway direction of Proposition 1.5, item 1.
- A potential with a second scale that selects alpha < 1: the localized family of the audit is the first candidate, with G(l m, l mu) computable from the exact secular equations of lemma 09.
- A fermion with both a fibre momentum and a base twist: on a nontrivial circle bundle (Witten spaces, file 02) the antiperiodic shift corresponds to a spin structure whose fibre circle does not lift the circle action. Which Witten spaces carry such a structure is an open item for the next job.

Part 11D, 02 witten spaces hypercharge quotient, referee pending

Witten spaces as the hypercharge quotient of the five-dimensional representation

Part of the 11D line of research (README). Theorem 2.1 and Propositions 2.2 to 2.4 and 2.6 are proved here, Theorem 2.1 and Proposition 2.6 conditionally as the referee notes say; Proposition 2.5 is a negative result in a stated scope, refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5. Attempted by Claude (Anthropic), 3 Nov 2026, with a remark from the steer of the same day on the mixing of u and d. Checks: checks_11d.py, sections A to D, Python with exact fractions. Referee pending.

Referee note (GPT-6.1 Terra, 4 Nov 2026). The exclusion in Proposition 2.5 has an independent analytic proof. The required-bidegrees lemma and the general assertion that u^c, d^c must have the same flux image require a common ten-dimensional chirality; the exclusion itself also holds without that extra assumption. See the referee report; other reviewed items are noted separately.

Referee note on Theorem 2.1 and Propositions 2.2, 2.4 (GPT-6.1 Terra). The quotient and tensor-centre calculations are independently checked. The cited M^{111} identification is conditional on the recalled fibration; the original quotient metric needs rescaling to be Sasaki-Einstein, and the action on holomorphic functions is dual to the action on U. See the report.

Referee note on Proposition 2.3 (GPT-6.1 Terra). The Gysin, Smith normal form and spin calculations are independently checked; the cited Kreck-Stolz paper was unavailable. See the report.

Setting

Witten's spaces M^{pqr} = G/H, with G = SU(3) x SU(2) x U(1) and H = SU(2) x U(1) x U(1), are the seven-dimensional spaces of least dimension with the isometry group of the Standard Model, so 4 + 7 = 11 (Witten 1981; LITERATURE.md §F). Castellani, D'Auria and Fré gave them Einstein metrics; Killing spinors exist only

for p = q, where the AdS4 vacuum has N = 2 and the space is M^{111} (Fabbri, Fré, Gualtieri, Termonia, hep-th/9903036). Fabbri et al. (hep-th/9907219, §4) write the cone over M^{111} as the Kähler quotient of C^5 by a circle acting with weights 2 on a triplet U and -3 on a doublet V, with D-term 2|U|^2 = 3|V|^2, and M^{111} as a circle bundle over P^2 x P^1. In later notation it is M^{3,2}.

These weights are six times the hypercharges of the 5-bar of SU(5). The rest of the file develops that observation.

Theorem 2.1 (M^{111} is the hypercharge quotient of the 5-bar)

Statement. Let 5bar = C^3 + C^2 carry the Standard Model action on the left-handed fields d^c in (3bar, 1)_{1/3} and L in (1, 2)_{-1/2}; write U for d^c and V for L, so 6Y = 2 on U and -3 on V. Let mu_Y = (2|U|^2 - 3|V|^2)/6 be the moment map of hypercharge, and

```
Y_{3,2} = (mu_Y^{-1}(0) ∩ S^9) / U(1)_Y.
```

1. Y_{3,2} is a smooth, simply connected seven-manifold, the circle bundle of O(3, 2) over CP^2 x CP^1 (up to orientation), and its cone is the Kähler quotient C^5 // U(1)_Y of Fabbri et al.
2. The cone is Calabi-Yau (Gorenstein, sum of weights 0) because tr Y = 0 on the 5-bar, that is, because U(1)_Y lies in SU(5).
3. The group acting effectively on Y_{3,2} through the quotient is (U(3) x U(2))/U(1)_Y. Its subgroup S(U(3) x U(2))/U(1)_Y is PSU(3) x SO(3), and the overall phase of U(5) supplies one further circle, the Reeb (R-symmetry) direction.
4. The functions on Y_{3,2} that are polynomial in U, V and their conjugates have U(1)_Y-charge zero; the holomorphic ones are spanned by the monomials of degree (3k, 2k) in (U, V), whose span as tensors is Sym^{3k}(3bar) x Sym^{2k}(2) and as functions its dual Sym^{3k}(3) x Sym^{2k}(2): for k = 1 the functions form the (10, 3) of Fabbri et al.

Proof. On mu_Y^{-1}(0) ∩ S^9 we have |U|^2 = 3/5 and |V|^2 = 2/5, a product of round spheres S^5 x S^3. The circle acts by (lambda^2 U, lambda^{-3} V); a fixed point would need lambda^2 = lambda^3 = 1, so lambda

= 1 and the action is free. The quotient of S^5 x S^3 by the full torus T^2 of phases of U and V is CP^2 x CP^1, and Y_{3,2} -> CP^2 x CP^1 is a principal bundle for T^2/U(1)_Y. A character lambda_U^m lambda_V^n of T^2 is trivial on U(1)_Y exactly when 2m - 3n = 0, so the characters of T^2/U(1)_Y are generated by (m, n) = (3, 2), whose line bundle is O(3) ⊠ O(2) up to the sign convention; this gives item 1 and item 4. Simple connectivity follows from the exact homotopy sequence of U(1) -> S^5 x S^3 -> Y_{3,2}. Item 2: the sum of weights is 3 * 2 + 2 * (-3) = 0. Item 3: an element (g, h) of U(3) x U(2) acts trivially on the quotient exactly when (g, h) = (lambda^2, lambda^{-3}), because mu_Y^{-1}(0) spans C^5. The intersection of SU(3) x SU(2) with U(1)_Y consists of lambda^6 = 1, generated by (e^{2 pi i/3} I_3, -I_2), the centre Z_3 x Z_2. ■

So the isometry group of the Witten space with Killing spinors, SU(3) x SU(2) x U(1) (Fabbri et al.), is the centralizer of hypercharge in U(5), divided by hypercharge. The SU(3) x SU(2) part is the Standard Model one; the remaining circle is the R-symmetry, and hypercharge itself has been quotiented. Witten assigned the U(1) of G to hypercharge; in this realization hypercharge is a different U(1) (03, Proposal 3.1).

Proposition 2.2 (uniqueness of the count 7)

Statement. (a) Among the quotients Y_{a,b} = (S^5 x S^3)/U(1), with weights b on C^3 and -a on C^2 and gcd(a, b) = 1, a, b > 0, the cone is Calabi-Yau only for (a, b) = (3, 2). (b) For a complex representation of dimension n and a free circle action, (S^{2n-1} ∩ mu^{-1}(0))/U(1) has dimension 2n - 3; this is 7 only for n = 5. Among representations of SU(5), with hypercharge as the circle, only the 5 and the 5-bar give a seven-dimensional quotient.

Proof. (a) The Calabi-Yau condition is 3b - 2a = 0. (b) Dimension count: 2n - 1 for the sphere, minus one for the moment map, minus one for the circle. The irreducible representations of SU(5) of dimension 5 are the 5 and the 5-bar. ■

Witten's count reads: 11 = 4 + (2 * 5 - 3). The seven internal dimensions come from the smallest representation

of SU(5), with hypercharge as the circle. For the 10 the quotient has dimension 17, and for the 5 + 10 it has 27.

Proposition 2.3 (topology of Y_{a,b})

Statement. Let Y_{a,b} -> B = CP^2 x CP^1 be the circle bundle with Euler class a h_1 + b h_2, a != 0. Then H^2(Y) = Z + Z/gcd(a, b), b_3(Y) = 0, H^4(Y) is a group of order a^2, cyclic when gcd(a, b) = 1, and Y is spin exactly when a is odd and b is even. For (a, b) = (3, 2): H^2 = Z, H^4 = Z/9, spin. The pullback H^2(B) = Z^2 -> H^2(Y) = Z is (x, y) -> 2x - 3y up to sign: the two divisor classes h_1 = [{l(U) = 0}] and h_2 = [{m(V) = 0}] go to 2 and -3, the values of 6Y on d^c and L.

Proof. The Gysin sequence H^{k-2}(B) -> H^k(B) -> H^k(Y) -> H^{k-1}(B) -> H^{k+1}(B), with the first map the cup product with c = a h_1 + b h_2 and H^*(B) = Z[h_1, h_2]/(h_1^3, h_2^2). In degree 2 the cokernel of 1 -> (a, b) is Z^2/(a, b). In degree 4 the map H^2(B) -> H^4(B) has matrix [[a, 0], [b, a]] in the bases (h_1, h_2), (h_1^2, h_1 h_2), with determinant a^2 and Smith invariants (gcd(a, b), a^2/gcd(a, b)); since H^3(B) = 0 this gives H^4(Y), and the same matrix is injective, so H^3(Y) = 0. For spin: w_2(B) = h_1 mod 2, and the kernel of H^2(B; Z/2) -> H^2(Y; Z/2) is spanned by c mod 2; so w_2(Y) = 0 exactly when (1, 0) = (a, b) mod 2. For (3, 2), Z^2/(3, 2) -> Z by (x, y) -> 2x - 3y is an isomorphism. ■ (Python: Smith invariants and the spin condition, section C.)

The product member CP^2 x S^2 x S^1 of Witten's family (a = b = 0) is not spin.

Proposition 2.4 (the Z_6 of the Standard Model)

Statement. (a) Every monomial in U and V, and every state of the exterior algebra Λ^*(5bar), satisfies 2t + 3d + 6Y = 0 mod 6, with t the triality (1 for 3, -1 for 3bar) and d = 2j mod 2. (b) The U(1)_Y-neutral monomials have t = 0 and d = 0. (c) Every tensor field on Y_{3,2} carries a representation of (U(3) x U(2))/U(1)_Y, on which SU(3) x SU(2) acts through PSU(3) x SO(3). So no tensor Kaluza-Klein mode on Y_{3,2}, on a background invariant under the displayed action, is a colour

triplet or a weak doublet; spinors are open.

Proof. (a) holds for U (t = -1, d = 0, 6Y = 2) and V (t = 0, d = 1, 6Y = -3) and is additive. (b) U^alpha V^beta is neutral when 2 alpha = 3 beta, so 3 | alpha and 2 | beta. (c) The effective group of Theorem 2.1 acts on the frame bundle and on every tensor bundle, and its intersection with SU(3) x SU(2) is the centre Z_6 (Theorem 2.1, item 3). ■

(a) is the statement that the Standard Model group is (SU(3) x SU(2) x U(1))/Z_6 = S(U(3) x U(2)). In the quotient realization the Z_6 is the kernel of SU(3) x SU(2) -> Isom(Y_{3,2}). States with fractional hypercharge are those charged under the quotient circle, and they cannot be functions or tensors on Y_{3,2}. For spinors the isometry group may lift to a double cover; whether fermionic Kaluza-Klein modes on Y_{3,2} can carry half-integer weak isospin or triality is open (next job, item 2 of the README).

Chiral zero modes on the base with abelian fluxes

Witten's argument makes Kaluza-Klein fermions on an odd-dimensional space vectorlike, for gauge fields acting by isometries. The even base B_6 = CP^2 x CP^1 escapes it: abelian fluxes on B_6 give chiral four-dimensional fermions through the index of the twisted Dirac operator. This section asks which fluxes give one generation.

Setting. Ten-dimensional Weyl fermions on M^4 x CP^2 x CP^1, each with definite charges under some U(1) factors with fluxes on B_6. Since CP^2 is not spin, a fermion is a spin^c spinor, Λ^{0,*} ⊗ K^{1/2} ⊗ N with N the flux line bundle of its charges, and the four-dimensional chiral content is the virtual representation H^*(O(alpha, beta)) with (alpha, beta) = (-3/2, -1) + pairing of the charges with the fluxes. The cohomology is H^0(O(alpha)) = Sym^alpha(3bar) for alpha >= 0, H^2(O(alpha)) = Sym^{-alpha-3}(3) for alpha <= -3, and zero otherwise on CP^2; on CP^1, H^0(O(beta)) = Sym^beta(2) for beta >= 0 and H^1(O(beta)) = Sym^{-beta-2}(2) for beta <= -2. Each Standard Model Weyl field must come from one ten-dimensional field, from the whole cohomology of one line bundle.

Lemma (required bidegrees). Writing P_s = (alpha + 3/2, beta + 1) for the pairing of the charges of field s (up to the common sign of charge and chirality):

field	P_s	note
Q (3, 2)	(-5/2, 2)	H^2(O(-4)) x H^0(O(1))
u^c, d^c (3bar, 1)	(5/2, 1)	H^0(O(1)) x H^0(O(0))
L (1, 2)	(+-3/2, 2)	
e^c, nu^c (1, 1)	(+-3/2, 1)	

The second entry is 2j + 1 and the first is +-5/2 on colour triplets and +-3/2 on singlets. (Python, section D, by enumeration of the line bundles with these cohomologies and the Riemann-Roch check.)

Proposition 2.5 (no generation from fluxes along hypercharge and B - L) *Scope note (4 Nov 2026, after the referee).* The statement on equal images and the linearity of 2j + 1 assume a common ten-dimensional chirality for the fields. The conclusion holds without it: in the span of 6Y and 3(B - L) the charges of L are -3 times those of Q, so their second pairings are P_2(L) = -3 P_2(Q), against |P_2| = 2 on both doublets.

Referee note on the third-charge search (GPT-6.1 Terra, 4 Nov 2026). The bounded search below has an independent analytic proof. The same argument extends to any real third charge and independent source chiralities; this stronger extension was checked by Claude Sonnet 5.5 (second check: the vectors y, b, p, w have rank 4 for every sign pattern), so it stands as a no-go in the same scope with two model identifiers. See the referee report.

Statement. In the setting above, with the canonical spin^c root and fluxes only in U(1) directions spanned by Y and B - L (any flux quanta), no choice reproduces one generation, with or without nu^c. More generally, the pairing must give u^c and d^c the same image, and 2j + 1 = (2, 1, 1, 2, 1, 1) on (Q, u^c, d^c, L, e^c, nu^c) must be a linear function of the fluxed charges.

Proof. The pairing M is linear in the charges, and M c_s = P_s with c_s the charges of the left-handed field s. With only Y and B - L fluxed, c_u - c_d = (-6, 0)

in the coordinates (6Y, 3(B - L)), so M vanishes on the hypercharge direction and P_s = 3(B - L)_s m for one vector m. The second entries 2 on Q and 1 on u^c would then need 3(B - L) to take values in the ratio 2 : 1 on (Q, u^c), against 1 : -1. In general, 2j + 1 is not in the span of 6Y = (1, -4, 2, -3, 6, 0) and 3(B - L) = (1, -1, -1, -3, 3, 3): from Q and u^c the coefficients are -1 and 3, and then d^c gives -5 against 1. ■ An exhaustive search adding a third integer charge z in [-3, 3]^6 to Y and B - L finds no solution either (Python, section D).

Scope and escapes. Abelian fluxes on CP^2 x CP^1 with the Fubini-Study complex structure; one ten-dimensional Weyl field per Standard Model field; canonical spin^c root; Dolbeault index as the chiral content. Known escapes: nonabelian bundles (an SU(2)_R doublet bundle), several ten-dimensional fields per four-dimensional field, a changed common twist, orbifolds and walls, and non-Kaluza-Klein (brane) states. The result points to the next proposition.

Proposition 2.6 (one generation from a universal charge, the grading and B - L) Referee note (GPT-6.1 Terra, 4 Nov 2026). The index calculation works with the stated cohomology convention. For the colour action of the earlier quotient, reverse the first-component coefficients to obtain the labelled colour representations; global flux quantization remains to be specified. See the referee report.

Statement. Let the fluxed charges be F (one on every field: a change of the common spin^c twist), chi (+1 on Q and L, -1 on u^c, d^c, e^c, nu^c) and B - L. Then one generation with nu^c is the exact index content for the pairing

```
P^{(1)} = c_0 chi + c_1 (B - L),   P^{(2)} = (3F +
↪  chi)/2,
```

with (c_0, c_1) = (-9/4, -3/4) or (-3/2, -3). The bidegrees are Q: (-4, 1); u^c, d^c: (1, 0); L: (-3, 1) or (0, 1); e^c, nu^c: (0, 0) or (-3, 0). Without chi, or without B - L, there is no solution.

Proof. Substitution, and the two failures by the span test (Python, section D2). ■

Reading.

1. u^c and d^c see the same line bundle O(1, 0), and so do e^c and nu^c. The flux treats them as SU(2)_R doublets with a common internal profile; hypercharge Y = T_3R + (B - L)/2 enters through T_3R, which carries no flux. This is the left-right pattern of Pati-Salam, produced here by the index.
2. chi is the finite grading gamma_F on the particle basis. The flux therefore couples to the grading, an axial abelian connection on B_6; in the language of the finite geometry this is a fluctuation proportional to gamma_F, the structure of the two gradings of lemma 23.
3. **Mass and charge bases.** Because u^c and d^c share one zero-mode profile, overlap integrals on B_6 cannot separate them; their masses come from data that break SU(2)_R (Higgs or wall couplings), and the basis that diagonalizes the masses is set by those data, independently of the basis in which the charges are diagonal. The steer of 3 Nov 2026 remarks that u and d should still mix, since mass eigenvalues are not charge eigenvalues, and that at the 9 label, with the left-handed interactions off, charge eigenvalues can coincide. Proposition 2.6 puts the coincidence in the flux charges: equal flux charges for u^c and d^c, unequal hypercharges through T_3R. At the 9 label the electroweak part of the algebra is removed (lemma 21, STRATEGY T2), and what distinguishes u^c from d^c in the light theory is then the vectorlike charge alone. Making this precise needs the SU(2)_R-breaking wall data and the 9D reduction of this index; listed as item 4 of the README.

What this file establishes, and the direction

Referee note (GPT-6.1 Terra, 4 Nov 2026). The M^{111} identification is conditional on the recalled fibration and metric normalization. The flux examples establish a particular construction and scoped obstructions; a general necessity for the grading flux remains open. See the wording audit.

The Witten space with Killing spinors is the hypercharge quotient of the 5-bar (Theorem 2.1), the count 7 is fixed by SU(5) (Proposition 2.2), and the Z_6 of the Standard Model is the kernel of the isometry action (Proposition

2.4). The price is that hypercharge is no longer an isometry, and no tensor Kaluza-Klein mode on an invariant background carries triality, weak isospin 1/2 or fractional hypercharge. Kaluza-Klein chirality on the base needs fluxes that see the grading and leave T_3R free (Propositions 2.5 and 2.6). Both results point to states outside the Kaluza-Klein spectrum of Y_{3,2} for the quarks and leptons: 03 takes them to be wrapped branes.

Part 11D, 07 dualities 9 11, referee pending

Dualities between nine and eleven dimensions, adapted to the spectral triple

Part of the 11D line of research (README, PLAN, task C). Attempted by Claude (Anthropic), 4 Nov 2026. Propositions 7.1 to 7.3 are proved here by hand; the string-theory statements of section 1 are recalled, not reread; the readings are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

1. The dualities (recalled)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The duality descriptions are conditional on the recalled sources. The M2 wrapping number and the auxiliary torus flux are separately specified; a map identifying them remains open. See the referee report.

1. **M-theory on T^2 and type IIB on S^1** (Schwarz, hep-th/9508143; Aspinwall, hep-th/9508154). The modulus tau of the torus is the IIB coupling and SL(2, Z) is IIB S-duality; the IIB radius scales as a negative power of the area A of the torus. M2-branes wrapped on T^2 are IIB momentum states, and momenta on T^2 are (p, q)-strings wound on S^1. Eleven dimensions at A -> infinity, ten (IIB) at A -> 0, nine in between.
2. **Sixteen supercharges, rank 17** (Polchinski and Witten, hep-th/9510169; Hořava and Witten, hep-th/9510209). The two heterotic strings on S^1 are T-dual through Wilson lines; their type I' form is

an interval with two orientifold planes O8 and sixteen D8-branes whose positions are the Wilson-line eigenvalues; at strong coupling, M-theory on an interval times a circle, with E8 walls.

3. **Sixteen supercharges, lower rank.** The CHL string (rank 9) and M-theory on the Möbius strip; a rank-1 component related to the Klein bottle (Aharony, Komargodski and Patir, hep-th/0702195); further components with discrete theta angles (Montero and Parra de Freitas, arXiv:2209.03361).
4. **Light-cone M-theory.** On a light-like circle with momentum N, M-theory is the U(N) quantum mechanics of nine Hermitian matrices (Banks, Fischler, Shenker and Susskind, hep-th/9610043; Susskind). On T^2 with a light-like component of C_3, the torus becomes noncommutative and SL(2, Z) acts by Morita equivalence (Connes, Douglas and Schwarz, hep-th/9711162).

2. What each one offers the finite geometry

duality	9	11	the finite triple
1	the nine-dimensional theory	large torus	the lowest Landau level of a torus with flux p: algebra M_p(C), J from charge conjugation with doubling (Proposition 7.3); towers = Landau levels; p = wrapped M2-branes = IIB momentum
2	type I' interval, or heterotic on S^1	Hořava-Witten on S^1	two sheets = two orientifold walls; D_F = Wilson lines = D8 positions (Propositions 6.1, 7.1, 7.2); A_F from Chan-Paton algebras; J from the orientifold

duality	9	11	the finite triple
3	rank-reduced nine-dimensional theories	M-theory on non-orientable surfaces	J as an orientation reversal; the reflection with antiperiodicity of Proposition 1.1 is the Möbius-type ingredient
4	nine transverse matrices, Spin(9)	large N	A_F as the commutant of block-diagonal matrices: stacks of 1, 2, 3 give C + M_2(C) + M_3(C); H_F in off-diagonal blocks; D_F = Γ^i ad(X^i), masses = separations

Duality 2 is the one with an exact meeting point with the corpus, through the angle of the top channel.

3. Proposition 7.1 (the angle of a channel lives on an interval)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The spectral quotient and endpoint multiplicities are independently checked. Theorem 22.1 uses the strict interior; the antiperiodic endpoint is a limit, with a degenerate window. Equality of one channel spectrum does not by itself identify full physical backgrounds. See the referee report.

Statement. In the parent of lemma 22, a channel with singular value σ of the Yukawa block and angle θ = l σ has the signed spectrum {(n pi + θ)/l, (n pi − θ)/l : n in Z} (Theorem 22.1). As a set, this depends only on the class of θ in R modulo the group generated by θ -> −θ and θ -> θ + pi, whose fundamental domain is [0, pi/2]. At the endpoints:

1. θ = 0: the spectrum is n pi/l with a zero mode of multiplicity two (periodic);
2. θ = pi/2: the spectrum is (n + 1/2) pi/l, each value twice (antiperiodic), two copies of the spectrum of the chiral interval of Proposition 1.1; it is the limit

X -> infinity of the wall data X = tan(θ), where the lowest eigenspace is degenerate, and in lemma 23 the point of antipodal gradings, where M_W = M_Z = 0 at fixed l with the fermion at pi/(2l).

Proof. The set is invariant under θ -> −θ (it exchanges the two progressions) and θ -> θ + pi (it shifts n). The orbit of θ meets [0, pi/2] in one point. The endpoint values are substitutions; for θ -> pi/2 the wall conditions Q_R(l) = X U_R(l), U_L(l) = −X^* Q_L(l) tend to U_R(l) = Q_L(l) = 0. The statements at θ = pi/2 are those of lemma 23 (Theorem 23.6 and the reading after Proposition 23.7). ■

So the parameter space of one channel of the finite Dirac operator, for a given parent, is an interval with two distinguished ends, and lemma 22's restriction l ||D_F|| < pi/2 is the choice of the fundamental domain.

4. Proposition 7.2 (the far end is the displaced D8-brane on the other wall)

Referee note (GPT-6.1 Terra, 4 Nov 2026). At $a = 1/2$ the full $SO(32)$ centralizer is $S(O(30) \times O(2))$, whose identity component is $SO(30) \times SO(2)$. The stated one-loop minimum uses the hypotheses of lemma 24 and $n_f > 0$; the D8 dictionary remains a proposal. (Second check.) In Spin(32)/Z_2 the component of determinant -1 sends the lift of W_{1/2} to its negative, so the heterotic centralizer is connected; a -> a + 1 holds for SO(32) matrices, and the lift has period 2. See the referee report.

Statement. Put a = θ/pi in [0, 1/2]. By Proposition 6.1 the channel spectrum is that of the states (30)_{±1} of SO(32) on a circle of radius R = l/pi with the Wilson line W_a = 1_30 ⊕ R(2 pi a). The interval [0, 1/2] is the moduli interval of one Wilson-line eigenvalue modulo a -> −a and a -> a + 1; the unbroken group is SO(32) at a = 0 and SO(30) x SO(2) for 0 < a <= 1/2; at a = 1/2 the holonomy is −1 on the plane of SO(2), W_{1/2} = diag(1_30, −1_2), and the (30)_{±1} tower is (n + 1/2)/R, the antiperiodic tower of Proposition 1.1. The one-loop minimum of lemma 24, Theorem 9, θ = pi/2, is the point a = 1/2.

Proof. Proposition 6.1 and Proposition 7.1 with θ = pi

a; the commutant of W_a in SO(32) for $0 < a < 1/2$ is SO(30) x SO(2), and at $a = 1/2$ its identity component is still SO(30) x SO(2), since only two eigenvalues are -1. Lemma 24, Theorem 9, item 1, places the minimum at $t = \sin^2 \theta = 1$. ■

Reading (proposal). In type I' the Wilson line W_a is fifteen D8-branes, with images, on one orientifold plane and the sixteenth at distance proportional to a from it (proportional only at constant dilaton; see 10, Proposition 10.1); $a = 1/2$ puts it on the other orientifold plane (recalled: hep-th/9711098, hep-th/0108135). The dictionary of file 06 then reads: the top angle is the position of the top's D8-brane on the type I' interval, the two ends of the fundamental domain are the two orientifold walls, and the one-loop vacuum of the flat family of lemmas 23 and 24 carries the top brane to the far wall, where the tower of the stack-brane strings is antiperiodic. In supersymmetric type I' the D8 positions are flat directions; the potential of lemma 24 belongs to the non-supersymmetric flat family, the setting in which a force on the brane is expected.

Under the strong-coupling duality of item 2 of section 1, the D8 position is a Wilson-line eigenvalue of the heterotic string on S^1, and in eleven dimensions an E8 holonomy around the circle on a Hořava-Witten wall. So the same datum, the top angle, is a brane position at 9 and a holonomy at 11.

5. Proposition 7.3 (a torus with flux, doubled, has the colour form)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The signs, kernel dimensions and generated matrix corner check out, with a spin-line twist for nontrivial spin structures. The displayed action is degenerate and nonunital on the doubled Hilbert space, so the claimed unital finite triple needs an extra scalar summand. See the referee report.

Statement. Let T^2 be a flat torus of area A with a spin structure, L a line bundle of degree $p >= 1$ with a connection of constant curvature, and Lbar = L^{-1} with the conjugate connection. On H = L^2(S ⊗ L) ⊕ L^2(S ⊗ Lbar) put D = D_L ⊕ D_Lbar, gamma = gamma_S ⊕ gamma_S, and $J(\psi, \chi) = (C\chi, -C\psi)$, where

C is the charge conjugation of the spin structure (antiunitary, C^2 = −1, C D = D C, C gamma_S = −gamma_S C, mapping S ⊗ Lbar to S ⊗ L). Let C^infinity(T^2) act by multiplication on the first summand only. Then:

1. J^2 = 1, J D = D J and J gamma = −gamma J, the signs of KO-dimension six, and the algebra satisfies order zero;
2. ker D = H^0(L) ⊕ H^1(Lbar) ≅ C^p ⊕ conj(C^p), with gamma = +1 on the first summand and −1 on the second; the other eigenvalues are the Landau levels, of order (p/A)^{1/2}, which go to infinity as A -> 0 at fixed p;
3. the compressions of the functions to the kernel generate M_p(C) on C^p ⊕ 0, so the window is the finite triple (M_p(C), C^p ⊕ conj(C^p), 0, J, gamma), with KO-6 signs and order zero: M_p(C) acts on the first summand and, through J, by its opposite on the second.

Proof. (1) J^2(ψ, χ) = (C(−Cψ), −C(Cχ)) = (ψ, χ) since C^2 = −1; JD = DJ and J gamma = −gamma J follow from the properties of C; order zero holds because the algebra acts on the first summand and J b J^{-1} on the second. (2) On a torus the spinor bundle is Λ^{0,*} up to a flat factor, so ker D_L = H^0(L) ⊕ H^1(L) = C^p ⊕ 0 by Riemann-Roch in genus one, and ker D_Lbar = 0 ⊕ H^1(Lbar) ≅ conj(C^p) by Serre duality; the Landau levels of a constant field of total flux p on area A scale as (p/A)^{1/2}. (3) The compression of a character e_{m,n} to H^0(L) is a nonzero multiple of a clock-and-shift Weyl operator U^m V^n with UV = e^{2 pi i/p} VU (Berezin-Toeplitz quantization of the torus; recalled; in the theta-function basis the compression is a Gaussian factor times the Weyl word, checked for p = 2 to 5 in the second check), and these operators span M_p(C). ■

The construction is the torus analogue of lemma 16 (a doubled parent with restricted algebra and a J covering the flip; Example 16.9 for the twistor case), and its window at p = 3 has the form of the colour sector of the Standard Model triple: M_3(C) on the quarks and its opposite on the antiquarks. Berezin-Toeplitz quantization of the torus is known; the KO-6 doubling is the corpus device.

Reading for duality 1 (proposal). The area A -> 0 at fixed flux keeps the window and sends the tower to infinity, the towerless limit; A -> infinity brings the Landau levels down. In M-theory on T^2, flux and wrapping number of M2-branes on T^2 are IIB momentum states, so the dimension p of the window counts IIB momentum on the nine-dimensional circle, and the SL(2, Z) of the torus acts on the family by Morita equivalence of the noncommutative tori of Connes, Douglas and Schwarz when a C_3 background is on. Whether p = 3 for colour has a reason in this picture is open.

6. A remark on duality 4 (known)

For U(N) matrices X^i (i = 1, ..., 9) that are block-diagonal, X^i = ⊕*a* *x_a^i* *1*{n_a} with distinct points x_a in R^9, the commutant of all X^i is ⊕*a* *M*{n_a}(C), and the fermion operator Γ^i ad(X^i) on the adjoint, with the sixteen-component spinor of Spin(9), vanishes on the diagonal blocks and has eigenvalues ±|x_a − x_b|, each with multiplicity 8 n_a n_b per ordered block (a, b), that is 16 n_a n_b per unordered pair. With stacks of sizes 1, 2, 3 the commutant is C + M_2(C) + M_3(C), the complex form of A_F, the off-diagonal blocks are bifundamentals as in the Krajewski diagrams of the finite geometry, and the masses are the distances between stacks; coincident stacks restore U(N), the analogue of massless W and Z, and large N is the eleven-dimensional limit. A quaternionic factor H needs an orientifold projection on the two-stack, and chirality needs more than the nine Hermitian matrices.

6b. The three labels in type I' variables (proposal)

Correction (4 Nov 2026). The length l of H and the type I' length are T-dual, and the limit at label 11 below ends at ten-dimensional SO(32). See 10, Proposition 10.3.

Hypothesis H has two parameters for the top channel: the interval length l and the angle θ = l m_t in [0, pi/2] (Proposition 7.1). In the type I' reading of Proposition 7.2 they are the length of the interval and the position of the top's D8-brane as a fraction of it. Then:

- **label 9:** θ = pi/2 at fixed l, the brane on the far

orientifold wall: the top at the tower scale pi/(2l) (lemma 22, Corollary 22.9: W, Z and top at the tower scale in the 10 -> 9 family), the antiperiodic tower, and the one-loop minimum of the flat family (lemma 24);

- **label 10:** $0 < \theta < \pi/2$, the brane between the walls, a window below the tower;
- **label 11:** $\theta \to 0$ together with $l \to \infty$, the joint family of 01 ($\theta_epsilon \sim epsilon\hat{}\{1 - alpha\}$): the brane returns to the stack, SO(32) is restored on that channel, and the interval opens.

In this reading the 9 and 11 endpoints are the two ends of the brane's path, the far wall and the stack, and the second scale required by Proposition 1.5 must fix how l grows as the brane returns. The flat one-loop potential pushes the brane to the far wall (label 9); reaching label 11 needs a term that pulls it back to the stack while the interval grows.

7. What this adds, and what remains open

Referee note (GPT-6.1 Terra, 4 Nov 2026). The brane and E_8-holonomy readings below are proposals. Proposition 7.3's displayed algebra action is nonunital; the corrected finite datum is in the referee report. Proposition 1.5 also permits flat directions.

Added: the angle of the top channel lives on an interval whose ends are the two orientifold walls of type I' (7.1, 7.2); the corpus's one-loop vacuum puts the top brane on the far wall, the SO(32) Wilson line diag(1_30, −1_2); the same datum is an E8 holonomy in the eleven-dimensional frame; a torus with flux gives the colour form of the finite triple with KO-6 signs (7.3).

Open: - A nine-dimensional spectral triple with sixteen D8 positions as the eigenvalues of D_F on the type I' interval, and its strong-coupling limit with E8 walls; the configuration 15 + 1 lies at the SO(32) end, while E8 needs seven D8-branes at an orientifold plane with divergent coupling. - The quaternionic factor H from a torus with even flux and a real structure of quaternionic type, and the full A_F from several tori or from a rank-three bundle with flux. - The lower-rank nine-dimensional components (duality 3), whose J-like orientation reversal may supply the real structure of the finite triple.

Part 11D, 08 orientifold sign, referee pending

The sign of lemma 06 as an orientifold type, and E8 walls from the configuration 15 + 1

Part of the 11D line of research (README, PLAN, task C, next step of 07). Attempted by Claude (Anthropic), 4 Nov 2026. Proposition 8.1 is proved (it adds a commutant statement to lemma 06); Proposition 8.2 is proved from Proposition 7.1; Remark 8.3 rests on recalled string-theory rules and on the dictionary of 06; readings are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

1. Setting

In type I' the two ends of the interval are orientifold planes. A stack of n Chan-Paton labels (k D8-branes with their images, n = 2k) on an orientifold plane of type O^- carries SO(n), and on one of type O^+ carries USp(n) (recalled; the physical O8 planes of the SO(32) web are O8^-). Lemma 06 factors a KO-6 finite triple into an odd one through a grading-reversing unitary involution U, commuting with the algebra and with D, with U J = s J U; s = −1 gives KO-5 (label 9) and s = +1 gives KO-7 (label 11), and the antiunitary j induced on H_+ satisfies j^2 = s and commutes with d^2 = D^2|{H+} (lemma 06, proof of Result 2 and the check of the electrodynamics data). STRATEGY T2c: with a realistic seesaw sector only s = +1 occurs.

2. Proposition 8.1 (the factorization sign fixes orthogonal or symplectic symmetry)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The antiunitary commutants are independently checked. They need not preserve d or the remaining data, and the Chan-Paton identification is a proposal. See the referee report.

Statement. Let E be an eigenspace of d^2 in H_+, of dimension n. The unitaries of E that commute with j

form O(n) if s = +1, and USp(n) if s = −1, in which case n is even.

Proof. j preserves E because it commutes with d^2. An antiunitary with j^2 = +1 is a real structure: E = E_R ⊗ C with E_R its fixed points, and a unitary commutes with j exactly when it preserves E_R, which gives O(n). An antiunitary with j^2 = −1 is a quaternionic structure: E is a right H-module of quaternionic dimension n/2, and the unitaries commuting with j form USp(n). ■

The even dimension for s = −1 is lemma 06's own statement (Kramers degeneracy of the singular values). The proposition adds the symmetry group.

Reading (proposal). The groups are those of Chan-Paton labels on an orientifold plane: s = +1 with O^-(SO), s = −1 with O^+ (USp). So the two odd factorizations of the modern triple, which carry the labels 11 and 9, have the form of the two kinds of orientifold planes. The seesaw sector selects s = +1 (T2c), which is the O^-type of the SO(32) web used in 06 and 07: the physical neutrino sector and the type I' walls of the dictionary agree in type. Two refinements from the second check: the symmetries of the pair (j, d) are smaller, O(n_+) x O(n_-) for s = +1 and U(n/2) for s = -1, since for s = -1 the antiunitary j swaps the +-mu eigenspaces of d; and there is a tension of labels, since here label 9 goes with O^+ while section 6b of 07 puts label 9 on a far O8^-wall (the two are the KO sign and the position of the top's brane, see 10).

3. Proposition 8.2 (wall channels come with doubled multiplicity)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The stated count is for one signed eigenvalue. The lowest $|D|$ multiplicity is $2k$ in the interior and at zero, and $4k$ at $\pi/2$; interpreting the signed count as Chan-Paton labels requires further data. See the referee report.

Statement. In the parent of lemma 22, k channels at a common angle θ contribute, at the eigenvalue θ/l of the window, an eigenspace of dimension k if 0 < θ < pi/2 and of dimension 2k at the ends θ = 0 and θ = pi/2.

Proof. By Proposition 7.1 each channel has the spectrum {(n pi ± θ)/l}; the value θ/l occurs once (n = 0,

sign +) for 0 < θ < pi/2, and twice at the ends: at θ = 0 as the zero mode of multiplicity two, at θ = pi/2 as (0 + pi/2)/l = (pi − pi/2)/l. ■

Reading (proposal). k D8-branes in the bulk carry k Chan-Paton labels and k D8-branes on an orientifold plane carry 2k, with their images; the parent reproduces this count, with the ends of the angle interval as the walls. With Proposition 8.1, a stack of channels at a wall has a symmetry of type SO(2k) or USp(2k), as a stack of D8-branes on an O8^- or O8^+ plane. The rank-reduced nine-dimensional theory with one O8^- and one O8^+ (duality 3 of file 07) would then be a parent whose two walls carry opposite signs s: one sheet of type 11 and the other of type 9.

4. Remark 8.3 (E8 walls from 15 + 1; conditional)

Correction (4 Nov 2026). The far-wall group is reached at the string-scale radius R = R_*, with the tower fixed, and only its charged states are D0-branes. See 10.

Referee note (GPT-6.1 Terra, 4 Nov 2026). The brane and slope arithmetic is independently checked conditional on the recalled rule. Wall enhancement requires tuning $z = 0$; for $7 + 7$ the two bulk positions must sum to L. The E_2 identification with the physical electroweak sector needs further data. The cited primary sources were unavailable to the referee; the slope rule and the E_{n+1} table were confirmed against Polchinski-Witten and Bergman-Gaberdiel-Lifschytz in the second check. See the referee report.

Conditions: the dictionary of 06 (fifteen D8-branes for the five light quarks in three colours, the sixteenth for the top) and the recalled rule that an O8^- plane with n <= 7 D8-branes on it, at infinite coupling, carries E_{n+1} (E_4 = SU(5), E_5 = SO(10), E_6, E_7, E_8), while n >= 8 admits no such point (Seiberg, hep-th/9608111; Bergman, Gaberdiel and Lifschytz, hep-th/9711098; recalled).

1. Dilaton bookkeeping: the dilaton is e^{phi} = z^{−5/6} with z piecewise linear along the interval (conformal coordinate; Polchinski and Witten, hep-th/9510169) and slope proportional to 8 − n_left, where n_left counts the D8-branes to the left (O8^-

carrying −8 units), so the coupling grows toward the wall with fewer branes; this reproduces the constant dilaton of 8 + 8, the SU(2) at the empty wall of the SO(32) point (the self-dual radius of the heterotic string) and 7 + 7 for E8 x E8 (extrapolated from the single-wall rule; neither Polchinski-Witten nor Bergman-Gaberdiel-Lifschytz states it). In the configuration 15 + 1 the wall with fifteen branes has no E point (15 > 7), and the coupling can diverge only at the far wall. With the top's brane on the far wall (the label-9 configuration of Proposition 7.2, n = 1) the far wall enhances to E_2 = SU(2) x U(1), the electroweak group; with the top's brane in the bulk it is empty and enhances to E_1 = SU(2). The light-quark wall keeps SO(30).

2. If branes move in colour triples (a light flavour is three D8-branes), a wall holds 3j branes, or 3j + 1 with the top's brane. The reachable wall groups are E_1, E_2 (the top's brane alone), E_4 = SU(5) (one flavour), E_5 = SO(10) (one flavour and the top's brane), E_7 (two flavours) and E_8 (two flavours and the top's brane); E_6 (five branes) is not reachable.
3. Hence an E8 wall needs the top's brane together with two light flavours, and two E8 walls are impossible with one top brane: E8 x E8 requires breaking a colour triple.

In this reading the top is what completes E8 at a wall (7 = 2 x 3 + 1), and with colour triples the proposed E8 x E7 arrangement has seven branes at one wall, six at the other and a bulk triple at x_b, while two E8 walls require relaxing the colour-triple rule (referee replacement): with that arrangement, the slopes are proportional to +1 on (0, x_b) and −2 on (x_b, L), so z vanishes at both walls when x_b = 2L/3. The E_n series also contains E_3 = SU(3) x SU(2) (two branes at a wall), which colour triples do not reach.

Reading (proposal). At strong coupling the wall that holds the top's brane in the label-9 configuration carries SU(2) x U(1), and the wall of the five light quarks in three colours carries SO(30). The electroweak group then appears at the top's wall, where the M-theory circle opens locally, and the quarks' flavour group at the other wall.

5. What remains open

- The bulk symmetry: k D8-branes in the bulk carry U(k), while Proposition 8.1 gives O or USp on every eigenspace of d^2; the relation between the window's degenerate levels in the bulk and U(k) needs the parent's own structures, not only j.
- The dilaton: a spectral-triple counterpart of the type I' dilaton profile (a coupling varying along the interval, with slope set by the brane charge to one side) and the infinite-coupling points of Remark 8.3. Proposition 9.1 of 09 writes the profile as a trace of a function of |D_F|, only implicitly after 10, Proposition 10.1, and the gradient is not the second scale in the scope of Propositions 10.5 and 10.6 there.
- The physical status of E8 x E7 with colour triples, and whether the top's brane at an E8 wall has a counterpart in the finite geometry.

Part 11D, 09 dilaton spectral and so32, referee pending

The type I' dilaton as a spectral function, the SO(32) of the EPJC paper, and two more dualities

Part of the 11D line of research (README, PLAN, task C). Attempted by Claude (Anthropic), 4 Nov 2026, after the steer of the same day (the coincident-brane route is overused; the type I' dilaton and the colour-flavour idea are worth pursuing; compare with the SO(32) of the EPJC paper). Proposition 9.1 is proved from the recalled type I' slope rule; Proposition 9.2 is proved from representation theory and the recalled orientifold projections; Proposition 9.3 is proved (linear algebra and Connes's sign table); the comparison of section 1 is read from the arXiv text of the paper through a summarizing fetch; the rest are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

Correction (same day): the first version stated the slope rule for e^{−phi}. The piecewise-linear function is z, with e^{phi} = z^{−5/6}; Proposition 9.1 holds as

stated with z in place of e^{−phi}, and infinite coupling is z = 0.

1. The SO(32) of A. Rivero, Eur. Phys. J. C 84 (2024) 1058 (arXiv:2407.05397)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The branching algebra is independently checked. The archived original abstract also states an additional chiral charge-+4/3 quark per generation. The full paper was unavailable for direct comparison, so its detailed attribution remains pending. See the referee report.

That paper uses the adjoint of SO(32) to classify the scalars of a supersymmetric Standard Model with three generations, as composites: pairs of quarks as colour triplets and quark-antiquark pairs as singlets and octets, with three extra scalars of charge +4/3. Its pattern is SO(2n) ⊃ SO(n_1) x SU(n_2) x SU(3) x U(1) with 2n = n_1 + 6 n_2, which for SO(32) and n_2 = 5 gives SO(2) x SU(5) x SU(3) x U(1): the 30 = 6 n_2 are the five light quarks u, c, d, s, b ("turtles") in three colours with conjugates, and the top is the non-participating "elephant". Its explicit route also passes through SO(32) ⊃ SU(16) x U(1), with 496 = 1 + 120 + 120bar + 255 (quarks from the 120s, leptons from the 255), and it mentions open strings, orientifolds and an SU(9) for M2 and M5.

Reading of the LaTeX source (Claude Sonnet 5.5, 4 Nov 2026; the earlier comparison used a summarizing fetch). The paper uses colour singlets for mesons and colour antitriplets for diquarks, and keeps only (1, 24, 1), (1, 15, 3bar) and (1, 15bar, 3) of its decomposition; colour octets and sextets lie outside it. The top is the favoured non-binding quark ("we have not a concrete argument"), and the postulates force the non-binding quark to be up-type. The chain is SU(16) x U(1) to SU(15) to SU(5) x SU(3), with SO(30) and SO(32) reached by "adding singlets". The identification of the top with the SO(2) factor, the O8^- plane and the Chan-Paton reading belong to these files, and "orientifold" in the paper means planar orientifold QCD. The footnote on the 84 + 12 split and SU(9) for M2 and M5 anticipates Proposition 21.3 and 04. The abstract specifies a chiral +4/3 quark per generation; the body has three

scalars of charge +4/3 and leaves their fermion partner open. The paper gives (15, 3bar) as the antisymmetric square of one fundamental without stating chirality, so the question of Proposition 9.3 stays open; its remark on odd-parity diquarks is consistent with that proposition. The paper's selection of an up-type quark by its postulates and lemma 21's by the spectral-action values are independent arguments with the same outcome.

Coincidence. The chain SO(32) ⊃ SO(2) x SO(30) ⊃ SO(2) x SU(15) x U(1) ⊃ SO(2) x SU(5) x SU(3) x U(1), the 30 as five light quarks in three colours, and the top as the factor SO(2), are the same as lemma 21 and the dictionary of 06. Lemma 21 records the branching 496 -> (435, 1) + (1, 1) + (30, 2) and leaves SO(32) unexplored; the EPJC paper explores it through composites, and the files 06 to 09 through type I' branes. What the 11D files add: the type I' realization of the SO(2) as the top's D8-brane and of the 30 as the Chan-Paton labels of fifteen D8-branes on an O8^- wall (06); the top angle as the brane's position and the one-loop vacuum on the far wall (Propositions 7.1, 7.2); the orientifold type O^- matched by the seesaw sign (Proposition 8.1); the far wall's enhancement to SU(2) x U(1) at infinite coupling (Remark 8.3); the dilaton (Proposition 9.1); the diquark counts as strings between the stack and its image, which need the O^- type (Proposition 9.2); and the real-structure condition under which these counts arise, as connection components along the interval and not as finite inner fluctuations (Proposition 9.3). In type I' language the paper's other chain, SU(16) x U(1), is the configuration with all sixteen D8-branes coincident in the bulk, and the chain with SO(2) x SO(30) is the configuration with fifteen on a wall and one apart; the second is the one the corpus selects. The SU(9) of M2 and M5 in the paper is the grading of e8 used in 04.

2. Proposition 9.1 (the type I' coupling function is a trace of a function of the finite Dirac operator)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The integration is independently checked conditional on the recalled slope rule. The trace is over sixteen selected channels, not automatically the full physical D_F Hilbert space; direct comparison with the cited source was unavailable to the referee. (Second check, against Polchinski-Witten.)

The formula z(L) = z(0) - c (8L - sum x_k) is exact in the positions; the dictionary x_k = (2L/pi) l m_k holds only for constant z, and in general theta_k = (pi/2) W(x_k)/W(L) with W = int z^{-1/3} (10, Proposition 10.1). In the profile 15 + 1 the endpoint values 7cL and 8cL stand, and theta_t = pi/4 gives 7.358 cL in place of 7.5 cL. See the referee report.

Setting. Type I' on an interval [0, L], in the conformal coordinate x, with sixteen D8-branes at x_1, ..., x_16 and O8^- planes at the ends. Recalled rule (Polchinski and Witten, hep-th/9510169): the dilaton is e^{phi} = z^{−5/6} for a function z that is continuous and piecewise linear in x, with slope c (8 − N(x)), where N(x) is the number of D8-branes to the left of x and c > 0; the coupling diverges where z vanishes. Dictionary (06, 07): x_k = (2L/pi) θ_k with θ_k = l m_k the angles of sixteen channels of the finite Dirac operator (fifteen light-quark channels with colour, and the top).

Statement.

1. z(x) = z(0) + c Σ_k min(x/2, x_k − x/2) for x in [0, L], so z is z(0) plus c Tr g_x(l |D_F|) over the sixteen channels, with g_x(θ) = min(x/2, (2L/pi) θ − x/2): a trace of a function of the finite Dirac operator, as the spectral action is.
2. At the far wall, z(L) = z(0) − c (8L − Σ_k x_k).
3. Infinite coupling at the far wall (z(L) = 0) holds exactly when z(0) = (2cL/pi)(4 pi − l Σ_k m_k): the coupling at the light-quark wall is fixed by the sum of the sixteen channel masses, the trace of |D_F| on these channels.

Proof. *∫0^x N(y) dy = Σ_k (x − x_k)+*, so z(x) = z(0) + c (8x − Σ_k (x − x_k)+) = *z(0) + c Σ_k (x/2 − (x − x_k)+)*, and x/2 − (x − x_k)_+ = min(x/2, x_k − x/2). Item 2 is x = L with Σ_k (L − x_k) = 16L − Σ x_k. Item 3 sets z(L) = 0 and substitutes x_k = (2L/pi) l m_k. ■

Correction (4 Nov 2026). The dictionary x_k = (2L/pi) theta_k is exact only for constant z; the map from angles to positions is nonlinear. See 10, Proposition 10.1.

Reading (proposal). With the light quarks near θ = 0, Σ_k m_k ≈ m_t and the condition is z(0) ≈ (2cL/pi)(4 pi − θ_t): 7cL at label 9 (θ_t = pi/2) and 8cL at label

11 (θ_t -> 0). The coupling therefore varies along the interval on a scale set by the spectrum of D_F, which is the kind of second scale that Proposition 1.5 requires to select a joint family; whether the type I' coupling can play that role in the corpus's flat family is the next question. Since the slope decreases with x, z is concave, so z > 0 on [0, L) once z(0) > 0 and z(L) = 0; positivity of z(0) gives l Σ_k m_k < 4 pi, a bound on the trace of |D_F| in units of the tower scale, met when only the top is heavy (θ_t <= pi/2).

3. Proposition 9.2 (diquarks as strings between the stack and its image)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The representation and charge counts are independently checked, conditional on the recalled projections. The shifts describe the specified gauge-field channels; Λ^2 of a line excludes that component, not all open-string states or strings to other branes. See the referee report.

Setting. V = C^5 ⊗ C^3, five light flavours in three colours, with Q = 2/3 on u, c and Q = −1/3 on d, s, b; the 30 of lemma 21 is V ⊕ Vbar. Fifteen D8-branes on an O8 plane carry the Chan-Paton space 30 (branes and images), with group SO(30) on an O8^- and USp(30) on an O8^+, and U(15) when the stack is moved off the plane (recalled).

Statement.

1. Under U(15), the adjoint of SO(30) is Λ^2V ⊕ Λ^2Vbar ⊕ (V ⊗ Vbar) and the adjoint of USp(30) is S^2V ⊕ S^2Vbar ⊕ (V ⊗ Vbar). The summands Λ^2V or S^2V and their conjugates are the strings between the stack and its image; V ⊗ Vbar are the strings with both ends on the stack.
2. Λ^2V = ($S^2$5, 3bar) ⊕ ($\Lambda^2$5, 6) and S^2V = ($S^2$5, 6) ⊕ ($\Lambda^2$5, 3bar).
3. In the colour antitriplet, Λ^2V has the charge counts (−2/3: 6, +1/3: 6, +4/3: 3) of lemma 21, and S^2V has (−2/3: 3, +1/3: 6, +4/3: 1). So the antisquark counts of three generations recorded in lemma 21 (6 of charge −2/3 and 6 of charge +1/3 in the 3bar) occur among the stack-image strings of an O8^- plane and not among those of an O8^+ plane.

4. With the angles θ_k = l m_k of 06, a string from channel i to the image of channel j has the spectrum (n pi ± (θ_i + θ_j))/l, and a string from channel i to channel j has (n pi ± (θ_i − θ_j))/l, n in Z.
5. A single brane has no string to its own image on an O8^- plane (Λ^2 of a line vanishes) and one on an O8^+ plane.

Proof. 1: 30 = V ⊕ Vbar as a U(15) module, Λ^2(V ⊕ Vbar) = Λ^2V ⊕ Λ^2Vbar ⊕ (V ⊗ Vbar) and likewise for S^2; the orientifold projection keeps the antisymmetric part for SO and the symmetric part for USp, and the mixed summand V ⊗ Vbar is the U(15) adjoint. 2: Λ^2(A ⊗ B) = (S^2A ⊗ Λ^2B) ⊕ (Λ^2A ⊗ S^2B) and S^2(A ⊗ B) = (S^2A ⊗ S^2B) ⊕ (Λ^2A ⊗ Λ^2B), with $\Lambda^2 3$ = 3bar. 3: split 5 = 2 ⊕ 3 by charge; $S^2 5$ = $S^2 2$ ⊕ (2 ⊗ 3) ⊕ $S^2 3$ has dimensions 3, 6, 6 with charges 4/3, 1/3, −2/3, and $\Lambda^2 5$ = $\Lambda^2 2$ ⊕ (2 ⊗ 3) ⊕ $\Lambda^2 3$ has dimensions 1, 6, 3 with the same charges. 4: Proposition 6.1 with charges (+1, +1) and (+1, −1) under U(1)_i x U(1)_j, whose holonomy is e^{2 pi i (a_i ± a_j)}, and θ = pi a. 5: dimensions. ■

Reading (proposal). (11, Theorem 11.3 and Proposition 11.4, qualify the selection below: the window sees the counts (3, 6, 1) also at s = +1, through the commutator with the masses, and parent scalars with the counts (6, 6, 3) need M_15(C) at both walls.) The diquark counts of lemma 21 and of the EPJC paper are the stack-image strings of an O8^- plane, the type that Proposition 8.1 associates with the factorization sign s = +1 selected by the seesaw sector (T2c); the O8^+ alternative gives the wrong number of up-type antisquarks. So the seesaw sign and the scalar counts select the same orientifold type. Two features follow from the projection. Diquark masses go with the sum θ_i + θ_j and meson masses with the difference θ_i − θ_j, so when the stack leaves the wall the diquarks become massive and the mesons stay massless: the diquarks are tied to the orientifold. And the top's single brane has no string to its own image, while its strings to the stack, the (30, 2), have masses near m_t (Proposition 6.1); in the EPJC paper the top's exclusion is an input, and here the O8^- projection with one brane for the top removes the top-top channel only (item 5); the top-to-stack strings (30, 2) remain, with masses near m_t, so the rest of the exclusion is a mass statement, as

in the paper.

4. Proposition 9.3 (the real structure decides between the two diquark counts)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The stated sign constraint excludes a nonzero antisymmetric block in the three KO labels. The signs permit all of S^2 V (second check: B ranges over exactly S^2 V for eps eps' = +1, with the diagonal block free); the algebra, grading and order conditions of a triple then select a subspace. Distinct-multiplet blocks remain related to their transposes. See the referee report.

Setting. E = V ⊕ Vbar with V as in Proposition 9.2, an antiunitary J on E that maps V onto Vbar and Vbar onto V (as particles and antiparticles are exchanged), J^2 = ε, and a Hermitian X on E with JX = ε' XJ, the condition that inner fluctuations obey in a KO-dimension with signs (ε, ε'). Let B be the block of X from Vbar to V, an element of V ⊗ V.

Statement.

1. B is symmetric (B in S^2V) if εε' = +1 and antisymmetric (B in Λ^2V) if εε' = −1.
2. In the KO-dimensions 5, 6, 7 of the finite part for the labels 9, 10, 11, (ε, ε') = (−1, −1), (+1, +1), (+1, +1), so εε' = +1: the diquark block of every J-compatible fluctuation lies in S^2V, whose colour antitriplet ($\Lambda^2$5, 3bar) has the counts (−2/3: 3, +1/3: 6, +4/3: 1).
3. For a connection on a circle whose real structure is complex conjugation (KO-dimension 1, (ε, ε') = (+1, −1)), the block lies in Λ^2V, whose colour antitriplet ($S^2$5, 3bar) has the counts (−2/3: 6, +1/3: 6, +4/3: 3) of lemma 21. The SO(32) Wilson line of Proposition 6.1 is of this type.

Proof. Choose a basis of V and let K: V -> Vbar be coordinatewise conjugation; J(v, w) = (ε K^{-1} w, K v) has J^2 = ε. Write X = [[A, B], [B*, D]]. The component of JX = ε' XJ that maps V to V gives ε K^{-1} B* = ε' B K, so B = εε' K^{-1} B* K^{-1} = εε' B^T in coordinates. Under the U(15)-equivariant identification of Hom(Vbar, V) with V ⊗ V, symmetric matrices form S^2V and antisymmetric ones Λ^2V. Items 2 and 3 use Connes's

sign table, ε = (+, +, −, −, −, −, +, +) and ε' = (+, −, +, +, +, −, +, +) for KO-dimensions 0 to 7, and the counts of Proposition 9.2. On the circle, D = −i d/dθ and J = conjugation give JD = −DJ, and a Hermitian connection A with JA = −AJ is i times a real antisymmetric matrix, an element of so(n), where the generator of R(2 pi a) lies. ■

Item 2 is a negative statement with the following scope: the blocks of J-compatible Hermitian operators between particles and antiparticles, for one multiplet V, in the three labels. It is the finite-geometry form of the symmetry of Majorana mass matrices, and of the Fermi-statistics constraint on a Lorentz-scalar bilinear ψ_i ψ_j of one Weyl multiplet without derivatives (recalled), under which the (15, 3bar) vanishes. Known escapes: item 3 (connection components along a circle or an interval), blocks between two distinct multiplets, which J does not constrain, and derivative couplings.

Reading (proposal). The counts of lemma 21 and of the EPJC paper, (6, 6, 3), belong to the antisymmetric block. In the corpus's setting this block arises as the component of an SO(30) connection along the circle or interval: the Wilson-line moduli of the type I' stack, whose stack-image part is Proposition 9.2. The finite triple alone, in any of the three labels, gives the symmetric block and the counts (3, 6, 1). So the composite scalars with the counts of lemma 21 point to the interval parent of lemma 22 and its gauge field, and the selection of O8⁻ in Proposition 9.2 concerns these connection components. This narrows the spin gap of Propositions 6.1 and 9.2: the Λ^2V states are scalars as circle components of a nine-dimensional gauge field, and they come with that field's vector and gaugino partners.

5. Two more dualities between 9 and 11 (recalled; proposals)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The E_1 alternatives need correction: the canonical-orthogonal root lattices distinguish $\mathbb{P}^1 \times \mathbb{P}^1$ from $dP_1 = \mathbb{F}_1$, and the usual local engineering assigns the latter to $\widetilde{E}_1$. The dP_2 and brane duality claims remain conditional on the unavailable primary sources. The Weyl action is a gauge relabelling of a holonomy, subject to the full background

data. See the report on the del Pezzo comparison and Weyl action.

1. **E-type walls and del Pezzo surfaces.** The E_{n+1} symmetry of an O8^- plane with n D8-branes at infinite coupling is the symmetry of the five-dimensional theory on a D4-brane probe, which M-theory realizes on a Calabi-Yau three-fold with a shrinking del Pezzo surface dP_{n+1} (Seiberg, hep-th/9608111; Morrison and Seiberg, hep-th/9609070; Intriligator, Morrison and Seiberg, hep-th/9702198). So the far wall with the top's brane (n = 1, E_2 = SU(2) x U(1)) has an eleven-dimensional description by a local dP_2, and the empty far wall (E_1 = SU(2)) by dP_1 or P^1 x P^1; the light-quark wall (n = 15) has none. The Witten base CP^2 x CP^1 of 02 is built from the same pieces (CP^2 = dP_0, and P^1 factors).
2. **Colour-flavour exchange as a T-duality.** In the heterotic string on S^1, the Weyl group of the gauge group acts on the Wilson lines as part of the T-duality group O(17, 1; Z). The colour-family transposition M of 05 is an element of the Weyl group of E8, so it is a T-duality of the nine-dimensional E8 x E8 string, and through Hořava-Witten a symmetry of the E8 holonomy on a wall in eleven dimensions. As a Weyl element it acts as a global gauge transformation, so by itself it relabels gauge-equivalent vacua; it exchanges colour and family physically only relative to a structure that tells them apart, such as the placement of H_F in 05. The four-family requirement of Propositions 5.1 and 5.4 remains.

6. Open

Referee note (GPT-6.1 Terra, 4 Nov 2026). The finite KO signs restrict the symmetric block without producing the full representation. Light interval scalar modes, their boundary conditions and the physical string dictionary remain to be constructed; the detailed EPJC comparison is pending.

- Whether the type I' coupling, as a spectral function of D_F (Proposition 9.1), supplies the second scale of Proposition 1.5 in the flat family of lemmas 23

and 24.

- The local dP_2 of the top's wall and its relation to the Witten base CP^2 x CP^1.
- Spin: in type I' the strings of Proposition 9.2 belong to nine-dimensional vector multiplets, with one real scalar each (the relative position), the same gap as in Proposition 6.1. By Proposition 9.3 a finite triple in the labels 9 to 11 gives S^2V, so the Λ^2V scalars must come from the connection along the interval; the next computation is the inner fluctuations of the parent of lemma 22 with an SO(30) algebra, and whether its interval component carries Λ^2V with the counts of lemma 21.
- Whether the (15, 3bar) of the EPJC paper could be read as bilinears of two distinct multiplets (the paper itself gives one fundamental with itself) (for example left-handed quarks and conjugates of right-handed ones arranged into one 30), which Proposition 9.3 does not constrain.

Part 11D, 10 dilaton and second scale, referee pending

The type I' dilaton against the second scale of Proposition 1.5

Part of the 11D line (README, PLAN). Attempted by Claude Opus 5.5 on 4 November 2026, after the merge of branch 11D into master, to answer the last item left open in 09: whether the dilaton can be the second scale that Proposition 1.5 requires. Sources read from the arXiv LaTeX: J. Polchinski and E. Witten, hep-th/9510169, section 3; O. Bergman, M. Gaberdiel and G. Lifschytz, hep-th/9711098, section 3. Numbers in Python (numpy, scipy, mpmath), scripts not kept by the rule of the project. The Codex quota is spent, so no Terra referee is available. An independent reading by Claude Sonnet 5.5, with the error and amendments below, is in the review. The amendments were rechecked by Claude Opus 5.5 before being applied, and the corrected text has not been read again.

Correction (4 Nov 2026). The first version counted 21 gauge generators with integer towers. Lemmas 23,

24 and 26 use sp(2) (+) su(4), with 25 generators: 16 with integer towers at theta = 0 and 9 with half-integer towers. The bracket for one generation is 41.3125, so the near-cancellation 64 - 63 stated in the first version came from the wrong count, and it is withdrawn. So is the small value theta_* = 0.202 that followed from it.

Outcome

1. **In the scope stated below, the dilaton gradient does not supply the second scale.** Along any path that opens the circle, its relative variation falls as 6/R^2 (Proposition 10.5). It enters as a power of alpha'/l^2, and that can matter only if the leading Casimir coefficient a_0 vanishes (Proposition 10.6).
2. **a_0 is now computed**, which was the open item of 01: 8 pi^2 a_0 = (3 zeta(5)/16)(4 n_D' - 3 (n_i - (15/16) n_h) - n_s - 5 n_g), with n_i and n_h the generators with integer and half-integer towers at theta = 0. For the corpus content it is positive, so at v = 0 the interval runs to large length, in the direction H wants.
3. **The dictionary of 06 to 09 needs three corrections** (Propositions 10.1 to 10.4). The map from angles to positions is nonlinear. The tower of Proposition 6.1 is a type I' winding state, so the length l of H and the type I' length are T-dual, and l -> infinity shrinks the type I' interval at fixed heterotic coupling. And the configuration 15 + 1 has a Romans mass in the bulk, which bounds length times coupling, so the classical profile has no eleven-dimensional regime.
4. **A scale-free action can still separate the window**, with a fixed small ratio instead of a vanishing one, if the light mass is exponentially small in a stabilized shape parameter (Remark 10.7), which amends the last sentence of Proposition 1.5.

The direction: the second scale must enter through a bulk energy depending on the electroweak order parameter, through a cancellation a_0 = 0, or through a fixed shape with an exponentially small ratio, as in lemma 29. These targets are stated in Proposition 10.6 and Remark 10.7. None is derived.

The dictionary against the source

Polchinski and Witten give, in conformal gauge, g = Omega^2(x) eta with Omega = C z^{-1/6}, e^phi = z^{-5/6}, and z = 3 C (B mu_8 - nu_0 x)/sqrt 2, with nu_0 = (N(x) - 8) mu_8 piecewise constant. This confirms the slope rule recalled in 08 and 09. We write z for this function up to its constant, normalized so that its slope is exactly 8 - N(x), which fixes the normalization in the formula for R below, with x in [0, L].

Proposition 10.1 (the map from angles to positions). Put W(x) = int_0^x z^{-1/3} dy. By Polchinski-Witten, equations (mass) and (wilmap), the Wilson-line eigenvalue of a brane at x_k is lambda_k = W(x_k) / (2 W(L)), that is

```
theta_k = pi lambda_k = (pi/2) W(x_k) / W(L),
```

and with L = 2 pi, in their units alpha' = 2, R^2 = (3/4) int z^{1/3} dx / (int z^{-1/3} dx)^2. So the linear dictionary x_k = (2L/pi) theta_k of Proposition 9.1 holds exactly when z is constant (8 + 8), and asymptotically for large R.

Proof. The string stretched from the wall to x_k has mass (1/(2 pi alpha')) int_0^{x_k} Omega^2 dx, since its area element is Omega^2 dx^0 dx^9, and Omega^2 is proportional to z^{-1/3}. A string winding from 0 to L and back has the mass 1/R. The ratio gives lambda_k. The formula for R is their equation (rad), reduced as in their equation (dil). ■

Consequence for Proposition 9.1: its items 1 and 2 hold as statements about z in terms of the positions x_k. Reading them as a trace of a function of |D_F| needs the map above, which depends on the whole configuration, so the coupling function is a trace only implicitly. Item 3 changes accordingly.

Proposition 10.2 (15 + 1 with infinite coupling at the far wall). Take fifteen branes at x = 0 and one at x_t, with z(L) = 0, and put rho = (z(x_t)/z(0))^{2/3} in [0, 1]. Then, with alpha' = 2,

```
theta_t = (pi/2) 8 (1 - rho) / (8 - rho),
 ↪  R_*^2 = 14 (8 - rho^2) / (8 - rho)^2.
```

So R_* runs from sqrt 2, for the top's brane on the stack,

to sqrt 7 / 2, for the top's brane on the far wall, which are the values (1/2)|n - 8|^{1/2} of Polchinski and Witten at n = 16 and 15. In units of the string length, R_/sqrt(alpha') lies in [0.935, 1]. In this chamber, R >= R_.

Proof. Integrate z^{1/3} and z^{-1/3} on the two segments, with slopes -7 and -8, and use z(x_t) = 8 (L - x_t). ■ Python agrees at x_t/L = 0.5, with theta_t = 0.5859 and R^2 = 1.9652, against 0.7854 for the linear dictionary.

So infinite coupling sits at the *smallest* radius of the chamber, at the string scale, and large R is large B, the weakly coupled region. Polchinski and Witten say this directly: "When B is large, the coupling e^{phi(x^9)} is small everywhere and the Type I' theory is weakly coupled; in this range the heterotic radius R is large".

Proposition 10.3 (T-dual lengths). The tower of Proposition 6.1, n/R, is a type I' winding tower: "a heterotic Kaluza-Klein state which is dual to a Type I' winding state" (Polchinski and Witten). With constant dilaton, the proper length L_p of the type I' interval and the radius satisfy L_p R_I = pi alpha' on the type I side, with alpha' that of type I'. So with l = pi R_I, measured in the type I' metric, l L_p = pi^2 alpha'. For the heterotic radius in the heterotic metric, l L_p = pi^2 alpha' sqrt(lambda_h), by Bergman, Gaberdiel and Lifschytz, with lambda_h the heterotic coupling.

Hence the walls of H, at distance l, and the O8 planes, at distance L_p, coincide only at the self-dual length. In particular, the reading "the interval opens" in section 6b of 07 has to be restated: at fixed heterotic coupling, l -> infinity shrinks the type I' interval, and it decompactifies the SO(32) circle.

The endpoint of that limit is the ten-dimensional SO(32) theory (heterotic or type I), since at fixed lambda_h the type I' coupling falls as 1/(sqrt(lambda_h) R_h). *Recalled, and supported in the two sources only by remarks of Bergman, Gaberdiel and Lifschytz at the constant-dilaton point:* the eleven-dimensional limit of the rank-17 moduli space is M-theory on an interval times a circle, with E8 x E8 at the walls, which by Remark 8.3 is out of reach with colour-respecting moves. If so, label 11 of

section 6b is not reached by theta -> 0, l -> infinity in the chamber 15 + 1.

Proposition 10.4 (Romans mass, length and coupling). On a segment with N branes to the left, the proper distance is linear in the inverse coupling:

```
L_p = (C_L l_s / |8 - N|) (e^{-phi_1} - e^{-phi_2}),
 ↪  C_L = 8 pi / 5,
```

with C_L from the standard normalization of the Romans mass, F_0 = (8 - N)/(2 pi l_s). So in the configuration 15 + 1, g(0) L_p <= C_L l_s / 7 for the total length. Near an infinite-coupling wall the coupling is g = C_L l_s / (|8 - N| s) at proper distance s, and the region g >= g_0 has length (3/4) (C_L / |8 - N|) g_0^{-4/3} in eleven-dimensional Planck units, which is 0.54 g_0^{-4/3} for N = 15. Therefore no region of the classical 15 + 1 background has both strong coupling and a length resolved in eleven-dimensional units.

Proof. Omega dx = C z^{-1/6} dz / (a C (8 - N)) integrates to a multiple of z^{5/6} = e^{-phi}. ■ This agrees with Bergman, Gaberdiel and Lifschytz, who conclude from the D0-brane bound states that they "expect neither to be applicable to the N != 0 backgrounds", meaning an M-theory interpretation or a matrix theory, and that "only the combination of eight D8-branes and an orientifold plane is known to arise in M-theory". Here their N is 8 - N. It holds wherever N != 8. The scope is the classical profile, which is not controlled where g is of order one, so this is a statement about the extrapolated background.

The far-wall electroweak group. Remark 8.3 places SU(2) x U(1) at the far wall at infinite coupling. By Bergman, Gaberdiel and Lifschytz, the states that become massless there are D0-branes at the wall, heterotic winding states. Let n be the number of branes at the strong wall, here n = 1 with the top's brane there, placed at their x^9 = 0 by reflecting the interval. Combining their equations (windingmass) and (D0mass), with N' = 8 - n, A = (B + 2 pi N')^{2/3} and b = B^{2/3},

```
M_D0 R = (N'/2) b / (A - b) = R^2 - R_*^2    (alpha'
 ↪  = 2),
```

which vanishes at B = 0, where R = R_*, and grows like R^2. Only W^+ and W^- can be such states: a neu-

tral Z is a Cartan combination, and the rank of a nine-dimensional theory with sixteen supercharges is fixed, so no massive neutral vector arises this way. So in the chamber 15 + 1 with the top's brane on the far wall, if the W^+- are these D0 states, they are massless at the string-scale radius, with the tower fixed, and they become heavier as R grows: M_W l rises with l. This is the fixed-tower path of the audit (its section 2.2), and away from it the W moves the wrong way for hypothesis H. So on one path, in this scope, the dictionary can place the electroweak group at the far wall, or the opening of H on the circle of Proposition 6.1, but not both. Escapes: other chambers; the W as a Wilson-line string between D8-branes, as in lemma 23; the top's brane in the bulk, with the far wall giving E_1.

The test

Proposition 10.5 (the gradient is a string-scale effect). With L = 2 pi and alpha' = 2, for large R, R^2 ~ 3 B / (8 pi), and the relative variation of z across the interval is at most 16 pi / B ~ 6 / R^2. So along a joint family, R_eps ~ eps^{-alpha}, this variation falls as eps^{2 alpha}, the map of Proposition 10.1 linearizes, and the dilaton is constant in the limit.

Proof. z is B plus a function of size at most 8 L = 16 pi. Insert in the formula for R. ■

So the corrections to any potential from the gradient are powers of alpha'/l^2. In the supersymmetric type I' background there is no potential at all: D8 positions, B and C are flat directions. A potential needs broken supersymmetry, as in the flat family of lemmas 23 and 24.

The coefficient a_0. Write 8 pi^2 l^4 V = A_0 + 3 F(2u) + (3/2) F((16/5) u) - n_f F(t), the potential of lemma 24 with its theta-independent part A_0 restored, where n_f counts twice the Dirac species at frequency one (lemma 26). The relative terms vanish at theta = 0, and each tower at theta = 0 is half of a circle tower, periodic or antiperiodic:

```
A_0 = (3 zeta(5) / 16) (4 n_D' - 3 (n_i - (15/16)
↪  n_h) - n_s - 5 n_g),
```

with n_D' = 2 N_D counting twice all Dirac species, of any frequency (for frequency zero the tower is the theta

= 0 tower), n_i and n_h the five-dimensional gauge generators with integer and half-integer towers at theta = 0, n_s the real bulk scalars with integer towers, and n_g = 1 if the five-dimensional graviton is included. This assumes that the bulk vacuum energy and the wall tensions are tuned away, since they carry the terms of order Lambda^5 l and Lambda^4.

Proof. A real degree of freedom on a periodic circle of length 2 l has V = -3 zeta(5)/(4 pi^2 (2 l)^4), and the interval tower n >= 0 gives half of this, -3 zeta(5)/(128 pi^2 l^4). In the normalization of lemma 24 this is -(3/16) zeta(5) per real degree of freedom. A half-integer tower has the factor -15/16 relative to an integer one, so a generator with mixed conditions contributes +3 (15/16) (3/16) zeta(5). For a fermion channel the sign is opposite, with four degrees of freedom per unit of n_f. The check: the shift to theta = pi/2 multiplies a tower by -15/16, so the difference is (31/16)(3/4) zeta(5) = 93 zeta(5)/64 = F(1), which is the value computed in Proposition 1.4. ■

In lemmas 23, 24 and 26 the gauge field is valued in sp(2) (+) su(4), with 25 generators. At theta = 0, sixteen have integer towers: 8 gluons, 3 W_L, 1 hypercharge combination with Neumann conditions at both walls, and the 4 odd generators O with Dirichlet conditions at both, whose interval components are the Higgs quaternion. Nine have mixed conditions and half-integer towers: 2 R^+-, 6 X and 1 orthogonal neutral direction. So n_i - (15/16) n_h = 121/16. For three generations, with 24 Dirac species, the bracket is 192 - 22.6875 = 169.3125 before gravity and bulk scalars, and A_0 = 32.92. So a_0 > 0: at v = 0 the length runs to infinity, in the direction of H, and the first half of the modulus target of the audit, a r^{-p} with a > 0 and p = 4, is present. For one generation the bracket is 41.3125.

Proposition 10.6 (when the gradient can be the second scale). Let V = l^-4 [G_0(theta) + (alpha'/l^2) G_1(theta) + ...], theta = l m, with G_0(0) = a_0 and G_0 = a_0 + b theta^2 + ... near 0.

1. If a_0 != 0, then at small m the leading balance is the scale-free one of Proposition 1.5: alpha = 1 or a runaway. The term in alpha'/l^2 changes neither.
2. If a_0 = 0, G_1(0) = a_1 > 0 and b < 0, there is a minimum at l^4 = 3 a_1 alpha' / (|b| m^2),

so alpha = 1/2: 1/l is the geometric mean of m and the string scale, up to constants, and m « 1/l « 1/l_s.

3. If supersymmetry is exact at theta = 0, then V = 0 at theta = 0 at all orders, a_1 = 0, and V = m^2 l^-2 H(theta, alpha'/l^2). Then l is fixed by H independently of m, or runs away, and no joint family results, unless the stationarity condition has a root interval (a flat direction), as in the counterexample to the universal form of Proposition 1.5.

So in the scope of a potential analytic in alpha'/l^2 around the flat family, with corpus content, the gradient is not the second scale. Item 2 would need a Bose-Fermi balance at theta = 0 that is not a supersymmetry at theta = 0, with about 169 further real bosonic degrees of freedom for three generations.

Proof. Items 1 and 2: differentiate; in item 2, V = b m^2/l^2 + a_1 alpha'/l^6 has V'' = 8 |b| m^2/l^4 > 0 at the stationary point. Item 3: V is even in theta and vanishes at theta = 0 for every l, so V = theta^2 l^-4 H = m^2 l^-2 H. Stationarity is -2 H + theta H_theta - 2 s H_s = 0 with s = alpha'/l^2, which at small m reduces to an equation independent of m, or has no root. ■

The flat family with the angle tied. For the top alone, n_f = 6: b is proportional to 54/5 - 6 > 0, and G > 0 on [0, pi/2] (Python: minimum 23.88, with A_0 = 32.92). So V has no stationary point with theta < pi/2, and l runs until theta leaves the fundamental domain of Proposition 7.1, where the top joins the tower; the periodically continued potential has stationary points only far outside the window, near theta = 8.2. For a universal mass at three generations, n_f = 48, G changes sign and there is one stable stationary point, theta_* = 1.009, with M_W/m = 0.685. For a universal mass at one generation, n_f = 16, theta_* = 1.031, with M_W/m = 0.653 (Python). Top-only content therefore has no vacuum in the window in this scope, and adding a graviton does not change the classification. The stationary angles are of order one, so these vacua do not separate the window.

A localized channel. For a channel with bulk mass M = -mu and c = mu l, the spectrum of lemma 9 gives the one-loop term

```
8 pi^2 l^4 V_loc = -2 int_0^infinity x^3 log(1 + ((w
↪  + c)/(w - c)) exp(-2 w)) dx,    w = sqrt(x^2 +
↪  c^2),
```

from the secular function cosh(w) - (c/w) sinh(w). At c = 0 it is -(45/64) zeta(5) = -0.729, and for large c it is -4 c^3 exp(-2 c) up to corrections. Corrections at large c are of relative order 1/c. In between it has a minimum, of depth -1.3246, at c = 1 to numerical precision (Python, mpmath). This concerns the interaction term alone: the bulk term, -(4/15) c^5 in dimensional regularization, and the wall term are dropped as renormalizations of the bulk energy and wall tension, and with the bulk term included the minimum disappears. So the interaction term of one fermion channel prefers c = 1, where the light pair merges with the tower (lemma 9.7). A large c_*, which is what a separated window needs, must come from other fields, such as the bosonic tower of the scalar that gives mu.

Other second scales

Remark 10.7 (amendment of Proposition 1.5, last sentence). On the scale-free path, m l is a fixed number. Proposition 1.5 says that on it the finite window never separates from the tower. That holds for theta_* of order one, which is what the flat family gives above. If m is exponentially small in a stabilized shape parameter, the ratio m/Delta is a small fixed number. On the disc bundle of lemma 29, fix M and let R grow: m = lambda_M / R and Delta = delta / R both tend to zero, with lambda_M / delta exponentially small, and the count at fixed energy has seven-dimensional growth. So a potential R^-4 G(M), with M stabilized at a large M_* and G(M_*) > 0, *gives the hierarchy of lemma 30 with alpha = 1 and the fixed ratio lambda_{M_}*/delta, with no second scale; in lemma 30 the ratio tends to zero. The cost is that R then has no vacuum at finite value. The physical point at finite v needs one more term, and the stabilization at large M_* is a target, since the interaction term of one fermion channel prefers c = 1.

Remark 10.8 (a bulk energy). The ansatz V = a r^-p + b eps^2 r^q, with a > 0, has a minimum exactly when p + q > 0 and q b > 0, and then alpha = 2/(p + q), so alpha < 1 needs p + q > 2: for p = 4, q > -2 with q != 0. On the flat family the angle is a holonomy (Corollary

22.2), which no local bulk operator sees: at large l its effect is the Casimir term, q = -2, the borderline case alpha = 1. So a term falling more slowly than m^2/l^2 must come from elsewhere, and the natural place is a local bulk energy in the electroweak order parameter: for q = 1, V = a/l^4 + b v^2 M^3 l gives l^5 proportional to 1/v^2, so alpha = 2/5, if the bulk energy vanishes at v = 0. That requires the order parameter to be a bulk field with a bulk vacuum energy, and it departs from D_F as a pure boundary condition. Whether the spectral action provides this is open. In the heat cutoff, the dependence of the potential on D_F is exponentially small at large l Lambda, by Poisson summation over the tower, of relative size 4 exp(-(l Lambda)^2), so that action has no such term at this order. The factor (3 - d_eff)/2 of lemma 25 multiplies the term quadratic in the gauge field, a different term.

Corrections to the earlier files

- 09, Proposition 9.1 and its reading: subject to Proposition 10.1.
- 07, sections 4 and 6b: Proposition 7.2 stands as a statement on the circle; the reading "at distance proportional to a" in section 4 is corrected by Proposition 10.1, and the reading "the interval opens" at label 11 by Proposition 10.3.
- 08, Remark 8.3 and its reading: the far-wall group is reached at R = R_*, with the tower at the string scale, by Propositions 10.2 and 10.4 and the section on the far wall. There is also a tension of labels: Proposition 8.1 pairs label 9 with O^+, and section 6b of 07 puts label 9 on a far O8^- wall. These are two different meanings of the label, the KO sign and the position of the top's brane.
- 06, section 2: "9, 10, 11" mixes KO labels with metric dimensions. In the string web, nine dimensions is finite R, and R -> 0 is T-dual to R -> infinity, whereas in H the finite interior is labelled 10 and l -> 0 is labelled 9.
- 05, Proposition 5.4: the conclusion is correct on rerun (Python, exact integers, all 192 configurations). The sentence "In every case S - N has d = 19 and d = 20 only, while S - T has d = 16 on the neutrino roots of one slot and d = 22 on those of the two

others" is wrong: S - N has only these values in 48 of the 192 configurations, and the other 144 show {13, 17, 19, 20}; the statement on S - T fails in 96 configurations, with the pattern 16, 16, 22. The values of d differ in every configuration, and that is what the proof needs.

Scope and escapes

Proposition 10.6 assumes a potential with a regular expansion in alpha'/l^2, around the flat family with its towers. It concerns the dilaton *gradient*. The following lie outside it: a dilaton potential from string tree level, in non-supersymmetric strings or with tadpoles, for which a term in positive powers of 1/g could grow with l at fixed v through the law of Proposition 10.4; a different frame relation between l and the four-dimensional Planck mass; wall-localized dynamics; and the other chambers of the moduli space, where 8 + 8 removes the Romans mass and E8 x E8 is available. The string-theory statements are read from the two sources for type I' and recalled for the moduli-space limits; only the first have been checked against the text. By AGENTS.md, a negative result needs an independent check by another model. This one has had a reading by Claude Sonnet 5.5, which found the generator count above, and no further reading after the corrections.

Part 11D, 11 so30 fluctuations, referee pending

Inner fluctuations of the interval parent with an SO(30) algebra

Part of the 11D line (README, PLAN). Attempted by Claude Opus 5.5 on 4 November 2026, to answer the question left in 09: whether the gauge field along the interval of lemma 22 carries the antisymmetric block with the counts of lemma 21, given that Proposition 9.3 excludes it from the finite triple alone. Numbers in Python (numpy), scripts not kept by the rule of the project. The Codex quota is spent, so no Terra referee is available. An independent reading by Claude Sonnet 5.5, whose amendments are applied below, is in the review. The corrected

text has not been read again.

Outcome

1. **Yes, under a condition on the walls.** With a real algebra M_30(R) on the light block, the fluctuation along the interval is an so(30) connection, and its block between particles and antiparticles lies in Λ^2 V, with the counts (6, 6, 3) in the colour antitriplet (Proposition 11.2). Massless scalars in exactly Λ^2 V + conjugate appear when both walls reduce the algebra to M_15(C), the commutant of quark number (Theorem 11.3).
2. **In lemma 22's own parent the massless scalars are absent.** Its algebra is free at x = 0, so the interval component has no zero mode, for any algebra (Theorem 11.3, item 2).
3. **A generator has a massless vector or a massless scalar, never both, on the interval**, in free five-dimensional Yang-Mills at trivial background. So the whole of Λ^2(30) of lemma 21 cannot be massless scalars with colour unbroken. The antisquark counts appear as scalars, while the slepton part (24, 1) + (1, 1) appears as vectors (Theorem 11.3, item 4).
4. **The finite window sees a symmetric shadow.** The connection acts on the window through its commutator with the masses, B_ij -> (m_i - m_j) B_ij, which maps the (S^2 5, 3bar) of Λ^2 V onto the (Λ^2 5, 3bar) of S^2 V, with counts (3, 6, 1) (Proposition 11.4). This reconciles Propositions 9.2 and 9.3. The kernel is the same-flavour diquarks.
5. **SO(30) has a precise meaning in the finite geometry:** it is the commutant of the real structure j = J U of lemma 06 on the massless light block, for the sign s = +1, and USp(30) for s = -1 (Proposition 11.1).

Setting

Let V = C^5 ⊗ C^3, with Q = 2/3 on u, c and Q = -1/3 on d, s, b. In the finite Hilbert space of the five light quarks, with colour, the chirality gamma = +1 subspace is

```
H_+ = {q_L} ⊕ {conj(q_R)} = V ⊕ Vbar,
```

of dimension 30: the left-handed Weyl fields of fifteen Dirac fermions. H_- = {q_R} ⊕ {conj(q_L)}, and J_F exchanges H_+ and H_-, with c = J_F restricted to H_+, as in lemma 22. Let U be a unitary, grading-reversing and J-related as in lemma 06, with U q_L = q_R and U conj(q_R) = s conj(q_L), so U J_F = s J_F U. Then j = U^-1 c is an antiunitary on H_+, with j^2 = s, j q_L = s conj(q_R) and j conj(q_R) = q_L. Here U only defines j: the representation pi below does not commute with U, so this is not a doubling in the sense of lemma 06, in which U commutes with the algebra.

The interval parent is that of lemma 22: D_X = E d/dx on L^2([0, l]; C^2) ⊗ (H_+ ⊕ H_-), with Q in H_+, U-components in H_-, and the wall conditions Q_R(0) = U_L(0) = 0, Q_R(l) = X U_R(l), U_L(l) = -X^* Q_L(l).

The block

Proposition 11.1 (the commutant of j). Let s = +1. In the basis V ⊕ Vbar with j(v, w) = (K^-1 w, K v), the skew-Hermitian a with j a j^-1 = a form so(30), and

```
a = [[alpha, B], [conj(B), conj(alpha)]],    alpha^*
 ↪  = -alpha,   B^T = -B,
```

so so(30) = u(15) ⊕ Λ^2 V, with u(15) the commutant of the complex structure I = (i, -i), the quark number. The Hermitian generator H = i a satisfies j H j^-1 = -H. For s = -1 the same computation gives usp(30) = u(15) ⊕ S^2 V.

Proof. Linear algebra, with Proposition 8.1 for the groups. For s = +1, j a j^-1 = a gives the block form, and a^* = -a gives alpha^* = -alpha and B^* = -conj(B), that is B^T = -B. Then j (i a) j^-1 = -i a. ■

So the SO(30) of lemma 21 is the symmetry of the massless block for the sign s = +1, the sign that Proposition 8.1 and STRATEGY T2c associate with label 11 and with the seesaw; and the counts of lemma 21 select the same sign:

	colour antitriplet	its charges (-2/3, +1/3, +4/3)	colour sextet
s = +1, Λ^2 V	(S^2 5, 3bar)	(6, 6, 3)	(Λ^2 5, 6)
s = -1, S^2 V	(Λ^2 5, 3bar)	(3, 6, 1)	(S^2 5, 6)

Fluctuations along the interval

Take on the light block D_F = 0, so X = 0, and let C_0, C_l be subalgebras of A = M_30(R), the real elements of End H_+. Let

```
A_l = {f in C^infinity([0, l], A) : f(0) in C_0,
 ↪ f(l) in C_l},    pi(f) = f on Q, 0 on the H_-
 ↪ components.
```

Proposition 11.2. Assume 1 lies in C_0 and C_l.

1. pi(f) preserves the domain, for every choice of C_0, C_l. Order zero and order one hold on this block.
2. The self-adjoint one-forms sum pi(f_i) [D_X, pi(g_i)] are exactly E a_x on the Q components, with a_x in C^infinity([0, l], so(30)), with no condition at the walls. The fluctuated operator is

   ```
   D_A = E (d/dx + a_x ⊕ U a_x U^-1).
   ```

3. J-hat A J-hat^-1 = E (U a_x U^-1 on H_-), with sign +, as KO-dimension six requires. The block of a_x between Vbar and V is antisymmetric. The two statements agree since E is antisymmetric: the block of the Hermitian operator E a on C^2 ⊗ E is symmetric, as Proposition 9.3, item 2 requires, with B antisymmetric and the factor E carrying the other sign. So the interval component evades Proposition 9.3 through the escape in its item 3.

For D_F != 0 on this block, order one fails in the finite window: [[D_F, pi(a)], J pi(b) J^-1] has the blocks -U b U^-1 M a from H_+ to H_-, and -a M^* U b U^-1 from H_- to H_+, which vanish for all a, b only if M = 0. The parent still satisfies order one (Theorem 22.3), but its wall algebra shrinks to the commutant of D_F in pi, which is zero for pi = (a, 0) with M invertible. So

this algebra describes the massless block, and the masses need a treatment without order one, as in lemma 19.

Proof. (1) The conditions Q_R = 0 and U_L = 0 at the walls are preserved by any operator acting on the H_+ index. [D_X, pi(g)] = E g' on Q, and J-hat pi(b) J-hat^-1 acts on the H_- components only, so both commutators vanish. (2) Given T, choose tau with support inside (0, l) and the same integral as T, which is a finite sum sum f_i g_i' of functions supported inside the interval. Take f = 1 and g = int_0^x (T - tau), which vanishes at both walls, so g lies in A_l and g' = T - tau. Self-adjointness of E T, with E skew, requires T skew. (3) J-hat E = E J-hat (direct from the formula for J-hat in Theorem 22.1), and c a c^-1 = U j a j^-1 U^-1 = U a U^-1. ■

Theorem 11.3 (zero modes of the interval component). Let k_0, k_l be the skew parts of C_0, C_l, and assume a five-dimensional Yang-Mills action for the fluctuation on a flat product with M^4, as in Proposition 22.8.

1. The gauge group is {g : I -> O(30), g(0) in K_0, g(l) in K_l}. The holonomy of a_x transforms as P -> g(l) P g(0)^-1, so the moduli are the double coset K_l SO(30) / K_0, and the massless four-dimensional scalars from a_x form

   ```
   the orthogonal complement of k_0 + k_l in
   ↪ so(30).
   ```

2. In lemma 22's algebra, C_0 = A, so k_0 = so(30), and there are no such scalars, for any algebra of this kind.

3. If C_0 = C_l = M_15(C), the massless scalars are exactly Λ^2 V ⊕ Λ^2 Vbar, with colour-antitriplet counts (6, 6, 3), and the massless vectors are u(15).

4. A generator in k_0 ∩ k_l has a massless vector and no massless scalar; a generator in the orthogonal complement of k_0 + k_l has a massless scalar and no massless vector; all others have neither. If the orthogonal projections onto k_0 and k_l commute (aligned walls), the towers are n pi/l in the first two cases, and (n - 1/2) pi/l for a generator in exactly one of k_0, k_l. For nonaligned walls the towers move: k_0 spanned by T_1 and k_l by cos(phi)

T_1 + sin(phi) T_2 give k l = n pi +- phi, which is the mechanism of lemma 23.

5. With colour unbroken at both walls, the (1, 8) of Λ^2(30) is a vector, so Λ^2(30) cannot consist entirely of massless scalars. In item 3, the (24, 8) + (24, 1) + (1, 8) + (1, 1) of lemma 21 are vectors, and the scalars are (15, 3bar) + (10, 6) + conjugates.

Proof. (1) A gauge parameter xi(x) with xi(0) in k_0, xi(l) in k_l shifts int a_x by xi(l) - xi(0), so the linear moduli are so(30)/(k_0 + k_l), and a constant a_x in the orthogonal complement is a zero mode of the quadratic form. (2), (3): substitution, with Proposition 11.1. (4) A_mu has Neumann or Dirichlet conditions at each wall according to membership in k_wall, and a_x the opposite, so the mode functions are cos(n pi x/l), sin(n pi x/l) or their half-integer versions. (5) Item 4, and the branching of lemma 21. ■

Scope of item 4. Free five-dimensional Yang-Mills at trivial background. Escapes: a circle without walls, nontrivial holonomy, brane-localized terms, further internal dimensions.

Note that the choice in item 3 is an input: the domain does not force the wall algebra to M_15(C), since by Proposition 11.2, item 1, every f preserves it at D_F = 0. It says that the walls conserve quark number and the bulk does not. On a circle, without walls, every generator has both a massless vector and a massless scalar, and the whole of Λ^2(30) appears, which is the nine-dimensional adjoint scalar used in 09. A wall condition, or the reflection-even algebra of Proposition 1.1, removes half. So the two readings differ: for the stack off the O8 plane, with holonomy I, diquarks are massive and mesons have massless scalars; on the interval with parity Ad(I), diquarks have massless scalars and mesons have none.

The shadow in the finite window

Proposition 11.4. Let s = +1, M = U Lambda with Lambda real for j and symmetric, and let a_tot = a ⊕ U a U^-1 be constant. Then:

1. [D_F, a_tot] has the off-diagonal blocks U S and S U^-1, with S = [Lambda, a] real and symmet-

ric, and [D_F, [D_F, a_tot]] = [Lambda, S] ⊕ U [Lambda, S] U^-1.

2. By Theorem 22.4, the compression of E a_tot to the window is -l^-1 int_0^l sin(x ad_{D_F})(a_tot) dx. For Lambda = diag(m_i) on flavours, on both V and Vbar, the block B_ij goes to

   ```
   - ((1 - cos(l (m_i - m_j))) / (l (m_i - m_j)))
    ↪ B_ij,    which is -(l/2)(m_i - m_j) B_ij
    ↪ to leading order,
   ```

   as a change of Lambda. The result is symmetric, as Proposition 9.3 requires of a finite J-compatible fluctuation.

3. For distinct masses, B -> (m_i - m_j) B_ij maps (S^2 5, 3bar) in Λ^2 V onto (Λ^2 5, 3bar) in S^2 V, with kernel the five same-flavour diquarks, and maps (Λ^2 5, 6) into (S^2 5, 6). In charges, (6, 6, 3) - (3, 0, 2) = (3, 6, 1). If the wall condition is imposed on the gauge unitaries g ⊕ U g U^-1, so that k_l is the commutant of Lambda in so(30), then with k_0 = u(15) the massless scalars of Theorem 11.3 are the same off-diagonal blocks. If it is imposed on pi(f) itself, as in the algebra of lemma 22, then k_l = 0 for invertible M.

4. For a universal mass, Lambda = m, the compression vanishes: S = 0, and M = m U is invariant under SO(30). Fifteen Dirac fermions of equal mass are thirty Majorana fermions of equal mass.

Proof. (1) Direct multiplication, using a_tot on H_- equal to U a U^-1. S is real and symmetric since a is real and antisymmetric and Lambda real and symmetric. (2) Item 1 gives sin(x ad)(a_tot) = (sin(x delta)/delta) ad(a_tot) on B_ij, with delta = m_i - m_j; integrate. (3) In the colour antitriplet, B is symmetric in flavour, and (m_i - m_j) B_ij is antisymmetric in flavour, hence total symmetric. Colour sextets are analogous. (4) S = 0. ■ Checked in Python on a small model, with V = C^3: reality, symmetry, the blocks of item 1 and their squares. Claude Sonnet 5.5 also checked the compression by direct quadrature of W^*(E a_tot)W, with five flavours in three colours, against Theorem 22.4 and the closed block (Python, 10^-15).

Reading. The commutator [D_F, a] is the form of the Higgs field in Connes' finite geometry, and E a_x is a Kaluza-Klein component along the interval. Theorem 22.4 (3), Q_l([D_X, f]) = [D_F, Q_l(f)], says that the second compresses to the first. Indeed E a_tot = [D_X, f_tot] with f(x) = (x - l) a, and f(l) = 0 lies in C_F. So in this parent, to first order and for the fixed compression, a finite Higgs one-form is the shadow of a gauge field along the interval, and it inherits the symmetry type of the finite triple by the commutator with masses. The moving window of Corollary 22.6 gives -l [D_F, h] and a quadratic term (lemma 19). This shadow is tangent to the O(30) orbit of Lambda, so it gives no radial mode, in agreement with No-go theorem 22.7; the odd generators of lemma 23 are absent here since M_30(R) maps H_+ to H_+. A generator commuting with the masses has no shadow. This explains the difference between the counts (6, 6, 3) of the parent and (3, 6, 1) of the finite triple, and it replaces the alternative O8^- against O8^+ of Proposition 9.2, for these blocks, by a parent and its window.

What remains open

- Whether M_15(C) at the walls can be derived from a boundary condition instead of chosen, for example by a wall term that conserves quark number.
- The treatment of nonzero masses, where order one fails on the block: the quadratic terms of lemma 19 for this algebra.
- The sleptons: on an interval the (24, 1) + (1, 1) are vectors. Scalars in these representations need components along further internal directions, such as the base of 02.
- The colour sextets (10, 6) come with the antitriplets, and nothing removes them in this construction.
- The relation to the top: the (30, 2) of SO(32), with the second factor as the top's SO(2).

Part R3 (referee report, lemmas 15, 16)

Referee report on lemmas 15 and 16 and on the T0 note assessment

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_15_16.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemma 15, lemma 16 and the assessment. Claude checked the two external citations against the arXiv LaTeX sources: the inner Leibniz condition in arXiv:1811.10843 and the commuting subalgebra C_F in hep-th/0610241, section 2.6.

We use the definitions and notation of Lemma 15 and Lemma 16. All calculations below are analytic; we ran no numerical computation.

The statements involving spectral propinquity require one correction to the supplied Farsi–Latrémolière source. Its displayed inner Leibniz condition is ill-typed as written. We state the correction under Theorem 15.3 and use it throughout.

Part 1. Statements and proofs

Lemma 15.1 Correct.

For self-adjoint e, its norm is the supremum of $|\varphi(e)|$ over states. The definition of extent therefore gives both norm inequalities. The Jordan-Leibniz inequality gives

$$L_E(e^m) \leq m\|e\|^{m-1} L_E(e)$$

by induction, and the polynomial estimate follows by the triangle inequality.

Theorem 15.2 Gap, closed here.

The commutativity argument in item 1 is correct. Bounded-Lipschitz lifts of two self-adjoint elements have uniformly bounded norms. Their Lie product maps to zero in every commutative approximating algebra, so Lemma 15.1 forces the Lie product of the two limit elements to vanish.

In item 2, the stated conditions on g do **not** imply that b_n stays within $3/8$ of an integer. They control neighborhoods of the half-integers, but permit values beyond the outermost integers.

We can repair the proof with the same constants. Choose $g : I \to [0,1]$ such that

$$g = 0 \quad \text{on } \bigcup_{j=0}^{k} [j - \tfrac{1}{8}, j + \tfrac{1}{8}], \qquad g = 1 \quad \text{outside } \bigcup_{j=0}^{k} (j - \tfrac{3}{8}, j + \tfrac{3}{8}).$$

A polynomial p with $\|p - g\|_I < 1/8$ has $\|p(a)\| < 1/8$. Lemma 15.1 then gives $\|p(b_n)\| < 1/4$ for large n, so the range of b_n lies in the union of the disjoint open $3/8$-neighborhoods. Each connected component of X_n maps into one neighborhood. At most k neighborhoods are reached, and a fixed one is missed along a subsequence. The polynomial q constructed in the note then gives the stated contradiction.

The clopen partition used at the start exists because components and quasicomponents agree in a compact Hausdorff space. Both conclusions follow.

Theorem 15.3 **Correct with amendment.** The estimates hold under the standard inner Leibniz definition. The supplied source's literal definition must be corrected before its tunnels can be checked as stated.

At source lines 602–604 (arXiv:2404.00240v3 LaTeX source), the inner product is required to lie in the domain of L_B, although that domain was defined on self-adjoint elements; the displayed inequality then names L_A. Literally, the first requirement fails for every nonzero complex Hilbert module: $\langle \omega, i\omega \rangle$ is a nonzero skew-adjoint element. The required condition is

$$\max\{L_B(\mathrm{Re}\langle \Theta, \Theta' \rangle), L_B(\mathrm{Im}\langle \Theta, \Theta' \rangle)\} \le 2\,\mathrm{DN}_P(\Theta)\mathrm{DN}_P(\Theta').$$

This is the condition in Latrémolière's Definition 2.2.

With that correction, the tunnel has the required properties:

- The top seminorm is Leibniz for Jordan and Lie products, has scalar kernel and a closed unit ball. Its centered unit ball is compact. Choosing the midpoint of the spectrum of a proves the quotient condition onto A; scalar lifts prove it onto $\mathbb{C}$. Its extent is at most ρ.
- The base seminorm on $\mathbb{C}\oplus\mathbb{C}$ is an L-seminorm with extent at most s.
- The module D-norm dominates the module norm and has a compact unit ball. The displayed modular estimate is valid without assuming P commutes with A.
- The inner-product calculation proves the stronger complex estimate

$$|\langle\xi,\xi'\rangle-\langle\eta,\eta'\rangle|\le 2s\,\mathrm{DN}_P(\xi,\eta)\mathrm{DN}_P(\xi',\eta'),$$

  which implies the corrected real-and-imaginary-parts condition.
- The lifts $\eta=cW\xi$ and $\xi=cW^*\eta$ attain the two D-norm quotients. The coordinate maps also preserve the left and right actions and inner products.

The pseudo-state choices in both directions are valid and are independent of t. Consequently they bound the **Hausdorff distance between families**, as required, by

$$g^{-2}+\rho+\Delta+\Delta/\varepsilon.$$

The conclusion follows by taking ε larger than $g^{-2}+\rho+\Delta+\sqrt{\Delta}$ and then decreasing ρ. The lower bound correctly includes the cap:

$$\Lambda_{\mathrm{spec}}\ge\min\left\{\frac{\sqrt{2}}{2},\frac{\delta}{2}\right\}.$$

Remark 15.4 **Error.** The uncapped lower bound is asserted for every $\varepsilon>0$, although spectral propinquity is at most $\sqrt{2}/2$.

The correct statement is

$$\min\left\{\frac{\sqrt{2}}{2},\frac{\varepsilon\delta_1}{2}\right\}\le\Lambda_{\mathrm{spec}}\le\frac{\varepsilon\delta_1}{2}+\frac{\varepsilon^2}{g_1^2}.$$

For example, take the periodic Dirac operator on a circle of length 1. It has zero modes and $\delta_1=1/2$. At

$\varepsilon = 4$, the asserted uncapped lower bound is $1 > \sqrt{2}/2$. The claimed leading asymptotic remains correct for sufficiently small ε.

Corollary 15.5 **Correct.**

For the stated sufficiently small ℓ, including $\ell < \sqrt{2}$ in the chosen normalization, the cap does not alter the lower bound. The interval diameter is ℓ, the light block is exactly $m_f\sigma_1$, and the next eigenvalue satisfies

$$g^2 \geq \mu(\ell)^2 + \frac{\pi^2}{\ell^2} = \frac{c(\ell)^2 + \pi^2}{\ell^2}.$$

Thus the upper correction is exactly bounded by $\ell^2/(\pi^2 + c^2)$, as displayed. The Connes–Lott lower bound has the correct factor $1/2$ and cap.

Corollary 15.6 **Correct.**

The boundary conditions are preserved by scalar Lipschitz functions, and the commutator seminorm is $\|f'\|_\infty$. The zero modes and positive and negative light modes in Theorem 12.7 have the correct total dimension $N_Q + N_U$. Matching them to the singular vectors of M_F gives $\Delta = O(\ell^2)$, including zero singular values. The gap is $O(\ell^{-1})$, so Theorem 15.3 gives the asserted $O(\ell)$ bound.

Theorem 15.7 **Correct with amendment.** The proof uses the corrected real-and-imaginary-parts inner Leibniz condition stated under Theorem 15.3.

The two base characters have distance below ε_n, since either is a state of the tunnel base algebra and the other is the sole state in its quotient image. Applying the corrected inequality to the real and imaginary parts proves the displayed $4\varepsilon_n$ inner estimate. Indeed, rotating one vector by a phase gives the stronger constant $2\varepsilon_n$; the stated constant is sufficient.

The upper multiplicity argument is valid: the Gram matrix of $N + 1$ target vectors has rank at most N, while the entrywise estimate gives

$$\|I - G\| \leq 16(N + 1)(1 + R)^2\varepsilon_n.$$

For concentration, Hausdorff separation supplies **one** pseudo-state matching the entire time family. Comparing its estimates at t and 0 gives the displayed $8C^2\varepsilon_n$ bound. Averaging over $0 \le t \le 1/\varepsilon_n$ gives

$$\int \left(1 - \operatorname{sinc} \frac{x-\lambda}{\varepsilon_n}\right) d\nu(x) \le 8C^2\varepsilon_n.$$

Since $|\operatorname{sinc} u| \le 1/2$ for $|u| \ge 2$, the stated tail bound follows.

For each eigenvalue of D_F, projecting the lifted vectors to its $2\varepsilon_n$-window changes them by $o(1)$; their Gram matrix remains invertible. Distinct eigenvalues have disjoint windows eventually. The lower counts and the preceding upper count give the exact total multiplicity N, after which the eigenvectors can be matched with operator error at most $2\varepsilon_n$. The gap to the remaining spectrum tends to infinity. If no spectrum remains, the gap is interpreted as infinite.

Corollary 15.8 **Correct.**

Faithfulness and finite-dimensionality of the limiting Hilbert space make the limiting algebra finite-dimensional. Theorem 15.2 then gives $\mathbb{C}$. On a circle of length ℓ, the constant modes of

$$-i\,\frac{d}{dx} + \operatorname{diag}(\lambda_1, \dots, \lambda_N)$$

carry the prescribed Hermitian matrix, while the other modes have absolute eigenvalue at least $2\pi/\ell - \|D_F\|$. The circle diameter tends to zero, so Theorem 15.7 applies. The construction concerns ungraded triples; additional real or even structures would impose their own compatibility conditions.

Proposition 15.9 **Correct with amendment.**

The symmetric spectral projection commutes with γ and J, and the displayed estimates hold for $D_B = WDW^*$. Two qualifications are needed for the subsequent claim about metric control.

First, if $\Delta > 0$, $W\omega$ need not be a pseudo-state for the graph norm of D_F. The normalized choice $cW\omega$, with

$c = (1+\Delta)^{-1}$, is a pseudo-state and adds at most Δ to the stated bounds.

Second, the induced γ_F, J_F satisfy the Dirac relations exactly with D_B, not necessarily with the independently chosen D_F. The defects with D_F are bounded by 2Δ. The case $\Delta = 0$ is exact; when $\Delta \to 0$, the defects vanish.

Finally, composition with an antiunitary produces an antilinear functional. The source's operational propinquity is defined for linear functionals, so convergence of J requires the extension acknowledged in the note.

Example 16(a) **Correct.**

At the lowest massless interval level, the two chiral profiles give

$$Q(f) = \mathrm{diag}(C_0 + C_1, C_0 - C_1).$$

The coefficients C_0, C_1 can be chosen independently, so the range and generated algebra are both $\mathbb{C} \oplus \mathbb{C}$.

Example 16(b) **Correct.**

The product-to-sum identities give the two displayed matrices. Their four coefficients $C_0, \ldots, C_3$ are independently recoverable, so the operator system has complex dimension 4.

For the two displayed functions, direct substitution gives

$$Q(f_1) = \left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, 0 \right), \qquad Q(f_2) = \left(0, \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \right).$$

Each coordinate projection of the generated algebra contains all symmetric 2×2 matrices, hence all of $M_2(\mathbb{C})$. The first displayed element generates a nonzero two-sided ideal supported on the first summand; simplicity of $M_2(\mathbb{C})$ makes that ideal the entire first summand. The second element does the same for the other summand. Therefore

$$A_F = M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).$$

Example 16(c) **Correct.**

For a uniformly distributed rank-one projector P_w on $\mathbb{C}^4$,

$$\mathbb{E}(P_w \otimes P_w) = \frac{I + \mathrm{Flip}}{20}.$$

Together with $\operatorname{tr}\Gamma_a = 0$, this gives $Q(x_a) = \Gamma_a/5$. Four of the gamma matrices generate $M_4(\mathbb{C})$. For the twisted Dolbeault operator, the kernel is $H^0(\mathbb{CP}^3, \mathcal{O}(1))$, of dimension 4; the higher cohomology groups vanish. Thus $d = 0$ on the selected subspace.

Proposition 16.1 **Correct with amendment.**

The order-zero equivalence is correct. The hypotheses imply $J_F Q(a)^* J_F^{-1} = Q(a)$, so order zero forces all compressed generators to commute. Conversely, if their generated algebra is commutative, the opposite image is the same algebra and order zero holds.

For the product assertion, specify that the fixed finite factor itself satisfies order zero and that the product real structure is the compatible one. Also, the hypotheses on J_0 alone do not assert $J_0^2 = \pm 1$ or a grading sign. Those relations must be added if J_F is called a real structure in the full sense.

Theorem 16.2 **Correct with amendment.** The Weyl-exponent assertion requires the original exponent to exist.

The graded flip is well defined on the complex tensor product. On each homogeneous summand it is the ordinary conjugate flip times a sign; it is therefore antiunitary. Direct application to homogeneous tensors gives

$$J^2 = 1, \qquad J\mathcal{D} = \mathcal{D}J, \qquad J\Gamma = \Gamma J.$$

For even a, $a^\circ = 1 \otimes (a^*)^c$. Hence order zero and order one hold, and

$$[\mathcal{D}, \pi(a)] = [D_0, a] \otimes 1.$$

The signs $(+, +, +)$ are the usual KO-0 signs.

An eigenbasis of D_0 decomposes $\mathcal{D}$ into finite-dimensional invariant blocks. On each block,

$$\mathcal{D}^2 = D_0^2 \otimes 1 + 1 \otimes (D_0^c)^2.$$

This proves essential self-adjointness on the algebraic tensor core and compact resolvent of the closure.

Writing $N_0(R)$ and $N_{\mathcal{D}}(R)$ for absolute-eigenvalue counting functions with multiplicity,

$$N_0(R/\sqrt{2})^2 \le N_{\mathcal{D}}(R) \le N_0(R)^2.$$

Thus, if $\lim_{R\to\infty} \log N_0(R)/\log R = p$ exists, the new exponent is $2p$. Compact resolvent alone does not guarantee such a limit: eigenvalue multiplicities can be chosen to make the logarithmic ratio oscillate.

Corollary 16.3 **Correct with amendment.** Assume $k \ge 1$, and distinguish the product projection from a spectral projection of $\mathcal{D}$.

The identities in the corollary and both order conditions hold for every even subalgebra $A_F \subseteq B(V)$. In particular,

$$J_F(d \otimes 1)J_F^{-1} = \gamma_V \otimes d^c, \qquad [D_F, a \otimes 1] = [d, a] \otimes 1.$$

The original compression has range $Q(A_0) \otimes 1$; the corollary uses the **generated** algebra $A_F \otimes 1$.

The projection $P_0 \otimes \overline{P_0}$ reduces $\mathcal{D}$, but need not equal $1_S(\mathcal{D})$ for any set S. For a counterexample, take an even finite D_0 with positive absolute eigenvalues

$$1, \quad 5, \quad 7, \quad \sqrt{73},$$

and let P_0 select the first three absolute values. The selected pair $5, 7$ and the mixed pair $1, \sqrt{73}$ both give squared product eigenvalue 74. Both reducing subspaces contain the eigenvalues $\pm\sqrt{74}$, so the product projection cannot be a spectral projection of $\mathcal{D}$. It is a joint cutoff in the two factors. In the kernel example used below, it *is* a spectral projection.

Corollary 16.4 **Correct.**

The displayed operators have signs $(+, +, -)$, the usual KO-6 signs. Both algebra actions are trivial on the last factor, so order zero and order one persist, and the Hilbert-space dimension is $2k^2$.

The signs alone do not establish the usual zero-dimensional orientability axiom. Every represented zero-cycle built from the left and opposite algebra actions acts as $T \otimes 1_2$, whereas the grading is $\Gamma_F \otimes \sigma_3$. No such zero-cycle can equal that grading.

Caveat 1: commutant **Correct.**

For

$$d = \begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix},$$

with B invertible, the relation $\mathrm{diag}(x, y)d = d\,\mathrm{diag}(x, y)$ is equivalent to

$$y = B^{-1}xB, \qquad [x, BB^*] = 0.$$

Invertibility entails equal dimensions of V_+ and V_-. In example (b), $B = \mathrm{diag}(\pi/(2\ell), 3\pi/(2\ell))$; its squared singular values are distinct. The commuting pairs are diagonal with the same entries in both summands, giving the stated $U(1) \times U(1)$.

Caveat 2: spectrum **Correct.**

The square of the doubled Dirac operator is the sum of the two squared factors, with multiplicities counted in the tensor product. In example (a), $D_F^2 = 2m^2 I$, so its eigenvalues are $\pm\sqrt{2}\,m$. Its commutator with the left copy of $\mathbb{C}^2$ still has norm $m|a - b|$, giving distance $1/m$.

Caveat 3: parent **Correct with amendment.**

For an ordinary compact spin example of dimension n, the underlying product Dirac operator has Weyl exponent $2n$, while the restricted algebra has the metric of one n-dimensional factor. The statement about the product manifold requires the charge-conjugation identification and algebra restriction specified in Part 2(iii). The discrepancy also means that the restricted triple is generally not the ordinary finite-rank twisted spin triple over the single factor.

Caveat 4: KO-6 **Correct with amendment.**

The Clifford point is an input to Corollary 16.4. The claim that obtaining the sign from a geometric parent is "open" should be scoped to the work checked here: the cited documents do not establish such a construction or a no-go theorem.

Caveat 5: dimension count **Correct.**

For $k = 4$, the dimension after adding the Clifford point is 32. Neither example produces the stated classified algebra or its representation. Moreover, the C^*-algebras generated in Lemma 16 are complex algebras; a quaternionic real summand requires additional real-algebra data.

Part 2. Interpretation

(i) Unbroken symmetry and the Standard Model Error, for the unrestricted claim. A metric spectral triple excludes nonscalar elements of its represented algebra that commute with D. That does not exclude every symmetry implemented on its Hilbert space.

For a counterexample within the spectral-propinquity domain, let $H = M_2(\mathbb{C})$ with Hilbert–Schmidt inner product, let $A = M_2(\mathbb{C})$ act by left multiplication, and let

$$D(x) = x - \tfrac{1}{2}\operatorname{tr}(x)I.$$

If $[D, L_a] = 0$, applying the commutator to I gives a scalar. Finite dimensionality then makes this a metric spectral triple. Yet every conjugation $x \mapsto uxu^*$, $u \in SU(2)$, commutes with D. With $Jx = x^*$, these are the transformations $L_u J L_u J^{-1}$. This example fails order one, which spectral propinquity itself does not require.

A narrower statement does hold for a **finite real first-order** triple. If $uJuJ^{-1}$ commutes with D, order one and $JD = \varepsilon' DJ$ imply that $a = [D, u]u^*$ commutes with u. Thus

$$u^n D u^{-n} = D - na.$$

Since D is bounded, $a = 0$. If the triple is metric, u is scalar and the induced transformation is trivial.

The Standard Model finite triple is outside the metric class. In the usual construction, the color action commutes with the finite Dirac operator; independently, the nonscalar subalgebra C_F is explicitly required to commute with it in Chamseddine–Connes–Marcolli, §2.6. The constant copy of such an element also obstructs the almost-commutative product. The original algebra is a real algebra, so applying a framework formulated for complex C^*-algebras also requires specifying a complexification.

(ii) Morita equivalence Error, as an argument that removes the scalar-algebra restriction. The Morita equivalence $C(K) \sim C(K, M_n(\mathbb{C}))$ is correct, but Morita equivalence does not make the two algebras isomorphic or identify their spectral propinquity limits. Theorem 15.2 remains a restriction on limits of the whole **scalar** algebra $C(K)$.

A vector bundle E can carry matrix-valued connections while $C(K)$ still acts by scalar multiplication on sections. Replacing the represented algebra by $\mathrm{End}(E)$ is an additional change of data. It may be a useful enlargement of T0, but it is not a consequence of the presence of E.

The product-collapse example in Farsi–Latrémolière (arXiv:2404.00240v3 LaTeX source) concerns an already noncommutative finite algebra B in $B \otimes C(K_\ell)$. Its finite triple must be metric, and the limiting Hilbert space has the additional factor $\ker D_K$, which must be nonzero. A one-dimensional kernel recovers the stated finite multiplicity. The example does not produce a noncommutative whole-algebra limit from scalar parent algebras.

(iii) The geometry of the doubled parent Correct with amendment.

For a compact even-dimensional spin manifold, an appropriate charge-conjugation identification of $\overline{H_0}$ with H_0 identifies the displayed $\mathcal{D}$ with the product Dirac operator on $X \times X$. The grading may acquire an overall sign under that identification. The real structure in Lemma 16 covers the exchange of factors and is not the usual pointwise spin real structure.

The spectral triple used in the construction represents only $C(X) \otimes 1$. It is therefore a product-manifold Dirac operator equipped with a restricted observable algebra. If "commutative Kaluza–Klein data" includes that restriction and the flip real structure, the description is accurate. If it means the ordinary spin triple of the total manifold with its full function algebra, or a finite-rank spinor twist over X, it is too broad. With the full product algebra, the generated compression is $A_F \otimes \overline{A_F}$, and the flip generally fails order zero when A_F is noncommutative.

(iv) What Lemma 16 advances Correct with amendment.

For every finite even representation (A, V, d, γ), the construction

$$V \longmapsto V \otimes \overline{V}, \qquad d \longmapsto d \otimes 1 + \gamma \otimes d^c$$

gives a real bimodule satisfying order zero and order one. It uses no pre-existing finite real structure. Consequently, those two order conditions follow formally from the chosen tensor-square representation; the construction does not recover an arbitrary prescribed finite H, D, J, or their multiplicities.

The examples supply more than that formal observation. In particular, the $\mathbb{CP}^3$ kernel example gives a simple tower-decoupling family under the note's generated-compression rule. Put

$$\mathcal{D}_\ell = \ell^{-1}\mathcal{D}$$

on the doubled parent. Its kernel is $V \otimes \overline{V}$, its complementary gap grows like $1/\ell$, and compression followed by algebra generation gives the constant finite data

$$A_F = M_4(\mathbb{C}), \qquad \dim H_F = 16, \qquad D_F = 0.$$

After adding the stated Clifford point, the dimension is 32 and the KO-6 signs hold. Here order one is automatic because the finite Dirac operator vanishes. The whole parent triple has scalar algebra in its spectral-propinquity limit, consistently with Lemma 15.

Thus Lemma 16 advances T0 for a specified truncation procedure, including a noncommutative finite algebra and an explicit bimodule. It does not establish the target under every meaning of ordinary Kaluza–Klein data: the parent algebra is restricted, the real structure flips factors, and algebra generation is an additional operation. For T0b, it gives examples with the requested signs and first-order relations, but leaves the desired algebra and representation, orientability, nonzero mass data, classification, and the beyond-first-order and twisted conditions unestablished. The Clifford point also fails the finite orientability condition described above. The assessment is right to reject an obstruction inferred from one metric; its description of a "concrete origin" should retain these qualifications.

(v) Prior sources Gap. We cannot identify a source that states either the full Theorem 15.2, including its component bound, or the exact graded-flip construction of Theorem 16.2. We make no originality claim.

Two constituent results are identifiable in the supplied source: Farsi–Latrémolière state the kernel collapse to $(\mathbb{C}, \ker D, 0)$ at lines 1231–1235 (arXiv:2404.00240v3 LaTeX source), and use the product Dirac formula at lines 1262–1267 (arXiv:2404.00240v3 LaTeX source). Those citations do not establish prior publication of either full statement.

Amendments and errors

- Correct the supplied source's inner Leibniz condition to use L_B on real and imaginary parts; Theorems 15.3 and 15.7 then have the stated estimates.
- Repair the choice of g in Theorem 15.2 as above. The theorem follows.
- Restore the cap in Remark 15.4.
- In Proposition 15.9, normalize the comparison pseudo-state when $\Delta > 0$, and distinguish the exact relations for D_B from the $O(\Delta)$ relations for D_F.
- Add the stated real-structure and product hypotheses to Proposition 16.1; assume a Weyl exponent exists in Theorem 16.2; distinguish a joint cutoff from a spectral projection in Corollary 16.3.
- Qualify the parent geometry and the scope of the claimed open KO-sign problem. Record the orientability limitation of the Clifford-point construction.
- Restrict the unbroken-symmetry claim to the conditions actually proved, and do not use Morita equivalence to erase the distinction between scalar and matrix-valued parent algebras.

Overall: 26 statements and interpretive items checked: 11 correct, 10 correct with amendment, 2 gaps, and 3 errors.

Part R8 (referee report, lemma 22)

Referee report on lemma 22

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_22.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 22.

We find two errors in the stated claims: the assignment of zero modes at nonzero wall coupling, and the identification of fixed compression with the first derivative of transport. The main interval construction and its compression formulas hold, subject to the amendments below. The numerical collider-exclusion claim remains unsupported.

Throughout, let $\ell > 0$, and write $D = D_F$, $\Gamma = \Gamma_F$, and $J = J_F$ when no confusion can arise.

1. Setting — Correct.

In bases related by c, the condition on M says that its matrix is complex symmetric. Let its Takagi factorization be

$$M = U\Sigma U^T.$$

Then

$$|M| = \overline{U}\,\Sigma U^T, \qquad V = UU^T$$

give $M = V|M|$, with V unitary and $V = cV^*c$. This construction includes the kernel: zero entries of Σ still have corresponding columns in the unitary U. The extension of the polar factor on the kernel need not be unique.

The formula

$$X = \tan(\ell|M|)V^*$$

and the reality condition $cX = X^*c^{-1}$ follow. Conversely, for every X satisfying that condition, the principal functional calculus gives the unique operator

$$D_F = \ell^{-1} \arctan \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \qquad \ell\|D_F\| < \pi/2.$$

Thus the claimed bijection is valid. The expansion $X = \ell M^* + O(\ell^3)$ assumes fixed M.

2. Theorem 22.1 — Correct with amendment.

The boundary subspaces are Lagrangian for the first-order boundary form, giving self-adjointness; compactness of the H^1-to-L^2 embedding gives compact resolvent.

Direct substitution verifies the stated signs and all three intertwining relations. The L- and R-components are orthogonal, while $R(x) = e^{x\Gamma_F D_F}$ is unitary, so W is an isometry.

For each singular value σ_k, with $\theta_k = \ell\sigma_k$, the signed spectrum is

$$\frac{n\pi \pm \theta_k}{\ell}, \qquad n \in \mathbb{Z},$$

with multiplicities added when values coincide. This gives the stated spectrum of $|D_X|$, the window, and the lower bound on the next level. The amendment is to state multiplicities, particularly when $\sigma_k = 0$: the zero mode then has multiplicity two per zero singular channel, and the two excited branches coincide.

For arbitrary signed momentum k, the scalar equation is

$$|\tan(k\ell)| = \sigma_k(X).$$

For the positive light mass $k = \sigma_k(D_F)$, its tangent is nonnegative and equals the corresponding singular value of X.

3. Corollary 22.2 — Correct.

On the uncoupled domain, multiplication by R maps the endpoint conditions to those of D_X, and

$$R^* D_X R = D_0 + E\Gamma_F D_F.$$

For one real channel, $\Gamma_F D_F = mF$, with F as defined in equation (21) of lemma 20. The sign and the placement of the spin matrix E agree.

4. Standard Model form and boundary types — Error.

The type assignment holds **at** $X = 0$: Q has type (a) and U has type (b). For $X \neq 0$, the wall conditions couple the fields, and the corresponding zero modes generally lift. For example, one channel with $X = t > 0$ has lowest masses $\pm\arctan(t)/\ell$ and no zero mode.

The correction is to say that the coupled modes continue from the type-(a) and type-(b) zero modes of the uncoupled problem.

One matrix X does encode the full finite operator, including Yukawa and Majorana data. Its entries need not have

only the original block pattern: the tangent introduces higher powers of D_F. For example, with real symmetric

$$M = \begin{pmatrix} a & b \\ b & 0 \end{pmatrix},$$

the corresponding entry of $X = \tan(\ell M)$ is

$$X_{22} = \frac{\ell^3}{3} ab^2 + O(\ell^5).$$

Thus the original zero block can acquire an induced entry.

5. Theorem 22.3 — Correct with amendment.

Assume explicitly that the finite algebra and its faithful even representation are unital. The two endpoint intertwining equations are equivalent to

$$[\pi(f(\ell)), D_F] = 0.$$

The commutant of a self-adjoint operator is closed under products and adjoints. Hence C_F, and consequently A_ℓ, are unital $*$-algebras.

For every $f \in A_\ell$, multiplication preserves the whole H^1 domain and

$$[D_X, \pi(f)] = E\pi(f')$$

there. The right side extends to a bounded operator on all of H_ℓ. The opposite action also preserves the domain: this follows from domain preservation by g^* and $J^\wedge D_X = D_X J^\wedge$. Finite order zero then gives

$$[[D_X, \pi(f)], \pi^\circ(g)] = E[\pi(f'), \pi^\circ(g)] = 0.$$

No finite order-one hypothesis is needed.

This agrees with Counterexample 4 of lemma 20: its parent has order one, while its two-point compression fails finite order one. The statement concerns this parent and its pointwise algebra; it does not assert that arbitrary compressions preserve either order condition.

The Standard Model identification of C_F also needs the usual nonzero Yukawa data and the specified representation. For example, if all Yukawa blocks vanish and only an algebra-commuting Majorana block remains, the commutant is larger.

6. Theorem 22.4 — Correct with amendment.

Both compression formulas, including the minus sign in $Q_\ell(ET)$, and the symmetric exponential average for constant a are correct. The four items hold with the stated constants. In particular, interpreting $\|a\|$ as $\|\pi(a)\|$,

$$\|Q_\ell(a) - \pi(a)\| \le \frac{\ell^2}{6}\|[D,[D,\pi(a)]]\| \le \frac{2}{3}\ell^2\|D\|^2\|\pi(a)\|.$$

The constants in item 4 are respectively 8/3 and 16/3.

For item 1, compactly supported bulk functions suffice to test every smooth matrix-entry density. Finite-dimensional linear algebra then gives equality of the two images. The constant function a itself need not belong to A_ℓ; the theorem supplies a domain-preserving function with the same compression.

The general warning about generated algebras is sound. It also has a simple analytic example. Let

$$M = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & M \\ M & 0 \end{pmatrix},$$
$$\pi(\mathbb{C}^3) = \{\mathrm{diag}(a,a) : a \text{ diagonal}\},$$

with $J = \left(\begin{smallmatrix} 0 & K \\ K & 0 \end{smallmatrix}\right)$. If P_1, P_2 are the first two diagonal channel projections, then on either graded block

$$[Q_\ell(P_1), Q_\ell(P_2)]_{12} = -\frac{\ell^2}{3} + O(\ell^4).$$

The original projections commute, whereas their compressed images generate a noncommutative algebra for sufficiently small nonzero ℓ.

The specific dimensions 15 and 30 remain **numerical claims**: the matrices and closure calculation were not retained, and the reported tolerance alone does not establish an exact rank. They should be accompanied by reproducible data and an exact or certified rank calculation. The general role of operator systems under spectral truncation is already developed in Connes and van Suijlekom, §1 and §3, arXiv:2004.14115v2.

7. Theorem 22.5 — Correct with amendment.

Every represented self-adjoint one-form is $E\pi(h)$, with $h^* = -h$. In fact **every smooth** such h arises, including

those with nonzero boundary value: set

$$b(x) = \int_{\ell}^{x} h(t)\, dt.$$

Then $b(\ell) = 0$, so $b \in A_\ell$, and $[D_X, b] = E\pi(h)$.

Thus imposing compact support would exclude some connection profiles, although it loses no endpoint holonomy in the identity component: any smooth path from 1 to such a unitary can be made constant near both endpoints.

The transport formula and wall condition are correct. The proof should separate its two cases. For a general w, $\pi(w)^* D_F \pi(w)$ need not commute with J_F, so the non-real part uses the spectral and isometry proof of Theorem 22.1 **without** its real-structure hypothesis. For the real fluctuation, $U_w = \pi(w) J_F \pi(w) J_F^{-1}$ commutes with J_F, and the claimed real window follows.

There is a useful stronger statement. Put

$$\tilde{u}(x) = u(x) w^*.$$

Then $\tilde{u}(\ell) = 1$, so $\tilde{u} \in A_\ell$. With

$$\widetilde{U} = \pi(\tilde{u}) J_F \pi(\tilde{u}) J_F^{-1},$$

direct differentiation and the endpoint condition give

$$D_X + A + J^\wedge A (J^\wedge)^{-1} = \widetilde{U} D_X \widetilde{U}^*.$$

Thus every scalar connection in this interval model is gauge equivalent to the original parent under its allowed endpoint gauge group.

8. Corollary 22.6 — Error.

The gauge-orbit comparison itself is correct. Chamseddine–Connes–van Suijlekom, §III.2, equations (10)–(11), arXiv:1304.7583v2 give the quadratic fluctuation and its gauge transformation. Here the correct substitution is $v = w^*$, so the moving window carries

$$U_v D_F U_v^* = U_w^* D_F U_w.$$

For the two-point example, if $w = \mathrm{diag}(e^{i\alpha}, e^{i\beta})$, the upper link is $m e^{-2i(\alpha-\beta)}$. It is $m e^{2i\psi}$ in lemma 19 when ψ is defined using $v = w^*$.

The error is the assertion that fixed compression is the first derivative of the transported window for the **same** connection. Let h be constant and replace it by εh. At fixed D,

$$\left.\frac{d}{d\varepsilon}\right|_0 D_{\text{transported}} = -\ell[D,h],$$

$$Q_\ell(Eh) = -\frac{\ell}{2}[D,h] + O(\ell^3).$$

For example, take $D = m\sigma_1$ and $h = i\sigma_3$. Their leading terms differ by a factor of two. The real fluctuation doubles both expressions in this example and retains the discrepancy.

The correction is to state that the displayed fixed-compression expression is a tangent commutator with its own weighted generator. The moving spectral projection contributes to the derivative of the transported operator. The comparison with lemma 14 is a common transport principle, with its global path qualification; lemma 13 computes a distinct fixed compression. The first open point of lemma 19 is answered only on the specified gauge orbits and with the moving window.

9. No-go theorem 22.7 — Correct with amendment.

For the stated one-interval, zero-bulk-mass parent, fixed X, even scalar inner one-forms, and the **moving spectral window**, the finite operator is unitarily conjugate to D_F. Its singular values are therefore fixed. The author's argument and the direct endpoint-preserving gauge transformation in item 7 give the same conclusion in this scope. A fixed compression can show changes in singular values at second order, so the window must be named.

For the usual Standard Model representation with nonzero Yukawa data, the normalized shifted Higgs quaternion runs through unit quaternions, a three-real-dimensional gauge orbit. These are gauge-orbit directions. Calling them physical or eaten modes requires a spacetime gauge action and its boundary conditions; the interval calculation alone supplies no such mode count. Degenerate data, including vanishing Yukawas, can have a smaller orbit.

The listed changes of wall coupling, admissible odd bulk

terms, bulk masses, or internal dimension lie outside the theorem's hypotheses and are directions for further constructions. Curvature in higher dimensions alone does not prove a radial mode. Likewise, a quantum effective action on the same fixed configuration space does not by itself add a radial coordinate; an additional composite or other degree of freedom would have to be identified.

10. Proposition 22.8 — Correct with amendment.

Evaluation shows that the unsymmetrized gauge one-form has boundary value in C_F. For the displayed masses, assume a flat product interval, a positive Yang–Mills kinetic form, the trivial background, unrestricted endpoint variation in the allowed directions, and no additional boundary kinetic or mass terms. Variation then gives Neumann conditions at 0, and Neumann on the wall Lie algebra and Dirichlet on its orthogonal complement at ℓ. The stated integer and half-integer towers follow.

The physical gauge Lie algebra is obtained from skew-adjoint algebra elements through the real gauge representation, with its kernel and any imposed unimodularity condition accounted for. For the usual Standard Model representation, $u(C_F)$ has dimension 10 before unimodularity, whereas photon plus gluons have dimension 9. The extra abelian direction must be addressed. The role of unimodularity and the hypercharge embedding is explicit in Chamseddine–Connes–Marcolli, §2.5, arXiv:hep-th/0610241.

The literature comparison is valid at the level of the boundary mechanism. The Neumann/Dirichlet towers appear in Csáki et al., §5.1, arXiv:hep-ph/0305237v1, and the fermion equation $\tan(m_n L) = M_D L$ is equation (5.6) of Csáki et al., arXiv:hep-ph/0310355v1. Their full gauge models have additional structure.

11. Corollary 22.9 — Correct with amendment.

Items 1, 2, 4 and 5 hold for fixed, orthogonal, J_F- and grading-compatible light and heavy channels, with $0 < \Theta < \pi/2$. For a general non-normal light block, write the upper wall block as

$$X_0 = \tan(\ell|M_0|)V_0^*,$$

or state the formula in terms of singular values. The light-channel coupling is $\ell M_0^* + O(\ell^3)$. Identifying it with a

four-dimensional Yukawa coupling remains an additional hypothesis.

For item 3, let P_L reduce $D_F(\ell)$, and suppose $D_L = D_F(\ell)|_{P_L H_F}$ is fixed. Then the precise statement for a constant observable is

$$P_L Q_\ell(a) P_L = \operatorname{sinc}(\ell \operatorname{ad}_{D_L})(P_L \pi(a) P_L),$$

and

$$\|P_L Q_\ell(a) P_L - P_L \pi(a) P_L\| \le \frac{2}{3}\ell^2 \|D_L\|^2 \|\pi(a)\|.$$

This remains true when $\pi(a)$ mixes light and heavy channels. The limiting light compression need not be a representation: products can pass through the heavy subspace.

The full observable estimate does **not** follow from Theorem 22.4, since $\ell\|D_F(\ell)\| \to \Theta$. An explicit counterexample uses

$$D_F(\ell) = \sigma_1 \otimes \operatorname{diag}(m, \Theta/\ell), \qquad \pi(a) = 1 \otimes \sigma_1,$$

with the commutative algebra generated by 1 and $\pi(a)$ and the compatible real structure $J = (\sigma_1 \otimes 1)K$. Then

$$Q_\ell(a) = \operatorname{sinc}(\Theta - \ell m)\, \pi(a),$$

which tends to $\operatorname{sinc}(\Theta)\pi(a)$, rather than $\pi(a)$.

The ratio $m_t/M_W = 2\Theta/\pi < 1$ is a result for the stated lowest-level, zero-bulk-mass fermion channel and strict-wall gauge spectrum. Bulk fermion masses or a changed gauge wall condition are possible directions beyond that family.

12. Numerical comparison for the interior — Gap.

For a separate flat model with Neumann conditions at 0 and a finite Robin wall mass,

$$t \tan t = s, \qquad s = \frac{g_4^2 v^2 \ell^2}{4},$$

where $g_4^2 = g_5^2/\ell$ in the usual flat normalization. The Z equation has $s/\cos^2\theta_0$. This agrees with the form and sign convention of Csáki et al., §5.2, equations (5.13) and (5.22).

For $g'^2/g^2 = 3/5$, the lowest-mode squared mass ratio increases from 5/8 to 1. A small **Python** calculation evaluated the equations at the stated rounded values: $M_W\ell = 1.03$ is consistent with the quoted ratio to the displayed precision, and the next W root lies between $3.58/\ell$ and $3.59/\ell$, about 280 GeV when $1/\ell \approx 78$ GeV. This is a numerical check; the monotonicity follows separately from the equation.

The top constraints are strict:

$$M_W\ell < \frac{\pi}{2\sqrt{8/3}} \simeq 0.962, \qquad M_W\ell < \frac{\pi}{2(m_t/M_W)} \simeq 0.732$$

for the measured ratio. The latter estimate is consistent with the 2025 PDG W, Z values and top-mass table.

The collider-exclusion sentence is a **gap**: a mass alone does not establish a search exclusion without the model's couplings, branching fractions, and an applicable experimental limit. It should be removed or supplied with that analysis. The finite wall scalar also changes the strict wall boundary problem; it is an extended model outside Proposition 22.8 and No-go theorem 22.7. Within the stated flat toy model, the incompatible top and gauge mass requirements support the limited conclusion; bulk masses, warping, and running are directions to investigate.

13. Input and derived statements — Correct with amendment.

The finite algebra, representation, operator, grading, and real structure are inputs. The construction derives the endpoint commutant for that input, the interval operator and its window, the stated compression map, and the transport formulas. The Yang–Mills spectrum additionally depends on the action and boundary assumptions in item 10. The exact window identifies D_F, J_F, Γ_F; the represented algebra is obtained through compression and need not be an exact representation of A_F.

Several ingredients have broader precedents. For any reducing finite spectral subspace, the identity

$$[D_F, W^*TW] = W^*[D_X, T]W$$

holds whenever T preserves the operator domain. Operator-system compression is treated generally in

Connes–van Suijlekom, §3. Boundary spectral geometry for the Standard Model was also studied by Chamseddine and Connes, §5, arXiv:1008.3980v1, and the interval fermion and gauge mechanisms occur in the Higgsless papers cited in item 10. The particular combination and formulas of lemma 22 can be stated explicitly; a wider novelty claim would need a fuller literature comparison.

Amendments and errors

- **Errors:** correct the boundary-type zero-mode statement at nonzero X and the first-derivative comparison between fixed compression and transport.
- **Gap:** support or remove the collider-exclusion claim.
- **Amendments:** state spectral multiplicities and unitality; qualify the Standard Model data, gauge quotient, and unimodularity; distinguish profiles from holonomies and real from non-real transport; scope the radial result to the moving window; provide reproducible data for the numerical algebra dimensions; state the light-block estimate and the additional assumptions of the Yang–Mills and finite-wall-scalar models.

Overall: 2 Correct; 8 Correct with amendment; 1 Gap; 2 Error.

Part R9 (referee report, lemma 23)

Referee report on lemma 23

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_23.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 23 and the note.

The boundary-spectrum formulas are correct under the stated Yang–Mills boundary conditions. The principal corrections concern the aligned-wall compression, the comparison with twisted spectral triples, the order of the holonomy double coset, and the bulk-mass extension of the one-loop potential.

The verdicts below count the lettered entries. The numerical checks use Python 3 with SciPy. They supplement

the analytic arguments and do not replace them.

1. Theorem 23.1 (statement and proof)

1a. Item 1 — Correct. Each wall condition is a maximal isotropic subspace for the boundary form. The resulting first-order operator is self-adjoint with compact resolvent, and the spin grading preserves its domain and anticommutes with it.

1b. Item 2 — Correct. Under the equal-dimension and no--1-eigenvalue hypotheses, the principal logarithm gives an anti-Hermitian K satisfying $\Gamma_0 K \Gamma_0 = -K$. The stated $D_F = \Gamma_0 K$, its singular values, and

$$\Gamma_1 = e^{lK} \Gamma_0 e^{-lK}$$

follow from the two-projection decomposition.

1c. Item 3 — Correct with amendment. The graph boundary condition and the converse hold on the stated branch $l\|D_F\| < \pi/2$. Define X explicitly as the $H_- \to H_+$ block of $\tan(lD_F)$. The spectral part of Theorem 22.1 applies without a real structure; its real-structure conclusion additionally needs the hypothesis in item 4. At an angle $\pi/2$, the principal logarithm and the isolated window used here cease to be unique.

1d. Item 4 — Correct. The antiunitary J_1 is an additional hypothesis. With $\widehat{J} = EJ_1$, the signs are $(+,+,-)$, and $J_F = \Gamma_0 J_1$ has the stated finite signs. In particular,

$$J_F \Gamma_1 J_F^{-1} = \Gamma_0 (J_1 \Gamma_1 J_1^{-1}) \Gamma_0 = -\Gamma_0 \Gamma_1 \Gamma_0.$$

The antilinearity of J_1 is compatible with $J_1 K J_1^{-1} = K$ on the principal branch.

2. Theorem 23.2 (statement and proof)

2a. Items 1 and 2 — Correct with amendment. Domain preservation is equivalent to commuting with the respective wall grading, and $[D, f] = Ef'$. The compression formula in item 2 requires the equal-dimension and branch hypotheses under which the isometry W of Theorem 22.1 exists. State those hypotheses at the item.

2b. Item 3, aligned walls — Error as stated for an arbitrary $*$-subalgebra P. A constant odd $a \in P$ generally fails the endpoint conditions and is an ambient

multiplier, rather than an element of the parent algebra. More seriously, $E_0(P)$ need not equal $P \cap \{\Gamma_0\}'$.

Let $H = \mathbb{C}^2$, $\Gamma_0 = \sigma_z$, and let P be the unital $*$-algebra generated by $\sigma_x + \sigma_z$. Then

$$E_0(P) = \operatorname{span}\{1, \sigma_z\}, \qquad P \cap \{\Gamma_0\}' = \mathbb{C}1.$$

The first space is the compression image, while the second is the even part of P. Compactly supported scalar profiles show that the image of the actual boundary algebra is $E_0(P)$.

The correction is to assume $\Gamma_0 P \Gamma_0 = P$. Then $E_0(P) = P^{\mathrm{ev}}$, and the aligned-wall conclusion holds. Continue to distinguish constant ambient multipliers from parent-algebra elements.

2c. Item 4 — Correct. Under the stated J_1 hypothesis and $[P, P^\circ] = 0$, the parent has a real structure and satisfies order zero and order one, even when P contains elements odd for Γ_0. The parent grading is $\hat{\Gamma}$, which commutes with those elements. Pointwise,

$$[\pi(f), J\pi(g)^* J^{-1}] = 0,$$
$$[[D, \pi(f)], J\pi(g)^* J^{-1}] = E[f', g^\circ] = 0.$$

2d. Comparison with Result 11 — Correct with amendment. The aligned-wall construction evades hypothesis 4 of Result 11: odd coordinate matrices are sent to zero rather than retained as $\Gamma_a/5$. Result 11 assumes an even parent, an inherited involutive grading, and retention of all those coordinate compressions simultaneously. Its stated escapes include restricting to the even algebra or changing the grading construction. The present construction points toward a matrix-valued parent with an inherited grading; it leaves the commutative-parent and retained-coordinate questions open.

3. Remark 23.3 (remark)

3a. Opposite action — Correct with amendment. For an odd **anti-Hermitian** h, the displayed signs for the unstarred conjugations $J_1 h J_1^{-1}$ and JhJ^{-1} are correct. The opposite representation used in order zero is $b^\circ = J_1 b^* J_1^{-1}$; state the star and the anti-Hermiticity assumption when giving signs. On the W-factor it incorporates $\sigma(b) = \gamma_W b \gamma_W$.

3b. Grand-symmetry twist — Error in the assertion that it is the same automorphism. The relevant references are:

- Agostino Devastato, Fedele Lizzi and Pierre Martinetti, "Grand symmetry, spectral action and the Higgs mass," *JHEP* 01 (2014) 042. The larger algebra and its reduction appear in §§3–5.
- Agostino Devastato and Pierre Martinetti, "Twisted spectral triple for the Standard Model and spontaneous breaking of the Grand Symmetry," *Mathematical Physics, Analysis and Geometry* **20** (2017), article 2. Its Eqs. (1.8)–(1.11) define the twisted commutator, first-order condition and automorphism; the publisher record confirms the publication details.

The latter authors call their ρ the "exchange of the left/right spinorial indices." It exchanges algebra summands. Conjugation by γ_W is inner on $M_2(\mathbb{H})$ and fixes its centre. Thus the full automorphisms have different algebraic actions. The correction is to describe a **structural analogy** between involutive automorphisms and compare the actual representations, opposites and first-order conditions before claiming an identification.

4. Theorem 23.4 and Example 23.5 (fluctuations and example)

4a. Connection removal — Correct with amendment. The displayed unitary identity follows from $u' = -hu$. It describes the perturbation $D + Eh$. For a fluctuation of a real triple, include the conjugate one-form:

$$D + Eh + \widehat{J}(Eh)\widehat{J}^{-1}.$$

In the example, the resulting generator has the form (L_F, R_F), as required for the stated action on $E \oplus E^*$. This also preserves the J_1 condition. The odd directions lie outside No-go theorem 22.7, whose hypotheses include an even algebra and a moving window.

4b. Holonomy and double cosets — Error in the order of the factors. With the convention $u(0) = 1$, $u(l) = w$, let $H_i = \mathrm{Stab}(\Gamma_i)$. A gauge transformation removing h has the form $u(x)c$, so it preserves both endpoints precisely when

$$c \in H_0, \qquad wc \in H_1, \qquad \text{equivalently} \qquad w \in H_1 H_0.$$

The holonomy quotient is $H_1\backslash G/H_0$, or $H_0\backslash G/H_1$ after replacing w by w^{-1}. The order matters for the stated w. Also restrict G to the unitary component reachable by a smooth path from the identity.

4c. Orbit, angles and polar coordinates — Correct with amendment. The orbit of quaternionic self-adjoint involutions with one-dimensional quaternionic $+1$ and -1 eigenspaces is

$$\mathbb{H}P^1 \simeq Sp(2)/(Sp(1)\times Sp(1)) \simeq S^4.$$

The wording "each grading with quaternionic eigenspaces" also includes ± 1; exclude those two cases. For $\Gamma_p = p_a\Gamma_a$, the eigenvalues of $\Gamma_q\Gamma_p$ are $e^{\pm id(p,q)}$, so the principal angles are $d(p,q)/2$.

The formula on E^* is correct: p' is the reflection of p through p_0, because

$$\Gamma_{p'} = \gamma\Gamma_p\gamma.$$

The quaternion direction q supplies three angular coordinates away from the poles. It is undefined at both poles, and at $\theta = \pi/2$ the finite logarithm and lowest-window choice are degenerate. The unquotiented orbit is S^4; quotienting the relative orientation by the stabilizer leaves the radial interval.

4d. Bulk-mass texture — Correct with amendment. For $X = \tan\theta\, 1$, the light-mass approximation from Theorem 12.6 has the scale

$$M_{\text{light}} \simeq \frac{\tan\theta}{l}\,\Phi_Q\Phi_U$$

when the light eigenvalues are small compared with the relevant tower thresholds. In the convention of Theorem 12.7, $\widehat{Y} = X/l = (\tan\theta/l)1$; a dimensionless identity matrix can be stated after factoring out that common scale. Positive localization parameters suppress the wall profiles. General finite-angle eigenvalues require the secular equation.

4e. Minimal composite Higgs comparison — Correct with amendment. The coset and the four real Higgs coordinates agree with those in Agashe, Contino and Pomarol, "The Minimal Composite Higgs Model," *Nuclear Physics B* **719** (2005), 165–187, §§2–3. Their Eq.

(6) identifies h/f_π with the sphere angle, corresponding here to 2θ. Their construction uses a warped AdS interval and additional fermion and boundary data. The shared coset alone does not identify the actions or potentials.

5. Theorem 23.6 (boundary conditions and theorem)

5a. Connector at one wall — Correct with amendment. The stated gauge boundary conditions form a consistent boundary-value problem with the connector treated as prescribed wall data. A real fermion connector can also be placed at one wall. For example, choose a unit vector e in the $+1$ eigenspace of Γ_1, put $f = J_1 e$, and set

$$K_c = |f\rangle\langle e| - |e\rangle\langle f|, \qquad \Gamma_{1,c} = e^{\eta K_c}\Gamma_1 e^{-\eta K_c}.$$

Then K_c anticommutes with Γ_1, commutes with J_1, and J_1 anticommutes with $\Gamma_{1,c}$. Restricting the wall algebra to the common stabilizer gives the connector conditions and retains the parent order conditions.

The amendment is to specify whether the connector is prescribed gauge-breaking data or a term in the fermion boundary operator. In the latter case, $\Gamma_{1,c}$ changes the affected fermion tower; the universal fermion product later in the lemma must be recalculated for that channel.

5b. Charged and neutral masses — Correct. An independent solution uses principal angles between the Neumann subspaces at the two walls. Put $s = \sin(2\theta)$. For each charged direction the squared sine of the angle is $s^2/2$. In the neutral sector the two Neumann directions are L' and $\sqrt{3/5}\,R' + \sqrt{2/5}\,B$; their projection onto the Dirichlet complement at the first wall has one nonzero squared singular value $4s^2/5$. Consequently

$$\sin^2(M_W l) = \frac{s^2}{2}, \qquad \sin^2(M_Z l) = \frac{4s^2}{5},$$

which give the stated tangent equations. This derivation also covers $\cos(2\theta) = 0$, where the divisions in the written proof need a limiting interpretation.

5c. Massless modes — Correct with amendment. For $0 < \theta < \pi/2$, the intersection has dimension nine: the photon and eight gluons. At each endpoint it has dimension twelve, with three additional electroweak zero

modes. The phrase "massless for all θ" is correct; state the endpoint enhancement explicitly.

5d. Normalization — Correct. With the stated fundamental trace forms, equal five-dimensional coefficients, flat profiles and no boundary kinetic terms,

$$g_3 = g_2 = \sqrt{\frac{5}{3}}\, g_Y, \qquad \frac{1}{e^2} = \frac{8}{3g_2^2}.$$

Indeed the hypercharge generator has squared trace norm $1 + 2/3 = 5/3$, while the photon generator has squared norm $1 + 1 + 2/3 = 8/3$. Here g_2 is the normalized four-dimensional zero-mode coupling.

5e. Items 1 and 2 — Correct. Expanding the sine equations gives $M_W^2 \sim 2m^2$, $M_Z^2 \sim (16/5)m^2$ as $\theta \to 0$. At $\theta \to \pi/2$, both displayed branches tend to zero and their squared ratio tends to $5/8$. The ratio itself is undefined at the endpoints.

5f. Item 3 — Correct. Let $a = \sin(2\theta)$. Then

$$\frac{M_W^2}{M_Z^2} = \left[\frac{\arcsin(a/\sqrt{2})}{\arcsin(2a/\sqrt{5})}\right]^2.$$

For $0 < a \leq 1$, this decreases with a, since $x/(\sqrt{1-x^2}\arcsin x)$ increases. Hence the midpoint is the global minimum, with exact value

$$\left(\frac{\pi/4}{\arctan 2}\right)^2 = 0.5032315\ldots.$$

The upper value $5/8$ is approached at the endpoints. This supplies the proof missing from the grid check.

5g. Item 4 — Correct. The ratio m/M_W increases from $1/\sqrt{2}$ to infinity. Python 3 with SciPy gives $\theta = 1.05888846\ldots$ for $m/M_W = \sqrt{8/3}$.

6. Comparison with lemma 18 and Proposition F

6a. Constants 2 and 16/5 — Correct with amendment. These are the coefficients in the lower bounds for the fully massive finite-action pole masses in Proposition 14 of lemma 18. They are not the small-x pole coefficients there: those are 3 and 24/5. The common 5/8 limit is a valid comparison, with different limiting

behaviour: the finite-action poles grow in the lemma 18 heavy limit, while the interval vector masses tend to zero at the antipodal endpoint.

Specify the trace or Hilbert–Schmidt norm in the commutator-norm claim. With the ordinary operator norm, the charged example gives m^2, rather than $2m^2$.

6b. Factor 3/2 — Correct with amendment. Proposition F gives the factor 3/2 for the stated leading finite spectral-action mass matrix. Comparing it with the assumed five-dimensional Yang–Mills quadratic form gives $3m^2$ versus $2m^2$ in the universal small-mass example. A calculation of the full parent spectral action, with its tower and boundary conditions, remains necessary for a direct matching claim.

7. Proposition 23.7 (heat trace)

7a. Heat trace and monotonicity — Correct. With $s = l\Lambda$,

$$\mathrm{Tr}_{\mathrm{channel}}\, e^{-D^2/\Lambda^2} = \frac{2s}{\sqrt{\pi}}\,\vartheta_3(\theta, e^{-s^2}).$$

The Jacobi product proves strict decrease on the interval, with zero derivative at the endpoints. At fixed $x = \theta^2/s^2$, its towerless limit is $2e^{-x}$. At $\theta = \pi/2$, the full lowest eigenspace is degenerate even though the heat trace is well defined.

7b. Escape from No-go theorem 9 — Correct with amendment. No-go theorem 9 assumes finite internal data, the positive-Laplace cutoff class, the literal flat-space spectral action, $A = 0$, and an allowed unconstrained ray $V = cV_*$, $c > 0$, without an added scalar potential. Its conclusion is strict decrease along that ray. The interval family changes the operator family and replaces the free ray by a compact relative-grading orbit with a tower. It therefore lies outside those hypotheses.

For the stated heat action, the antipodal point is a minimum of this family. Added potentials, constraints, other cutoff classes and quantum effects remain separate possibilities. The result points toward analysing the full tower and any additional scalar terms.

8. One-loop potential (products and potential)

8a. Three products — Correct. The product for a principal angle α, normalized at $\alpha = 0$, is

$$1 + \frac{\sin^2 \alpha}{\sinh^2 x}.$$

Using the fermion angle θ and the two gauge angles from 5b gives all three displayed products.

8b. Fermion coefficient — Correct with amendment. With the conventional four-dimensional Dirac determinant and each physical species counted once, $n_f = 2N_D$; one colour-triplet Dirac top contributes six. The universal example gives masses to both members of each weak doublet. For one complete generation it contains eight physical Dirac species, hence $n_f = 16$, before any connector modification; three identical generations give $n_f = 48$. A top-only $n_f = 6$ potential therefore needs a separately specified fermion sector or a complete accounting of spectator channels.

8c. Potential and endpoint curvatures — Correct with amendment. The logarithmic integrand and the vector multiplicities are correct under the stated one-loop assumptions. The curvature sentence omits the prefactor on its left-hand side. Let

$$I = \int_0^\infty \frac{x^3\,dx}{\sinh^2 x} = \frac{3}{2}\zeta(3), \qquad J = \int_0^\infty \frac{x^3\,dx}{\cosh^2 x} = \frac{9}{8}\zeta(3).$$

The precise expansions are

$$8\pi^2 l^4[V(\theta) - V(0)] = (10.8 - n_f)I\,\theta^2 + o(\theta^2),$$

and, with $\epsilon = \pi/2 - \theta$,

$$8\pi^2 l^4[V(\tfrac{\pi}{2} - \epsilon) - V(\tfrac{\pi}{2})] = (10.8I + n_f J)\epsilon^2 + o(\epsilon^2).$$

For $n_f \geq 0$, the latter curvature is positive.

8d. Numerical absence of an interior minimum — Gap in any conclusion beyond the sampled grid. The recorded computation is explicitly a grid statement, and its original code was not retained. An independent Python 3/SciPy calculation of the derivative found one sign change for each tested $n_f = 1, 2, 3, 4, 6, 8$; for $n_f = 8$, the stationary angle is approximately 0.3382. The known integral $I = 3\zeta(3)/2$ agreed to 5×10^{-16} in absolute error,

and a 1% perturbation failed the same check. A grid does not exclude additional stationary points between samples, so the claim about absence of an interior **local** minimum for all parameters in the displayed massless family remains open in the range $0 < n_f < 10.8$.

Two stronger statements follow analytically from the displayed potential:

- For every $n_f > 0$, $\theta = \pi/2$ is the unique global minimum on $[0, \pi/2]$. The gauge logarithms are nonnegative, and the negative fermion logarithm is minimized when $\sin^2\theta = 1$.
- For $n_f \geq 10.8$, the potential is strictly decreasing throughout the interior. To see this, let $t = \sin^2\theta$ and
$$\mathcal{J}(z) = \int_0^\infty \frac{x^3\,dx}{\sinh^2 x + z}.$$
The derivative, apart from the positive overall factor, is
$$(1-2t)\Big[6\mathcal{J}(2t(1-t)) + \frac{24}{5}\mathcal{J}\Big(\frac{16}{5}t(1-t)\Big)\Big] - n_f\mathcal{J}(t).$$
For $t \leq 1/2$, both arguments of $\mathcal{J}$ in the bracket are at least t, and $\mathcal{J}$ decreases. For $t > 1/2$, both contributions have the required sign.

These conclusions concern the displayed massless flat products, the stated gauge content and boundary conditions, and one-loop order. They direct attention to changed form factors or additional terms for an interior vacuum.

8e. Leading-logarithm form — Correct with amendment. The formal first term is $B\sin^2(2\theta) - A\sin^2\theta$, with $A, B > 0$, and any interior stationary point of that expression is a maximum. The logarithm expansion is not uniform at $x = 0$; the next term has an infrared-divergent integral. Present the expression as a formal leading approximation and use the full integral for conclusions about its global shape.

8f. Bulk-mass scan — Error as a finite one-loop potential for general scanned masses. Let $a = c_Q + c_U$. The stated fermion factor satisfies
$$G(x) = 1 - \frac{a}{x} + \frac{a^2}{2x^2} + O(x^{-3}),$$

so, for $t = \sin^2\theta$,

$$\log(1 + t(G-1)) = -\frac{ta}{x} + \frac{t(1-t)a^2}{2x^2} + O(x^{-3}).$$

Multiplication by x^3 gives angle-dependent ultraviolet divergences whenever $a \neq 0$. A finite quadrature value then depends on a regulator and subtraction prescription; the reported scan does not establish a regulator-independent potential.

The exceptional case $c_U = -c_Q$ has a common bulk mass on the two mixed components and

$$G(x) = 1 + \frac{w^2}{x^2 \sinh^2 w},$$

which has the required exponential decay. Keeping the flat gauge products is consistent as a **separate tree-level gauge-sector assumption**. For a coupled $Sp(2)$ gauge theory, unequal masses on components mixed by the gauge group require gauge-invariant completion or explicit breaking data. The correction is to specify that completion, the regulator and allowed wall counterterms, then recompute the potential.

8g. Escapes and spinor comparison — Correct with amendment. A frequency-two fermion contribution, modified gauge form factors, wall potentials and wall terms are possible changes to the stated flat calculation; each requires its own model and coefficients. For the comparison with Agashe–Contino–Pomarol, their §2.1, Eqs. (19)–(23), uses spinorial $SO(5)$ fermions and permits an approximate potential

$$\alpha \cos(h/f_\pi) - \beta \sin^2(h/f_\pi).$$

With $h/f_\pi = 2\theta$, the displayed leading flat expression has this angular form with $\alpha = A/2$ and $\beta = -B$, up to a constant. Their fermion form factors can also give the sign and competition needed for an interior minimum. The flat scan therefore has its stated scope; it does not exclude the spinorial composite-Higgs construction.

9. Measured values and open questions (comparison)

9a. Numerical comparison — Correct with amendment. Using the 2026 Particle Data Group values $m_t = 172.60$ GeV, $M_W = 80.3625$ GeV and $M_Z =$

91.1879 GeV gives $m_t/M_W \simeq 2.1478$ and $M_W^2/M_Z^2 \simeq 0.7767$. The top entry is a direct-measurement mass and its definition should be kept distinct from a specified pole-mass scheme. PDG quark table, p. 2; PDG gauge-boson table, pp. 1–2.

Python 3 with SciPy gives $\theta \simeq 1.1626$, $M_W l \simeq 0.5413$, $1/l \simeq 148.5$ GeV and the model ratio $M_W^2/M_Z^2 \simeq 0.5814$. For the separate wall-scalar model of lemma 22, fitting the measured vector ratio gives $M_W l \simeq 1.0282$; its lowest no-bulk-mass fermion channel then has $m/M_W < 1.528$. Thus the stated mismatch holds for those flat, tree-level families with the specified lowest-level and coupling assumptions. Bulk masses, warping, running and a combined wall/Wilson-line model remain directions to examine.

9b. Open questions — Correct with amendment. The listed questions remain appropriate. Add the real connector's contribution to the fermion tower, a gauge-invariant and renormalized bulk-mass potential, the degenerate antipodal window, and the scalar kinetic normalization needed to identify a canonically normalized Higgs field.

Amendments and errors

- Assume $\Gamma_0 P \Gamma_0 = P$ for the aligned-wall even-algebra claim, and distinguish ambient constant multipliers from elements of the boundary algebra.
- Replace the claimed equality with the grand-symmetry twist by a comparison of the involutions and their first-order conditions.
- Reverse the holonomy factors for the convention used: $w \in H_1 H_0$, with quotient $H_1 \backslash G / H_0$.
- Specify the real fluctuation, the connector's role, the altered fermion tower, endpoint degeneracies and trace norms.
- Add the missing $1/l$ scale and light-mode approximation to the bulk-mass texture; add $1/(8\pi^2 l^4)$ to the potential curvatures.
- Restrict the unproved absence of local minima to the sampled grid, and replace the generic bulk-mass scan by a calculation with a stated regulator, counterterms and gauge completion.

Overall: 11 Correct; 19 Correct with amendment; 1 Gap;

Part R10 (referee report, lemma 25)

Referee report on lemma 25

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_25.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 25 and the note.

The leading constant-profile mass formula is supported after the amendments below. The principal mathematical error is the extra factor of s in the displayed heat-trace derivative. The identification with a physical lowest gauge mode requires additional boundary and spectral assumptions.

The calculations in this report are analytical; no numerical code was run. Numerical values below are taken from the cited records.

1. **Proposition 25.1 — Correct with amendment.**

   The unfolding in lemma 25 is correct after multiplying it by $1/\sqrt{2}$. It maps the H^1 domain of D_0 onto the periodic H^1 domain on the circle. With

   $$D_F = \begin{pmatrix} 0 & M^* \\ M & 0 \end{pmatrix},$$

   the connection term satisfies

   $$E\Gamma_F D_F : \quad \Psi_Q \mapsto M^*\Psi_U, \qquad \Psi_U \mapsto M\Psi_Q.$$

   Thus the unfolded operator is $-i\partial_x + D_F$, with signed spectrum $n\pi/l + \mu_i$.

   The amendment is to name the frame. Corollary 22.2 states

   $$D_X = R(D_0 + E\Gamma_F D_F)R^*, \qquad R(x) = e^{x\Gamma_F D_F}.$$

   An even T that is constant **after transport** unfolds to $1 \otimes T$. In the original frame it is $R(x)TR(x)^*$, which generally varies with x. Conversely, a constant original-frame T becomes R^*TR

after transport and generally does not unfold to $1 \otimes T$. For broken T, it also need not preserve the original coupled wall domain. The transported field generally has an odd bulk component and therefore requires an admissible enlarged gauge algebra; it is not automatically a field of the even algebra in lemma 22.

There is a separate error in the holonomy sentence at line 31. With the stated convention, put $U = e^{-i\pi\Gamma_0/4}$. The correct relation is

$$\Gamma_1\Gamma_0 = e^{2l\Gamma_0 D_F} = Ue^{2ilD_F}U^*.$$

Conjugation by Γ_0 instead sends e^{2ilD_F} to its inverse. The circle holonomy can acquire the inverse sign under the opposite parallel-transport convention.

2. **Theorem 25.2 — Correct with amendment.**

The exact kernel at line 39 and its quadratic coefficient have the stated signs. Lemma 18 states its theorem for a finite internal trace; an extension to this infinite tower must be supplied. Here it is available: V and T split into finite-dimensional blocks indexed by n. Applying the finite theorem to a symmetric finite set of blocks and passing to the limit is justified by Gaussian bounds, uniform for D_F in a bounded neighborhood of zero. The same bounds justify differentiation and Taylor expansion under the sum.

Let $t = 1$, $d = \mu_i - \mu_j$, $b = u(1-u)$, and

$$E(u) = e^{-[(1-u)(k+\mu_i)^2 + u(k+\mu_j)^2]}.$$

An exact integration-by-parts form of the $q = 0$ bracket is

$$d^2 \int_0^1 E(u)\left[1 - b(2k + \mu_i + \mu_j)^2\right] du.$$

Its expansion begins

$$d^2 e^{-k^2}\left[1 - \frac{2}{3}k^2 + \frac{2k^3 - 5k}{3}(\mu_i + \mu_j) + O(\|\mu\|^2)\right].$$

The displayed cubic term is odd in k and cancels between n and $-n$. Similarly, the q^2 bracket begins

$$e^{-k^2}\left[\frac{2}{3}-\frac{2k}{3}(\mu_i+\mu_j)+O(\|\mu\|^2)\right],$$

and its linear term cancels. The grading symmetry gives the same evenness for the summed expression.

The proof should include these cancellations and remainder terms. At fixed $s=l\Lambda$, for fixed nonzero Hermitian T, the stated $O(\|D_F\|^3/\Lambda)$ mass remainder is valid but can be sharpened to

$$M^2_{\text{Taylor}}=\left(\frac{3}{2}-\langle k^2\rangle\right)\frac{\|[D_F,T]\|^2_{\text{HS}}}{\|T\|^2_{\text{HS}}}+O\left(\frac{\|D_F\|^4}{\Lambda^2}\right).$$

Here $k_n=n\pi/(l\Lambda)$ is dimensionless. Equations presented later without a remainder must be labeled leading-order formulas.

3. **Spectral dimension — Error.**

   The factor at line 60 is wrong. Direct differentiation gives

   $$\boxed{\langle k^2\rangle=\frac{1}{2}\frac{d\log S}{d\log s}=\frac{s}{2}\frac{d\log S}{ds}.}$$

   The extra s multiplying the logarithmic derivative must be deleted. The subsequent coefficient $(3-d_{\text{eff}})/2$ uses the corrected identity.

   The spectral-dimension interpretation is correct for the **unshifted unfolded circle**, equivalently the paired Neumann-plus-Dirichlet interval trace. It should not be called the spectral dimension of an arbitrary interval boundary problem. Indeed, their separate heat traces are

   $$S_N=\frac{S+1}{2},\qquad S_D=\frac{S-1}{2},$$

   and have different logarithmic derivatives. For nonzero D_F, the actual parent heat trace contains $e^{-t(k_n+\mu_i)^2}$; S is the unperturbed trace used in the leading expansion.

4. **The two limits and covariance — Correct with amendment.**

The limiting factors 3/2 and 1 are correct. The exponential corrections have polynomial prefactors. More precisely,

$$d_{\text{eff}}(s) \sim \frac{4\pi^2}{s^2} e^{-\pi^2/s^2} \quad (s \to 0),$$
$$d_{\text{eff}}(s) = 1 - 4s^2 e^{-s^2} + O(s^2 e^{-2s^2}) \quad (s \to \infty).$$

State the order of limits: first take the small-D_F/Λ coefficient at fixed s, then take either limit.

The five-dimensional covariance argument is valid on the unfolded circle for this operator and these insertions. With Hermitian connections the curvature convention gives $F_{5\mu} = i[D_F, A_\mu]$, up to the sign convention for the covariant derivative; its norm gives the quoted commutator quotient.

On a general space with boundary, the bulk local coefficient alone does not rule out boundary contributions. In the present calculation the full operator with the specified transported fields is unitarily equivalent to the circle problem, so its trace has the required cancellation. State that specific reason rather than a general boundary assertion.

5. **Corollary 25.3 — Correct with amendment.**

The coefficients agree with Remark 16 of lemma 18 and with the CCM mass relation, Eq. (4.16) at leading order. With the literal particle-and-antiparticle trace,

$$Q_{11} = 12, \qquad K_{11} = \sum_f N_f m_f^2,$$

which gives $(3 - d_{\text{eff}})/24$. For the top alone, $K_{11} = 3m_t^2$, giving the displayed ratio and

$$y_t = \frac{2g}{\sqrt{3}} \quad (d_{\text{eff}} = 0), \qquad y_t = \sqrt{2}\, g \quad (d_{\text{eff}} = 1).$$

The first is also CCM's top-dominance approximation, Eq. (5.27).

The neutral result $M_W^2/M_Z^2 = 5/8$ follows **to this order**, with the common trace and normalization. "Massless directions are unaffected" should say that

their mass coefficient remains zero; their kinetic coefficients can change.

The twelve doublets have the same tower factor S in the stipulated flat, bulk-mass-free model, including when only the top has nonzero Dirac mass. With bulk masses used to localize light fields, their spectra and overlap matrix elements change. The kinetic coefficient must then be recomputed from all eigenstates and overlaps; it cannot generally be obtained by assigning the same S to every doublet. Light zero modes can still contribute even when their excited modes are suppressed. For an explicit localization mechanism and profiles, see Adachi, Kurahashi and Maru, arXiv:1404.4281v1, Eqs. (2.10)–(2.11).

6. **General cutoffs — Correct with amendment.**

The weighted formula is correct for the positive Laplace class and moment hypotheses in lemma 18, Eq. (4). It is a leading small-D_F formula. A precise version is

$$S_\nu(s) = \int S(s/\sqrt{\tau})\, d\nu(\tau),$$
$$d_\nu = \frac{d\log S_\nu}{d\log s} = 2\langle \tau k^2 \rangle_\nu,$$

so that the coefficient is $(3 - d_\nu)/2$. Under those moment assumptions, dominated convergence gives the stated endpoints. The crossover and its corrections depend on ν.

Replace "general cutoffs" by "positive Laplace cutoffs satisfying (4)." A general smooth spectral cutoff, including a nonconstant cutoff flat near zero, need not belong to that class.

7. **Scope and the lowest mode — Gap.**

The small-angle comparison with Theorem 23.6 is correct. For its *universal* fermion mass $m = \theta/l$, the $d_{\text{eff}} = 1$ formula gives

$$M_W^2 = 2m^2, \qquad M_Z^2 = \frac{16}{5} m^2$$

at leading order. Expanding the exact Yang–Mills

formulas gives, respectively,

$$2m^2 - \frac{4}{3}m^4 l^2 + \cdots, \qquad \frac{16}{5}m^2 - \frac{64}{75}m^4 l^2 + \cdots.$$

Lemma 25 directly computes a Taylor quotient for a specified constant profile. For the unrestricted unfolded circle, translation invariance separates its zero Fourier sector from nonzero Fourier sectors; one must still diagonalize the generator matrix within that sector. Calling the result the first nonzero term of a **lowest-mode mass** also requires an admissible gauge field, a positive kinetic form and an isolated lowest branch. Boundary constraints or additional terms can alter the profile problem. Theorem 23.6 establishes the comparison for its own Yang–Mills boundary problem; it does not establish the full finite-cutoff spectral-action gauge spectrum.

Add these hypotheses to Scope, together with the transported-frame qualification, the distinction between Taylor and pole masses, and the fact that the calculation is about a prescribed background rather than a demonstrated vacuum. "First order in D_F" refers to the mass; the displayed mass **squared** starts at quadratic order.

8. **Reading — Correct with amendment.**

The factor 3/2 in Proposition F is the finite-action coefficient relative to the Yang–Mills commutator quotient in the stated trace. The crossover explains that coefficient's two limits in the present constant-profile calculation. Proposition F also contains a bounded-fermion versus tower-scale obstruction and overlap conditions; those do not follow solely from the factor 3/2. The reading should retain their separate hypotheses.

The direct-measurement figure uses the PDG 2026 top listing, p. 1, 172.60 ± 0.27 GeV, and the PDG 2026 *W* listing, 80.3625 ± 0.0077 GeV. The top value is a direct-measurement Monte Carlo mass, with a scheme interpretation discussed by PDG.

The remembered 163 GeV is supportable with a scheme and scale: Yoon and Peskin,

arXiv:1810.12352v3, §3.3 and Eq. (23) use approximately $\overline{m}_t(\overline{m}_t) = 163$ GeV for a 173 GeV pole mass. They also quote about 147 GeV at 2 TeV. Dividing either running mass by an on-shell W mass is only a formal orientation, rather than a common-scale matching test.

The proposed matching scale of order $30M_{\mathrm{KK}}$ and its running are outside lemma 25. The cited steer note labels its calculation unrefereed, uses remembered tower coefficients, and reports that the three couplings do not meet at one point. Retain the figure only as a conditional estimate from that separate model.

9. **Literature — Gap.**

The endpoint $m_t = 2M_W$ is already present in flat five-dimensional gauge–Higgs literature. Cacciapaglia, Csáki and Park, *Fully Radiative Electroweak Symmetry Breaking*, arXiv:hep-ph/0510366v2, §5 explicitly state the tree-level relation for a suitable rank-four representation and discuss the assumptions on the top zero mode. Adachi and Maru, arXiv:1804.06012v2, Eq. (3.23) obtain a limit approaching $2M_W$ with their specified brane mixings. These are representation-dependent mechanisms; they do not establish the lemma's spectral-action crossover.

Ordinary universal-extra-dimension models need separate specification. For example, Cordero-Cid et al., arXiv:1108.2926v4, §2 formulate the five-dimensional Standard Model with separate Higgs and Yukawa sectors; universal propagation alone does not impose $m_t = 2M_W$.

The inspected primary records do not exhibit the particular $(3 - d_{\mathrm{eff}})/24$ coefficient or its theta-function crossover. We cannot claim novelty on that basis. The search used web-index queries for spectral action, heat kernels, compactification and gauge–Higgs mass relations, including a 2026 query; it followed the 2014 Adachi–Kurahashi–Maru reference to the 2006 paper. A systematic forward-citation search, author recency sweep and broader database search remain open.

10. **Paper's final sections — Error.**

    The sum-rule section of boundary_condition_note.tex carries the leading result into an unqualified displayed equality. Add the small-mass remainder and the transported-frame, trace and constant-profile assumptions. In particular, the towerless top-only Taylor expansion already contains the nonzero correction

$$M_W^2 = \frac{3}{8} m_t^2 - \frac{23}{128} \frac{m_t^4}{\Lambda^2} + \cdots$$

    in lemma 18, Eq. (19). Thus the leading equality must not be stated as an exact finite-mass result. Apply the same qualification to the abstract and to "throughout" for the neutral ratio.

    The statement that running "would be a power law" also requires a specified effective theory, thresholds and renormalization prescription. The subsequent **Status** section contains separate claims based on lemma 24 and its literature search; those lie outside lemma 25 and were not audited here. Provenance should be updated if this report is recorded.

Amendments and errors

- Correct the factor in the heat-trace derivative and identify the unperturbed paired interval or circle trace.
- Name the transported frame for T, correct the holonomy conjugation, and state the admissible gauge algebra and wall conditions.
- Add the finite-block convergence argument, odd-k cancellations and leading-order remainders.
- Qualify the covariance, cutoff, neutral-ratio and lowest-mode statements by their hypotheses.
- Cite the top-mass schemes and prior gauge–Higgs literature; treat the crossover's priority as open pending a wider search.
- Amend the paper and abstract to state leading-order results, and keep the separate running and Status claims within their own scopes.

Overall: 0 Correct; 6 Correct with amendment; 2 Gap; 2 Error.

Part R11 (referee report, lemma 24)

Referee report on lemma 24

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_24.md. Reproduced as delivered, with links made relative. The amendments are collected at the end of lemma 24.

The main operator-window construction and the new one-loop argument are correct in their stated families. The principal gap is the statement of convergence of compression maps for general observables as the interval shrinks: the varying source spaces need an identification, and different interval profiles have different limits.

This report gives one verdict for each of the ten requested items. The checks are analytic; no numerical computation was used.

1. **Result 1 — Correct.** In Result 1, $C_i^2 = \Pi$, $(\operatorname{sgn} A)^2 = 1-\Pi$, and the cross terms vanish. Thus both χ_i are self-adjoint involutions and anticommute with E. They commute with $\hat{\gamma}$, so the boundary domain is preserved by the grading.

   Let $B = SA/r$. At 0, the complementary boundary condition selects $B < 0$; at ℓ, it selects $B > 0$. The reversal makes the boundary term in (3) nonnegative and gives (2). The finite kernel conditions are maximal isotropic and give the interval operator of lemma 23.

   The H^1 realization is correct. Decomposition into eigenspaces of A gives self-adjoint finite-dimensional interval problems. On their direct sum, (3) controls both q' and Aq; ellipticity of A identifies the graph domain with the stated H^1 domain. Rellich compactness then gives compact resolvent. The projection Π reduces both the expression and the domain.

   Differentiation verifies (4). The principal-logarithm hypothesis gives $\ell\|D_F\| < \pi/2$, and the interval spectrum together with (2) gives (5). Thus, when $\Delta_K/r > \|D_F\|$, the displayed embedding is the isolated finite window.

2. **Result 2 — Correct with amendment.** With the endpoint convention made explicit, condition (6) gives the claimed local self-adjoint elliptic realization. Write G_0 at $x = 0$ and G_ℓ at $x = \ell$, or write G_i at $x = i\ell$ for $i \in \{0, 1\}$. The present "at $x = i$" leaves the index convention implicit.

   For $\chi = ZG$, the adapted tangential principal symbol exchanges the two χ-eigenspaces because $[G, \sigma_A] = 0$. The normal symbol exchanges them as well. Bär–Ballmann, Corollary 4.18 and Theorem 4.11 give the relevant ellipticity and self-adjointness criteria. Their Example 4.20 assumes anticommutation with the full adapted operator; (6) supplies the principal-symbol condition needed for Corollary 4.18.

   Equation (7) is correct for the stated maximal smooth scalar $*$**-algebra**. Preservation by both f and $\bar{f}$ makes the boundary subspace reducing, hence gives commutation with χ_i. Commutation with $C_i^2 = \Pi$ then forces multiplication by f_i to preserve $H^0(\mathbb{CP}^3, \mathcal{O}(1))$. Those sections have no common zero, so f_i is holomorphic and therefore constant.

   The local hypothesis is restrictive. On the unaugmented Dolbeault spinor bundle, the six-dimensional complex Clifford representation is irreducible. A fibre endomorphism commuting with every $\sigma_A(y, \xi)$ is scalar; a smooth self-adjoint involution of this form is consequently $+1$ or -1 on connected K. Additional fibres and other local elliptic boundary projectors are directions outside this sufficient criterion.

3. **Result 3 — Correct with amendment.** Equations (9)–(11) follow as stated. Equation (9) proves that $g^2 = 1$ is equivalent to $[G, \Pi] = 0$. Scalar multiplication commutes with G, giving (10); the complex-linear commutant of the generated $M_4(\mathbb{C})$, or of $M_2(\mathbb{H})$ acting on $\mathbb{C}^4$, is $\mathbb{C}1$. Hence the self-adjoint involution g is ± 1.

   The auxiliary example also checks: $Q_K(y_a) = \Gamma_a^{\text{section}}/5$, and the Clifford traces give $\operatorname{tr} g^2 = 16/5$. Applying (9) and taking the largest eigenvalue gives

the displayed $2/\sqrt{5}$ lower bound on boundary-trace mixing.

Amend the marked-point sentence at lines 236–244 to say that the **natural field displayed there** supplies no separately marked point. A local field can encode a chosen point: with the auxiliary bundle, $G_p(y) = p_a \Gamma_a^{\mathrm{aux}}$ is a local constant field whose coefficients contain p. The restriction (11) concerns the unchanged four-section window and an actual involution obtained by compression. A coupled window, extra fibres, or spectral walls remain directions to examine.

4. **Result 4 — Correct with amendment.** Inserting W_ℓ into the scalar multiplier gives (12), including its independence of r. For real S^4 functions, (15) follows from quaternionic Hermiticity and the six independent matrices $1, \Gamma_a$. When the walls are aligned,

$$\mathcal{E}_p(\Gamma_a) = p_a \Gamma_p,$$

so direct scalar compression gives (13). Generating $M_2(\mathbb{H})$ first and then taking its Γ_p-centralizer gives (14).

For (16), state explicitly that the two Hom spaces and the two product-factor scalar algebras of Example 16.9 have been supplied. The V-side left action on the second Hom space commutes with its right grading, giving

$$\mathbb{R} \oplus \mathbb{R} \oplus M_4(\mathbb{C}).$$

The endpoint argument proves equality of the **finite-dimensional linear images**: interior-supported scalar functions separate the same matrix-entry densities. It does not by itself identify the parent algebras, their norms, gauge groups, or one-forms. The restriction $\mathbb{R} \oplus \mathbb{R}$ applies to real S^4 scalars, aligned walls, and direct compression. Intermediate algebra generation and other windows remain available constructions.

5. **Result 5 — Gap.** Equations (17)–(20) are correct under their displayed hypotheses. In particular, the complementary resolvent in (17) has norm at most

r/Δ_K; at fixed D_F, the interval tower also decouples as $\ell \to 0$. The Schur estimates (18) hold with the stated positive g_r. Equation (19) holds for an x-independent ambient multiplier, and (20) follows from the even part of the conjugation series.

The final assertion about convergence of **the compression maps** needs a common domain and a specified collapse map. For example, take aligned walls, $D_F = 0$, and the admissible scalar functions

$$f_\ell(y, x) = 6(x/\ell)(1 - x/\ell).$$

Both endpoint values are zero, but $\Phi_\ell(f_\ell) = 1$. Thus a general family of observables cannot be replaced by its value at the collapsing endpoint.

One exact correction is to fix $F(y, s)$ on $K \times [0, 1]$, set $f_\ell(y, x) = F(y, x/\ell)$, and state

$$\Phi_\ell(f_\ell) \longrightarrow \mathcal{E}_0 Q_K \left(\int_0^1 F(\,\cdot\,, s)\, ds \right).$$

For x-independent ambient multipliers, the limit is $\mathcal{E}_0 Q_K$, as follows from (19). Nonconstant x-independent functions on K are ambient multipliers; they do not themselves obey the spectral-wall endpoint restriction. Result 4's equality of finite images supplies representatives with the required endpoint values.

The rate discussion after (18) also needs a proof or a strengthened assumption: the two displayed rv_r conditions alone do not control the denominator g_r. A simple sufficient addition is

$$r(b_r + L) \le (1 - \varepsilon)\Delta_K$$

for some fixed $\varepsilon > 0$. The stated $b_r = O(1/\ell)$ case can instead be justified from the stronger complementary estimate

$$|\mathcal{D}_{r,Q}| \ge \left(\frac{\Delta_K^2}{r^2} + \frac{\pi^2}{4\ell^2} \right)^{1/2},$$

obtained by applying the one-ended Poincaré inequality to the positive and negative B-spectral components. Together with $g_r > 0$ and fixed L, this supplies a complementary gap of order $1/r$.

6. **Result 6 — Correct.** The table in Result 6 separates the chosen data from the algebraic consequences. The proposed $\widehat{J} = E\mathcal{J}_1$ has the stated signs and induces $J_F = \Gamma_0 J_1$ when all the displayed commutation and anticommutation hypotheses hold. Those are sufficient conditions; the note does not construct the required tangential extension or establish its order-zero property.

   The quaternionic j commutes with the Γ_a, so j alone fails the required anticommutation with Γ_p. At the finite level, if the two gradings are Γ_p, Γ_q, a unit $n \perp p, q$ gives $J_1 = \Gamma_n j$ with the required square and anticommutation. Extending that choice to the stated parent remains a separate task.

   The connector calculation gives the displayed algebra after choosing the two complex lines and nonzero coupling. A marked quaternionic line supplies neither of those complex lines by itself.

7. **Result 7 — Correct.** For No-go theorem 22.7, the endpoint-preserving unitary $v = u\,u(\ell)^*$ implements the fluctuation and preserves the original domain. The corresponding real unitary works under the stated order-zero condition. Consequently the moving spectral window is unitarily equivalent to the original one, with the same singular values. The displayed fixed-compression example also has the stated sign and can change singular values at second order.

   The conclusion concerns one interval, fixed wall coupling X, zero bulk mass, scalar inner fluctuations from the even algebra, and the moving spectral window. It is an exact spectral statement without an action or loop expansion. Variable wall coupling, odd connection directions, suitable bulk masses, further internal connection data, and additional fields lie outside those hypotheses. The result directs the search for a radial variable toward those constructions; it does not decide the full fluctuation space of the $K \times I_\ell$ parent.

8. **Lemma 8 and Theorem 9 — Correct with amendment.** The proof of Lemma 8 checks. The

displayed formula for b'' is positive for $x > 0$, since $\sinh x \cosh x > x$. Log-sum-exp is convex and increasing in each argument, so the integrand in (21) is jointly log-concave. Its integral is finite for every u, and Prékopa's Theorem 6, p. 342 applies. Differentiation under the integral gives $0 < e < 1$ and nondecreasing e.

There is also a separate analytic check of the key monotonicity, using a **covariance argument**. Let

$$\rho(s) = \frac{(\operatorname{arsinh}\sqrt{s})^3}{2\sqrt{s(1+s)}}, \qquad L(v) = \log \rho(e^v).$$

The substitution $s = \sinh^2 x$ writes $\mathcal{J}(z) = \int_0^\infty \rho(s)/(s+z)\, ds$. For $v = 2\log\sinh x$,

$$L'(v) = \frac{3\tanh x}{2x} - \frac{1+\tanh^2 x}{2},$$

which is strictly decreasing: $x\operatorname{sech}^2 x - \tanh x < 0$. With $z = e^u$, let the probability density in y be proportional to $e^y \rho(e^{u+y})/(1+e^y)$. Then

$$e(e^u) = \mathbb{E}\left[\frac{1}{1+e^y}\right],$$
$$\frac{d}{du} e(e^u) = \operatorname{Cov}\left(\frac{1}{1+e^y}, L'(u+y)\right) > 0.$$

Both functions in the covariance are strictly decreasing. This check uses the same integral but does not use the log-concavity or Prékopa argument.

For Theorem 9, the differential inequality for R_c follows from $a = ct(1-t) \geq t$, $e(a) \geq e(t)$, and $e(t) < 1$. Dominated convergence gives

$$H(0+) = 54/5, \qquad H(1/2-) = 0.$$

Hence H is strictly decreasing and all three conclusions follow. The independent covariance check above verifies the essential monotonicity; the final classification uses the derivative calculation (23).

Amend the theorem to state $\theta \in [0, \pi/2]$ explicitly. At $\theta = \pi/2$, the determinant has the stated continuous endpoint value, while the isolated principal-logarithm window is degenerate. The no-interior-minimum conclusion assumes the displayed flat,

massless, one-loop determinant, its gauge multiplicities and connector wall condition, frequency-one fermions, no fermion-connector modification, and no additional angle-dependent term. Modified form factors, fermion sectors, bulk masses, wall terms, and the uncomputed K-tower effective action remain directions for an interior vacuum.

9. **Result 10 — Correct with amendment.** For real finite opposite masses, identity (17) of this result's discussion—namely equation (10)'s displayed G-identity at lines 658–663—makes the fermion contribution strictly decreasing in t. With the separately assumed flat gauge terms and $n_f > 0$, $t = 1$ is the unique global minimum.

   For $a = c_Q + c_U \neq 0$, the displayed expansion of G and of the logarithm is correct. State that the divergent contribution has nonzero fermion multiplicity and is considered without cancellation by additional sectors. A regulator, subtraction and specified physical completion are needed to draw a conclusion from the general-mass potential.

   The opposite-mass Python scan in lemma 23 covers the listed finite sets of masses and multiplicities and 721 sampled angles. It gives a numerical result for those samples; it does not exclude a local minimum between samples or at unsampled parameters. General masses with a specified renormalized completion and changed gauge form factors remain open directions.

10. **Consequence for T0 and T0b, and the note — Correct with amendment.** The consequence keeps the central questions open: the construction supplies an operator and grading window with prescribed spectral walls, while the coupled local scalar window, compatible reality, second sheet and connector require further data. The no-go statements do not close either T0 or T0b.

In the note, replace "higher modes" in the radius-collapse sentence by **higher K-modes**. At fixed ℓ, the interval tower remains; its decoupling uses the subsequent $\ell \to 0$ limit. Also identify the exact window there as the operator-and-grading window, since the required parent

reality structure is conditional.

At the one-loop statement, state $0 < n_f < 54/5$ explicitly for the unique interior maximum and retain the determinant assumptions listed in item 8. When this review is recorded in the repository, update the provenance sentence at line 185 to identify the review and Astra 6's one-loop proof.

Amendments and errors

- Make the local-wall endpoint indices explicit and distinguish Bär–Ballmann's principal-symbol criterion from its stronger full-operator example.
- Qualify the marked-point statement to apply to the displayed natural field.
- State the product-factor inputs for (16) and the scope of equality of compression images.
- Define the source identification and averaging map for interval-collapse convergence; add a denominator estimate to the mixing-rate proof.
- State the angle interval, endpoint degeneracy, and complete determinant hypotheses for Theorem 9 and the paper note.
- Scope the bulk-mass scan to its sampled parameters and state the nonzero-multiplicity assumption for the ultraviolet divergence.
- Update the note's radius-collapse and provenance wording when the review is recorded.

No error was established in the displayed equations. Item 5 has the stated gap in the general compression-map conclusion.

Overall: 3 Correct; 6 Correct with amendment; 1 Gap; 0 Error.

Part R15 (Terra referee, lemma 28)

Referee report on lemma 28: Proposition 28.13 and the corrected numerics

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_28.md. Reproduced as delivered, with links made relative. The corrections are applied in lemma 28 and [the script] (tools/lemma28_radial.py, removed on 4 Nov 2026 by the

rule that checking scripts are not kept; editorial change of the link only). The three items marked Error concern statements in the file: the identification of the boundary channels with a singular vector, a mislabelled table column, and the scope of the Correction and Outcome paragraphs. The matrix of Proposition 28.13 and its closed form are confirmed.

Count: 3 errors; 3 correct with amendment; 0 correct without amendment. The count refers to the six numbered items.

1. Slice operator — Correct with amendment.

With Hermitian Clifford generators $c_j c_k + c_k c_j = 2\delta_{jk}$, the spin connection is

$$\nabla_i = E_i - \frac{1}{4}\Gamma_{ijk} c_j c_k, \qquad \Gamma_{ijk} = \langle \nabla_{E_i} E_j, E_k \rangle.$$

Indeed, the commutator of the connection matrix with c_j gives $\Gamma_{ijk} c_k$. The Koszul expression in [the script] (tools/lemma28_radial.py, removed on 4 Nov 2026 by the rule that checking scripts are not kept; editorial change of the link only) has the correct order of indices. The $\mathfrak{h}$-part of a bracket is omitted in these reductive-frame coefficients; the H-action remains necessary in the equivariance condition.

For $s_v(g) = F\lambda(g)^{-1}v$, differentiation along $g\exp(te_i)$ gives $-FV_i$, with the appropriate scale divisor. The pullback connection contributes $+E_i F/\mathrm{scale}_i$ in the vertical directions. Its sign in the script is consistent with the spin term.

The displayed radial form is correct for the script's $h = 1$ metric after absorbing the half-density. For the general metric in Theorem 28.9, let

$$u = h^2 \rho\, \psi.$$

Before this conjugation the radial derivative is

$$iZ\left(\partial_x + 2\frac{h'}{h} + \frac{\rho'}{\rho}\right).$$

After it, there is no extra h'/h or ρ'/ρ term, and the precise operator is

$$T = iZ\partial_x + \frac{A_0}{h(x)} + \frac{A_1}{\rho(x)} + \frac{\rho(x)}{h(x)^2}A_2 + i\phi(x)\tau.$$

A collar modification of the connection adds $-s(x)A_E/\rho(x)$. Thus the notation $D_6(\rho(x))$ in Result 28.5 needs the $h = 1$ qualification. The file should also distinguish its Hermitian Clifford convention from the skew-Clifford $c(dr)$ used in Theorem 28.9.

2. Exact tests — Correct with amendment.

The stated anticommutation and Casimir identities have the right form:

$$\{A_0, A_1\} = 0, \qquad D^2_{S^4,\tau_{\mathrm{tw}}} = \mathrm{Cas}_G(V) + \frac{3}{2} - \mathrm{Cas}_K(\tau_{\mathrm{tw}}).$$

Here τ_{tw} denotes the relevant K-twist; it should be distinguished from the Higgs matrix τ. Python reproduced the printed A_0^2 spectra and anticommutator norms in all four representations tested by the script. The zero in the A_0^2 spectrum on the **20** by itself is not a core zero mode of the full normal problem.

For the Fubini–Study metric with base S^4 of radius 1, the fibre radius is $\rho = 1$. The homogeneous strict nearly Kähler metric has $\rho = 1/\sqrt{2}$. More generally these radii are h and $h/\sqrt{2}$. The squashing conventions and the two special metrics are described in Appendix A.1, especially (A.1).

At $\rho = 1$, $j(j+4)$, $j = 0, 1, \ldots$, is the correct D^2 eigenvalue on $(0,0)$-forms with values in $\mathcal{O}(1)$. It is not the spectrum of the *entire* $\mathcal{O}(3)$-twisted spinor bundle: the **20** also contains $D^2 = 10$. The last sentence of the Correction paragraph needs that restriction. In addition, the script prints the Casimir and Fubini–Study spectra; it does not assert their expected numerical values.

The tests do not separately determine all three coefficients. For example, as an algebraic test of the listed identities, replacing

$$A_1 \mapsto tA_1, \qquad A_2 \mapsto A_2 + (1-t)A_1$$

preserves $\{A_0, A_1\} = 0$ and the tested line-connection operator at $\rho = 1$. The curvature construction supplies an additional test of A_2. With $a, b = 0, \dots, 3$ and $\alpha = 4, 5$, Koszul's formula gives

$$A_2 = -\frac{i}{8} \sum_{a,b,\alpha} C^{\alpha}_{ab} c_a c_b c_\alpha.$$

The nonzero C^{α}_{ab} for $a < b$ are $C^5_{02} = -1$, $C^4_{03} = -1$, $C^4_{12} = 1$, and $C^5_{13} = -1$. An in-memory Python calculation compared this contraction on the full eight-dimensional spinor space with the script's ρ-coefficient. The Frobenius residual was 4.8×10^{-16}; reversing the coefficient gave residual 2.83. This check shares the script's Lie-bracket input, while using a separate curvature contraction. It is a numerical check of the displayed algebraic derivation.

3. Proposition 28.13 — Error.

The finite wall matrix, its kernel, and its trace calculation are correct for the channel convention used in that calculation. The error is its identification of those channels with the right singular vector for the boundary condition stated in Theorem 28.9.

For the stated $\hat{\Gamma} = \mathrm{diag}(1, -1)$ and $\hat{\Gamma} Z = -1$, the boundary domains are

$$Zu = -u \quad \text{for } T, \qquad Zv = +v \quad \text{for } T^*.$$

Let $T_\eta = iZ\partial_x + D_6 + i\eta\phi\tau$, with $\phi > 0$, and write a wall profile as $e^{q(x-\ell)}$, $q > 0$. The leading equations give

$$q = -\eta\phi Z\tau \quad \text{for } T_\eta, \qquad q = +\eta\phi Z\tau \quad \text{for } T^*_\eta.$$

Consequently, with the **positive mass and boundary condition of Theorem 28.9**, the right wall channels have $Z = -1, \tau > 0$, and their fibre trace is $+w$ in the script's $\hat{t}$ convention. The $Z = +1, \tau < 0$ channels of Proposition 28.13 give $-w$; they describe the adjoint wall problem for the reversed mass, or a problem with the

opposite wall condition. Python reproduced $+2/3$ and $-2/3$, respectively, at $\rho = 1$. Changing the mass sign alone cannot make a right T-domain vector satisfy $Z = +1$.

For the matrix calculation, let $E_{s,t;j}$ be the rank-one map from input basis vector j to spinor basis vector s tensored with twist basis vector t, using the zero-based bases of the script. Choose

$$(f_0, f_1, f_2) = (E_{2,2;2}, E_{2,3;3}, E_{4,2;3}),$$
$$(g_0, g_1) = \left(E_{3,2;3}, \frac{E_{0,2;0} + E_{6,2;1}}{\sqrt{2}} \right).$$

Direct Clifford contraction, with the stable-profile overlaps, gives

$$\begin{pmatrix} 1/\rho & -3/(2\rho) & -(\rho + 1/\rho) \\ -i\sqrt{2} & 0 & -i\sqrt{2} \end{pmatrix}.$$

Reordering and changing unit phases gives the displayed $A(\rho)$. The entry $3/(2\rho)$, for example, contains the $\sqrt{3}/\rho$ vertical connection matrix element and the $\sqrt{3}/2$ overlap. Python/SymPy reproduced this matrix using exact symbolic arithmetic from the bracket and Clifford data.

Its kernel is as stated. The $-3/2$ channel and one $-1/2$ channel have input support $(1)_{-1}$, while the remaining channel has support $(1)_{+1}$; their output spaces are orthogonal. This yields

$$w(\rho) = \frac{2(\rho^2 + 2)^2}{9 + 2(\rho^2 + 2)^2},$$

including $w(0) = 8/17$, $w(1) = 2/3$, and $w(\infty) = 1$. For horizontal radius h, replace ρ^2 in this formula by ρ^2/h^2.

The line-preserving connection replaces the first entry of the first row by zero. Its remaining block has the stated sum and product of squared singular values, is invertible for $\rho > 0$, and has squared singular values $3, 6$ at $\rho = 1$. The Fubini–Study cohomology statement in the proposition has the stated scope.

For fixed positive radius, the overlap matrix is the correct **leading formal wall operator**. Since its kernel on

the **4** is one-dimensional and separated from the other two singular values, the next correction is $O(M^{-1})$ in the fixed-block expansion and does not change its limiting kernel. A uniform global reduction still needs proof, as the file acknowledges.

The sentence calling 1.863 and 2.789 "the next values on the 4" needs to say **wall values**. In a Python radial calculation at $\phi_0 = 40$, $\ell = 3$, the levels were approximately 1.865 at the wall, 2.000 at the core, and 2.789 at the wall.

4. Result 28.5 — Error.

The Python rerun reproduced the main tables. The origin exponents are

$$\pm 1, \quad \pm 2 \text{ with multiplicity three}, \quad \pm 3.$$

For $\phi_0 = 3, 4, 5$, the measured decay rates in the first table agree with the corresponding constant-coefficient values $0.8557, 1.1514, 1.4762$ to the stated precision. At fixed ρ, the slow eigenvalues have the reported expansion

$$\kappa = \frac{\phi}{2} - \frac{2}{\rho} + O(\phi^{-1}).$$

For example, Python gave 30.0663 at $(\phi, \rho) = (64, 1)$ and 31.0625 at $(64, 2)$. The exponent comparison is numerical evidence for the displayed profiles; it is not a general tunnelling theorem.

The table has a numerical mesh floor, but its size depends on parameters and the mesh. At $(\phi_0, \ell) = (6, 4)$, Python gave lowest singular values $0.00548, 0.00401, 0.00355$ for $N = 2500, 5000, 10000$. These values do not establish a zero continuum value. Also, the script uses three-point quadrature for non-polynomial coefficients. Rayleigh quotients would give rigorous upper bounds with the exact quadratic form; the quadrature results are numerical approximations to those bounds.

The substantive table error is the heading **"sup of the density ratio."** The script computes

$$\frac{\max_x |\operatorname{tr}(B(x)\widehat{t})|}{\max_x \operatorname{tr} B(x)},$$

not the supremum of the pointwise ratio. At $\ell = 3$, Python gave:

ϕ_0	Ratio of maxima reported	Pointwise maximum of $\lvert \operatorname{tr}(B\hat{t})\rvert / \operatorname{tr} B$
4	0.00648	0.849
5	0.00197	0.856
6	0.000473	0.858

The pointwise maximum occurs at the outer boundary, where the core density is small. The integrated density values -0.005707, -0.001836, and -0.000455 reproduce the table. Their ratios to λ^2 are empirical observations for these parameters, not a proved $O(\lambda^2)$ asymptotic. They support the hypothesis used by Proposition 28.16 for the numerical modes; numerical nonzero values alone do not prove the exact finite-mass algebra. At $\phi_0 = 16, 40$, the reported integrated values are around 10^{-11}, so their signs in these calculations do not establish nonvanishing.

For the other representations, the zero multiplicities on $\mathbf{1}, \mathbf{5}, \mathbf{10}, \mathbf{14}, \mathbf{35}$ follow from the central character. Python reproduced the reported wall/core identifications on $\mathbf{16}, \mathbf{20}, \mathbf{40}$, including core values close to $\sqrt{5}, \sqrt{8}, \sqrt{12}$. For $\mathbf{64}$, an additional in-memory Python construction gave a small-radius wall value approaching $\sqrt{15}$. A scan of **401 radii** in $[0.1, 20]$ found no zero on any of the four representations; this supports a sampled statement and does not prove nonvanishing over the whole interval.

Two profile qualifications matter. The reported $\rho(x) = \tanh x$ metric is not exactly a product near the finite wall. The displayed tanh collar switch is nonzero, though tiny, at $x = 0$; an actual smooth extension with the pullback connection near the core needs a cutoff that vanishes on a neighbourhood of zero. The rounded collar values therefore support the limiting model and do not establish exact finite-mass equality.

5. Correction, Outcome, and Conjecture — Error.

In the Correction paragraph, the old sign error and the withdrawal of the affected numerical claims are sup-

ported. The statement that **every number** was wrong is too broad: the previous Result 28.5 already listed the same origin exponents, and some displayed values agree after rounding. The Astra report says that its referee did not reproduce the old boundary-value numbers and that its proofs did not use them; this review did not repeat those proofs. The final sentence also needs the $(0,0)$-form restriction described in item 2.

For the four Outcome items:

1. The compression-map mechanism has the stated scope for the **prescribed** windows of Theorem 28.21.
2. The four core states are established for the normal model and its induced core operator. The displayed boundary weight is a finite wall-model result; convergence of an actual spectral mode remains conditional on the missing analysis. The boundary-sign and h-scale amendments above also apply.
3. The isolated window remains a Conjecture. Besides coercivity, the normal reductions, quasimode errors, and matching must be controlled.
4. The asserted exact finite-mass algebra is stronger than the numerical evidence. The table supports quaternionic-symmetry breaking at the resolved sample points, conditional on the continuum nonvanishing required by Proposition 28.16. It does not establish the exact algebra at every mass reported.

Conjecture 28.6 is compatible with Counterexample 28.17 when it asserts convergence of **compression maps**: convergence of those maps does not imply convergence of their generated algebras. The conjecture needs the boundary-sign and horizontal-scale conventions above. Its exponential statement in ℓ also needs a specified family of metrics and mass profiles with a uniform positive decay rate; Theorem 28.9 permits more general profiles.

6. Concentration literature — Correct with amendment.

Let $\mathcal{A} = i\Phi$ and let $\sigma(\xi)$ be the symbol of the self-adjoint Dirac operator. Since Φ acts on the twist factor, it commutes with the Clifford symbol, and

$$\mathcal{A}^*\sigma(\xi) = \sigma(\xi)^*\mathcal{A}.$$

Thus this is a concentrating pair in the sense of Maridakis, arXiv:1510.07004v1, Definition 2.1 and Lemma 2.2. The commutator $[D, \Phi]$ is order zero, and the doubled self-adjoint operator satisfies the corresponding anticommutation condition. On a closed manifold, the concentration principle applies to suitable low modes near the singular set of Φ.

The stronger spectral-separation theorem of Maridakis, arXiv:2110.11654v2, §2, Assumption 3 and the Spectral Separation Theorem has an additional hypothesis that fails for this spin-3/2 mass. At the core, its required direction-independent leading endomorphism would have to represent $(n \cdot J)^2$ for every normal unit vector n. These matrices depend on n: in a $J_3 = 3/2$ state, the expectations of J_3^2 and J_1^2 are $9/4$ and $3/4$. An anisotropic extension of that theorem, or a separate core analysis, is therefore needed. Prokhorenkov–Richardson, arXiv:math/0307251v3, §§2–3 supplies the related symbol condition and treats its stated localization setting.

Adding the wall requires boundary estimates and the stable boundary model. The closed-manifold energy argument acquires a boundary term: on a half-line,

$$\|(\partial_t + M)f\|^2 = \|f'\|^2 + M^2\|f\|^2 - M|f(0)|^2,$$

and e^{-Mt} is a boundary-localized zero mode although the mass has no zero there. A proof for this problem must combine the core analysis with that boundary layer, control the $O(M^{-1})$ reduction errors, and establish coercivity on the complement uniformly across representations.

For the source order, the arXiv LaTeX-source endpoints were requested first, but the available reader could not open their compressed responses and direct network retrieval failed. We then read the cited sections of the three arXiv HTML versions identified above. We did **not** read the LaTeX sources or PDFs of those three papers. The separate Fubini–Study-radius citation above was checked in its PDF appendix.

Part R16 (Terra referee, lemma 29)

Referee report on lemma 29, by Terra 6.1

Model gpt-6.1-terra, effort max, read-only sandbox, run on 4 November 2026 with the prompt terra_check_29.md. The report below is verbatim, except that local absolute links were rewritten as relative links. No Python was run by the referee.

Assessment by Claude Opus 5.5, reading it against the text. Item 10: correct, the definition of A_eff as a chiral map made K(rho) impossible as written, while the displayed operator was the intended one; corrected. Sign of the fibre part: correct, -2/3 from the kernel vector (-2, 1, 1), with channel weights 4, 1, 1 and fibre values -1, -1, +1; the squared values (45 +- 3 sqrt 33)/8 were recomputed by hand. Items 2, 4 to 9, 11, 12: applied. Item 9 also supplies the density B_w, checked at w = 1 and against the trace. Item 13: the identifiers were checked on arXiv. A referee that ran no computation and reports no counterexample has found no error in case (b); that is weaker than a proof by a second author, and the corrected case (a) has not been read again.

The principal error is in case (a): the displayed definition of A_{eff} makes Condition $K(1)$ impossible as written. There is also a sign error in the reported fibre part: at $\rho = 1$ it is $-2/3$. Both have identifiable corrections; neither supplies a counterexample to the proposed eight-state count.

The case-(b) coercivity argument survives the main cut-off and projection checks, subject to the domain, gauge and constant clarifications below. No Python calculation was run. The table was compared with the stated limiting values, but its computations were not independently reproduced.

1. **CONFIRMED — Proposition 29.1.** For the connection metric with $h = 1$ and $\rho = r$, the normal fibres are flat and totally geodesic. In a horizontal/vertical orthonormal frame, the difference between the Levi-Civita connection and the split connection is the O'Neill tensor. Its curvature component is linear in y, and its spin contraction gives,

in the convention of Proposition 29.1,

$$R = -\frac{i}{8} \sum_{a,b,\alpha} \langle [\tilde{e}_a, \tilde{e}_b], f_\alpha \rangle c(e_a) c(e_b) c(f_\alpha).$$

Thus $T_M = T_0 + R$ exactly on the tube. There is no additional first-order term for this metric and the pull-back connection. In the moving radial frame this is rA_2; the apparent polar mean-curvature term belongs to the ordinary vertical Dirac operator.

The bracket normalization used in the radial system reduces A_2, on the nontrivial horizontal chirality, to a unitary conjugate of

$$\tfrac{1}{2}(s_1 \otimes \sigma_1 + s_2 \otimes \sigma_2),$$

with s_i Pauli matrices. Its norm is 1; it vanishes on the other horizontal chirality. Hence $\|R\| \le |y|$.

The common principal connection causes no extra derivative of $\widehat{Q}$. In the equivariant-map realization, the horizontal fields differentiate the G argument and $\widehat{Q}$ acts by the same operator on the fibre at every g. Diagonal K-invariance of the actions on y, the normal spinors and E makes this construction descend. Individual horizontal derivatives have a frame index that transforms under K; the contracted D_H is equivariant.

With $c(f_\alpha) = \gamma_5 \sigma_\alpha$, the relevant algebra is

$$D_H \widehat{Q} = \widehat{Q}^* D_H, \qquad D_H \widehat{Q}^* = \widehat{Q} D_H.$$

It gives both displayed sums of squares and commutation of D_H^2 with the two normal squares. Finally,

$$\Sigma_7|_K = ((2,1) \oplus (1,2)) \otimes (1,2) = (2,2) \oplus (1,3) \oplus (1,1),$$

so the stated spin representation is correct.

2. **AMENDMENT — Proposition 29.2.** The spectrum and kernel conclusions follow, with a distinction between two projections. The projection P_W onto the normal K-representation $W_0 = (1,3) \oplus (1,1)$, extended over *both* base chiralities,

commutes with D_H. The projection onto the actual fibre kernel of $\widehat{Q}$ also selects the chirality $\gamma = -\epsilon$:

$$P_0 = P_{\gamma=-\epsilon} \otimes P_W.$$

This projection need not commute with D_H. Rather, $D_H P_0 = P_1 D_H$, where P_1 projects onto the fibre kernel of $\widehat{Q}^*$, and P_0 commutes with D_H^2. The proof should use P_0 and the latter commutation for the normal spectral gap.

For the stated normalization, the twisted round-S^4 formula is

$$D_H^2 = C_2^G(V) + \tfrac{3}{2} - C_2^K(j), \qquad C_2^K(j) = 2j(j+1).$$

The twist has $j = 0, 1$. Only representations with odd central character occur. The least G-Casimir is $5/2$ on the 4; the next is $15/2$ on the 16. The $j = 1$ zero on the 4 occurs in the base chirality of type $(1, 2)$, exactly the chirality selected by (H4). The adjoint normal kernel selects the opposite base chirality, where that zero is absent. The $j = 0$ part on the 4 gives the next squared value 4.

On noncompact N, the calculation should first be stated on compactly supported smooth sections and extended through the closed graph forms. The doubled model is essentially self-adjoint on that core. The normal estimate $(y \cdot J)^2 \geq |y|^2/4$ gives compact normal resolvent; the compact base and the commuting squared decomposition give the required discrete model spectrum. No boundary condition is imposed at infinity.

3. **CONFIRMED — Lemma 29.3.** Coupling spin $1/2$ to spin $3/2$ gives

$$\sigma \cdot J = F^2 - \tfrac{3}{4} - \tfrac{15}{4},$$

so its eigenvalues are $3/2$ and $-5/2$. For each y, rotational invariance gives eigenvalues $\pm|y|/2, \pm 3|y|/2$ for $y \cdot J$; hence the squared lower bound is pointwise valid. Consequently

$$\frac{M^2}{4}\||y|u\|^2 \leq \|T_0 u\|^2 + \frac{5M}{2}\|u\|^2,$$

which gives exactly the stated constants after taking square roots. The adjoint estimate follows in the same way. The positive μ_1, rather than its numerical value, is all that the subsequent proof needs.

4. **AMENDMENT — Lemma 29.4 and IMS.** Define C_Φ explicitly as a bound on the *Clifford contraction*

$$\left\| \sum_i c(e_i) \nabla_{e_i} (f\, n \cdot J) \right\|_{\mathrm{op}} .$$

This removes the ambiguity in writing only $|\nabla(f\, n \cdot J)|$. With that convention the bulk estimate follows from expansion of $(D + i\Phi)^*(D + i\Phi)$.

The IMS step is valid. The derivatives of all three cutoffs are supported in a compact annulus away from both ends; the bulk estimate controls their error by $CM^{-2}(\|T_M u\|^2 + \|u\|^2)$. Dropping the nonnegative $\|T_M \chi_b u\|^2$ is allowed because the same bulk estimate separately controls $\|\chi_b u\|$. Scalar radial cutoffs preserve both wall domains.

The comparison with the model is also correct: extension by zero puts $\chi_c u$ in the model domain, and

$$u \perp \chi_c \ker T_0 \quad \Longrightarrow \quad \Pi(\chi_c u) = 0.$$

The analogous wall orthogonality is exact.

5. **AMENDMENT — Lemma 29.5.** Parthasarathy's formula gives the displayed result. In the standard $SU(4)$ normalization, the central weight of the $\mathcal{O}(k)$ twist is represented by

$$\tfrac{k}{4}(1, 1, 1, -3),$$

whose squared length contributes $3k^2/8$ to the Casimir. The scalar curvature of the stated Fubini–Study metric is 12, giving $R/8 = 3/2$. Thus

$$D_k^2 = C_2(L) + \tfrac{3}{2} - \tfrac{3}{8} k^2.$$

The invariant unitary connection on each line is the Chern connection: the difference of two invariant line connections is an invariant one-form, and the H-isotropy representation has no trivial summand.

The eligibility argument works. Each of the two $SU(4)$ representations of dimension 20 below the relevant Casimir threshold restricts to $Sp(2)$ as $4 \oplus 16$. The $\mathcal{O}(3)$ slice contains the $Sp(2)$ representation 4 only once, and that occurrence is already the cohomological kernel; neither 20 can therefore occur. In the charge convention of the earlier report, $\mathcal{O}(3)$ has charge -3. The slice weights give its 4 once in $Z = +1$ and zero times in $Z = -1$, proving the kernel chirality without a numerical mode-position argument.

The correction is to "$\kappa_w = \sqrt{3}$, for all V": $\sqrt{3}$ is the **global uniform gap**, attained on the 4. It is not the lowest nonzero value on every isotypic component; the stated lowest wall value on the 16, for example, is $\sqrt{5}$.

6. **AMENDMENT — Lemma 29.6.** In radial parallel gauge the sum of squares and Robin analysis are correct. With $q = -\tau$, the free Robin zero channel is

$$(Z = -1, q > 0) \quad \text{for } T_M, \qquad (Z = +1, q > 0) \quad \text{for } T_M^*.$$

The first-order form $d - m|q|$ has the normalized zero profile $\sqrt{2m|q|}\, e^{m|q|(x-l)}$, and its orthogonal complement has spectrum starting at m^2q^2. The boundary condition kills the cross-term boundary pairing.

The collar gauge should be stated. For example, add $itq\, dr$, with a cutoff supported in the collar, to a line-preserving connection. This is invariant, unitary and satisfies (H3). In the original radial trivialization it changes the Dirac operator by $-tZq$; on the $q = 3/2,\ Z = +1$ line,

$$T^*_{M,\mathrm{new}}(pg) = -\tfrac{3}{2} t\, pg.$$

Radial parallel transport removes this term and restores the displayed profile. Thus the geometric estimate persists, while the displayed formula and trial state require that gauge or the corresponding parallel-transport factor.

The charge calculation in item 5 establishes the relative core/wall sign. The index discussion in Remark

29.12 gives a consistency check; it does not by itself identify the local chirality. If the $\mathcal{O}(3)$ zero mode instead lay in $Z = -1$ with $q > 0$, pg would be a wall zero mode for T_M, contradicting the positive lower bound in part 1 of Lemma 29.6.

7. **AMENDMENT — Theorem 29.7 and Lemma 28.14.** Lemma 28.14 is correct. If the trial range has dimension eight and $\|\widehat{D}U_M\| < \delta$, its spectral projection onto $(-\delta, \delta)$ has dimension at least eight. If that projection had dimension greater than eight, it would contain a nonzero vector perpendicular to the trial range, contradicting the complementary bound. The spectral theorem then gives the stated angle estimate.

   With the projection and collar-gauge clarifications above, the case-(b) IMS argument supplies the hypotheses for each fixed $0 < \delta < \sqrt{3}$. The core trial defect is $O(M^{-1/2})$: Lemma 29.3 gives the required first moment even without invoking Gaussian decay. The wall defect is exponentially small in radial parallel gauge. No counterexample to the case-(b) count was found.

8. **AMENDMENT — Proposition 29.8.** The tunnelling identity is correct for a singular pair with real nonnegative λ. For a real scalar weight preserving the domain, the quadratic-form identity

$$\|T_M(\xi u)\|^2 = \lambda^2 \|\xi u\|^2 + \|c(d\xi)u\|^2$$

   is also correct.

   The Agmon proof needs two explicit choices. First, choose the IMS cutoffs so that the region where the exponential weight varies is covered by the bulk estimate, including the transition near the collar. Second, the final cutoff loses the integral across its support. Apply the weight with a smaller parameter, for example $\eta' = \eta/2$, and choose the support width s so that

$$(1-\eta')\left(I - \int_{r_*}^{r_*+s} f\,dr\right) \geq (1-\eta)I,$$
$$I = \int_{\varepsilon}^{l-a} f\,dr.$$

This supplies the exponent stated in the proposition. The absorption uses $\lambda \to 0$, supplied by the trial states.

The factor $1/2$ is justified uniformly across channels by $(n \cdot J)^2 \geq 1/4$; it does not require every channel to have eigenvalue $1/2$. For the subsequent $l \to \infty$ statement, require $M \geq M_\eta$ and uniform bounds on the endpoint geometry, cutoffs, connection and positive bulk mass. Translation invariance in the bulk alone should not stand in for those uniform hypotheses.

9. **AMENDMENT — Theorem 29.9.** The compression estimate follows from Theorem 28.15 after choosing equivariant unitary identifications of the varying window spaces with the model copies of 4, and after the applicable window theorem is established.

   With the coordinate convention of lemma 28,

$$Q_B(a) = \int_{S^4} a(x)(1 - x \cdot \Gamma)\, d\mu(x),$$
$$Q_B(x_a) = -\Gamma_a/5.$$

   In case (b), $B(z) = 4p_z$, with $p_z \leq (1-x\cdot\Gamma)/2$. This base-coordinate convention accounts for the sign when comparing with the coordinates in lemma 10.

   More explicitly, if p_z is the coherent line selected by the $q = 3/2$ channel and p_{jz} its quaternionic partner, the case-(a) limiting density is

$$B_w(z) = 2((1 + w)p_z + (1 - w)p_{jz}).$$

   It gives $Q_Y(x_a) = -\Gamma_a/5$ and the stated factor w on the y_T components. Since $w > 0$, the 5- and 10-components are present; together with independent endpoint restrictions they give the claimed real algebra. This is a limit-map statement.

10. **ERROR — Proposition 29.10 and Theorem 29.11 as written.** At the definition of A_{eff}, P contains only $Z = -1$ states and Y only $Z = +1$ states. Therefore

$$F = \Pi_Y(A_d + A_o)P : P \longrightarrow Y$$

is a chiral operator. It is not the self-adjoint operator subsequently described. If regarded as an operator on the ambient space, it kills all of Y; if regarded as $P \to Y$, its kernel lies in $Z = -1$. In either reading, the stated Condition $K(\rho)$, requiring its kernel to be exactly a 4 in $Z = +1$, cannot hold.

The correction is to define the full operator on $P \oplus Y$ by

$$\mathcal{A}_{\text{eff}} = \begin{pmatrix} 0 & F^* \\ F & 0 \end{pmatrix},$$

and impose $K(\rho)$ on this operator. The matrix $A(\rho)$ of Proposition 28.13 represents $F^* : Y \to P$ on the 4; F is its adjoint.

With this correction, the proposed block argument has the required structure. The profile overlap is $\sqrt{3}/2$. The stable-to-stable block has norm $\sqrt{3}/\rho$, so its contribution has norm $3/(2\rho)$; the full perturbation instead has $\|A_o\| = 2/\rho$, as required in the elimination estimate. The complementary block has inverse norm at most $(m/2 - \|A_o\|)^{-1}$ for large m, and its Schur correction is bounded by

$$\frac{\|A_o\|^2}{m/2 - \|A_o\|}.$$

On $V \neq 4$, the corrected effective operator at $\rho = 1$ is bounded below by $\sqrt{5} - 3/2$. On the 4, the nonzero squared singular values of $A(1)$ are

$$\frac{45 \pm 3\sqrt{33}}{8},$$

and its one-dimensional multiplicity kernel lies in $Z = +1$. This proves the intended corrected $K(1)$. The quoted 1.863 is the value on the 4; the displayed argument gives the smaller uniform lower bound $\sqrt{5} - 3/2$ off that representation. The cutoff trial state must use the exact corrected half-cylinder kernel; its finite 4-block equation gives exponential decay.

There is a separate sign error in the reported fibre part. At $\rho = 1$, the kernel vector is $(-2, 1, 1)$, and

the signed fibre trace is

$$\frac{-4-1+1}{6} = -\frac{2}{3}.$$

In general it is $-w(\rho)$. The positive $w(\rho)$ is its magnitude.

11. **AMENDMENT — Compatibility of (H1)–(H4).** The hypotheses are compatible. A smooth positive ρ can equal r near zero and be constant on a disjoint collar; take $h = 1$. Likewise, $f = r$ near zero and $f = f_0 > 0$ on the collar gives $f(r) = r\,b(r^2)$ for a smooth positive b. The metric and mass extend smoothly across the zero section. Since X deformation retracts to S^4, its relevant $H^2(-;\mathbb{Z}_2)$ vanishes and the oriented tangent bundle is spin. Invariant Hermitian connections can be interpolated with a radial cutoff.

    The chirality convention should be explicit. If $\tau = n \cdot J$, then (H4) gives $q = -\gamma_b \tau$; the displayed convention $q = -\tau$ uses $\gamma_b = +1$. The collar expression also uses the radial gauge described in item 6.

12. **ERROR — Outcome and scope.** Outcome item 2 and Theorem 29.11 give the signed fibre part as $+2/3$; it is $-2/3$. The assertion that $K(1)$ holds also requires the corrected effective-operator definition in item 10. The nearest supported statement is the intended case-(a) result after that correction and the accompanying Schur-complement argument.

    Remark 29.12 should say that the opposite-sign **end-model modes** lie in the indicated kernels. At finite mass the full operator can have a nonzero tunnelling singular value, so those localized states need not lie in the exact kernels of T_M or T_M^*.

    The scope paragraph otherwise identifies the restrictions on metrics, the connector and finite-mass algebras. "Uniformity over representations" follows here from the stated full-space estimates and ellipticity of the corrected fixed boundary operator; it should not be attributed to the fixed-manifold setting alone. Away from $\rho = 1$, finiteness of the possible kernel representations follows from ellipticity,

while an explicit representation cutoff for a practical finite check remains unwritten.

The displayed **Python** table is consistent, to its shown precision, with the stated limits: in particular 2.2361, 2.8284 and 3.4641 agree with $\sqrt{5}, \sqrt{8}, \sqrt{12}$, and the larger $M = 32$ wall discrepancies are of the stated mesh scale. The table was not rerun. The quoted $\mu_1 \approx 1.02$ is a Python approximation; the two supplied tools do not themselves reproduce that oscillator calculation.

13. **AMENDMENT — Prior work.** General forms of two key identities already appear in the literature:

    - Maridakis, arXiv:1510.07004v1, Lemma 2.2 gives the concentrating-pair square and its zeroth-order cross term, which includes the bulk identity used here. Prokhorenkov–Richardson, arXiv:math/0307251v3, §§2–3 treats the earlier Witten-deformation localization framework for the stated isolated-singularity setting.
    - Kaad–van Suijlekom, arXiv:1710.03182v1, Proposition 18 gives a horizontal/vertical Dirac factorization with an explicit curvature contraction. Its stated proper-submersion and dimension hypotheses differ from this seven-dimensional noncompact model; it is a close structural ancestor of Proposition 29.1.
    - Bismut–Lebeau, *Complex immersions and Quillen metrics*, §§8–9, DOI 10.1007/BF02699352, develops rescaled normal operators and block estimates for complex immersions. Kottke, arXiv:0909.5661v2, Theorem 11 gives the families Callias index in terms of the positive eigenbundle; it supplies an index, rather than this boundary spectral window.
    - Maridakis, arXiv:2110.11654v2, §2 proves a spectral-separation theorem under additional assumptions. Its Assumption 3 requires a direction-independent leading endomorphism; the spin-3/2 mass here has direction-dependent $(n \cdot J)^2$. Prokhorenkov–

Richardson, arXiv:math/0610160v2, §3 treats equivariant Witten localization and normal models.

The targeted search covered Callias operators on bundles, submersion factorization, Bismut–Lebeau localization, equivariant Witten deformation, boundary chirality and a 2026 recency query. The cited sections were opened. Taubes's 1996 paper, DOI 10.4310/jdg/1214459411, was identified through Maridakis's reference chain, but its original full text was inaccessible in this search. Citation chains and the 298-page Bismut–Lebeau paper were not exhaustively reviewed. The search found prior general identities; it does not establish whether the specific eight-state theorem has appeared elsewhere.

The corrections in order of severity are:

1. Replace the definition of A_{eff} and $K(\rho)$ by the full chiral block, then restate the case-(a) proof.
2. Correct the signed case-(a) fibre part to $-w(\rho)$, in particular $-2/3$ at $\rho = 1$.
3. Specify radial parallel gauge, the model domains and the distinction between P_W and the one-chirality fibre-kernel projection.
4. Complete the Agmon cutoff argument with an η-margin and state the uniform hypotheses for varying l.

Part R17 (Terra referee, lemma 30 and the joint-limit note)

Referee report on lemma 30, the note and the corrected case (a) of lemma 29, by Terra 6.1

Model gpt-6.1-terra, effort max, read-only sandbox, run on 4 November 2026 with the prompt terra_check_30_note.md. The report below is verbatim, except that local absolute links were rewritten as relative links. No Python was run by the referee.

Assessment by Claude Opus 5.5, reading it against the text. All items applied, with these checks. Item 7: cor-

rect; the bulk interpolation allowed by (H3) need not keep j_E parallel. The fix was checked by hand: j_E (n . J) j_E^-1 = -n . J gives j_E P_q j_E^-1 = P_{-q}, so j_E is parallel for the line-preserving connection and for real affine interpolation with the pull-back connection, which is the interpolation in tools/lemma28_radial.py, and C then commutes with i Phi_M as well as with D. The unconditional bound eta_t = O(M^{-1/2}) is adopted, with its proof written out. Item 4: correct; lemma 24 has V = C^4 with gradings of type (2, 2). Item 9, first point: the counterexample to the implication from the hierarchy alone is correct. Item 5: the sign in Proposition 29.10 and the missing projection estimate are applied; the Weyl constant has the rank 64 = 8 x 4 x 2. Item 1: the gap is an equality, since the next level is (pi - theta_max)/l. Item 13: the identifiers not already in the record were checked on arXiv. The source-level attributions to Parthasarathy, Bismut-Lebeau and Kirchberg stay unread, as the referee says. This was the last Codex job available in the weekly quota, so these corrections have had no further independent reading.

The main mathematical gap is in Proposition 30.7: the allowed case-(b) connections need not preserve the charge conjugation used in the proof. With an added compatibility hypothesis, the weighted argument gives the displayed exponential bound. A fresh reading of the corrected case (a) also found a remaining sign error in Proposition 29.10.

The interval and disc-bundle counting arguments otherwise support the stated joint limits, subject to the amendments below. No Python calculation was run in this review.

Part A: lemma 30

1. **AMENDMENT — Theorem 30.1.** The compression constant 2/3, the gap bound and both counts agree with Theorems 22.1 and 22.4. Add $l_* > 0$, $E_* > 0$, and the algebra and representation assumptions needed to define $Q_\varepsilon(a)$ and $\pi(a)$. Alternatively, state the compression estimate for a bounded even operator a, using its operator norm.

   There are $r/2$ singular values counted with multi-

plicity, and each gives two signed arithmetic progressions. These remain r **progressions with multiplicity** when singular values coincide or vanish. In particular, a zero singular value contributes two zero modes. Thus the moving-window count is r, and the fixed-energy leading term is

$$\frac{2rl_*E}{\pi}\varepsilon^{-\alpha}.$$

For small ε, the displayed lower bound for the first discarded level is in fact an equality:

$$\Delta_\varepsilon = \frac{\pi}{l_\varepsilon} - \varepsilon\|D_*\|.$$

2. **AMENDMENT — Corollary 30.2.** The two endpoint pairs are correct for the **universal fully massive family**, with both members of every Standard Model doublet assigned the same Dirac mass and no Majorana mass. In that family the two commutator quotients are $2m^2$ and $16m^2/5$; multiplying them by $(3 - d_{\text{eff}})/2$ gives $(3, 24/5)m^2$ as $d_{\text{eff}} \to 0$ and $(2, 16/5)m^2$ as $d_{\text{eff}} \to 1$.

   Lemma 25 concerns the leading **Taylor quotient** for the heat-cutoff spectral action, with an even vector generator constant in the transported frame and an enlarged admissible gauge algebra. The interval poles belong to the separately specified five-dimensional Yang–Mills problem. The coefficient comparison holds at leading order; it does not identify the full spectral-action pole problem with that Yang–Mills problem.

   The path is correct:

$$s = l_\varepsilon E_\varepsilon = l_* E_* \varepsilon^{\beta-\alpha} \longrightarrow 0, \qquad \frac{m_\varepsilon}{E_\varepsilon} \longrightarrow 0.$$

   At fixed $\Lambda > 0$, $s \to \infty$ and $m_\varepsilon/\Lambda \to 0$. Lemma 25 states its expansion first at fixed s; a joint-limit use should add a uniform remainder argument, or state explicitly that the displayed endpoints concern the leading coefficient. Varying the action cutoff is an additional prescription, as the corollary says.

3. **AMENDMENT — Proposition 30.3.** The root and asymptotics follow from Theorem 9.4 and

Lemma 9.5. For $c > 1$, the first oscillatory root satisfies

$$\pi < k_1 l < \frac{3\pi}{2}, \qquad \frac{\Delta}{\mu} = \sqrt{1 + \left(\frac{k_1 l}{c}\right)^2} \longrightarrow 1.$$

The two tail identities follow from $\mu = \kappa \coth s$ and $m = \kappa / \sinh s$.

The localized family needs a **new length**, since its length cannot also be the $l_* \varepsilon^{-\alpha}$ of Theorem 30.1. State

$$l_\varepsilon^{\mathrm{loc}} = \frac{(1-\alpha)\log(1/\varepsilon) + c_0}{\mu_* \varepsilon^\alpha},$$

with $\mu_* > 0$ and $0 < \alpha < 1$. The reuse of l_ε is also misleading in the note.

4. **ERROR — Corollary 30.4 for arbitrary finite data.** The cylinder in lemma 24 has $\ker A \simeq \mathbb{C}^4$, with wall gradings having two positive and two negative directions. It does not realize an arbitrary dimension r from Theorem 30.1. For example, finite data with $H_+ = H_- = \mathbb{C}^3$ and $D_* = 0$ satisfy the interval hypotheses and have $r = 6$; the stated cylinder has a four-dimensional tangential kernel.

   Restrict the corollary to the compatible $r = 4$ data, or construct and state an enlarged tangential bundle for other dimensions. The gap estimate then follows from lemma 24. The seven-dimensional count needs an additional Weyl argument: Dirichlet and Neumann bracketing of the squared product operator give the same leading term as $l_\varepsilon, r_\varepsilon \to \infty$, proportional to $l_\varepsilon r_\varepsilon^6 E^7$. Lemma 24's gap estimate alone does not prove that asymptotic.

5. **AMENDMENT — Theorem 30.5 and corrected case (a).** For a fixed shape, it is legitimate to substitute $M = M_\varepsilon \to \infty$ into Theorem 29.7 or 29.11. Proposition 29.8 supplies M_η, C_η independent of M after the shape and η are fixed. Substitution gives $\lambda_M / R_\varepsilon \le C_\eta \varepsilon$. No uniformity in a varying shape is needed for this family.

   The domain of $\widehat{D}_M$ is independent of M: the wall projector is fixed and MV is bounded, with $\|V\| \le (3/2) \sup f$. Min–max applied to the singular values gives the displayed comparison of the

counting functions of $|\widehat{D}_M|$ and $|\widehat{D}_0|$. The local self-adjoint elliptic boundary problem has the usual leading Weyl law. Counting both signs and complex multiplicity,

$$C_X = \frac{64\,\omega_7\ \mathrm{Vol}_g(X)}{(2\pi)^7},$$

since the seven-dimensional spinor bundle has complex rank 8, E rank 4, and the operator is doubled. The estimate $M_\varepsilon/R_\varepsilon \to 0$ makes the bounded perturbation negligible at the counting threshold. The physical first discarded level also tends to zero: its shape-unit value is at most $s_9(\widehat{D}_0) + cM_\varepsilon$.

Dilation is exact under the canonical spinor identification and the L^2 unitary factor $R^{-7/2}$. With physical radial coordinate $s = Rr$, the potential is

$$\frac{M}{R}\, f(s/R)\, n\cdot J.$$

A radial half-density normalized with respect to ds acquires the corresponding $R^{-1/2}$ factor. The compression estimate uses C^1 norms in fixed shape coordinates. Add $0 < \delta$, $E_* > 0$, and state that if $m_\varepsilon = 0$, zero has total multiplicity eight.

Fresh check of case (a). The corrected chiral definition of A_{eff}, the profile overlap $\sqrt{3}/2$, the perturbation bound $3/2$, and the squared nonzero singular values

$$\frac{45 \pm 3\sqrt{33}}{8}$$

are consistent. The displayed matrix at $\rho = 1$ has kernel $(-2, 1, 1)$, giving signed fibre part $(-4 - 1 + 1)/6 = -2/3$. However, Proposition 29.10 still says that the fibre part tends to $+w(\rho)$. It must say $-w(\rho)$. Theorem 29.11 has the correct sign.

The adjoint estimate in Proposition 29.10 also needs one explicit step. Orthogonality to the *exact* half-cylinder kernel does not immediately give orthogonality to $\ker F^*$. The Schur-complement graph estimate gives $\|P_{\ker T_M^*} - P_{\ker F^*}\| = O(1/m)$; inserting this projection estimate supplies the required $O(1/m)$ loss. With that addition, no further error was found in the corrected case-(a) block argument.

6. **CONFIRMED — Proposition 30.6.** Normalized angular Haar measure and Schur's lemma give

$$Q(f) = \frac{4}{\int \operatorname{tr} B} \int\!\!\int f(r,g)\lambda(g)^{-1} B(r) \lambda(g)\, dg\, dr.$$

Quaternionic conjugation exchanges b_+ and b_-, so $\|B - P_{\mathbb{H}} B\| = |b_+ - b_-|/2$. This gives $2\eta_t$. Each compression of a real f with $\|f\|_\infty \le 1$ is a contraction; telescoping against its j-conjugate gives $2k\eta_t$ for a product of k such compressions.

7. **ERROR — Proposition 30.7 as stated for all allowed connections.** The proof assumes $CT_M = T_M C$ throughout the interior. Setting 29.0 permits any invariant unitary interpolation in the bulk; it does not require that interpolation to preserve j_E.

This is a failure of a stated premise in the proof. For an allowed connection that is pull-back on a bulk annulus, put $q = n \cdot J$ there and add the invariant skew-Hermitian connection form

$$i\, h(r)\, \nabla^0(q^2),$$

with real h supported inside the annulus. It has no radial component and leaves both end hypotheses intact. Since $j_E q^2 j_E^{-1} = q^2$, antiunitarity sends this added form to its negative. Consequently j_E is not parallel and C need not commute with the resulting twisted Dirac operator. This example shows the failure of the asserted symmetry; it does not by itself establish a counterexample to the exponential estimate.

Add $\nabla^E j_E = 0$, or another explicit hypothesis proving the same global symmetry. Under that hypothesis, the other checks work as follows.

- $C^2 = j^2 = -1$, so $F \mapsto CFj^{-1}$ is an antilinear involution on $\mathrm{Hom}_G(\mathbf{4}, -)$. On the core it preserves D_H, the full fibre term $i\widehat{Q}$, and R. With the conventions of Proposition 29.1, it anticommutes with $\widehat{Q}$ itself, because it is antilinear.
- The phase can make $\langle \chi_c F_0, u \rangle$ real, and then $\Pi(\chi_c \zeta) = 0$. The qualification by χ_c is essential.

- On $r \le l - a$, $g + \tilde{g} = G$. The cut-off $\xi\zeta$ vanishes near the wall and belongs to the required domain. The bulk first-order term has a positive η-margin; the core estimate uses the projection condition. The derivative of the wall cut-off is supported where $g = G$, so its contribution is $O(e^{-G})$. Thus the proposed argument gives $\|\xi\zeta\| = O(e^{-G})$.
- Pointwise,

$$|b_+ - b_-| \le (\|F\| + \|CF\|)\|\zeta\|.$$

  Weighted Cauchy–Schwarz on $r \le l - a$, and the L^2 tail bound beyond $l - a$, both give $O(e^{-2G})$.

Therefore, **conditional on global C-preservation**, the proof gives

$$\eta_t \le C_\eta \exp\left[-(1-\eta)M \int_{2e}^{l-a} f\, dr\right].$$

The integral starts at $2e$, whereas Proposition 29.8 displays the tunnelling bound starting at e; the stated exponent is therefore not literally twice that displayed tunnelling exponent. A smaller core cut-off can be chosen if the comparison is to use the same starting radius.

Under the original general connection hypotheses, the window estimate still gives $\eta_t = O(M^{-1/2})$: choose an equivariant normalized core model fixed by the local real structure and use the L^2 distance to that model from Theorem 29.7. Finally, an exponential upper bound expressed in M does not imply $O(\lambda^2)$ without a lower bound on λ.

The Python samples cited at lemma 30, line 106 vary the length at fixed mass and use profiles outside Setting 29.0. They support a qualitative comparison and do not test the stated $M \to \infty$ theorem.

8. **AMENDMENT — Remark 30.8.** The vector-sector and connector obligations are accurately identified. The gauge formula assumes a fixed microscopic coupling, a constant internal Yang–Mills profile and the stated normalization; then $g_4 \propto R^{-7/2}$.

For the toy potential, require $a, b, p, q > 0$. Its unique minimum is

$$r_\varepsilon = \left(\frac{ap}{bq}\right)^{1/(p+q)} \varepsilon^{-2/(p+q)},$$

and $p + q > 2$ gives $0 < \alpha < 1$. The complete-manifold sentence just before the remark also needs qualification: global charge conjugation gives a real structure on the multiplicity space, but the provided argument does not establish that this space is one-dimensional for **every** M or for every completion. Exact quaternionicity of a selected core copy follows if that multiplicity is one.

Part B: the note

9. **ERROR — Several statements exceed their supplied hypotheses.** The following corrections are needed in joint_limit_note.tex.

   - **Hierarchy, lines 40–44.** The hierarchy alone does not imply that the fixed-energy count diverges. For a compact-resolvent counterexample, take signed spectrum

     $$0, \quad \pm\varepsilon^\alpha, \quad \pm 1, \quad \pm 2, \ldots$$

     and $E_\varepsilon = \varepsilon^\beta$, $\alpha < \beta < 1$. The hierarchy holds with one retained zero mode, while $N_\varepsilon(E) = 3$ for fixed $0 < E < 1$ and small ε. State divergence as a separate property proved for the specific families.
   - **Interval and cylinder, lines 50–79.** Add the representation assumptions for the compression bound, separate the localized length from the angle-family length, and restrict the unmodified cylinder to compatible four-dimensional kernel data, as in items 1, 3 and 4. State the scope of the heat-cutoff Taylor quotient from item 2.
   - **Disc-bundle hypotheses, lines 83–90.** Define $S = G \times_K (1, 2)$ and its pull-back in $E = S^3(S)$; state the round S^4 radius, smoothness and positivity of ρ and f, $f_0 > 0$, and $Sp(2)$-invariance of the connection. The displayed metric with $\rho = r$ does encode flat nor-

mal fibres near the core; the note also explicitly states constant f and the Fubini–Study radius on the collar. Case (a) in lemma 29 assumes the pull-back connection throughout, whereas the note only specifies it near the two ends. A bulk interpolation may be treatable by the bulk estimate, but that extension needs to be stated and proved.

- **Sign and localization, lines 87–90.** The sign selects the core kernel of the **end model**. At finite M, the full T_M can have nonzero tunnelling singular value. Describe the two four-dimensional subspaces by their proved proximity to the cut-off core and wall models, rather than as exact states at the two ends.
- **Core proof, line 99.** The gap of the first-order fibre operator is of order $\sqrt{M}$; the gap of its square is of order M. D_H exchanges the relevant base chiralities and need not preserve the actual one-chirality fibre kernel. The argument uses the normal-mode projection extended over both chiralities, or the commutation of the actual kernel projection with D_H^2.
- **Bulk and wall, lines 101–110.** Replace the variable-f lower bound by an integral bound or a bound using the minimum of f on the support. Restrict the asserted wall gap to the relevant stable $q > 0$ channels: the full line sum also includes the $\mathcal{O}(-3)$ zero modes. The case-(a) value $-w(\rho)$ is the limiting effective-wall value, with a finite-M correction.
- **Tunnelling and joint limit, lines 128–162.** Define the integration endpoints and the fixed cut-offs e, a, rather than writing only "over the bulk." State the equivariant identifications and the shape-coordinate C^1 norm in the compression estimate. Include the zero-mass multiplicity qualification from item 5.
- **Defect, lines 166–175.** Add the global charge-conjugation compatibility from item 7. Replace "no component on the model kernel" by $\Pi(\chi_c\zeta) = 0$, and distinguish the two action integrals in the tunnelling and defect estimates. The assertion that only the wall breaks C also needs the added connection hypothesis.

- **Bimodule, line 185.** Specify "32 complex-dimensional bimodule with a real structure," equivalently 64 real dimensions before taking a real fixed subspace.
- **Status, line 189.** Lemma 29 case (b) had a Terra referee report; the corrected case-(a) text had not had a second reading. Proposition 30.7 and Theorem 30.5 were new.

10. **AMENDMENT — Physics framing and named sources.** The abstract acknowledges that the disc-bundle vector poles remain open, and the body says that **4** is pseudoreal. Put both qualifications next to the headline claim: the disc-bundle result treats a **fermion** window, and no calculation identifies its tunnelling scale with M_W or M_Z. The title should describe a target electroweak limit or a fermion joint limit. The auxiliary $\widehat{\Gamma}$-grading and the zero virtual $Sp(2)$-index do not establish the chiral Standard Model spectrum.

The reading of Witten's October 1983 preprint, pp. 7–8 and 43–44, is consistent at the stated level. Witten discusses vectorlike quantum numbers under the specified geometric gauge assumptions and states equivariant cobordism invariance of the character-valued index. Here the extending object is the **sum** of the four line summands; the individual $\mathcal{O}(k)$ do not all extend. The signed indices are $(4, 0, 0, -4)$. This index permits the spectral pairing and does not prove the window.

Maridakis, arXiv:2110.11654v2, §2, Assumption 3 requires a direction-independent leading endomorphism. For spin 3/2, $(n\cdot J)^2$ depends on n as an endomorphism, even though its eigenvalues are direction-independent. The paper also works on a closed manifold, so the boundary layer requires separate analysis.

The Bryant–Salamon qualification is correct. Karigiannis–Lotay, arXiv:2002.06444v3, Eq. (3.36) and Theorem 3.6 gives a complete asymptotically conical G_2 metric on the same total bundle, with a radial horizontal scale and non-flat fibre metric outside Setting 29.0. Homothetic enlargement of the fixed shape has a flat pointed limit at a fixed interior point; a fixed boundary point gives a half-space limit.

11. **AMENDMENT — Citations and companion drafts.** The internal interval references support the exact spectrum and the qualified Taylor coefficient. The note should preserve the companion draft's explicit distinction between a leading Taylor quotient and a pole calculation. towerless_intervals.tex treats the relation between the KO labels and geometric limits as a proposal; the joint note's introduction presents that relation more definitively and should retain the proposal status.

Connes–van Suijlekom, §1 supports the operator-system discussion. Replace "generated algebras do not converge when compression maps do" at line 146 by "need not converge"; constant families already show that convergence can occur.

The Kirchberg citation is apt for the **untwisted** $R/3$ consistency check in complex dimension three. It does not by itself prove the twisted $\mathcal{O}(k)$ spectrum. Kirchberg's 1990 paper, introduction and Theorem 2, explicitly gives the odd-dimensional bound and identifies $\mathbb{CP}^3$ as a limiting case. The Parthasarathy and Bismut–Lebeau entries have matching bibliographic identities in the Annals record and Numdam record, respectively; their full original texts could not be checked directly in this review, so the precise source-level attribution remains unchecked here. The numerical normalization in lemma 29 must be justified in the argument.

Baptista, arXiv:2506.09126v2, §1 and Eq. (1.2), supports the distinction between Killing and non-Killing metric modes in its stated Kaluza–Klein setting. It does not calculate the vector sector on this disc bundle.

12. **AMENDMENT — Table I.** Every displayed $M = 32$ row matches lemma 29's Python table. The first **16** entry is the core value and the second is the wall value; on the **20**, the first is the wall value and the second is the core value. Adding those labels would make the comparison easier to assess.

The smallest computed **4** value is floor-dominated at the reported meshes. Its mesh dependence does not determine whether the continuum value vanishes. The stated

0.005 wall discrepancy is a Python mesh observation from the source, not an independently established error bound. The caption should attribute the limit values to the end models and their spectral comparison: the window theorem by itself gives the low-window count and gap, not all individual excited limits.

13. **AMENDMENT — Prior results and comparison.** Several mechanisms in the note have established precedents that an expert would expect to see distinguished from the new combination.

- Kaplan, hep-lat/9206013v1, Eqs. (0) and surrounding discussion, derives chiral defect localization. Shamir, hep-lat/9303005v1, §§1–3, treats boundary fermions and the finite fifth direction. These precede the interval localization mechanism.
- Csáki et al., hep-ph/0305237v1, §§2 and 6, derives gauge symmetry breaking by interval boundary conditions. Cacciapaglia–Csáki–Park, hep-ph/0510366v2, §5, obtains $m_t = 2M_W$ at tree level for a suitable rank-four representation. Neither source establishes the lemma-25 crossover.
- Maridakis, arXiv:1510.07004v1, Lemma 2.2, gives the concentrating-pair square identity. Kaad–van Suijlekom, arXiv:1710.03182v1, Proposition 18, gives a horizontal–vertical Dirac factorization with a curvature contraction. Their settings differ from the stated seven-dimensional boundary problem.
- Kottke, arXiv:0909.5661v2, §5, gives a families Callias index theorem under asymptotically conic hypotheses. It supplies index information under those hypotheses; the finite boundary-window estimate is a separate question.
- Iso–Kitazawa–Suyama, arXiv:2111.06508v2, §§1 and 4–5, follows massive Kaluza–Klein gauge modes that become massless during symmetry restoration in flux compactifications. The mechanism and geometry differ from the present joint family.

Witten's closed-geometry index restrictions and the boundary construction have different hypotheses; no contradiction was found in the checked records. These targeted checks do not establish priority for the eight-state theorem.

Corrections in order of severity

1. Add global charge-conjugation compatibility to Proposition 30.7 and the note, or supply a different proof for the allowed case-(b) interpolations. Retain the unconditional $O(M^{-1/2})$ defect estimate until then.
2. Remove the claimed implication from the hierarchy alone to divergent fixed-energy counting; retain the separate counts proved for the named families.
3. Correct Proposition 29.10's signed fibre part to $-w(\rho)$, and add the projection estimate for the adjoint wall bound.
4. Restore the disc-bundle hypotheses in the note, restrict the cylinder data, and give the localized interval family its own length.
5. Qualify the electroweak, chirality, Taylor-mass and finite-versus-limit compression claims in the title, abstract and theorem statements.

Part R18 (assessment of the Astra audit)

Assessment of the audit of commit b2ddf98

The user supplied on 4 November 2026 an outside audit of the repository at b2ddf98, titled "conneskk: audit and a route from the massless electroweak limit to eleven dimensions", kept verbatim in 2026-11-03_astra_audit_d11.md. The file does not state its authorship; the 11D line, merged on 4 November, attributes it to Astra 6, and we follow that. This note records what was checked and what was done. Provenance: Claude Sonnet 5.5 recomputed all items, read-only, in Python and with pdftotext, and Claude Opus 5.5 rederived by hand the items marked below, against the source lemmas. The user asked for attention to detail with the reports of smaller models, so the column "by hand" says which verdicts do not rest on such a report alone.

Verdicts

Audit section	Claim	Verdict	By hand
1	Withdrawal of the alleged normalization error at boundary_condition_note.tex:53	Correct: both components have the factor	Sonnet only
2.1	D^2 = 2 lambda^2 on the 64-dimensional product, P_- D P_- = 0,		[P_-, D]
2.2	The massless limit must specify a path; theta -> 0 at fixed l opens no tower	Correct, and adopted as STRATEGY T14	Yes
2.3	ind D_{O(k)} = (k^3 - k)/6, values 4, 0, 0, -4, total zero	Correct	Yes; Witten 1983, pp. 43-44, gives the reason
2.4	Supplement omits lemma 17	Correct; lemmas 01, 02, 04 are also omitted, and 17 is added	Yes
2.4	docs/ last refreshed at 4ae5b43; T13 steers only to D = 9; docs/index.html has no seven-dimensional entry	Correct	Sonnet only
2.4	PDF overflow, 81 of 478 and 13 of 263 pages	Depends on an unstated tolerance: Sonnet found 113 and 15 at tolerance 0, 83 and 15 at 2 pt, 12 mobile pages above 5 pt, with maximum overhang 8.7 pt, through pdftotext -bbox	Sonnet only

Audit section	Claim	Verdict	By hand
3	Finite window and opening tower on the interval, equations (1) to (4)	Correct; the statement needs Delta_eps defined, and the references are Theorems 22.1, 22.4. Now Theorem 30.1	Yes
3.1	Vector poles and M_W^2 = 2 m^2, M_Z^2 = (16/5) m^2; coefficients 3 and 24/5	Correct, the latter for the fully massive family of lemma 18, Proposition 14. Moreover M^2 = ((3 - d_eff)/2)	
4	Localized family, equations (6) to (8), and the table	Correct. Now Proposition 30.3	Yes; table rows by Sonnet
5.1	Equation (9), Weyl exponent on the cylinder	Correct. Now Corollary 30.4	Yes
5.2	Dilation, conditional scaling of X	Correct; the condition is removed in Theorem 30.5 for the class of lemma 29, pending its referee	Yes
6	Bound (13) with constant 2, and the telescoping	Correct, once “defect” means	

Audit section	Claim	Verdict	By hand
6, (14)	Target: eta_t <= C (lambda/ delta)^2 + errors	Stated as a target there. Proposition 30.7 proves the form exp(-2 G) for the upper bound exp(-G) on lambda; a bound in terms of lambda itself would need a lower bound on lambda, which is open	Yes
6.1	Table of eta_t at L = 2, 3, 4, 5	Rows reproduced by Sonnet (Python, same solver, so a reproduction with a new observable)	Sonnet only
7.1	Toy potential and its minimum	Correct as an elementary statement; underived, as the audit says	Yes
Appendix A	M^{111} data: c_1 = 3h + 2u, dim H^0(O(3,2)) = 30; six arXiv identifiers	Correct on the items checked	Sonnet only

No error in a displayed equation of the audit was found.

What was done

- Lemma 30 takes the propositions that are proved, with proofs redone, and adds Theorem 30.5 and Proposition 30.7.
- Item 2 of the audit's list (the shape-unit estimate for X, starting from the line-preserving collar) is

lemma 29, which was already in progress when the audit arrived, and goes further: case (a) at the Fubini-Study radius as well.
- STRATEGY gains T14, the joint limit, parallel to T13, as the audit asks.
- paper/build_supplement.sh gains lemma 17, and lemmas 28 to 30 once their referees are applied.

What was not adopted

The note proposed in section 8 is being written at the request of the user, as paper/joint_limit_note.tex. The M^{111} comparison stays a comparison, and no W, Z -> 0 family links it to the finite geometry. The layout repair of the supplement is recorded and has not been done.

Part R19 (independent reading of 11D files 10 and 11)

Independent reading of 11D/10 and 11D/11, by Claude Sonnet 5.5

Model Claude Sonnet 5.5, effort max, read-only, run on 4 November 2026 because the Codex quota was spent. The subagent was refused a file write, so it returned the report as text, and the verdicts and corrections below are that text, verbatim, filed here by Claude Opus 5.5. Its scripts were not kept, by the rule of the project that checking scripts are not saved; the report names what each computed.

Assessment by Claude Opus 5.5, reading it against the text, as the user asked for reports of smaller models. Correction 1 was checked against lemma 23, line 180 ("Sp(2) x SU(4), of dimension 25") and its Theorem 23.6, and recomputed: the bracket is 169.3125 for three generations and 41.3125 for one, and tools/d11_needed_numbers.py, corrected, reproduces theta_* = 1.009, 1.031 and G(pi/2) = 23.88. The near-cancellation 64 - 63 in the first version of 11D/10 was an artifact and is withdrawn. Correction 2 was checked against the conclusions of Bergman, Gaberdiel and Lifschytz, and the Planck length formula was rederived. Correction 3: the rank argument for the Z was accepted on reasoning. Correction 5 is the mechanism of

lemma 23. Correction 7: stationarity at c = 1 was recomputed (mpmath). Corrections 6, 9 to 12 were read and applied. Item 3's rank-17 statement stays marked as recalled, since neither source proves it. This reading is by a smaller model and no further independent reading followed the corrections.

A. Verdicts

1. Prop 10.1: CONFIRMED. AMENDMENT: z must have slope exactly 8 - N(x) on [0, 2 pi].
2. Prop 10.2: CONFIRMED.
3. Prop 10.3: AMENDMENT (metric of l). The 10D SO(32) endpoint is CONFIRMED. The rank-17 eleven-dimensional-limit statement: COULD NOT CHECK from the two sources (only BGL remarks support it).
4. Prop 10.4: linearity, C_L = 8 pi/5, bound, strong-coupling statement CONFIRMED. AMENDMENT on the inference "no eleven-dimensional lift".
5. Far wall: formula and limits CONFIRMED. AMENDMENT: only W^+ and W^- are D0 states; the closing sentence needs its scope.
6. Prop 10.5: CONFIRMED.
7. a_0: method, signs, normalization, F(1) check CONFIRMED. ERROR in the generator count: 21 integer towers against 16 integer + 9 half-integer in lemmas 23, 24, 26.
8. Prop 10.6: items 1, 2 CONFIRMED; item 3 CONFIRMED in scope (no proof given); n_s = 129 ERROR (inherits item 7).
9. Tied angle: classification CONFIRMED; the quoted numbers inherit the item 7 error.
10. Localized channel: CONFIRMED. AMENDMENT: dropped bulk and wall terms.
11. Remark 10.7: AMENDMENT (wording). Remark 10.8: heat-cutoff statement CONFIRMED, no conflict with lemma 25. AMENDMENT: q > -2, not q > 0.
12. Corrections list: AMENDMENT (the S - T sentence of Prop 5.4 is also false; 07 section 4 omitted).
13. Prop 11.1: CONFIRMED. AMENDMENT: factor s in j q_L; s = +1 implicit in 11.4.

14. Prop 11.2: CONFIRMED with three AMENDMENTS (proof choice of f; second order-one block; finite window against parent).
15. Thm 11.3: items 1, 2, 3, 5 CONFIRMED; item 4 AMENDMENT (aligned subalgebras); the rule "vector or scalar" holds in a stated scope.
16. Comparison paragraph: CONFIRMED.
17. Prop 11.4: CONFIRMED.
18. Reading after 11.4: AMENDMENT (first order, fixed compression); no conflict with lemmas 19, 22, 23.

B. Corrections in order of severity

1. ERROR (file 10: Outcome 2, a_0 section, Prop 10.6 last paragraph, tied-angle paragraph). Lemmas 23, 24, 26 use sp(2) + su(4), 25 generators: 16 integer-tower (8 gluons, 3 W_L, 4 O with (D,D), 1 hypercharge combination) and 9 half-integer-tower (2 R^+-, 6 X, 1 orthogonal neutral; (N,D)). File 10 counts 21 integer towers (the Neumann generators at x = 0). Correct: 8 pi^2 a_0 = (3 zeta(5)/16)(4 n_D' - 3(121/16) - n_s - 5 n_g). Brackets: 169.3125 (3 generations; file: 129), 41.3125 (1 generation; file: 1), 36.3125 (with graviton; file: -4), n_s = 169.3 for a_0 = 0 (file: 129). A_0 = 32.918 and 8.032. Tied-angle values: universal 3 generations theta_* = 1.009, M_W/m = 0.685 (file 0.877, 0.876); universal 1 generation theta_* = 1.031, M_W/m = 0.653 (file 0.202, 1.394, "small because A_0 is small", an artifact); top only G(pi/2) = 23.88, no stationary point. The sign a_0 > 0 and the top-only classification survive. The single-weight formula is valid for integer towers only. Taking all 25 as integer would give 117 and -11. The script carries the same n_v = 21. The Outcome writes n_f where n_D' is meant.
2. AMENDMENT (Prop 10.4). "Known fact that massive IIA has no lift" has no reference. The source is BGL (introduction): type I' with N != 0 "has no simple M-theory interpretation"; conclusions: "we expect neither … applicable". The classical profile is not controlled at g of order 1, so the proposition establishes the statement for the extrapolated profile. Add: g(0) L_p <= C_L

l_s/7 holds exactly for the total length (the "small change on the last segment" is unnecessary), and the region g >= g_0 has eleven-dimensional length (3/4)(C_L/|8-N|) g_0^{-4/3} l_11.

3. AMENDMENT (far wall). Only W^+ and W^- are D0 states. A neutral Z is a Cartan combination, and the rank of a 9D theory with 16 supercharges is fixed, so no massive neutral vector arises. Scope of the closing sentence: chamber 15 + 1, top's brane on the far wall, W^+- as D0 states. Escapes: other chambers, W as a Wilson-line D8-D8 string (lemma 23), top's brane in the bulk (far wall E_1). Add M_D0 R = R^2 - R_*^2 (alpha' = 2).
4. AMENDMENT (Prop 10.3). l L_p = pi^2 alpha' holds for l = pi R_I measured in the type I' metric. For the heterotic radius in the heterotic metric, l L_p = pi^2 alpha' sqrt(lambda_h) (BGL (hettypeI), (simpleSduality)). alpha' is the type I' alpha'. Mark paragraphs 2 and 3 as recalled; there is no proof block.
5. AMENDMENT (Thm 11.3 item 4, Outcome 3). The towers need commuting projections onto k_0 and k_l. Counterexample: k_0 = span T_1, k_l = span(cos phi T_1 + sin phi T_2) gives kl = n pi +- phi. List escapes: circle, nontrivial holonomy, brane-localized terms, extra dimensions, nonaligned subalgebras.
6. AMENDMENT (Prop 5.4 correction). The sentence "S - T has d = 16 on the neutrinos of one slot, 22 on the two others" fails in 96 of 192 configurations (pattern 16, 16, 22). Add 07 section 4 ("distance proportional to a") to the list.
7. AMENDMENT (localized channel). The bulk term (-(4/15) c^5 in dimensional regularization) and the wall term are dropped. "Prefers c near 1" refers to the interaction term only. The stationary point is c = 1 exactly. Large-c corrections are relative 1/c.
8. AMENDMENT (Remarks 10.7, 10.8). "Fixed ratio lambda_{M_*}/delta" instead of "realizes the hierarchy of lemma 30". A joint family with alpha < 1 needs p + q > 2 (q > -2 for p = 4, q != 0). Write "the potential" in the heat-cutoff sentence.
9. AMENDMENT (Prop 11.2). The choice f = T, g = x 1 gives only wall-compatible T; use f = 1 with g' = T - tau (tau interior-supported, same integral)

plus tau = sum f_i g_i'. Add the second order-one block -a M^* U b U^-1. Order one fails in the finite window; the parent satisfies it (Thm 22.3), and its wall algebra shrinks to the commutant of D_F in pi (zero for pi = (a, 0), M invertible).

10. AMENDMENT (Prop 11.1, 11.4). j q_L = s conj(q_R). Prop 11.4 requires Lambda = s j Lambda j^-1, so it is stated for s = +1. pi is not U-invariant, so U only defines j (not a lemma 06 doubling). Move the sextet out of the "colour antitriplet" column.
11. AMENDMENT (Reading after 11.4). The statement is first order with the fixed compression (-(1/2)[D_F,h]). The moving window gives -1[D_F,h] and the quadratic term (Cor. 22.6, lemma 19).
12. Minor: normalization of z; define n (n = 1, N' = 7) and the reflection to BGL's x^9 = 0; Prop 10.6 item 3 needs its own proof; heading "what would work"; ellipsis in the PW quotation; the folded potential has stationary points near theta = 8.2, outside the window.

C. Derivations, in summary

The subagent's derivations are reproduced in its scripts: s1 (items 1, 2, 5, 6, mpmath), s2 (item 7, mpmath), s3 (items 8, 9, numpy, scipy), s4, s4b, s4c (item 10, mpmath), s5, s5b (item 12, exact integers), s6 (items 13, 14, 17, numpy, scipy), s7 (item 11, mpmath), all Python 3.14. It also reported: no first-person singular, possessive or "not Y; it is Z" constructions in the two files; Prop 10.3 mostly recalled with no proof; "known fact" without a reference; the evaluative heading of Remark 10.8; the PW quotation dropping e^{phi}. These are corrected in the files.

Part R21 (second check of the Terra referee of the 11D line)

Second check of the Terra referee of 11D files 01 to 09, by Claude Sonnet 5.5

The user asked, on 4 November 2026, for a second check of each evaluation in the Terra referee report, with Claude Sonnet 5.5, since no other model had quota left.

The checks ran in five batches, one at a time, each redoing the step on which a verdict turns and checking the records. The subagents were refused file writes, so each returned its report as text; the reports are filed below verbatim, each followed by the assessment of Claude Opus 5.5, who rechecked the claims against the text and the source lemmas before applying them. Scripts are in the session scratchpad and were not kept in the repository.

Batch 1: files 01 and 02

Report A. Verdicts

1. Prop 1.1 and joint family: PARTLY. Map, domains, grading, spectrum (k+1/2)pi/l, r window states, gap and N(E)=2 r l E/pi redone and correct. Missing: the joint family lives on the periodic circle (B1).
2. Prop 1.4: AGREE. F(1)=93 zeta(5)/64, J(0)=3 zeta(3)/2 and slope (54/5-n_f)J(0) reproduced; normalization matches lemmas 23, 24.
3. Prop 1.5: AGREE. Counterexample G=theta^4 and small-angle claim hold; true-determinant roots of theta G'-4G are isolated (1.0094 at n_f=48, 1.0309 at n_f=16, A_0 from 11D/10). Remark 10.7 and Terra amend different sentences: no conflict.
4. Prop 2.5: AGREE. Cohomology table enumerated for common and independent chirality, both colour conventions: no flux in span(6Y, 3(B-L)) works, with or without nu^c; c_L=-3 c_Q forbids |P_2|=2 on both doublets.
5. Prop 2.6: AGREE. Exactly the two stated solutions; on P(3bar) the first components reverse; none without chi or B-L, for either chirality hypothesis.
6. Thm 2.1, Props 2.2, 2.4: AGREE. Freeness, Euler class (3,2), Z_6, congruence, transverse Einstein constants 10, 10, factor 5/4. M^{111} identification: COULD NOT CHECK (no repository source); label "conditional" is honest.
7. Prop 2.3: AGREE. Smith forms for nine pairs; (3,2): H^2=Z, H^3=0, H^4=Z/9, spin. Kreck-Stolz: COULD NOT CHECK.

Ledger and README rows match all seven verdicts

(status and checkers). Recorded as sonnet_5_5, this check gives d11_flux_nogo a third identifier, in its stated scope.

B. Corrections, by severity

1. 11D/01 line 36 ("the parent unfolds to -i d/dx + D_F on this circle (lemma 25)") and line 44 ("In the circle language of Proposition 1.1"): Proposition 25.1 unfolds the lemma-22 parent to the periodic circle. Write "unfolds to -i d/dx + D_F on the periodic circle of length 2l; theta interpolates between it (theta=0) and the antiperiodic circle of Proposition 1.1 (theta=pi/2); the joint-family window is the n=0 mode".
2. 11D/01 Prop 1.5: line 5 says "amended at its statement below", yet heading (line 88) and lines 92-97 are unchanged. Replace line 97 by "If the roots of theta G'-4G are isolated, a stationary branch has l=theta_/m *(alpha=1; Remark 10.7 for small theta_)*. A root interval (G=theta^4 near 0) makes every alpha<1 stationary and selects none." The universal wording also stands in STRATEGY.md line 77, INDEX.md line 497, 11D/06 line 63, paper/connes_kk_dictionary_note.tex line 112.
3. 11D/02 lines 62-77: lemma table, Statement ("the same image", "2j+1 … linear") and first two proof sentences need "common ten-dimensional chirality"; add the chirality-free proof (|M_2(c)|=2 on Q and L, c_L=-3 c_Q). STRATEGY.md line 77 still reads "pending check".
4. Stale headers: 11D/01 line 3, 11D/02 line 3 ("Referee pending"; "pending an independent check"; "Theorem 2.1 … proved here" against ledger "conditional").
5. Prop 2.4(c): ledger d11_z6, 11D/02 lines 50, 52, 101 and STRATEGY.md line 77 say "no bosonic Kaluza-Klein mode" without the Terra scope; add "tensor modes on a background invariant under the displayed action; spinors open".
6. 11D/02 line 26 "Sym^{3k}(3bar) x Sym^{2k}(2): for k = 1 this is the (10, 3)": Sym^3(3bar)=10bar; write Sym^{3k}(3) for functions. Line 60: add "convention for base P(3); for P(3bar) use (c_0,c_1)=(9/4,3/4) or (3/2,3)".
7. 11D/01 line 50 "(on the circle of Proposition 1.1,

fibre C^2, …)": write "the scalar circle gives 2 l E/pi; fibre H_F gives 2 r l E/pi". Prop 1.1 item 3: "even f(|x|), Lipschitz at the fixed points".

8. 11D/10 Prop 10.6 item 3 ("no joint family results") needs the Prop 1.5 scope (V=0 or tuned H leaves flat directions).

C. Derivations

B1: one channel, walls Q_R=0, U_L=0 at 0 and Q_R=X U_R, U_L=-X Q_L at l, X=tan(theta): the determinant vanishes at l*lambda = n pi +- theta, so n pi at X=0 (zero mode), (n+1/2) pi as X tends to infinity. B2: (theta G'-4G+4A_0)/theta^2 tends to -2c, c=(54/5-n_f)J(0). B3: left-handed fields come from left-handed modes (pairing +(alpha+3/2, beta+1)) or conjugated right-handed modes (pairing minus that); common chirality gives P_2=+(2j+1), independent chirality +-(2j+1). Third charge z: coefficients (-1,3), d^c gives -5 against 1; no vector of 117,649 passes both components. Scripts (Python): s1_oneloop.py, s2_flux.py, s3_topology_spectra.py, s4_periodicity.py, s5_third_charge.py, s6_unfolding_map.py.

Assessment Checked against the sources by Claude Opus 5.5. Correction 1: lemma 25, Proposition 25.1, is stated on the circle with periodic conditions, while Proposition 1.1 concerns the chiral interval, which is antiperiodic; applied in 11D/01 and in the dictionary note. Correction 2: G = theta^4 gives V = m^4, constant in l, so every l is stationary; applied, with the heading changed and the INDEX link updated. Correction 3: in the span of 6Y and 3(B - L), Q has charges (1, 1) and L has (-3, -3), so the chirality-free argument holds; added as a scope note. Corrections 4, 5, 7 (first part) and 8: applied. Correction 6: the tensor and function spans are now distinguished; the colour-convention point for Proposition 2.6 was already in the referee note of 11D/02. The Lipschitz point is in the referee note of 11D/01. The ledger records sonnet_5_5 as a second checker on the ten rows of files 01 and 02.

Batch 2: files 03 and 04

Report Second check of Terra sections on 11D files 03 and 04 (Claude Sonnet 5.5; Python 3 scripts). Ledger

and README rows otherwise match Terra.

A. Verdicts 1. Prop 3.2: AGREE. Odd Lambda^k(5bar) = SM Weyl fields, even = conjugates, X = 3, -1, -5. 2. Prop 4.1: AGREE. Lemma 21 sums rebuilt from SM charges; among 12-state whole-species blocks only u, c, t and the three neutrinos keep (5/8, 8/3). 3. Prop 3.8, Cor 3.9: AGREE. All 17 nine-dimensional SU(3) representations decomposed; self-conjugate ones are 9(1), 3+3bar+3(1), 8+1; centre trivial on Y_{3,2}. 4. Props 4.2, 4.3: AGREE with both amendments. None of the 17 Lambda^3 equals F1 or F2; only 9(1), 3+6(1), 3bar+6(1) give {1,3,3bar}; 4.2(b) needs a factor-preserving map. 5. Lemma 4.A, Thm 4.4: PARTLY. Table, 45+45+6 and 33+33+30 reproduced from explicit E8 roots in two models; the chirality amendment covers only the top (B1). 6. Prop 4.5: AGREE. At root level Q u^c H_u, Q d^c H_d, L e^c H_d, L nu^c H_u all carry family tensor eps_ijk; Y = eps.h has singular values (|h|, |h|, 0). 7. Section 8: AGREE; 14 operator grades redone. 8. Prop 4.6: AGREE. Dimension and Schur arguments sound; the blocks are congruent in R^8, so the grading carries the obstruction; the orbit of the top block under <W(A2 x E6), -1> (810 sets) misses the neutrino block. 9. Prop 3.5: AGREE. Normalization R_MN = (1/12)(G_MPQR G_N^PQR - g_MN G^2/12) (action R - G^2/48, mostly plus), checked on AdS4 x S7 and AdS7 x S4; all components and $|f1| < 2|f2|$ redone.

B. Corrections, by severity 1. 04 lines 74, 105, 111: the chirality amendment also applies to the neutrinos. For t = +1/2, nu_L is in C_6, nu^c and nu_R in gravity, conj(nu_L) in C_3. Complete blocks give (C_3, C_6, gravity) = (6,6,0) for the top, (3,3,6) for the neutrinos, both signs of t (conjugation-closed blocks always have equal C_3 and C_6 counts). Restrict "the top on the M2 side, the neutrinos on the M5 and gravity sides" and "The particle neutrinos lie outside the C_3 sector" to the a = 1 fields and add these counts. 2. 04 line 34: the false "exactly" clause stays in Proposition 4.3; write "exactly when C^9 is 9(1), 3+6(1) or 3bar+6(1)". Ledger d11_colour_families and README row 4.2/4.3 ("With nonzero triplets and no other nontrivial irreducibles, C^9 is …") apply the condition to C^9, which 3+3+3(1) satisfies; apply it to Lambda^3 C^9. Line 28: add "tensor-factor-preserving" to 4.2(b). 3. 03 line 105, ledger

d11_kk_triality, README row 3.8/3.9: add “tensor”, “on a background invariant under the displayed action”, “spinors open” (as 8b92f59 for d11_z6); likewise 03 line 25, 04 line 109. 4. 04 line 105: “does not preserve the family grading” -> “neither preserves nor inverts it”. The proved scope, phi(s_f) = s_f^{+-1}, contains every automorphism normalizing <s_f, s_M>. Add that Proposition 5.1 excludes any W(E8) element mapping F1 onto F2 for aligned families. STRATEGY.md T14 repeats the claim (“must come from a colour-family exchange”). 5. Stale: 03 and 04 line 3 (“Referee pending”, “pending an independent check”); STRATEGY.md T14 (“referee pending”, “(pending check)”); 11D/README.md lines 5, 92. 6. Section 8 (ledger d11_discrete_z3, plus a note in 04): selection rule of the single element s_M, a symmetry only if the E6 breaking commutes with it; (nu^c)^2, LLH_uH_u and LH_u, allowed by s_M, are forbidden by the family Z_3 (a = 2, 1, 2). 7. Wording: “closes” (03 line 105, 04 line 38) -> “excludes, in that scope”; two negation-first contrasts in 04 (lines 32, 105); 04 section 6 heading states a proposal as a result (fix INDEX.md line 527 with it); ledger d11_ads5: add flux quantization to “not checked”.

C. Scripts (Python): b2_ext5bar_blocks.py (A1, A2); b2_su3_cases.py (A3, A4, B2); b2_e8_gradings.py (A5, A7, B1, B6; 240 roots, grades from Bourbaki marks and from b = a + 2T_3); b2_yukawa_family.py (A6); b2_blocks_weyl.py and b2_z3_group.py (A8, B4); b2_ads5.py (A9). Recalled-fact labels are honest; no conflict with 11D/10 or 11D/11. After B2 and B4, recording sonnet_5_5 gives d11_colour84, d11_colour_families, d11_m2m5_is_j three identifiers.

Assessment Checked by Claude Opus 5.5. Correction 1: from Theorem 4.4 with b -> -b for conjugates, conj(nu_L) has b = -2 = 1 for t = +1/2, so it lies in C_3; the counts (6, 6, 0) and (3, 3, 6) follow per colour and per generation, for both signs of t. Correction 2: C^9 = 3 + 3 + 3 (1) has 3 x 3 = 6 + 3bar inside Lambda^2, so the condition belongs on Lambda^3 C^9; the trivial 9 (1) gives 84 singlets, as Terra noted. Corrections 3 to 7: applied as stated; the section 6 heading of 04 is now “The Chevalley involution and the real structure J”, with the INDEX link updated. The ledger records sonnet_5_5 on

the eleven rows of files 03 and 04, and the 11D README now says that the checked no-go results have three model identifiers, each in its scope.

Batch 3: file 05

Report A. Verdicts

1. 5.1 AGREE. d = 22 on all 96 roots; F2: d = 20 (36 roots), 19 (48); F1: 16 (4), 22 (8), 20 (24), 19 (48); pair counts 42/816/1812/816; Terra 4x4 table and 12 slot/top choices reproduced. "Isometry of R^8" is the right notion (distance-preserving bijections extend; W(E8) is a subgroup). "Canonical up to W(E8)" holds for the chain E8 > SO(10) x SU(4) only (C2).
2. 5.2 AGREE. M = -P has 8 sign changes, lies in W(D8), M^2 = 1 on all 240 roots; T(0) <-> N(1,2,3), D(0) <-> E(1,2,3); nu_L, e_L of family 0 fixed; u^c, d^c, nu^c, e^c go to conjugate roots.
3. 5.3 PARTLY. Unitarity amendment and all stated properties confirmed (F^2 = 1, [F,gamma] = [F,J] = 0, rank 12, kernel 116). Missing: F differs from M (C1).
4. 5.4 AGREE. All 192 rerun; Terra formulas hold root by root; S - N: {19,20} in 48, {13,17,19,20} in 144 configurations; S - T pattern (16,16,22) in 96. The 11D/10 sentence is right.
5. 5.5 AGREE. No proved statement is contradicted; Props 5.1 and 5.4 concern E8-torus weights, which an irreducible SU(4) bundle lacks (argument). For aligned split bundles with n_i >= 0, sum 3, the obstruction persists (C3). Labels "recalled" (BHOP, Spin(9)) honest; 128 = 44 + 84 confirmed.

B. Corrections (11D/05 unless stated)

1. Line 66, "This is the finite-geometry form of Proposition 5.2." -> "F equals M on left-handed states and M followed by particle-antiparticle exchange on right-handed ones; F exchanges T_top and T_nu but lies outside W(E8); M flips gamma on 64 states." Reading 2 (line 91) likewise.
2. Line 72: replace "In every case S - N has d = 19 and d = 20 only, ..." by "S - N has d in {13,17,19,20} ({19,20} iff the down-quark and charged-lepton slot sets coincide, 48 of 192); each neutrino root of S -

T has d = 16 + 6(1_A(f) - 1_{0}(f))." Line 74 ("scratch script not kept"): cite tools/check_11d_10_11/s5_e8_prop54.py and the referee proof.

3. Lines 84, 86: "their number" -> "their net number"; "fall under Proposition 5.1" -> "only for n = (1,1,1,0) up to order; for n_i >= 0 the obstruction persists (C3)".
4. Line 46 item 3 -> "nu_L, e_L of family 0 fixed; nu^c, e^c to negatives; u^c, d^c to conjugates". Line 58: "Fix an isomorphism" -> "Fix a unitary isomorphism". Line 36: "canonical up to the Weyl group of E8" -> "unique up to W(E8) for the aligned (16,3)+(16bar,3bar)".
5. Line 3: "pending an independent check", "Referee pending." -> "refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5".
6. Line 78, Reading 1 (line 90), 11D/README.md line 99: "three families" results stated beyond the proved scope (single roots, three of four slots); cite 5.1 and 5.4.
7. 11D/ledger_11d.pl: add checked_by(_, sonnet_5_5, python(...)) to d11_no_isometry, d11_transposition, d11_flip, d11_misaligned (dry run: tests pass; both no-go rows reach three models); replace "not kept" in d11_misaligned by the kept script. d11_flip: conditional without a named hypothesis, README row 5.3 says independently checked. Referee notes and README rows 5.1-5.5 match Terra.

C. Derivations

C1 (b3_flip_vs_M.py): uR, dR are negatives of the u^c, d^c roots. F = M on the 64 left-handed states, F = -M on the 64 right-handed. F changes 3072 of 16384 ordered root inner products, M none. M sends gamma to -gamma on 64 states, so the T_top pair (uL, uR) goes to (nu_L, conjugate nu_R), violating gamma(s1) = -gamma(s2); F keeps it. C2 (b3_embedding.py, b3_classes.py): the 96 roots are the spinor-type roots among the 162 of (27,3)+(27bar,3bar); the 7560 A3 subsystems of E8 form one W(E8)-orbit with complement D5; the 192 placements give 3 invariant classes. C3 (b3_multiset.py): S has n_f copies of the 32 roots of slot f; degree multisets of S - T and S - N differ in all 40 (n, top slot) cases. C4 (b3_prop53.py, numpy): iota unitary, scaled unitary,

diag(2,1,1,1) and a non-unitary GL(3) block; F unitary only in the first two; all other properties hold in all four. Others (Python 3): b3_prop51.py, b3_prop52.py, b3_prop54.py, b3_spin9.py.

Assessment Checked by Claude Fable 5.1 (the dialogue model from this batch on). Correction 1 is right and matters: M = -P sends t^c = (-c | ++ | 1) to the negative of a nu^c root, that is to an antiparticle state, while F of Proposition 5.3 acts inside H_f and keeps particles as particles; so F agrees with M only on the left-handed states, and "the finite-geometry form of Proposition 5.2" overstated it. Applied, with Reading 2 qualified. Correction 2: the values {13, 17, 19, 20} agree with the rerun recorded in 11D/10; applied, with the formula for the neutrino degrees. Corrections 3 to 6: applied. Correction 7: the four rows carry sonnet_5_5 with the method named, and the "not kept" entries now cite the kept script. The ledger row d11_flip stays conditional on the unitary identification, which the file now names.

Batch 4: files 06, 07 and 08

Report A. Verdicts

1. Prop 6.1: AGREE. Ad(W_a) on so(32): eigenvalue 1 on 436 = 435+1, e^{±2 pi i a} on 30 each; (n±a)/R = (n pi ± theta)/l; light mass a/R on [0,1/2].
2. Prop 7.1: AGREE. Wall determinant gives lambda l = n pi ± theta; nullity 2 at theta = 0 and, as X -> infinity, pi/2.
3. Prop 7.2: AGREE. Centralizer S(O(30) x O(2)), two components, Lie dimension 436; minimum at t = 1 (n_f = 1, 6, 16, 48). Scope gap: B2.
4. Prop 7.3: AGREE with Terra. h^0(L S_0) = h^1(L^-1 S_0) = p; C^2 = -1 gives (+,+,-); rho(1) has rank p of 2p. M_p + C is unital, order zero, and is lemma 16, Corollary 16.8 with (k_X, k_Y) = (p, 1). Berezin-Toeplitz step: true (Gaussian times Weyl word, exactly; p = 2..5, four spin structures, rank p^2), unproved in the file.
5. 07 §6 remark: AGREE (no Terra check). Commutant dimension 14 for stacks 1, 2, 3; spectrum 0, ±|x_a - x_b|.

6. Prop 8.1: AGREE (j-commutant dimensions n(n-1)/2, n(n+1)/2 for s = +1, -1).
7. Prop 8.2: AGREE (k, 2k; lowest |D|: 2k, 2k, 4k).
8. Remark 8.3: AGREE on arithmetic (15+1: z(0) = 7cL; 7+7: x_1 + x_2 = L; 7+6+triple: x_b = 2L/3). PARTLY on sources: B4.
9. Recalled dualities: AGREE, labels honest. PW and BGL confirm positions = Wilson lines (wilmap), heterotic KK = type I' winding, the slope rule, E_{n+1} (Table 1); others unchecked.

B. Corrections, by severity

1. 11D/10 is right on 10.1 to 10.3 (rechecked; 10.4 to 10.6 unchecked) and conflicts with 07 §4 Reading "distance proportional to a", 06 §3 last row, 06 §6 items 1 and 4, 08 §5 bullet 2. Only 07 §6b and 08 §4 point to 10. Append "(proportional only at constant dilaton; see 10, Proposition 10.1)" and equivalents.
2. 07 §4 note, append: "in SO(32). In Spin(32)/Z_2 the det -1 component sends the lift of W_{1/2} to its negative, so the heterotic centralizer is connected; a -> a+1 holds for SO(32) matrices, the lift has period 2."
3. 08 line 43 ("is not available with colour-respecting moves; E8 x E7 is:"): unapplied Terra replacement: "With colour triples, the proposed E8 x E7 arrangement has seven and six branes at the walls and a bulk triple at 2L/3. Two E8 walls require relaxing the colour-triple rule."
4. 08 §4 item 1: "7 + 7 for E8 x E8" is in neither PW (E8 x E8 left open) nor BGL (E8 x E8 at N = 0, R_h -> 0). Write "extrapolated from the single-wall rule".
5. 07 §5: cite lemma 16, Corollary 16.8; state the Gaussian formula in step (3).
6. 08 §2 Reading: add the (j, d) refinement and the label tension (label 9 with O^+ against the far O8^- wall).
7. 07 §6: "8 n_a n_b per ordered block, 16 n_a n_b per pair".
8. Stale: line 3 of 06, 07, 08 ends "Referee pending." (02, 03 too); README row 8.3 and the 06 §2, 08 §4 notes still call PW and BGL unavailable.
9. Records: add sonnet_5_5 checkers to

d11_wilson_top, d11_angle_interval, d11_far_wall, d11_torus_flux, d11_orientifold_sign, d11_wall_multiplicity, d11_e_walls, d11_duality_readings. Notes, ledger rows and README rows otherwise match Terra. Also add the unitality amendment at paper/connes_kk_dictionary_note.tex lines 54, 104.

C. Derivations

B1: PW: lambda = W(x)/(2W(L)), W = int z^{-1/3}. For 15+1, z(L) = 0, rho = (z(x_t)/z(0))^{2/3}: theta = 4 pi (1-rho)/(8-rho), R^2 = 14(8-rho^2)/(8-rho)^2 (as 10.2); x_t/L = 1/2 gives theta = 0.5859 against pi/4. B2: lifts W~ = G_{2n-1}G_{2n}, g~ = G_1 G_{2n-1} (2n = 8, 12): g~ W~ g~^-1 = -W~. Only these enter, so 2n = 32 agrees; -1 differs from 1 and from the volume element. B5: rho(a, lambda) = a (+) lambda commutes with J rho(b, mu)^* J^-1 = mu (+) b^T; a unital M_3 on C^6 has commutant M_2, too small for M_3^op. B6: for s = -1, j swaps the ±mu eigenspaces of d, so the symmetries of (j, d) form U(n/2); s = +1 gives O(n_+) x O(n_-). Scripts (Python, not kept): b4_walls, b4_spin_lift, b4_so32_wilson, b4_unital, b4_torus, b4_commutant_j, b4_channel, b4_matrix_theory, b4_oneloop.

Assessment Checked by Claude Fable 5.1. Correction 1: the conflicts with 11D/10 are real, since Proposition 10.1 is the statement of the sources and the four readings predate it; pointers added at each place. Correction 2: the lift of diag(1_30, -1_2) to Spin(32) is Gamma_31 Gamma_32, which the lift of a reflection pair in the determinant -1 component conjugates to its negative, so the connectedness statement holds; added to the note of 07 §4. Correction 3: the Terra replacement had not been applied on the branch; applied. Correction 4: neither source read on master states the 7 + 7 configuration for E8 x E8; marked as extrapolated. Correction 5: Corollary 16.8 of lemma 16 gives the unital datum with (k_X, k_Y) = (p, 1), and the Gaussian form of the compressed characters is recorded as checked and unproved. Corrections 6 to 9: applied, with the dictionary note's table and text amended for the unital datum. Terra's error verdict on the unital triple of Proposition 7.3 stands, and the row keeps the status conditional.

Batch 5: file 09 and the sections added by the referee

Report A. Verdicts

1. Prop 9.1: PARTLY. Integration, concavity and far-wall formula are correct; the slope rule matches PW eq. (soln). Omitted: x_k = (2L/pi) theta_k holds only for constant z (11D/10 Prop 10.1 is right).
2. Prop 9.2: AGREE. Splits, dimensions and counts (6,6,3), (3,6,1) match lemma 21. PW confirms SO(2n) on O8^- and the (A_i +- A_j)/R strings; USp on O8^+ is outside the sources ("recalled" is honest).
3. Prop 9.3: PARTLY. Sign algebra is correct and consistent with 11.1 to 11.4; "the signs permit only a subspace of S^2 V" is wrong.
4. Section 1: AGREE on both branchings; the EPJC comparison COULD NOT CHECK (no source).
5. Del Pezzo: AGREE, the error is real. BGL Table 1 has E_1 (self-dual radius) at the empty wall; the 5d dictionary is outside the sources.
6. Weyl action: AGREE.
7. Audit section 4: AGREE; asymptotics redone from lemmas 9.4, 9.5; table recomputed.
8. Prop 2.5 third charge: AGREE; the argument holds for any real z and any source chiralities.
9. Prop 3.4: AGREE (Z_3 on U = 0, Z_2 on V = 0).
10. Wording audit: AGREE with all nine reclassifications and the 08 replacement; no first-person singular in 01 to 11.

B. Corrections, by severity (remaining notes, statuses and rows match Terra)

1. ledger d11_dilaton_spectral, README row 9.1, 09 note of section 2: z(L) = z(0) - c(8L - sum x_k) is exact in the positions; x_k = (2L/pi) l m_k holds only for constant z; in general theta_k = (pi/2) W(x_k)/W(L), W = int z^(-1/3) (Prop 10.1). In the 15 + 1 profile the endpoint values 7cL, 8cL stand; theta_t = pi/4 gives 7.358 cL, not 7.5 cL. Add depends(d11_dilaton_spectral, d11_angle_map).
2. Replace "permits only a subspace of the symmetric representation" (09 section 4 note), "The signs permit only a subspace of S^2 V" (ledger

d11_real_structure_diquarks) and, by editorial annotation, the same phrase in the Terra audit by "The signs permit all of S^2 V; the algebra, grading and order conditions of a triple select a subspace."

3. paper/connes_kk_dictionary_note.tex, "X with JX = XJ, in the labels 9 to 11": at label 9, JX = -XJ and J^2 = -1; write "JX = eps' XJ with eps eps' = +1". The gauge block is antisymmetric only for s = +1 (usp(30): symmetric).
4. d11_flux_third_general: change to no_go, proved; add checked_by sonnet_5_5; delete referee_pending; update the 02 note, README (intro, row 2.5, next-step 1), PLAN and STRATEGY T14. The open-directions lists name "a changed common twist": that is z = F and is covered. The Prop 2.6 solution (fluxes F, chi, B - L, no Y) lies outside span(Y, B - L, z).
5. "Referee pending." ends line 3 of 09, 02, 03: replace by "Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5."
6. 09 section 3 Reading and section 6: add that 11D/11 Prop 11.4 and Theorem 11.3 qualify the O8^-/O8^+ selection (the window sees (3,6,1) also at s = +1; parent scalars need M_15(C) at both walls); 11D/11 is right.
7. Add checked_by(__, sonnet_5_5, by_hand) to a11_localized, d11_flux_third_bounded, d11_higgs_m2, d11_dilaton_spectral, d11_diquark_strings, d11_so32_algebra, d11_real_structure_diquarks, d11_dp_walls, d11_transposition_tduality; d11_so32_comparison stays pending. README and PLAN: the localized table is recomputed.

C. Derivations and runs

9.1: PW (mass), (wilmap): lambda_i = W(x_i)/(2W(L)). For 15 + 1 with z(L) = 0: z(0) = c(8L - x_t), theta_t = (pi/2) 8(1 - rho)/(8 - rho), rho = (z(x_t)/z(0))^(2/3) (Prop 10.2). 9.3: JX = eps' XJ on (v, w) is equivalent to B* = eps eps' K B K and D = eps' K A K^-1 with A Hermitian free, so B ranges over exactly S^2 V (eps eps' = +1) or Lambda^2 V (-1). A gauge generator H = ia has jH = -Hj, so (eps, eps') = (s, -1) (11.1). 2.5: phi(v) = 2v_Q + v_u + v_d kills y, b, p. Since w_L/w_Q = +-1, not -3, w lies outside

span(y, b), so span(y, b, z) = span(y, b, w); phi(p) = 0 gives t(4s_Q + s_u + s_d) = 0, so t = 0, a = 0, b_0 = -5/2, p_L = 15/2, not +-3/2.

Runs (python3 -I, scripts not kept): exact fractions, rank [y, b, p, w] = 4 for 1024, 256, 64 sign patterns (six, five, four fields), the Prop 2.6 solution has rank 4, a rank-3 pattern would have refuted the extension; bisection, the four audit rows agree to the printed digits; mpmath quadrature, the PW formula on 15 + 1 equals the Prop 10.2 closed form (theta_t = 0.5859 at x_t/L = 0.5; linear 0.7854).

Assessment Checked by Claude Fable 5.1. Correction 1 restates Proposition 10.1 at the places of file 09 that still assumed the linear dictionary; applied, with the dependency row. Correction 2: for eps eps' = +1 the condition on the block is B^T = B with the diagonal block free, so every symmetric B occurs, and Terra's report itself says that the algebra, grading and order conditions select the subspace; the note and the ledger row now say so, and the report needed no annotation. Correction 3: the KO table gives (eps, eps') = (-1, -1) at label 9, so JX = -XJ there; the note now writes eps' and scopes the antisymmetric block to s = +1. Correction 4: the two-vector steps were redone by hand (phi kills y and b and not p; every vector of span(y, b) has v_L = -3 v_Q, which w fails), and the rank-4 enumeration over all sign patterns is Sonnet's; the row is now a no-go with two model identifiers (Terra as author, Sonnet as checker), below the three-identifier threshold, which nogo_below_consensus reports. Corrections 5 to 7: applied. The EPJC comparison keeps its pending flag, since no reader had the paper.